# Rank-six fusion rings: categorification criteria and models through multiplicity nine

17 October 2026

**Abstract**

We study complex categorifiability of the 25421 rank-six fusion rings of multiplicity at most nine in the *Fusion Ring Census*. Arithmetic and sphericalization criteria reduce unrestricted existence to exact finite problems in central induction and module categories. A common framework combines restriction Gram matrices, character and Galois symmetry, pivotal rotation, projection to fusion subcategories, and simultaneous product moments. Signed dimension fields, trace saturation, and Galois orbits of topological spins exclude all 8 809 rings of exact multiplicity nine. Positive constructions include three neutral components of conformal extensions, identified by exact tensor reconstruction. The resulting classification has 25362 generally excluded rings, 57 realized rings, and 2 unresolved rings; the exact multiplicity bands $1, 2, 3, 5, 6, 7, 9$ are complete. Every exclusion and model identification has an exact reproducible certificate. For the two unresolved rings, exact reductions determine one complete rank-26 modular candidate and leave three possible Morita-dual based rings, respectively. All-order indicator identities and central phase masses reduce the latter ring's retained center profiles to 640 and force eight identical spin restriction columns. Odd cyclic Clifford factorization constrains all odd rotations for the first ring; its 1144 tested central-target blocks pass. Every tensor functor from a hypothetical categorification of either residual ring is fully faithful, and neither Morita class contains a braided representative. For the multiplicity-eight residual, the twist-one simples in the center are exactly the six constituents induced from the unit. Both rings remain unresolved: none of these necessary data is a categorical construction or a general nonexistence proof.

**Research status. AI-generated research · independent verification pending.** Treat this manuscript and its accompanying research data with caution: they have not been independently verified by a human mathematician or formalized in Lean.

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# 1 Classification results and scope

A fusion ring is a finite-rank based ring with nonnegative integral structure constants, a distinguished unit, and duality satisfying Frobenius reciprocity. We work with full ordered multiplication tensors; in particular, commutativity is not part of the definition. The multiplicity is the maximum of all structure constants, including the unit coefficients. A categorification means a fusion category over $\mathbb{C}$, without an assumed pivotal, spherical, or unitary structure. We classify existence for a ring, not equivalence classes of categories with that ring.

Vercleyen–Slingerland constructed a substantial low-rank database [44]. Its rank-six portion is complete through multiplicity four. The multiplicity-one categorification problem was settled by Liu–Palcoux–Ren [29]. The unrestricted rank-four classification [12] gives fifteen Grothendieck rings without a multiplicity restriction. The finite rank-four and rank-five projects [13, 11] develop arithmetic, sphericalization, and central-induction certificates. At rank six, however, neither the ring nor a signed quotient of its sphericalization can simply be assumed commutative. This is the principal structural change addressed here.

All positively identified rings and their models appear in Table 3; its structure column is part of the classification statement, not merely a description of a chosen presentation. In particular, Schopieray's category [43, Example 1] realizes $R^{(6)}_{2,124}$ spherically, whereas [43, Theorem 1] excludes *every* pseudo-unitary categorification of that same ring. Thus general or spherical existence cannot be identified with unitary existence, even in rank six and multiplicity two. The electronic ledger records general, pivotal, spherical, pseudo-unitary and unitary existence separately for every tensor. The source identifications are detailed in Section 6.4. The paper is organized by mathematical operation rather than by individual exceptions. Section 2 fixes the source census, its scope, and the tensor conventions used by every automated test. Section 3 reduces arbitrary categorifications to the appropriate arithmetic and spherical cases. Section 4 presents one central-induction framework, with precisely scoped refinements rather than separate ad hoc tests. Section 5 supplies module and Morita constraints. Section 6 treats models by construction, including central-boson quotients, equivariantizations, relative centers, and neutral components of conformal extensions. Section 7 specifies exhaustive certificate checking and the remaining frontier.

In particular, the positive catalogue includes the extended Haagerup category $\mathcal{EH}_1$ [4, 22, 6] and Izumi's $C_5 + 5$ near-group category [25], not just the constructions derived from them. The mixed fermionic quotient of Proposition 6.17, the sign-swap equivariantization

of Theorem 6.18, and the relative-center construction of Corollary 6.20 give three additional full-tensor identifications. None is inferred merely from a dimension match or from survival of the necessary criteria.

**Theorem 1.1** (Source census [39])**.** *The published* Fusion Ring Census*, at the revision fixed in Section 2, contains exactly* 25421 *based-isomorphism classes of rank-six fusion rings of multiplicity at most nine. The numbers at exact multiplicities* $1, \ldots, 9$ *are*

$$39,\ 154,\ 384,\ 872,\ 1582,\ 2768,\ 3814,\ 6999,\ 8809.$$

*All three possible duality types and all noncommutative cases are included. No categorifiability filter is applied. Completeness of this counting input is the result of the cited repository; the present paper addresses categorification of its tensors.*

**Theorem 1.2** (Categorification status through multiplicity nine)**.** *In the census of Theorem 1.1, the accompanying exact certificates exclude* 25362 *rings from general complex categorification. There are* 57 *positively identified rings:* 56 *have unitary categorifications, whereas* $R^{(6)}_{2,124}$ *has a spherical categorification but no pseudo-unitary categorification. These positive sets and the general exclusions are pairwise disjoint. The remaining* 2 *rings are not decided for general existence by the present criteria and bounded searches. Exclusions of pivotal, spherical or pseudo-unitary categorifications are recorded separately when general existence remains open. Their distribution is given in Table 1.*

*Among the* 577 *rings of multiplicity at most three, precisely* 46 *admit general, pivotal, or spherical categorifications; the other* 531 *are generally excluded. Exactly* 45 *of the positive rings admit pseudo-unitary or unitary categorifications; the sole exception is* $R^{(6)}_{2,124}$*. At exact multiplicity three, all* 384 *rings are decided: three have unitary models and* 381 *are generally excluded. In particular, Theorem 5.25 excludes* $R^{(6)}_{3,215}$*, the last residual from that stratum. The completed exact-multiplicity bands are* $1, 2, 3, 5, 6, 7, 9$*. The other tested bands retain the residuals listed in Table 1 and the electronic ledger.*

| Exact multiplicity | Rings | Excluded | Unitary | S, not PU | Undecided |
|---:|---:|---:|---:|---:|---:|
| 1 | 39 | 18 | 21 | 0 | 0 |
| 2 | 154 | 132 | 21 | 1 | 0 |
| 3 | 384 | 381 | 3 | 0 | 0 |
| 4 | 872 | 866 | 5 | 0 | 1 |
| 5 | 1582 | 1581 | 1 | 0 | 0 |
| 6 | 2768 | 2765 | 3 | 0 | 0 |
| 7 | 3814 | 3812 | 2 | 0 | 0 |
| 8 | 6999 | 6998 | 0 | 0 | 1 |
| 9 | 8809 | 8809 | 0 | 0 | 0 |
| Total | 25421 | 25362 | 56 | 1 | 2 |

Table 1: Complete counting census; categorification is complete at exact multiplicities $1, 2, 3, 5, 6, 7, 9$ and partial in the other displayed bands. The column S, not PU denotes a spherical realization with no pseudo-unitary categorification. All excluded rings are ruled out generally.

**The low-multiplicity residuals.** Every ring at exact multiplicity 2 is decided. Every ring at exact multiplicity 3 is decided. At exact multiplicity 4 the unresolved records are

$$R^{(6)}_{4,155}.$$

The only unresolved rings are $R^{(6)}_{4,155}$ and $R^{(6)}_{8,1660}$; their structural reductions are given in Section 7.3. Every one of the 8 809 rings of exact multiplicity nine is generally excluded by the initial sieve and the completion in Section 7.6. Theorem 4.39 completes the exact multiplicity-five band. Theorem 6.21 identifies three rings by unitary conformal extensions and complete tensor reconstruction.

For reproducible identifiers, retain the unit as the zeroth basis element, use the exported representatives in the generator's fixed coordinate conventions, and sort their full tensors in row-major lexicographic order at each exact multiplicity. This order is a convention on the exported representatives, not a claim that each is the lexicographic minimum under all 120 relabellings. Distinctness up to based isomorphism is supplied by the cited census. The representatives and identifiers through multiplicity eight are fixed in the registry. The multiplicity-nine stratum uses the same row-major convention. The source-to-registry correspondence is checked in all 216 coefficients for every tensor. The $j$th tensor at multiplicity $m$ is denoted $R^{(6)}_{m,j}$. The complete tensor, rather than this identifier, is the input to every obstruction. A separate crosswalk records matches to the ancillary database of [44]; its indices play no role in a proof.

All reductions are applied to full tensors. The cheap arithmetic filters are retained because they avoid unnecessary induction searches; the stronger blocks are used only with their stated hypotheses. The rank-four and rank-five results are cited rather than reproved. This is a unified implementation of sufficient exclusion certificates, not a claim that the criterion set is logically minimal.

**Completing multiplicity three.** Theorem 5.25 generally excludes $R^{(6)}_{3,215}$. Proposition 3.27 first reduces arbitrary categorification to pseudo-unitary categorification by an exhaustive signed sphericalization calculation. The full two-sided Morita-action Gram calculation then forces the involution to interchange the two module structures on each of its fixed simple objects. The resulting local anticommutation relation forces their third Frobenius–Schur indicators to equal $-1$. Exhaustive exact separation of the central-induction branches contradicts these two requirements. Thus no pivotal or positivity hypothesis remains in the exclusion.

The higher-multiplicity exclusions use the spectral module, rank-three descent, and order-six Morita–grading criteria of Section 5, the biquadratic trace and Casimir-parity criterion of Proposition 3.8, and the formal-codegree trace-ideal theorem and ramification criterion of Section 3, the all-orders Gauss obstruction of Proposition 3.11, and the higher-dimensional and quartic stabilizer-trace extensions of Section 5. The completed signed and character-induction searches are recorded in the ledger.

The sequential exclusion table in Section 7.1 counts each removed tensor once. Structural and arithmetic tests precede finite induction and signed searches. A tensor is declared positive only after identifying its complete based ring with a categorical construction; an unfinished search or a surviving necessary system leaves it unresolved. The lower-rank categorification results are used only in their proved ranges.

# 2 Census input and tensor conventions

## 2.1 One authoritative counting source

All counting data are taken from the public repository *Fusion Ring Census* [39], pinned to revision

`2a340871242879665fa7ba2b01fa15b8bdeaada3`.

The source files are `results/census.json` and `results/rank6/tables.txt.gz`. They declare a complete, verified rank-six census through multiplicity nine. Its 25421 tensors include all

16612 classes through multiplicity eight and 8809 classes of exact multiplicity nine. Rank includes the tensor unit; multiplicity is the largest structure coefficient, and every duality type is included.

The paper trusts this census as its counting input. It supplies neither another census generator nor a claim to have independently enumerated the same domain. The repository contains the enumeration algorithms, coverage arguments, completion records, and tensor checks supporting Theorem 1.1. A valid and duplicate-free tensor list alone would not establish exhaustive coverage; that claim is supplied by the source census, not by the import operation below.

### 2.2 Frozen data and reproducible identifiers

A copy of the source tensor export is retained with the classification code, so a classification check has no live-network dependency. Its compressed SHA-256 is

`0e7552e17620452f237e94751c40cd93b0757a466168629db8cc1bf0b9451d00`,

and its uncompressed SHA-256 is

`be377c8114d4c0a532b79850efd588090c1dfb5ae3ffc847ecf3eadb97f88ab3`.

These hashes identify the input bytes; they are not substitutes for the source's completeness argument or for categorification certificates.

Every ring is represented by its full ordered tensor $N = (N_{ij}^k)_{0 \le i,j,k < 6}$, with $b_0 = \mathbf{1}$. The multiplicity is $\max_{i,j,k} N_{ij}^k$. Noncommutative inputs are retained without symmetrization. A based-ring isomorphism is a unit-fixing basis permutation preserving every structure constant. Consequently, comparison with a model uses the complete tensor, not only dimensions, a fusion graph, or a numerical fingerprint.

The identifier $R_{m,j}^{(6)}$ is local to this paper's frozen tensor registry. Representatives and identifiers through multiplicity eight are fixed by the certificate registry. The exact-multiplicity-nine tensors are placed in row-major lexicographic order using the exported representatives. This order need not give the lexicographically smallest representative under all basis permutations. Import matching records the correspondence between the source representatives and the registry. Every mathematical criterion receives the tensor rather than the identifier, so relabelling cannot alter its conclusion.

The import checks all 25421 tensors for the fusion-ring axioms and examines all 120 unit-fixing permutations of each tensor, for a total of 3050520 canonicalization comparisons. It finds no repeated based-ring class. The explicit source-line and basis-permutation correspondence is retained in the supplement. These checks establish validity, distinctness, and agreement with the certificate registry; exhaustive enumeration remains the separate input described above.

### 2.3 Counting, exclusions and constructions

The source census answers a finite counting question about based rings. A categorification exclusion answers a different question about complex fusion categories. We therefore keep the census fixed while applying necessary criteria and matching models. The resulting ledger covers every imported tensor exactly once; accepted negative certificates, positive constructions and unresolved records are disjoint.

The exact prefix $39,154,384,872$ agrees with the complete rank-six range of [44]. We use no external categorifiability flag as a proof. The classification code may reconstruct arithmetic, tensor or isomorphism data required by a criterion, but it does not repeat the census search. The counting proposal in `oeis/` derives directly from the same source counts and has no dependence on whether a categorification has been found.

# 3 Arithmetic and categorical reductions

## 3.1 Conventions and imported criteria

Write $N_{ij}^k$ for the coefficient of $b_k$ in $b_i b_j$ and $i^*$ for duality. The matrices used in the software have entries $(N_i)_{jk} = N_{ij}^k$. Consequently they give the right-handed matrix convention

$$N_i N_j = \sum_k N_{ji}^k N_k.$$

This distinction is harmless for commuting matrices, but essential when checking noncommutative signed tables. Frobenius reciprocity gives $N_{i^*} = N_i^{\mathsf{T}}$. The positive dimension character is denoted $d_i = \mathrm{FPdim}(b_i)$ and $D = \sum_i d_i^2$.

The complexified based ring is a semisimple $*$-algebra. For an irreducible representation $E$ of dimension $r_E$, let $f_E$ be its formal codegree, in the convention of [37]. The regular Casimir matrix

$$C_R = \sum_i N_i N_{i^*}$$

has eigenvalue $r_E f_E$ with multiplicity $r_E^2$. It is only in the commutative case that all these multiplicities are one and its characteristic roots are themselves the formal codegrees.

We use the following established necessary conditions. Their precise applications are certified by polynomial, prime, dimension-field, and signed-table witnesses.

**Proposition 3.1** (Established inputs)**.** *For a complex fusion categorification, the following hold.*

*(i) Every formal codegree is a d-number [37]. If $p(t) = t^s + a_1 t^{s-1} + \cdots + a_s$ is its minimal polynomial, this is expressed by $a_s^i \mid a_i^s$ for $1 \le i \le s$.*

*(ii) For a commutative Grothendieck ring, character values are cyclotomic integers. The polynomial factorization and finite-field witnesses used here are those of [13, Section 2]; the commutativity hypothesis is checked before this test is used.*

*(iii) A fusion subring gives a fusion subcategory. Its Frobenius–Perron dimension divides that of the ambient category in the ring of algebraic integers. A subring of rank two with $x^2 = 1 + mx$, $m \ge 2$, is excluded by [36].*

*(iv) Weakly integral fusion categories are pseudo-unitary [15]. For the two-Frobenius–Perron-dimension reduction we use the Galois-conjugacy result of [41, Lemma 3.1]. It is a reduction to a pseudo-unitary Galois conjugate, not a claim that every originally given spherical structure is positive.*

*(v) The zero-spectrum obstruction of [29, Theorem 2.14] is a general categorification obstruction. Its finite hypothesis is tested directly on the full ordered tensor.*

*(vi) The spherical dimension, normalized norm, sign-object moment, and low-induction criteria are those of [13, Sections 2,3,6,7]. Adjoint parity follows from the Fuchs–Runkel–Schweigert theorem in [60, Theorem 3.9]: in any pivotal fusion category, a self-dual simple occurring with odd multiplicity in $X^* \otimes X$ has second indicator $+1$. The finite joint relative-trace criterion is the named theorem of [11].*

For completeness, the finite tensor hypothesis in input (v) is as follows. Put $\langle ab, cd \rangle = \sum_k N_{ab}^k N_{cd}^k$. There must be nine indices $a, b, c, u, v, w, x, y, z$ such that

$$N_{ua}^w N_{vu}^b N_{vw}^c N_{xz}^a N_{bx}^y N_{yz}^c > 0, \quad N_{ba}^c = 1, \quad \sum_k N_{ux}^k N_{v^*y}^k N_{wz^*}^k = 0,$$

and each of the two sets

$$\{\langle vu, ca^*\rangle, \langle bu^*, cw^*\rangle, \langle v^*b, wa^*\rangle\}, \qquad \{\langle bx, cz^*\rangle, \langle yx^*, ca^*\rangle, \langle b^*y, az^*\rangle\}$$

contains 1. Theorem 2.14 of [29] excludes any such tensor. The certificate stores the nine indices and the checker reevaluates all these expressions. No associator is assumed to be unitary or pivotal.

Input (v) is especially inexpensive because it needs just a small integer witness. We do not import the negative part of any integral classification table or quadratic-family classification as a list of forbidden labels.

**Corollary 3.2** (Certified lower-rank descent)**.** *Suppose a rank-six fusion ring contains a based fusion subring of rank four and multiplicity at most* 32*, or of rank five and multiplicity at most* 17*. If that subring is not based-isomorphic to one of the positive rings in the respective main theorem of [13, 11], the rank-six ring is not categorifiable over* $\mathbb{C}$*.*

*Proof.* A hypothetical categorification would contain the fusion subcategory on the prescribed basis subset. Apply the cited complete lower-rank classification. The test enumerates unit-containing dual-closed subsets, checks product closure, and compares the restricted tensor with every positive model under every unit-fixing permutation. No negative labels are read. □

### 3.2 A certificate of a nonabelian character field

**Theorem 3.3** (Noncommuting field automorphisms)**.** *Let* $R$ *be a commutative fusion ring. Suppose an irreducible factor* $f \in \mathbb{Q}[t]$ *of the characteristic polynomial of some multiplication matrix admits polynomials* $a, b \in \mathbb{Q}[t]$ *such that*

$$f(a(t)) \equiv f(b(t)) \equiv 0 \pmod{f}, \qquad a(b(t)) - b(a(t)) \not\equiv 0 \pmod{f}.$$

*Then* $R$ *has no complex fusion categorification.*

*Proof.* In a categorification, the eigenvalues in question are values of one-dimensional characters of $R \otimes \mathbb{C}$ and are cyclotomic integers by Proposition 3.1(ii). Hence $K = \mathbb{Q}[t]/(f)$ embeds into an abelian cyclotomic extension of $\mathbb{Q}$. Every subfield of an abelian Galois extension is Galois with abelian Galois group. Substitution by $a$ and $b$ defines unital endomorphisms of the field $K$; they are injective, hence surjective by finite dimensionality. The last displayed condition says that these automorphisms do not commute, a contradiction. □

Factoring $f$ over $\mathbb{Q}[t]/(f)$ is allowed in discovery, but is unnecessary in verification: irreducibility, divisibility, two substitutions and one nonzero remainder suffice. In the present census the test excludes $R^{(6)}_{2,119}$, with the degree-six factor

$$f(t) = t^6 - 3t^5 - 2t^4 + 9t^3 - 5t + 1.$$

The two substitution polynomials and their nonzero commutator are stored in the certificate and recomputed from this polynomial. This test is a stronger certificate generator for the existing cyclotomicity condition, not a new categorical hypothesis.

The same automated test excludes $R^{(6)}_{6,1891}$. A multiplication matrix has the irreducible factor

$$t^6 - 11t^5 + 5t^4 + 38t^3 - 5t^2 - 11t - 1,$$

and the certificate supplies two noncommuting automorphisms of its degree-six field. Their defining substitutions and nonzero commutator are checked by exact polynomial reduction. Thus this case is decided by the uniform arithmetic criterion without an induction search.

### 3.3 Positive dimensions after a tensor-sign change

**Lemma 3.4** (Tensor-sign Galois reduction)**.** *Let $\mathcal{C}$ be spherical and let $\sigma$ be an embedding of its simple-dimension field into $\mathbb{R}$. Put $\eta_i = \operatorname{sgn}(\sigma(\dim X_i))$. If*

$$\eta_0 = 1, \qquad N_{ij}^k > 0 \Longrightarrow \eta_i \eta_j = \eta_k, \tag{3.1}$$

*then $\mathcal{C}$ has a pseudo-unitary Galois conjugate after changing its spherical structure by a tensor automorphism of the identity.*

*Proof.* Extend $\sigma$ to the field of definition and conjugate the categorical structure, as in the Galois reduction of [13]. The scalars $\eta_i$ define a tensor automorphism of the identity precisely because of (3.1). They agree on duals and square to one, so twisting the pivotal structure preserves sphericality. The resulting dimensions are $|\sigma(\dim X_i)| > 0$. A positive ring dimension character is the Frobenius–Perron character, so this spherical structure is positive and the conjugate is pseudo-unitary. □

This lemma must not be replaced by arbitrary basis sign changes: the support condition (3.1) is essential. The spherical checker now tests it at every real embedding of a remaining character component. Combined with the complete signed-sphericalization tests of Section 3.4, it supplies general exclusion certificates for $R_{2,37}^{(6)}$, $R_{2,57}^{(6)}$ and $R_{3,204}^{(6)}$. The new step concerns spherical character components; the exhaustive signed analysis is still required before a conclusion is promoted to general nonexistence.

### 3.4 Signed sphericalization at rank six

Let $\mathcal{C}$ be an arbitrary fusion category. Its canonical sphericalization $\widetilde{\mathcal{C}}$ has a sign object $\varepsilon$ of dimension $-1$ and two lifts of every simple object, interchanged by tensoring with $\varepsilon$; see [15, 38]. Choose a unit-preserving set of lifts. Passing to

$$T = K_0(\widetilde{\mathcal{C}})/(\varepsilon + 1)$$

gives integral signed structure constants satisfying

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{3.2}$$

They retain associativity, the unit, duality, and Frobenius reciprocity. All these statements concern a necessary quotient of an actual sphericalization; conversely, an admissible signed table need not have a categorification.

The sphericalization arguments in [13, 11] choose the lift of each self-dual object with positive second indicator. We keep this normalization. In particular, the square-channel congruences are modulo four rather than merely modulo two whenever that normalization applies. Non-self-dual lifts are chosen compatibly with duality. These restrictions are part of the cited normalization theorem, not heuristic reductions of a search.

#### 3.4.1 An exhaustive oriented parameterization

Set $t_{abc} = T_{ab}^{c^*}$. In the absence of commutativity, identify only the orbits generated by

$$(a,b,c) \longmapsto (b,c,a), \qquad (a,b,c) \longmapsto (c^*, b^*, a^*). \tag{3.3}$$

These are the cyclic and dual-reflection consequences of reciprocity. An arbitrary transposition is not allowed. Unit and duality coefficients are fixed first. Give each remaining orbit its interval in (3.2), impose the applicable square-channel congruences, and substitute into all ordered associativity equations.

**Proposition 3.5** (Exhaustive signed search)**.** *The finite search just described contains the signed quotient of every hypothetical categorification, after the indicated lift normalization. If all its associative tables are rejected by necessary signed sphericalization criteria, the original ring has no complex categorification.*

*Proof.* The quotient construction gives (3.2) and (3.3). Thus every quotient occurs in the finite product of integer domains. Every pruning operation is a necessary consequence of an equation over those domains. At a leaf the full ordered associativity identities are checked. Exhausting the product therefore exhausts every possible normalized quotient. A genuine category would supply one of the remaining tables and one of its spherical dimension characters, so rejection of all tables and all such characters is contradictory. □

The implementation checks machine-integer bounds before native computation and does not interpret overflow or timeout as exhaustion. The exact verifier reconstructs the parameter domains and the complete list of surviving signed tables; it does not merely recheck a collection of stored rejected examples.

### 3.4.2 When the signed algebra must be commutative

Let $d = (1, d_1, \dots, d_5)$ be the dimension character of the selected lifts, $K = \mathbb{Q}(d_1, \dots, d_5)$, and $\delta = \sum_i d_i d_{i^*}$. Inflating a representation of $R$ to $K_0(\widetilde{\mathcal{C}})$ doubles its formal codegree. Since the sphericalization has dimension $2\delta$, central induction gives

$$\delta/f_E \in \mathcal{O}_K, \qquad f_E \in K \quad (E \in \mathrm{Irr}(R \otimes \mathbb{C})). \tag{3.4}$$

Indeed, [38, Theorem 2.13] gives a central simple of dimension $\delta/f_E$, and its restriction is an integral combination of the selected lifts and their negatives. Its dimension is a nonzero algebraic integer in $K$, proving both assertions.

**Proposition 3.6** (Rank-six commutative descent)**.** *Suppose $R$ is commutative of rank six and has a hypothetical categorification. Its signed quotient $T \otimes \mathbb{C}$ is commutative if any of the following tensor-checkable conditions holds:*

*(a) all six basis elements are self-dual;*

*(b) exactly two basis elements are self-dual;*

*(c) a formal codegree has degree greater than two;*

*(d) the formal codegrees generate two different quadratic subfields;*

*(e) zero has multiplicity less than four in* $\det(tI - C_R) \bmod 2$.

*Proof.* The coefficient-of-unit pairing of a signed quotient is still positive definite on the distinguished basis after complex conjugation. Its regular matrices are adjoint under duality. Hence the algebra is semisimple. It has a one-dimensional dimension representation. If it is noncommutative of dimension six, its only possible Wedderburn type is

$$\mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{3.5}$$

The Galois orbit of the dimension character has at most two elements, so $[K : \mathbb{Q}] \leq 2$. Equation (3.4) excludes (3.5) in cases (c) and (d).

In case (a), duality is the identity on the signed basis. Being also an anti-automorphism, it forces commutativity. For (b), consider duality as a complex-linear involutive anti-automorphism of (3.5). The spherical dimension character is fixed, so both one-dimensional blocks are fixed. On $M_2(\mathbb{C})$, an involutive anti-automorphism is of orthogonal or symplectic

type, with linear trace 2 or $-2$. The total trace is therefore 4 or 0. In the distinguished basis it is the number of fixed basis elements, namely 2, a contradiction.

Finally the unique two-dimensional representation of (3.5) is Galois invariant. Its formal codegree is a rational algebraic integer, hence an integer. The regular Casimir of $T$ has the even eigenvalue $2f_E$ with multiplicity four. Thus its characteristic polynomial is divisible by $t^4$ modulo two. But (3.2) implies $C_T \equiv C_R \pmod 2$, contradicting (e). □

This proposition is a reduction, not a noncategorification criterion by itself. Once its hypotheses are verified, the commuting-character methods of [13, 11] apply to every signed branch. Otherwise we use Proposition 3.5. If a noncommutative table of type (3.5) remains, its one-dimensional characters are computed as the common kernel of the matrix commutators. That space has dimension two. The two-dimensional block and its formal codegree are then recovered from the regular Casimir with the multiplicities in (3.5). Each decomposition identity is checked exactly.

### 3.5 Biquadratic spherical traces

The following test uses sphericalization to obstruct a fusion category without assuming that it itself admits a spherical structure. Throughout this subsection the input ring is commutative, self-dual, and of rank six. Write $N_i$ for its regular multiplication matrices and $T_0 = \mathrm{Tr}(\sum_i N_i^2)$.

**Lemma 3.7** (Dimension-field containment)**.** *If the input ring is categorifiable, there is a commutative integral signed based algebra $A$ of rank six, with symmetric multiplication matrices $M_i$, such that $M_i \equiv N_i \pmod 2$ and $|(M_i)_{jk}| \le (N_i)_{jk}$. It has a dimension character $d = (1, d_1, \dots, d_5)$ in a totally real field $K$ of degree at most six. If $\Delta = \sum_i d_i^2$, every original formal codegree $f$ belongs to $K$, and $\Delta/f$ is an algebraic integer in $K$. The same statement holds for every formal codegree of $A$.*

*Proof.* Sphericalization has two lifts of each simple and a central sign object $\epsilon$ of dimension $-1$. Each lift of a self-dual simple is self-dual: its two possible lifts have opposite nonzero dimensions, whereas a dual has the same dimension. Thus its Grothendieck ring is commutative. Quotienting by $\epsilon + 1$ gives $A$ and the asserted bounds and parity. Its regular matrices are real symmetric, hence simultaneously diagonalizable with real characters. Evaluation at the dimension character is a field quotient of $A \otimes \mathbb{Q}$, proving $[K : \mathbb{Q}] \le 6$. An original character extends to sphericalization with $\epsilon = 1$ and formal codegree $2f$. The induction theorem for formal codegrees [38, Theorem 2.13] produces a central simple object of dimension $2\Delta/(2f) = \Delta/f$. Forgetting it to sphericalization expresses this dimension as an integer combination of the numbers $\pm d_i$. The same argument applies to signed characters with $\epsilon = -1$. □

Suppose the original formal-codegree fields generate $E = \mathbb{Q}(\sqrt{p}, \sqrt{q})$, where $p, q > 1$ are squarefree, $p \equiv 1 \pmod 4$, and their quadratic field discriminants are coprime. The lemma forces $K = E$, since $4 \mid [K : \mathbb{Q}] \le 6$. Write $d_i = (\alpha_i + \beta_i\sqrt{p} + \gamma_i\sqrt{q} + \delta_i\sqrt{pq})/4$. Orthogonality of the four distinct Galois-conjugate characters gives

$$16 + \sum_{i=1}^{5} \alpha_i^2 = p \sum_{i=1}^{5} \beta_i^2 = q \sum_{i=1}^{5} \gamma_i^2 = pq \sum_{i=1}^{5} \delta_i^2.$$

Consequently $\mathrm{Tr}_{E/\mathbb{Q}}(\Delta) \in pq\mathbb{Z}_{>0}$. The two remaining signed formal codegrees are greater than one, since all are positive and their reciprocals sum to one. Since the signed Casimir trace is at most $T_0$,

$$\mathrm{Tr}_{E/\mathbb{Q}}(\Delta) < T_0 - 2. \tag{3.6}$$

**Proposition 3.8** (Finite biquadratic obstruction)**.** *Assume in addition that the Frobenius–Perron formal codegree has minimal polynomial $X^2 - 5nX + 25n$, where $n = s^2 + 4 = k^2\ell$ and $\ell \in \{q, pq\}$. Every categorification supplies an integral element*

$$\theta = \frac{\Delta}{5k\sqrt{\ell}} = \frac{x + y\sqrt{p} + z\sqrt{q} + w\sqrt{pq}}{4}$$

*subject to all the following finite tests:*

1. *$\Delta$ is totally positive, its trace satisfies* (3.6)*, and $pq$ divides that trace. Put $m = z$ if $\ell = q$ and $m = w$ if $\ell = pq$. Then $m > 0$ and $\mathrm{Tr}(\Delta) = 5k\ell m$.*
2. *$\mathrm{Norm}(\theta)$ divides $\ell^2 m^4$.*
3. *Put $1 - \mathrm{Tr}(1/\Delta) = A/B$ in lowest positive terms. For some positive integer $j$, the last two signed codegrees have sum $Aj$ and product $Bj$, both roots greater than one, and sum at most $T_0 - \mathrm{Tr}(\Delta)$. If their quadratic polynomial is irreducible, $Bj \mid (Aj)^2$ and its quadratic field embeds into $E$.*
4. *The product of this quadratic polynomial and $\prod_{\sigma: E \hookrightarrow \mathbb{R}} (X - \sigma\Delta)$ is congruent modulo two to the original Casimir characteristic polynomial.*

*If the finite set is empty, no complex fusion category realizes the ring.*

*Proof.* For $u = (s + \sqrt{n})/2$ one has $u^2 - su - 1 = 0$ and $D = 5\sqrt{n}\,u$. Thus $\theta = u\Delta/D$ is integral by Lemma 3.7. The integral basis is the product of the integral bases of the two quadratic fields, because their discriminants are coprime. If $(e, h) = (1, 2)$ for $q \equiv 1 \pmod 4$, and $(e, h) = (0, 1)$ otherwise, its congruences are

$$\begin{aligned} hw &\equiv 0 \pmod 2, & h(z - w) &\equiv 0 \pmod 4, \\ y - ew &\equiv 0 \pmod 2, & x - y - ez + ew &\equiv 0 \pmod 4. \end{aligned}$$

Total positivity supplies the strict coordinate bounds $|w| < m/\sqrt{p}$, $|x| < m\sqrt{q}$, $|y| < m\sqrt{q/p}$ when $\ell = q$; and $|x| < m\sqrt{pq}$, $|y| < m\sqrt{q}$, $|z| < m\sqrt{p}$ when $\ell = pq$. Together with the trace bound these give the finite search. The global dimension is a $d$-number by [37, Theorem 1.2]. For a nonzero $d$-number $a$ in a Galois field, $\mathrm{Tr}(a)^4/\mathrm{Norm}(a)$ is integral: $\mathrm{Tr}(a)/a$ is integral and one takes its norm. Applying this to $\Delta$ gives the norm divisibility in (2). The two complementary codegrees are Galois stable, so their sum and product are integers. The reciprocal identity gives $(Aj, Bj)$; the quadratic $d$-number test and Lemma 3.7 give (3). Finally $M_i \equiv N_i \pmod 2$ implies the congruence in (4). □

The tensor-only implementation `new_biquadratic_trace.py` excludes `R6_7_3813`, `R6_8_6940`, `R6_8_6973`, and `R6_8_6998`. Their fields are respectively $\mathbb{Q}(\sqrt{5}, \sqrt{193})$, $\mathbb{Q}(\sqrt{5}, \sqrt{34})$, $\mathbb{Q}(\sqrt{17}, \sqrt{257})$, and $\mathbb{Q}(\sqrt{5}, \sqrt{13})$. For `R6_8_6973`, the trace is $2570m$, the trace bound gives $1 \le m \le 2$, and the lattice condition requires $17 \mid m$. For `R6_8_6998`, the only remaining complementary pair has polynomial $X^2 - 2350X + 2500$; its product with the four dimension codegrees is $X^6$ modulo two, whereas the original Casimir polynomial is $X^6 + X^2$.

### 3.6 Formal-codegree trace ideals

The field-containment lemma admits a useful consequence even when the signed dimension field has not been identified. For an algebraic integer $f$, write

$$\mathfrak{t}(f) = \mathrm{Tr}_{\mathbb{Q}(f)/\mathbb{Q}}(f\mathcal{O}_{\mathbb{Q}(f)}) = \tau(f)\mathbb{Z}, \qquad \tau(f) > 0.$$

Conjugating $f$ does not change this integer ideal.

**Theorem 3.9** (Formal-codegree trace-ideal obstruction)**.** *Let $R$ be a commutative self-dual rank-six fusion ring, with regular matrices $N_i$ and Casimir trace $T_0 = \mathrm{Tr}(\sum_i N_i^2)$. Choose any collection of its formal codegrees $f$ and put $L = \mathrm{lcm}_f\, \tau(f)$. If $R$ is categorifiable, the signed sphericalization has a dimension field $K$ and global dimension $\Delta$ satisfying*

$$0 < \mathrm{Tr}_{K/\mathbb{Q}}(\Delta) \in L\mathbb{Z}, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\Delta) \le T_0.$$

*In particular, $L > T_0$ excludes every complex fusion categorification. No pivotality assumption or local-field isolation is required.*

*Proof.* Lemma 3.7 gives $F = \mathbb{Q}(f) \subseteq K$ and $\Delta/f \in \mathcal{O}_K$. The relative trace of an algebraic integer is integral. Therefore

$$\mathrm{Tr}_{K/F}(\Delta) = f\,\mathrm{Tr}_{K/F}(\Delta/f) \in f\mathcal{O}_F.$$

Transitivity of trace gives the asserted divisibility for every selected $f$. The embeddings of $K$ give distinct characters of the signed algebra, with formal codegrees the corresponding conjugates of $\Delta$. All its formal codegrees are positive. Thus their total, the signed Casimir trace, is at least $\mathrm{Tr}_{K/\mathbb{Q}}(\Delta)$. The signed coefficient bounds give $\mathrm{Tr}(\sum_i M_i^2) = \sum_{i,j,k}(\iota_{ij}^k)^2 \le \sum_{i,j,k}(N_{ij}^k)^2 = T_0$. □

The implementation uses every rational and quadratic original codegree. For a rational integer $f$, $\tau(f) = f$. For a quadratic codegree $f = (A + b\sqrt{q})/2$, with $q > 1$ squarefree, the integral basis gives

$$\tau(f) = \begin{cases} \gcd(A, (A+bq)/2), & q \equiv 1 \pmod 4, \\ \gcd(A, bq), & q \equiv 2, 3 \pmod 4. \end{cases}$$

Omitting higher-degree factors weakens the test and creates no exclusion. The calculation depends only on the complete multiplication tensor.

For `R6_8_6997`, the rational formal codegrees include 15, while the FP codegree has minimal polynomial $X^2 - 6145X + 30725$ and trace ideal $6145\mathbb{Z}$. Hence $L = 18435 > T_0 = 6173$, giving a general exclusion. The same test supplies independent exclusions of `R6_7_3812`, `R6_8_6234`, `R6_8_6509`, `R6_8_6940`, and `R6_8_6973`.

**Corollary 3.10** (Signed saturation from a trace ideal)**.** *For an unordered triple of nonunit basis labels with coefficient $a \ge 2$, let $w$ be its number of ordered permutations, and put*

$$\eta = \min w(a^2 - (a-2)^2),$$

*where the minimum runs over those triples. If $0 \le T_0 - L < \eta$, every signed sphericalization coefficient has absolute value equal to its original nonnegative coefficient. All remaining choices are finite sign choices subject to associativity.*

*Proof.* Parity forces any decrease of absolute value to be at least two. Full symmetry forces the same decrease on all $w$ ordered permutations. Such a decrease loses at least $\eta$ from the Casimir trace, whereas Theorem 3.9 permits loss at most $T_0 - L$. The unit coefficients do not change in the signed quotient. □

For `R6_8_6520`, one has $L = 6145$, $T_0 = 6165$, and $\eta = 24$. Thus its signed quotient is saturated. Exhaustive exact signed associativity leaves one table up to the certified sign gauges; that table is gauge-equivalent to the original ring. The non-FP character orbits have zero entries and cannot be dimension characters. Consequently every categorification would have a pseudo-unitary Galois conjugate. Proposition 3.11 excludes that possibility, completing the general noncategorifiability proof.

The authoritative implementation is `new_trace_ideal.py`. It checks all arithmetic from the tensor, replays signed enumeration when used, and rejects altered certificates and omitted signed tables. Its control report uses the frozen initial forty residual tensors and all fifty-four established positive models; the latter are all admitted by this necessary test.

### 3.7 An all-orders quadratic Gauss obstruction

The trace-ideal reduction also gives a complete exclusion at multiplicity eight. The argument below treats every possible central restriction column and every root-of-unity order; it does not impose a numerical conductor cutoff.

Write $R(m)$ for the commutative self-dual rank-six tensor on $1, a, b, c, d, e$ whose nonunit coefficients equal $m$, except that the four unordered triples

$$(a,a,b),\ (a,a,c),\ (a,b,b),\ (b,b,c)$$

have coefficient $m-1$, and

$$(c,d,d),\ (c,e,e),\ (d,d,e),\ (d,e,e)$$

have coefficient $m+1$. Coefficients are invariant under permutations of the three indices, and the unit coefficients are fixed by self-duality. Set

$$n = 25m^2+4, \quad U = \frac{5m+\sqrt{n}}{2}, \quad j = (1,1,1,1,1)^t, \quad \delta = (-1,-1,0,1,1)^t.$$

The FP dimensions are $(1, U-1, U-1, U, U+1, U+1)$, and $D = 5(U^2+1) = (5n+25m\sqrt{n})/2$. The other five characters are the quadratic conjugate of the FP character and, with $p = (1+\sqrt{5})/2$, $q = (1-\sqrt{5})/2$,

$$(1,p,q,-1,0,0),\ (1,q,p,-1,0,0),\ (1,0,0,1,-p,-q),\ (1,0,0,1,-q,-p).$$

The corresponding formal codegrees are $D, D', 5, 5, 5, 5$. These identities, character completeness, and the full tensor are checked exactly by `software/new_gauss_family.py`.

**Proposition 3.11** (Quadratic Gauss family obstruction)**.** *The tensor $R(7) = R^{(6)}_{8,6520}$ has no pseudo-unitary categorification. Consequently the trace-ideal signed reduction excludes every complex fusion categorification of this tensor. More generally, the same exact verifier gives a necessary criterion for $R(m)$, $m \geq 3$: whenever its exhaustive column enumeration has the families described below, it requires*

$$12m^2+2 \ \geq\ \frac{\varphi(f)mb}{2c}, \qquad n = b^2 q_0, \quad f = \operatorname{disc}\mathbb{Q}(\sqrt{q_0}), \quad c = \sqrt{f/q_0},$$

*where $q_0$ is squarefree.*

*Proof.* Assume a pseudo-unitary categorification and use its positive spherical structure. If $v_Y$ is the restriction column of a central simple, then $\sum_Y v_Y v_Y^t = G$, where $G = \sum_i N_i^2$. The unit column of $G$ is $(6, 5mj+\delta)^t$, and its nonunit block is

$$5I + (25m^2+1)jj^t + 5m(j\delta^t + \delta j^t) + \delta\delta^t.$$

By central induction, $I(1)$ contains six distinct simples with dimensions $1, z = U^2, t = U^2+1, t, t, t$ and twists one [38, Theorems 2.5 and 2.13]. The symmetry of the central $S$-matrix gives, for character-labelled low columns,

$$\dim(Y_\chi)\chi(v_\psi) = \dim(Y_\psi)\psi(v_\chi).$$

Applying this with the quadratic-conjugate character and the four codegree-five characters determines the $z$ column as $(1, mj)^t$: indeed the conjugate dimension of a $t$ column is $t' = t/z$, so reciprocity gives each small character value as $zt'/t = 1$. The complete character table then determines the column uniquely. Write the remaining four columns as $(1, mj + w_\nu)^t$. Their dimensions and the unit column of the Gram identity give

$$j^t w_\nu = 0, \qquad \delta^t w_\nu = 1, \qquad \sum_\nu w_\nu = \delta.$$

On the three-dimensional subspace perpendicular to $j$ and $\delta$, $G$ restricts to $5I$. Taking the trace of its residual Gram inequality therefore gives $\sum_\nu \|w_\nu\|^2 \le 16$. Each squared norm is at least two, so every $w_\nu$ has squared norm at most ten and coordinates in $[-3, 3]$.

The exhaustive enumeration has 32 individual deviations, 59 unordered quadruples satisfying the sum and norm conditions, and 11 satisfying a labelling by the four small characters and modular reciprocity. Exactly three have positive-semidefinite residual Gram matrix. Their rows of deviations are

$$W_1 = \begin{pmatrix} -1 & 0 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 0 & 1 \end{pmatrix}, \qquad W_2 = \begin{pmatrix} -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & 0 \end{pmatrix},$$

$$W_3 = \begin{pmatrix} -1 & 0 & 1 & 1 & -1 \\ -1 & 1 & -1 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 0 \end{pmatrix}.$$

All enumeration bounds have just been proved; no low column is selected in advance.

A remaining column is $(0, v)^t$, with $v \in \mathbb{Z}^5_{\ge 0} \setminus \{0\}$. Let $Q$ be the nonunit residual Gram matrix. The necessary conditions are

$$Q - vv^t \succeq 0, \qquad Q - vv^t \ge 0 \text{ entrywise}, \qquad D/\dim(v) \text{ algebraically integral.}$$

The last condition follows because $D^2/\dim(v)^2$ is a formal codegree of the modular center. All three residuals annihilate $(1, 1, -4, 1, 1)^t$. Hence the first, second, fourth, and fifth coordinates determine the third. The bounds $v_i \le \lfloor \sqrt{Q_{ii}} \rfloor$ and

$$|v_a - v_b| \le \lfloor \sqrt{Q_{aa} + Q_{bb} - 2Q_{ab}} \rfloor, \qquad |v_d - v_e| \le \lfloor \sqrt{Q_{dd} + Q_{ee} - 2Q_{de}} \rfloor$$

make the enumeration finite and complete. Gram positivity is checked with exact rational Schur complements, or equivalently by the kernel condition and $v_J^t Q_J^{-1} v_J \le 1$ on a full-rank principal submatrix. For a quadratic number $x + y\sqrt{n}$, integrality is equivalent to integral trace $2x$ and integral norm $x^2 - ny^2$.

For $m = 7$ the counts at the bounded, Gram-compatible, and integral-quotient stages are respectively

$$(511, 58, 1), \qquad (6147, 282, 9), \qquad (511, 58, 1).$$

The outer branches admit only $j$, although $Q$ has rank two, and are therefore impossible. The complete middle list is

$$j, \quad j \pm (e_a - e_b), \quad j \pm (e_d - e_e), \quad 7j - e_i + e_k \quad (i \in \{a, b\},\ k \in \{d, e\}).$$

For the general verifier, the accepted list may also include the first five types with $j$ replaced by $2j$ and the last four types with $m$ replaced by $2m$; any additional high type causes the criterion to abstain.

Put $u = e_a - e_b$, $v = e_d - e_e$. Exact Gram identities are

$$\delta^t Q \delta = 32, \quad \delta^t Q j = 80m, \quad u^t Q u = v^t Q v = 8, \quad \delta^t Q u = \delta^t Q v = u^t Q v = 0.$$

If $n_m, n_{2m}$ count the two possible cross-column families, the first two equations give $n_m + n_{2m} = 8$ and $n_m + 2n_{2m} = 8$. Thus exactly eight cross columns occur, all with coefficient $m$. They saturate both $u, v$ squared forms, excluding all offset types. The three vanishing mixed forms force each of the four columns $B_{ik} = mj - e_i + e_k$ to occur twice. Subtracting them leaves

$$Q - 2\sum_{i,k} B_{ik} B_{ik}^t = (12m^2 + 2) j j^t.$$

Thus the number of uniform columns, with a column $2j$ counted four times, is $L = 12m^2 + 2$.

Let $\Theta_{ik}$ be the sum of the two twists belonging to $B_{ik}$, and let $\Lambda$ be the sum of uniform twists with these weights. It is a sum of exactly $L$ roots of unity. The first twist moment of $I(X)$ vanishes for nonunit $X$ [38, Theorem 2.5], so, putting $w = 5U$, the five equations combine as

$$w\Lambda j + t\sum_{i,k} \Theta_{ik} B_{ik} = -mzj - t(4mj + \delta).$$

Pairing with $\delta$ gives $\sum \Theta_{ik} = -2$; pairing with $j$ gives

$$w\Lambda = -mz - 2mt, \qquad \Lambda = -\frac{m^2 + m\sqrt{n}}{2}.$$

For completeness the required root-of-unity bound holds for all orders. For $K = \mathbb{Q}(\sqrt{q_0})$ with positive fundamental discriminant $f$, let $c = \sqrt{f/q_0}$. The coefficient of $\sqrt{q_0}$ in the normalized trace to $K$ of a single root of unity has absolute value at most $c/\varphi(f)$. If the root's cyclotomic field does not contain $K$, the trace is rational. Otherwise its order is divisible by $f$. Averaging over the kernel of the primitive quadratic character expresses the irrational part as an induced quadratic Gauss sum divided by $\varphi(N)$. This sum is zero unless the extra factor $N/f$ is squarefree and coprime to $f$; in the nonzero case its magnitude is $\sqrt{f}$. The formula follows directly by Chinese remaindering: excess exponents at primes of $f$ cancel over lifts, and the coprime factor contributes its Ramanujan sum. As $\varphi(N) \geq \varphi(f)$, the stated bound follows. Normalized trace and the triangle inequality then require

$$L \geq \frac{\varphi(f)}{c} \, |[\sqrt{q_0}]\Lambda| = \frac{\varphi(f) m b}{2c}.$$

For $m = 7$, $n = q_0 = f = 1229$, $b = c = 1$, and $\varphi(f) = 1228$. The forced sum of 590 roots is $-(49 + 7\sqrt{1229})/2$, but its required length is at least $1228 \cdot 7/2 = 4298 > 590$. This contradiction proves the pseudo-unitary exclusion. The separate complete signed trace-ideal certificate supplies a pseudo-unitary Galois conjugate of any hypothetical categorification, proving the general exclusion. □

The executable certificate is tensor-based and invariant under relabelling of the five nonunit objects. Its normal and optimized-Python replay reconstructs every low and high column, rejects missing branches and altered arithmetic. The frozen 40-input continuation scan applies the criterion to every input; only $R^{(6)}_{7,3582}$ and $R^{(6)}_{8,6520}$ match this tensor family, and both are excluded in the pseudo-unitary scope. The independent trace-ideal reduction makes the latter exclusion unrestricted. The scan of all 16612 tensors through multiplicity eight checks the

parameter-three through parameter-seven members uniformly:

| tensor | forced root count | Gauss lower bound |
|---|---|---|
| $R^{(6)}_{4,803}$ | 110 | 342 |
| $R^{(6)}_{5,1500}$ | 194 | 400 |
| $R^{(6)}_{6,2554}$ | 302 | 1440 |
| $R^{(6)}_{7,3582}$ | 434 | 1344 |
| $R^{(6)}_{8,6520}$ | 590 | 4298 |

The table records a separate application of the same necessary criterion to all five rings.

### 3.8 Ramification and rational-codegree trace bounds

This criterion applies to arbitrary complex fusion categorifications. It does not assume a pivotal structure and does not enumerate selected dimension characters. Let $R$ be commutative and self-dual of rank six, and write $T_0 = \sum_{i,j,k}(N_{ij}^k)^2$.

**Proposition 3.12** (Ramified dimension field and Casimir trace)**.** *Suppose the FP global dimension is $D = a + b\sqrt{q}$, where $q > 1$ is squarefree, $q \equiv 2, 3 \pmod 4$, and $a, b \in \mathbb{Z}$, $b > 0$. Suppose also that*

$$R/2R \cong \mathbb{F}_2[z]/(z^2) \times \prod_j \mathbb{F}_{2^{e_j}}.$$

*Let $\mathcal{L} \subseteq \mathbb{Z}^2$ be the lattice of pairs $(S, B)$ for which*

$$\frac{2qaS - 2qbB}{a^2 - qb^2}, \qquad \frac{2aB - 2qbS}{a^2 - qb^2}, \qquad \frac{2qS}{c}, \qquad \frac{2B}{c} \quad \text{are integers}$$

*for every rational original formal codegree $c$. Let $v$ be the positive generator of the projection of $\mathcal{L}$ onto its first coordinate. If*

$$4qv + 5 > T_0,$$

*then $R$ has no complex fusion categorification.*

*Proof.* Sphericalize a hypothetical categorification and quotient its Grothendieck ring by the central sign object plus the unit. The result is a rank-six integral signed algebra $A$ with multiplication coefficients $t_{ij}^k$ satisfying

$$t_{ij}^0 = \delta_{ij}, \qquad |t_{ij}^k| \le N_{ij}^k, \qquad t_{ij}^k \equiv N_{ij}^k \pmod 2.$$

The two lifts of each original self-dual simple have opposite nonzero spherical dimensions, so duality cannot interchange them. Consequently the regular matrices $M_i$ of $A$ are real symmetric and commute. The dimension character has values $d_i$ in a totally real field factor $K$ of $A \otimes \mathbb{Q}$, and $\Delta = \sum_i d_i^2$ is the global dimension of the original category.

For every original formal codegree $f$, its pulled-back character on sphericalization has formal codegree $2f$. The central-induction theorem [38, Theorem 2.13] therefore gives a central simple of dimension $\Delta/f$. Forgetting it expresses this dimension as an integer combination of the $\pm d_i$. Thus $\Delta/f \in \mathcal{O}_K$, and in particular $D \in K$ and $\mathbb{Q}(\sqrt{q}) \subseteq K$. This also recalls the dimension-field argument of Lemma 3.7.

Since $A/2A = R/2R$, lift the displayed primitive idempotents to $A \otimes \mathbb{Z}_2$. A finite free factor with finite-field reduction is finite etale over $\mathbb{Z}_2$, so its generic fiber is unramified. The only remaining factor has rank two. But $\mathbb{Q}_2(\sqrt{q})$ is ramified for $q \equiv 2, 3 \pmod 4$, and every component of $K \otimes \mathbb{Q}_2$ contains this field. No such component can be supported on an unramified factor. Hence $[K : \mathbb{Q}] \le 2$, and therefore $K = \mathbb{Q}(\sqrt{q})$.

Now $\mathcal{O}_K = \mathbb{Z}[\sqrt{q}]$. Write $d_i = \alpha_i + \beta_i\sqrt{q}$ for $1 \le i \le 5$, and set $S = \sum_i \beta_i^2 > 0$ and $B = \sum_i \alpha_i\beta_i$. The two conjugate dimension-character vectors are orthogonal, because the $M_i$ are commuting real symmetric matrices. It follows that

$$1 + \sum_i \alpha_i^2 = qS, \qquad \Delta = 2qS + 2B\sqrt{q}, \qquad \Delta + \Delta' = 4qS.$$

Integrality of $\Delta/D$ gives the first two conditions defining $\mathcal{L}$. Integrality of $\Delta/c$ gives the other two for every rational codegree $c$. Therefore $S \ge v$.

The signed Casimir $\sum_i M_i^2$ has the six signed formal codegrees as its eigenvalues. The two dimension characters contribute $4qS$. Each of the other four codegrees is strictly greater than one: equality would force every nonunit character value to vanish, contrary to the unit coefficient in $x_i^2$. Since the trace is integral,

$$4qv + 5 \le \mathrm{Tr}\Big(\sum_i M_i^2\Big) = \sum_{i,j,k} (t_{ij}^k)^2 \le T_0,$$

contradicting the asserted strict inequality. □

**Corollary 3.13.** *The rings $R^{(6)}_{7,3812}$, $R^{(6)}_{8,6234}$, and $R^{(6)}_{8,6509}$ have no complex fusion categorifications.*

*Proof.* For all three rings,

$$q = 226, \qquad D = 2260 + 150\sqrt{226}, \qquad D^2 - 4520D + 22600 = 0.$$

The following table lists their exact local factors and trace data. Here $B_2 = \mathbb{F}_2[z]/(z^2)$ and $D'$ is the conjugate of $D$.

| Ring | Algebra modulo two | Formal codegrees | $v$ | $T_0$ |
|---|---|---|---|---|
| $R^{(6)}_{7,3812}$ | $B_2 \times \mathbb{F}_4 \times \mathbb{F}_2^2$ | $D, D', 3, 15, 5, 5$ | 15 | 4548 |
| $R^{(6)}_{8,6234}$ | $B_2 \times \mathbb{F}_4 \times \mathbb{F}_2^2$ | $D, D', 3, 15, 5, 5$ | 15 | 4548 |
| $R^{(6)}_{8,6509}$ | $B_2 \times \mathbb{F}_2^4$ | $D, D', 3, 3, 15, 15$ | 15 | 4556 |

In particular $\Delta/15 \in \mathcal{O}_K$ and $\Delta = 452S + 2B\sqrt{226}$ imply $15 \mid S$, because $\gcd(452, 15) = 1$. Thus the signed Casimir trace is at least $904 \cdot 15 + 5 = 13565$, exceeding each displayed $T_0$. □

The finite local checks exhaust all 64 elements modulo two, find every primitive idempotent, and verify orthogonality and completeness of the factor decomposition. Every nonzero element of a claimed field factor is checked to have an inverse; the remaining rank-two factor has exactly one nonzero square-zero element. The lattice step is recovered from exact congruences, not floating-point bounds. The three certificates require neither a signed-table search nor a pseudo-unitary exclusion.

The exact certificates and their tensor-based replay are indexed by `software/new_ramified_signed.py` and `research/round2_higher/ramification_application.json`. The independent field-containment exclusion of $R^{(6)}_{8,6940}$ overlaps the biquadratic criterion and is counted only once.

### 3.9 Local character fields and quadratic codegree saturation

The next criterion strengthens the signed Casimir bounds by determining the dimension field before enumerating possible formal codegrees. Write $A$ for the signed algebra supplied by Lemma 3.7, $K$ for its dimension character field, and $\Delta$ for its dimension codegree. Thus every

original or signed formal codegree $f$ lies in $K$, and $\Delta/f$ is an algebraic integer. In particular this statement concerns the sphericalization of an arbitrary categorification; it does not assume a pivotal structure on that category.

**Proposition 3.14** (Two local ways to isolate the dimension field)**.** *Let $R$ be commutative and self-dual of rank six, and suppose an original formal codegree generates $E = \mathbb{Q}(\sqrt{q})$, where $q > 1$ is squarefree. Either of the following conditions implies $K = E$ and that the other four signed formal codegrees are rational integers greater than one:*

1. *$q \equiv 5 \pmod 8$, and $R/2R$ is a product of finite fields with exactly one even-degree factor, of degree two;*
2. *$R/2R$ has three local factors: $\mathbb{F}_8$, a rank-two local algebra, and $\mathbb{F}_2$.*

*In the second case the conclusion uses cyclotomicity of characters of a categorifiable fusion ring.*

*Proof.* The dimension-field containment lemma gives $E \subset K$. Because the regular matrices of $A$ are symmetric, $A \otimes \mathbb{Q}$ is a product of totally real number fields. Its primitive local idempotents at two lift from $A/2A = R/2R$ to $A \otimes \mathbb{Z}_2$. A finite free local factor with finite-field special fibre is finite étale: its relative differentials vanish by Nakayama, and it is flat. It is therefore the unramified extension with that residue degree. Equivalently one may apply the classification of finite étale algebras over a henselian local ring; see the Stacks Project, Tag 04GK.

In the first case $E \otimes \mathbb{Q}_2$ is the unramified quadratic field. Every completion of a number field containing $E$ has even degree over $\mathbb{Q}_2$. Since $K$ is a field quotient of $A \otimes \mathbb{Q}$, its completions must use only the unique even-degree local factor. That factor has degree two, so $K = E$. If a codegree belonging to another character field $F$ were irrational, it would generate $E$, because it already belongs to $K = E$. Then $F$ would require another even-degree local factor, which is impossible. All remaining codegrees are therefore rational.

In the second case the lifted rank-three factor has generic fibre the unramified cubic extension of $\mathbb{Q}_2$. Let $F$ be the global character field containing this local factor. A signed character extends to a character of the sphericalization with the sign object sent to $-1$. Hence its values are cyclotomic by [15, Theorem 8.51]; in particular $F/\mathbb{Q}$ is abelian and Galois. All its completions at two have the same local degree, here three. Thus $[F : \mathbb{Q}]$ is three or six. Degree six would require two degree-three local factors, whereas the remaining lifted blocks have ranks two and one. Hence $F$ is cubic and uses the entire rank-three block. The remaining rational algebra has dimension three. The field $K$ contains the quadratic field $E$, so it lies in this remainder and has degree two. Thus $K = E$. The other character fields are $F$ and $\mathbb{Q}$; their intersections with $E$ are rational. The remaining codegrees, which lie in these intersections, are rational as claimed.

In both cases rational formal codegrees are algebraic integers. They are positive, and the sum of reciprocals of the six signed codegrees is one. Each is consequently greater than one. □

The second argument does not assert that the whole two-adic order is étale. In particular a rank-two order with dual-number reduction may have split generic fibre. Only its rank and its separation from the cubic block are needed.

**Proposition 3.15** (Finite quadratic spectrum bound)**.** *Suppose $K = E = \mathbb{Q}(\sqrt{q})$ and the complementary four codegrees are rational, as in Proposition 3.14. Assume $q \equiv 1 \pmod 4$ and the original Frobenius–Perron codegree has the form $D = c\sqrt{q}\,u$, where $c$ is a positive integer and $u$ is an integral unit of norm $-1$. Let $T = \operatorname{Tr}(\sum_i N_i^2)$. Every categorification supplies*

*integers $x, y$ and $f_1 \le \cdots \le f_4$ satisfying the following finite system:*

$$\Delta = \frac{c(qy + x\sqrt{q})}{2}, \qquad x \equiv y \pmod 2, \quad y > 0, \quad x^2 < qy^2, \quad cqy \le T - 5,$$

$$n = qy^2 - x^2, \qquad n \mid 4qy^2, \qquad \sum_{i=1}^{4} \frac{1}{f_i} = 1 - \frac{4y}{cn} > 0,$$

$$f_i > 1, \qquad f_i \mid \gcd\left(\frac{c(qy - x)}{2}, cx\right), \qquad cqy + \sum_{i=1}^{4} f_i \le T.$$

*In addition $\Delta/f$ is integral for every original rational codegree $f$, and the polynomial*

$$\left(z^2 - cqy\, z + \frac{c^2 qn}{4}\right) \prod_{i=1}^{4} (z - f_i)$$

*is congruent modulo two to the original Casimir polynomial. If the system is empty, $R$ is not categorifiable. Otherwise its least total trace bounds the possible signed Casimir deficit.*

*Proof.* Since $\Delta/D$ is integral, $\theta = \Delta/(c\sqrt{q}) = u\Delta/D$ is integral. The integral basis of $E$ gives $\theta = (x + y\sqrt{q})/2$ with the stated parity. Total positivity of $\Delta$ gives $y > 0$ and $x^2 < qy^2$. Its two dimension characters contribute trace $cqy$; the other four codegrees exceed one. The signed Casimir trace is an integer at most $T$, giving the displayed finite bound.

The global dimension is a $d$-number by [37, Theorem 1.2]. Here

$$\mathrm{Tr}(\Delta) = cqy, \quad \mathrm{N}(\Delta) = \frac{c^2 qn}{4}, \quad \mathrm{Tr}(1/\Delta) = \frac{4y}{cn}.$$

The quadratic $d$-number divisibility gives $n \mid 4qy^2$, and the reciprocal-codegree identity gives the reciprocal equation. The integral-basis coordinates of $\Delta$ are $c(qy - x)/2$ and $cx$. Thus $\Delta/f_i$ is integral precisely when $f_i$ divides their greatest common divisor. Divisibility by original rational codegrees follows separately from induction. Finally the signed and original matrices are congruent modulo two, so their Casimir characteristic polynomials have the same reduction. Every enumeration range is finite. The case $x = 0$ is retained: $K$ can be quadratic even when $\Delta$ itself is rational. □

**Theorem 3.16.** *The rings $R^{(6)}_{8,6524}$, $R^{(6)}_{8,6525}$ and $R^{(6)}_{8,6999}$ have no complex fusion categorifications.*

*Proof.* For the first two rings, $q = 149$, $c = 9$,

$$D = \frac{6705 + 549\sqrt{149}}{2}, \qquad T = 6741.$$

Their reductions at two have field degrees $(2, 3, 1)$ and $(2, 1, 1, 1, 1)$, respectively. Proposition 3.14(1) applies. The finite spectrum test has $1 \le y \le 5$ and leaves only

$$y = 5, \qquad x = \pm 61, \qquad (f_1, f_2, f_3, f_4) = (9, 9, 9, 9).$$

Both traces equal 6741.

For $R^{(6)}_{8,6999}$, $q = 41$, $c = 18$,

$$D = 3690 + 576\sqrt{41}, \qquad T = 7416.$$

The local factors are $\mathbb{F}_8$, the dual numbers of rank two, and $\mathbb{F}_2$. Proposition 3.14(2) applies. The bound is $1 \le y \le 10$; the only survivors are

$$y = 10, \qquad x = \pm 64, \qquad (f_1, f_2, f_3, f_4) = (9, 9, 9, 9),$$

again with total trace $T$.

In all three cases the signed Casimir trace is the sum of the squares of the signed structure constants. Since each absolute value is bounded by its original coefficient, equality of traces forces equality of every absolute value. The remaining unrestricted sign-only associativity systems are exhausted exactly. Every solution is a dual-compatible sign gauge of the original tensor. After this gauge, the lift fusion rules are the original positive ones, and the recorded dimension-character tests leave only the Galois orbit of the positive character. Any categorification would therefore have a pseudo-unitary Galois conjugate.

The pseudo-unitary possibility is excluded by the exact `global_codegree_moment` certificates, checked independently on the full input tensors. For the first two rings the obstruction is visible already in the integral span of simple dimensions. They are $1, u-1, u, u, u, u$, where $u = (37+3\sqrt{149})/2$ and $u^2 - 37u + 7 = 0$. The codegree 9 would require a central simple of dimension

$$D/9 = (61u - 11)/3 \notin \mathbb{Z} + \mathbb{Z}u,$$

which cannot be the dimension of an integral restriction to the original category. This also gives a direct check on the final induction obstruction in those two cases. □

The tensor-only generator and exact verifier are `software/round3_arithmetic.py`; all local algebras, surviving spectra, sign gauges, and complete general certificates are recorded in `research/round3/arithmetic/`. The independent enumeration in `research/round3/models/arithmetic_audit.py` reproduces the first pair of saturated spectra without an optimizer or the generator's recursive divisor enumeration.

### 3.10 Noncommutative signed spectra

The preceding field-isolation argument has a different replacement when the signed quotient is forced to be noncommutative. This follows either from reduction modulo two or from a separate exhaustive exclusion of the commutative signed sector. It is essential in this case to keep the representation multiplicities.

**Proposition 3.17.** *Let $R$ have rank six. Suppose either that its reduction modulo two is noncommutative, or that every commutative signed quotient permitted by sphericalization has been excluded by necessary conditions. Suppose its two linear characters form a real quadratic orbit over $E = \mathbb{Q}(\sqrt{q})$, and let $D, D'$ be their formal codegrees and $c$ that of its unique two-dimensional representation. For any categorification there are $\Delta \in E$ and $h \in \mathbb{Z}_{>2}$ satisfying*

$$\frac{1}{\Delta} + \frac{1}{\Delta'} + \frac{2}{h} = 1, \qquad \mathrm{Tr}_{E/\mathbb{Q}}(\Delta) + 8h \le T, \qquad T = \sum_{i,j,k} (N_{ij}^k)^2.$$

*Both embeddings of $\Delta$ are positive, $\Delta$ is a d-number, and*

$$\Delta/D, \quad \Delta/D', \quad \Delta/c, \quad \Delta/h \quad \textit{belong to } \mathcal{O}_E.$$

*For fixed $R$ these conditions are a finite exact search.*

*Proof.* The negative-sign quotient of sphericalization is now an oriented signed algebra $A$. Its signed basis is orthonormal for the coefficient-of-unit inner product, and duality gives adjoints of left multiplication. Thus its regular representation is a finite dimensional $*$-algebra with positive inner product and is semisimple. In the first case parity preserves a nonzero commutator; in the second case the commutative alternative is excluded separately. In either case dimension six and the existence of a linear character give

$$A \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

There are exactly two linear characters. The spherical dimension character is real and dual invariant, and its field $K$ therefore has degree at most two. Induction applied to the original characters still gives $\Delta/f \in \mathcal{O}_K$. In particular $D \in K$, so $K = E$. The signed linear codegrees are $\Delta, \Delta'$.

The unique two-dimensional representation is preserved by Galois action. Its formal codegree $h$ is consequently rational and an algebraic integer, hence an integer. The formal-codegree reciprocal identity weights it by its representation dimension two. On the four-dimensional regular matrix block the Casimir has eigenvalue $2h$, so that block contributes $8h$ to the regular trace. This proves both displayed identities, with the trace inequality following from the coefficient bounds. Positivity of the two linear codegrees and the reciprocal equation imply $h > 2$. Induction from the negative-sign character gives $\Delta/h$; induction from the original characters gives the other quotients. The usual global-dimension $d$-number theorem applies to $\Delta$.

For example, when $q \equiv 1 \pmod 4$, write $\Delta = (a + b\sqrt{q})/2$, with $a \equiv b \pmod 2$. Then $0 < a \leq T - 24$ and $qb^2 < a^2$. These bounds are finite. For each pair $(a, b)$ the reciprocal equation determines

$$h = \frac{2}{1 - a/\,\mathrm{N}(\Delta)}.$$

Testing integrality, positivity, the four quotients, and the quadratic $d$-number condition completes the exact necessary search. □

**Theorem 3.18.** *The ring $R^{(6)}_{6,170}$ has no complex fusion categorification.*

*Proof.* Its original codegrees are $148 \pm 24\sqrt{37}$ and 4, with representation dimensions $1, 1, 2$, and $T = 328$. The preceding finite system leaves exactly

$$\Delta = 148 \pm 24\sqrt{37}, \qquad h = 4, \qquad \mathrm{Tr}(\Delta) + 8h = 328.$$

Thus every signed coefficient has its original absolute value. The complete oriented sign-only associativity enumeration has 1631 search nodes and 16 solutions; all sixteen are dual-compatible sign gauges of $R$. No commutativity or indicator normalization is imposed. Under each gauge the two possible dimension characters are Galois conjugates of the positive one, so every categorification would have a pseudo-unitary Galois conjugate.

Multiplicity-weighted central induction has 146 candidate restriction columns for the nontrivial linear character and 25 for the two-dimensional representation. Their complete compatibility search yields nineteen low frames. Every frame violates an ordinary exact Gram or moment condition. The pseudo-unitary possibility is therefore impossible, completing the general exclusion. □

The implementation is `software/round3_noncommutative.py`, with arithmetic spectra, all oriented sign tables, the nineteen induction branches, and replays in `research/round3/noncommutative/`. For rings with commutative reduction modulo two, the first hypothesis of Proposition 3.17 is unavailable. Their entire commutative signed sector must be treated before the noncommutative spectrum calculation supplies a general reduction.

**Theorem 3.19.** *The rings $R^{(6)}_{7,200}$ and $R^{(6)}_{8,491}$ have no complex fusion categorifications.*

*Proof.* Both reductions modulo two are commutative. First impose commutativity on the unrestricted oriented signed coefficient problem, identifying opposite coefficient variables and intersecting their original parity intervals. If an opposite original coefficient is zero, its identified signed coefficient is forced to zero. This exhausts all commutative signed quotients without an indicator normalization.

The complete enumerations have respectively 176 and 208 signed tables, forming 11 and 13 dual-compatible sign-gauge orbits. Their character algebras have respectively 27 and 53 components. Each component has a zero dimension, unequal dimensions on dual objects, or an exact norm-divisibility obstruction to induction. The latter uses the original formal codegrees, whose quadratic polynomial is $z^2 - 1560z + 1872$, and the matrix-block codegree 12. The regular Casimir eigenvalue of that block is 24; it must not replace the formal codegree in the induction test. The recorded norm obstructions occur at primes 3 or 13. Consequently no categorical signed quotient is commutative.

Proposition 3.17 now applies. For both rings its finite system has exactly

$$\Delta = 780 \pm 216\sqrt{13}, \qquad h = 12, \qquad \mathrm{Tr}(\Delta) + 8h = 1656 = T.$$

The resulting sign-only searches have 4927 and 6463 nodes, respectively, and sixteen solutions each. Every solution is a dual-compatible gauge of the original tensor. Thus any categorification has a pseudo-unitary Galois conjugate. In each ring the multiplicity-weighted induction problem has ten complete low frames, all excluded by exact ordinary Gram or moment conditions. This excludes the pseudo-unitary possibility and proves the claim. □

The separate commutative-sector verifier is `software/round3_commutative_sector.py`; its complete certificate is included in each general certificate in `research/round3/noncommutative/`. Both general replays also reconstruct the noncommutative arithmetic and induction calculations. Removing the sector certificate is explicitly rejected by the verifier.

### 3.11 Dimension-block Fourier bounds and saturated twist support

Let $K_1 = \mathbb{Q}(d_i)$, $K_0 = \mathbb{Q}(D)$ and $e = [K_1 : K_0]$ for a spherical dimension function. For a character $\chi$ of $\mathrm{Gal}(K_0/\mathbb{Q})$, let $f_\chi$ be its primitive Dirichlet conductor. Schopieray's dimension-block theorem [52, Theorem 5.1] gives the additional necessary bound

$$\left| \sum_{\sigma \in \mathrm{Gal}(K_0/\mathbb{Q})} \frac{\chi(\sigma)}{\sigma(D)} \right|^2 \leq \frac{1}{e^2 f_\chi}. \tag{3.7}$$

Its equality case supplies a restriction on individual twists, rather than only on their averages. Here $S$ denotes the center's modular $S$-matrix normalized by $D$, and $T$ is its matrix of actual twists.

**Proposition 3.20** (Saturated conductor support)**.** *Suppose equality holds in* (3.7)*. Inflate $\chi$ to $G_1 = \mathrm{Gal}(K_1/\mathbb{Q})$, and let $w_\chi = \sum_{\sigma \in G_1} \chi(\sigma) e_{A_\sigma}$, where $A_\sigma$ is the low central summand indexed by the dimension character $\sigma(d)$. The cyclic subrepresentation generated by $w_\chi$ in the canonical modular representation of $Z(\mathcal{C})$ factors through $\mathrm{SL}_2(\mathbb{Z}/f_\chi\mathbb{Z})$. Consequently, for a central simple with restriction column $v$,*

$$L_\chi(v) := \sum_{\sigma \in G_1} \chi(\sigma)\sigma\left(\frac{\sum_i v_i d_i}{D}\right) \neq 0 \quad \Longrightarrow \quad \theta_Z^{f_\chi} = 1. \tag{3.8}$$

*If several saturated characters give nonzero coefficients, the twist order divides the greatest common divisor of their conductors.*

*Proof.* We extract the equality information from the local compression calculation in [52, Lemma 4.1]. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z}/\ell^a\mathbb{Z})$, let $U$ be its upper unipotent subgroup, and let $P$ project onto $U$-fixed vectors. For a nontrivial torus character $\alpha$ of conductor $\ell^c$, the universal permutation representation on primitive columns realizes $\mathrm{Ind}_U^\Gamma \mathbf{1}$. Its $\alpha$-space of $U$-invariants splits into the one-dimensional big-cell space and the small-cell space. The compression $PsP$

maps the big-cell line to the small-cell space and maps the latter back to a line. Both nonzero singular values are $\ell^{-c/2}$. Hence every vector attaining this norm belongs to the span of the big-cell vector and the adjoint image of the opposite-character big-cell vector.

Writing a primitive column as $(p, q)$, these two vectors are, up to nonzero scalar factors,

$$\alpha(q)[q \text{ is a unit}], \qquad \alpha(p)[q \equiv 0 \pmod{\ell^c}].$$

Both are pullbacks of functions on primitive columns modulo $\ell^c$. Their entire $\Gamma$-cyclic spans therefore factor through reduction modulo $\ell^c$. Every irreducible representation with $U$-invariants embeds equivariantly and isometrically, up to a scalar, into this permutation representation. Thus an irreducible constituent containing a nonzero norm-attaining vector itself factors through that reduction.

For the trivial local character the bound is 1. Equality implies that $sw$ is $U$-fixed as well as $w$. The vector $w$ is therefore fixed by both upper and lower unipotents. These generate $\Gamma$, so its cyclic representation is trivial. Decompose a general unitary representation orthogonally into irreducibles, and then use the Chinese remainder tensor decomposition at the prime divisors of its level. Equality in the product norm bound forces every occupied local factor to attain its own bound. The preceding conclusions show that a norm-attaining vector generates a representation of level dividing $f_\chi$.

The proof of the dimension-block bound estimates the matrix coefficient $\langle w_{\bar\chi}, PSPw_\chi\rangle$ by $f_\chi^{-1/2}\|w_\chi\|^2$. Equality in the bound forces equality in this compression-norm estimate and in Cauchy–Schwarz. The preceding argument applies to $w_\chi$. In particular $(T^{f_\chi} - I)Sw_\chi = 0$. The known dimension-character rows give $S_{Z,A_\sigma} = \sigma(d_Z/D)$, so the $Z$-coordinate of $Sw_\chi$ is precisely $L_\chi(v)$. This proves (3.8). The greatest-common-divisor assertion follows by intersecting root-of-unity groups. □

For the remaining rings, the nontrivial equality conductors are 12 for $R^{(6)}_{5,1204}$ and $12, 24$ for $R^{(6)}_{8,1660}$; $R^{(6)}_{4,155}$ has strict inequalities. The exact implementation computes (3.8) from the dimensions and each restriction column.

**Proposition 3.21** (Universal moments of a saturated real character)**.** *In Proposition 3.20, suppose $\chi$ is real and put $f = f_\chi$, $\lambda = \sum_{\sigma\in G_1}\chi(\sigma)/\sigma(D)$. Let $M = \mathrm{Ind}_B^{\mathrm{SL}_2(\mathbb{Z}/f\mathbb{Z})}\chi$, where $B$ is the upper triangular subgroup and its character is trivial on unipotents and equals $\chi$ on the diagonal torus. Write $S_f, T_f$ for the standard unitary operators on $M$, and $P_\chi$ for projection onto its unipotent-fixed, torus-$\chi$ space. Let the columns of $K$ span the subspace on which $P_\chi S_f P_\chi$ has eigenvalue $\lambda$. For every exact matrix identity*

$$\sum_{n=0}^{f-1} c_n K^* S_f^* T_f^n S_f K = 0,$$

*the following necessary moment equation holds:*

$$\sum_{n=0}^{f-1} c_n \sum_Z |L_\chi(v_Z)|^2 \theta_Z^n = 0. \tag{3.9}$$

*Proof.* By the preceding proposition, the cyclic representation generated by $w_\chi$ factors through this finite group. It embeds equivariantly and isometrically into $M$. Indeed, in an isotypic summand $\pi\otimes\mathbb{C}^m$, the generating vector lies in $\pi^{B,\chi}\otimes\mathbb{C}^m$; cyclicity forces $m \le \dim \pi^{B,\chi}$. Frobenius reciprocity gives exactly this latter multiplicity in $M$. Complete reducibility and invariant inner products make the embedding isometric.

Since $\chi$ is real, equality in Cauchy–Schwarz in the preceding proof says $P_\chi S w_\chi = \lambda w_\chi$. Thus its embedded vector lies in the column space of $K$. Applying the displayed matrix

identity to that vector gives $\sum_n c_n \langle Sw_\chi, T^n Sw_\chi\rangle = 0$. The simple-coordinate expansion of this inner product is exactly (3.9). □

The signed permutation model on the projective line, its projectors, equality eigenspace and complete matrix-relation kernel are constructed by `research/round5/unification/equality_modrep.py`. This uses the full equality eigenspace, so it does not select an unproved irreducible constituent or a particular vector within it. There is also a direct spectral support test. If $E_r$ is the orthogonal projector onto the $\zeta_f^r$-eigenspace of $T_f$, compute

$$K^* S_f^* E_r S_f K = \frac{1}{f} \sum_{n=0}^{f-1} \zeta_f^{-rn} K^* S_f^* T_f^n S_f K.$$

If this matrix is zero, positivity implies $E_r S_f K = 0$. Thus $L_\chi(v_Z) \neq 0$ forbids $\theta_Z = \zeta_f^r$. This exact discrete Fourier calculation can remove phase variables before the moment equations are solved.

**Corollary 3.22** (Different saturated conductors are orthogonal)**.** *Let $\chi, \psi$ be saturated characters with distinct primitive conductors $f_\chi, f_\psi$. Then for every integer $n$,*

$$\sum_Z \overline{L_\chi(v_Z)} L_\psi(v_Z) \theta_Z^n = 0. \tag{3.10}$$

*Only the exponents $0 \leq n < \gcd(f_\chi, f_\psi)$ are needed.*

*Proof.* Every irreducible constituent in the cyclic span of $w_\chi$ contains a nonzero unipotent-fixed vector of torus character $\chi$. Its local congruence level is at most the conductor exponent, by Proposition 3.20, and cannot be smaller: otherwise this torus character would factor through a smaller modulus. The constituent therefore has exact conductor level $f_\chi$. Constituents belonging to distinct conductors cannot be isomorphic. Orthogonal complete reducibility of the unitary modular representation now makes the two cyclic subspaces orthogonal. In particular $\langle Sw_\chi, T^n Sw_\psi\rangle = 0$, which is the displayed formula. Every nonzero summand has both Fourier coefficients nonzero, so its twist order divides both conductors. The expression is therefore periodic with period their greatest common divisor. □

## 3.12 Different ideals and a split quadratic dimension field

The remaining dense quadratic profile requires a refinement that also controls possible larger signed dimension fields.

**Theorem 3.23** (Different-ideal obstruction)**.** *Let $K$ be the dimension-character field of the commutative signed algebra obtained by sphericalization of a commutative self-dual fusion ring. Its global dimension $\Delta$ belongs to the different $\mathfrak{D}_{K/\mathbb{Q}}$.*

*Proof.* Write $d_i$ for the signed basis dimensions, and let $e_K \in K \otimes K$ be the separability idempotent. Orthogonality of distinct Galois-conjugate characters gives

$$(\Delta \otimes 1) e_K = \sum_i d_i \otimes d_i \in \mathcal{O}_K \otimes \mathcal{O}_K.$$

For $a$ in the codifferent, contract the second factor by $x \mapsto \mathrm{Tr}(ax)$. The right side is integral, and the left side is $\Delta a$. Thus $\Delta \mathfrak{D}^{-1} \subseteq \mathcal{O}_K$, which is the assertion. □

For a local extension with ramification index $e$ and different exponent $d$,

$$\mathrm{Tr}(\mathfrak{p}^k) = p^{\lfloor (k+d)/e \rfloor} \mathbb{Z}_p.$$

In particular, every tamely ramified odd prime divides the trace of the different. A cyclic cubic extension of conductor 9 has $e = 3, d = 4$ and contributes 9. We use these standard local trace formulas and the conductor–discriminant formula; see Serre, *Local Fields*, Chapters III and VI, and Milne, *Class Field Theory*, Chapter V, Theorem 3.27.[1]

**Proposition 3.24.** *A commutative self-dual rank-six fusion ring with formal codegrees*

$$9, 9, 9, 9, \qquad 3690 \pm 576\sqrt{41},$$

*Casimir trace* 7416*, and reduction* $\mathbb{F}_2^4 \times \mathbb{F}_2[\epsilon]/(\epsilon^2)$ *satisfies the following necessary condition for categorifiability: its signed Casimir trace is* 7416*. Consequently every signed coefficient has its original absolute value.*

*Proof.* Put $E = \mathbb{Q}(\sqrt{41})$. The dimension-field containment lemma gives $E \subseteq K$, $\Delta/9 \in \mathcal{O}_K$, and

$$\frac{\Delta}{18\sqrt{41}} \in \mathcal{O}_K, \qquad 738 \mid \mathrm{Tr}_{K/\mathbb{Q}}(\Delta), \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\Delta) \leq 7416.$$

Indeed $3690 + 576\sqrt{41} = 18\sqrt{41}(32 + 5\sqrt{41})$, and the last factor is an integral unit of norm $-1$. Cyclotomicity makes $K/\mathbb{Q}$ abelian Galois, of degree 2, 4, or 6.

If $[K : \mathbb{Q}]$ is 4 or 6, then 2 splits completely in $K$. The four etale rank-one blocks of the completed signed order have generic fibre $\mathbb{Q}_2$; its only other block has rank two. A Galois number field using a completion of degree two would require at least two such local components, whereas only one rank-two block is available. The same argument excludes ramification at 2. It allows rational idempotents inside the non-etale block and does not assume that block is maximal.

Theorem 3.23 and the trace bound restrict ramified primes outside $\{3, 41\}$ to 5 and 7, and forbid their simultaneous occurrence, which would require trace at least $35 \cdot 738$. The complete list of possible larger fields follows from elementary Dirichlet-character arithmetic:

1. A biquadratic field $E\mathbb{Q}(\sqrt{m})$ requires $m > 1$ squarefree, positive, prime to 41, supported on $\{3, 5, 7\}$, and $m \equiv 1 \pmod 8$. The only such product is 105, already forbidden by its primes 5 and 7.

2. A cyclic quartic character with square the quadratic character of conductor 41 is a quartic character modulo 41 times a quadratic character of positive odd discriminant $m$. Since $2^{10} \equiv -1 \pmod{41}$, splitting at 2 requires $m \equiv 5 \pmod 8$. The permitted products leave exactly $m = 5, 21$. The conductors are $205, 861$, and the possible dimension traces are respectively $3690, 7380$ and 5166.

3. A degree-six field is $EL$ with $L$ cyclic cubic. Its conductor is 7, 9, or 63: neither 5 nor 41 admits a cubic character, and a nontrivial cubic character at 3 has conductor 9. The prime 2 is inert in the conductor-7 and conductor-9 cubic fields. Only conductor 63 remains, with dimension trace 5166.

There are no additional prime-power conductors: characters of order dividing 4 or 6 are trivial on the higher prime-power unit groups at primes other than 2 or 3; the relevant quadratic odd characters have squarefree conductor.

In the sextic case, coprimality of the discriminants gives $\mathcal{O}_{EL} = \mathcal{O}_E \otimes \mathcal{O}_L$ and a product of differents. Write

$$\Delta = 18\sqrt{41}\,\frac{a + b\sqrt{41}}{2}, \qquad a, b \in \mathcal{O}_L, \quad a \equiv b \pmod 2.$$

[1]The latter reference is available at `https://www.jmilne.org/math/CourseNotes/CFT.pdf`.

Positivity at both embeddings of $E$ over each real embedding of $L$ makes $b$ totally positive, with $\mathrm{Tr}_{L/\mathbb{Q}}(b) = 7$. At 7, the cubic different is $\mathfrak{p}_7^2$, and 18 is a unit. The different theorem therefore forces $b \in \mathfrak{p}_7^2$. Hence

$$49 \le \mathrm{Norm}(b) \le (7/3)^3 < 49,$$

a contradiction.

The cyclic quartic cases require a short exact lattice enumeration. For $M = 5, 21$, set

$$\alpha^2 = \frac{M(41 + 5\sqrt{41})}{2}, \qquad \beta^2 = \frac{M(41 - 5\sqrt{41})}{2}, \qquad \alpha\beta = 2M\sqrt{41}.$$

The defining polynomial is $X^4 - 41MX^2 + 164M^2$, and an integral basis is

$$1, \quad \frac{1 + \sqrt{41}}{2}, \quad \frac{1 + \sqrt{41} + 2\alpha}{4}, \quad \frac{1 - \sqrt{41} + 2\beta}{4}.$$

The checker verifies its integral multiplication, cyclic Galois action, complete splitting at 2, and discriminant $41^3M^2$. The preceding field list and the conductor–discriminant formula prove maximality. For $\theta = \Delta/(18\sqrt{41}) = \sum_j z_j b_j$ and trace-dual basis $b_j^*$,

$$z_j = \mathrm{Tr}\left(\Delta \frac{b_j^*}{18\sqrt{41}}\right).$$

At fixed positive trace $T$, this lies between $T$ times the smallest and largest conjugates of $b_j^*/(18\sqrt{41})$. Exact algebraic sign comparisons certify a finite integer box for every $z_j$. Imposing the trace and total positivity leaves $28, 216, 16$ candidates at traces $3690, 7380, 5166$, respectively. Every candidate fails a recorded $d$-number divisibility. The supplementary certificate contains all 260 candidates and the exact coefficient witnessing each failure.

It remains to consider $K = E$. Write $\Delta = 18(41y + x\sqrt{41})/2$. Then $x \equiv y \pmod 2$, $1 \le y \le 10$, $x^2 < 41y^2$, and

$$41y^2 - x^2 \mid 164y^2.$$

The last condition is the quadratic $d$-number test. The four complementary codegrees form either four rational integers, two rational integers and a quadratic pair, or two quadratic pairs in $E$. Rational codegrees divide the integral content of $\Delta$. A quadratic pair has polynomial $X^2 - SX + P$, with $S^2 - 4P = 41r^2$, $P \mid \mathrm{Norm}(\Delta)$, $P \mid S^2$, both roots greater than one, and both dividing $\Delta$ integrally. The total trace bounds $S$ and $r$. The reciprocal identity $\sum_f 1/f = 1$ completes the finite test. Exactly seven dimension candidates reach this stage; none admits a quadratic complementary pair. The only complete spectra have $y = 10$, $x = \pm 64$, and four codegrees 9. Their total trace is 7416, proving the proposition. □

The full tensor of `R6_8_6526` satisfies these hypotheses. Exact saturated signed associativity leaves one table up to the certified sign gauges, equivalent to the original tensor. Every non-FP original character orbit fails the arithmetic dimension-character tests. Thus a categorification would have a pseudo-unitary Galois conjugate, which the existing global-codegree moment certificate excludes. The resulting *general* exclusion is implemented in `software/round3_split_arithmetic.py`, with canonical certificate `research/round3/arithmetic/R6_8_6526_general.json`.

### 3.13 The dimension order and signed central traces

The positive-dimension induction system has two extensions that are useful when an exhaustive sphericalization search leaves a non-positive spherical character. The first sharpens an arithmetic condition without adding any central variables. The second retains the algebraic part of the same trace system, but discards every inequality that depends on positivity of categorical dimensions.

### 3.13.1 The actual order of simple dimensions

Let $d = (1, d_1, \ldots, d_{r-1})$ be a dimension character under consideration, and let $K = \mathbb{Q}(d_i)$ and $\delta = \sum_i d_i d_{i^*}$. The finitely generated group

$$\mathcal{O}_d = \sum_i \mathbb{Z} d_i \subset K$$

is an order: it contains 1, consists of algebraic integers, and is closed under multiplication because the $d_i$ satisfy the multiplication equations with integer coefficients. It need not be the maximal order of $K$.

**Proposition 3.25** (Dimension-order obstruction)**.** *If a fusion ring $R$ has a spherical categorification with simple dimensions $d_i$, then for every formal codegree $f$ of $R$,*

$$f \in K, \qquad \delta/f \in \mathcal{O}_d. \tag{3.11}$$

*The same conclusion holds for the dimensions of selected lifts in the canonical sphericalization of an arbitrary categorification of $R$. Thus the order-membership test may be applied to each dimension-character branch of a complete signed-sphericalization search.*

*Proof.* In the spherical case the central simple $A_E$ attached to an irreducible representation of $R$ with codegree $f$ has dimension $\delta/f$, by [38, Theorem 2.13]. Its forgetful image is an object of the original category. Its dimension is therefore a nonnegative integral linear combination of the $d_i$, in particular an element of $\mathcal{O}_d$. Since $\delta$ is nonzero, this also puts $f$ in $K$.

For an arbitrary categorification use its sphericalization, of dimension $2\delta$. An irreducible representation of $R$ lifts through the forgetful quotient to one with formal codegree $2f$; both selected and opposite basis lifts contribute to that codegree. The same central induction theorem gives dimension $(2\delta)/(2f) = \delta/f$. All simple dimensions of the sphericalization are among $\{d_i, -d_i\}$, so forgetting this central object gives an integral linear combination of the selected dimensions. This proves the assertion without assuming that the selected lifts themselves form a fusion subcategory. □

The test is finite and exact. Express the $d_i$ in a rational power basis of $K$, clear one common denominator, and put their coordinate lattice in Hermite normal form. Membership of $\delta/f$ is then an integer linear-algebra question. A negative certificate can equivalently give a rational linear functional $\ell : K \to \mathbb{Q}$ satisfying

$$\ell(d_i) \in \mathbb{Z} \quad \text{for all } i, \qquad \ell(\delta/f) \notin \mathbb{Z}.$$

The checker verifies the defining polynomial of $f$, reconstructs the lattice and the functional, and checks both displayed conditions. A failure on one character component is not a noncategorification proof unless all possible sphericalization and character branches are closed. In particular a failed character of a positively realized ring cannot be promoted to an exclusion of that ring.

For illustration, the complete certificate for $R^{(6)}_{4,375}$ has four oriented signed tables, all gauges of the original tensor. Its nonzero spherical components have order $\mathbb{Z}[\sqrt{5}]$ and dimensions

$$\delta = 70 + 30\sqrt{5} \quad \text{or} \quad \delta = 30 - 10\sqrt{5}.$$

The codegree 4 gives quotients outside that order in both cases. The supplied functional certificates, not this identifier or this displayed calculation, are the inputs to the automated verification.

### 3.13.2 The algebraic trace system without positive dimensions

Suppose a spherical character takes values in a real field $K$ of degree at most two, has nonzero simple dimensions, and satisfies $d_i = d_{i^*}$. Write $\delta = \sum_i d_i^2$. The restriction vectors of central simples remain nonnegative integral vectors, even when some categorical dimensions are negative. Consequently the Gram identity (4.2), its finite entry bounds, and the formal-codegree low-column equalities remain valid. They do not use an inequality between categorical dimensions.

For a commutative input let $\chi_a$ be a character label of a low simple $Z_a$ and put $\beta_a = \delta/f_a$. If $v_W$ is a candidate restriction vector, define

$$d_W = d \cdot v_W, \qquad H_{aW} = \beta_a \chi_a(v_W), \qquad \Lambda_W = (H_{aW}/d_W)_a.$$

Assume that the full ring character table is defined over $K$. The character correspondence of Theorem 4.9 gives $H_{aW} = S_{Z_a,W}$, with $S$ unnormalized. The following formulation uses no absolute value of $d_W$.

**Theorem 3.26** (Signed central-induction obstruction)**.** *A spherical categorification with the specified character supplies a nonnegative integral solution of the following finite necessary system. The unknowns count high central restriction vectors and their joint relative twist-trace types. They satisfy the Gram equations, the signed identities*

$$\sum_W (v_W)_i d_W \theta_W^j = \delta \nu_j(b_i), \tag{3.12}$$

*and the cyclic-operator bounds on the right-hand indicators. Low columns have twist one and dimension $\beta_a$. They admit at least one complete character labelling satisfying modular reciprocity.*

*All coordinates of $\Lambda_W$ are algebraic integers. Their joint Galois types have balanced multiplicities, separately for each relative twist-trace type. An absent Galois type forbids the corresponding restriction vector. Finally, if $P$ is the normalized trace from a common cyclotomic field to $K$, then*

$$\sum_W P(\theta_W) H_{aW} H_{bW} = \delta S_{Z_a,Z_b} \tag{3.13}$$

*for every pair of low labels.*

*Exhaustive failure of the system for all bounded low columns, all compatible character labellings, and all permitted second indicators excludes the specified spherical character. Exact rational Farkas certificates suffice for failure of the nonnegative rational relaxation.*

*Proof.* Adjunction proves the Gram identity with nonnegative restriction multiplicities. Central induction supplies the low dimensions and their unit multiplicities exactly as in (4.3); neither statement assumes pseudo-unitarity. The trace formula [32] proves (3.12) with the actual, possibly negative, dimensions. Bounds obtained as traces of finite-order cyclic operators on ordinary complex vector spaces do not assume positivity of categorical dimensions. Only these algebraic bounds, and not the positive low-mass inequalities, are retained here.

The finite relative-trace list is justified by the exponent-prime support theorem [5] and the same relative-trace calculation used in (4.7), with $\delta$ in place of $D$. The high columns are bounded by the positive integral Gram matrix, not by the sign of $d_W$. Their additional algebraic-integrality and $d$-number conditions follow from the spherical modular center as in the proof of Theorem 4.3. Columns of dimension zero are impossible.

Each $\Lambda_W$ is the restriction of the simultaneous character of integer fusion matrices of the center to the labelled low objects. It therefore consists of algebraic integers, and its full Galois orbit occurs with the corresponding multiplicities. For the normalized center representation

$S/\delta, T$, the permutation of columns sends $\theta_W$ to $\sigma^2(\theta_W)$; see [5, Theorem 2.9]. A square fixes the real field $K$ of degree at most two, so normalized relative traces of every twist power are unchanged. This proves the asserted balances of joint eigenvalue types. It does not identify $d_W$ with its absolute value, nor impose the positive dimension-orbit rule (4.25).

The center has Gauss sum $\delta$. Its modular relation, together with the twist-one property of low objects, is $\sum_W \theta_W S_{Z_a,W} S_{W,Z_b} = \delta S_{Z_a,Z_b}$. All coefficients other than the twists lie in $K$. Taking $P$ proves (3.13). The actual central simples give integer counts satisfying all the displayed equations. Expanding them in a rational basis of $K$ and relaxing to nonnegative rational counts can only enlarge the solution set. Farkas' lemma therefore proves the last assertion. □

**Exact scope of the implementation.** The routine first checks the complete signed character table, its multiplication equations, duality, nonzero dimensions and field degree. It enumerates low columns using Gram bounds and exact dimension equalities, without calling a positive-dimension low-mass test. It then uses signed versions of the inherited indicator rows and adds joint spectrum and low modular rows. In particular it does not reuse the absolute-value dimension balances of Proposition 4.10. Only the second-indicator rows depend on the chosen signs, and each such row depends on one sign. The checker reconstructs the all-minus and all-plus endpoint systems and selects the exact normalized row for each sign choice. Independent mixed-sign reconstructions check this optimization; every separating-vector inequality is still verified. The second indicators of self-dual objects are both allowed unless an independently checked condition forces their sign.

For $R^{(6)}_{4,446}$ the remaining non-positive component is

$$d = (1, 1, -1, -1, 2 - \sqrt{6}, 2 + \sqrt{6}), \qquad \delta = 24.$$

There are 290 low matrices. Reciprocity leaves two, each with two character labellings; all 32 choices of second indicators are retained for each labelling. The 128 exact separation certificates exclude these systems. The positive character component has the preceding character-induction obstruction. The full oriented search has sixteen gauge tables, so the assembled certificate excludes general categorification, not just this displayed spherical structure.

### 3.13.3 Application and structure-specific conclusions

Component-level rejection and general nonexistence are different conclusions. For example, rejecting one character of $R^{(6)}_{4,174}$ does not exclude that ring. Complete proof trees for the order and signed framework include

$$R^{(6)}_{4,375}, \quad R^{(6)}_{4,446}, \quad R^{(6)}_{5,1507}, \quad R^{(6)}_{5,1508}, \quad R^{(6)}_{5,1582}, \quad R^{(6)}_{6,2767}, \quad R^{(6)}_{7,1907}.$$

For the last five, shorter norm and moment certificates close the exhaustive signed alternatives, and these are retained rather than credited to a stronger criterion unnecessarily. The finite applications are specified by full tensors and exact proof trees.

The shorter pseudo-unitary certificates for $R^{(6)}_{7,3582}$ and $R^{(6)}_{7,3812}$ are intermediate ingredients. Their unrestricted existence branches are closed by the ramified signed reductions and Casimir trace criteria above; both rings are generally excluded.

## 3.14 Sphericality and Galois reduction at multiplicity three

**Proposition 3.27.** *Every complex fusion category with based ring $R^{(6)}_{3,215}$ admits a spherical structure and has a pseudo-unitary Galois conjugate. Consequently general categorifiability of this ring is equivalent to pseudo-unitary categorifiability. This reduction alone asserts neither existence nor nonexistence; the final obstruction is proved in Theorem 5.25.*

*Proof.* Apply the exhaustive oriented signed-sphericalization construction of Section 3.4, using every coefficient in its full parity interval and imposing no positive-indicator or modulo-four normalization. There are 1480 associative signed quotients. The 32 unit-preserving sign gauges partition them into 63 orbits; one orbit consists of the 32 gauges of the original fusion tensor. On representatives of the other 62 orbits there are 271 rational character components. Of these, 228 have a zero simple dimension and are impossible. For each of the other 43, the norm of the prospective global dimension has 17-adic valuation zero. The original ring has a formal codegree with minimal polynomial $t^2 - 272t + 1088$, whose norm has 17-adic valuation one. Thus the signed-sphericalization norm-divisibility condition rejects every remaining component. No nongauge quotient can occur for an actual category.

For completeness, a gauge quotient gives sphericality of the original category, not merely compatibility with spherical dimensions. If $T_{ij}^k = \eta_i\eta_j\eta_k N_{ij}^k$, replace the chosen lift of $X_i$ in the sphericalization by its tensor product with the sign object when $\eta_i = -1$. The new signed quotient has $T = N$. The sum and difference of the multiplicities of the two lifts of $X_k$ in a tensor product are then both $N_{ij}^k$, so the opposite lift has multiplicity zero. These chosen lifts therefore generate a full tensor subcategory of the sphericalization which projects tensor-equivalently onto the original category. Transport its spherical structure along that equivalence.

The four rational characters of the original ring have a zero entry, so none is a spherical dimension character. The only remaining character orbit is

$$(1, 1, d, d, d-1, d+1), \qquad d^2 - 8d - 1 = 0.$$

One of its two real embeddings is the Frobenius–Perron character, $d = 4 + \sqrt{17}$. Galois conjugation to that embedding gives a spherical category with positive Frobenius–Perron dimensions, hence a pseudo-unitary category. Conversely a pseudo-unitary categorification is a complex fusion categorification.

The complete unrestricted signed enumeration has 274655 search nodes and is frozen in `research/new_frontier/frontier_spherical_reduction.json`. The companion Python script recomputes all 1480 quotients and the complete sign orbits, verifies the original gauge orbit, checks all 271 nongauge character components and their arithmetic obstructions, and reconstructs the original dimension-character orbit using exact arithmetic. The older normalized twelve-table enumeration is retained separately as a regression control; it is not needed for this proof.

□

# 4 One central-induction certificate framework

The following subsections are blocks of one restriction-and-trace system. The Gram data and representation multiplicities are fixed first. Character labels, reciprocity, Galois orbits, indicators, and modular identities then add equations or remove impossible columns from that same system. Each block states its own scope: Frobenius–Perron positivity is used only in the pseudo-unitary blocks, whereas the signed block retains negative spherical dimensions. A Farkas vector certifies infeasibility only after all low frames, character assignments, and permitted indicator choices have been exhausted. No particular block is mandatory when a shorter inherited certificate already proves the required exclusion.

### 4.1 Central induction with representation multiplicities

Assume in this section that $\mathcal{C}$ is pseudo-unitary, with its positive spherical structure. Let $F : Z(\mathcal{C}) \to \mathcal{C}$ be forgetful and $I$ its adjoint. Put

$$B_{ji} = [FI(b_j) : b_i] = \sum_{a,k} N_{aj}^{k} N_{k,a^*}^{i}. \tag{4.1}$$

This is the induction Gram matrix. In a commutative ring it agrees with the regular Casimir. In a noncommutative ring one must use (4.1), not silently substitute the Casimir.

For a central simple $Z$, write $v_Z = ([F(Z) : b_i])_i$. Adjunction gives

$$B = \sum_{Z \in \mathrm{Irr}\, Z(\mathcal{C})} v_Z v_Z^{\mathsf{T}}. \tag{4.2}$$

The summands of $I(\mathbf{1})$ are indexed by irreducible representations $E$ of $R \otimes \mathbb{C}$:

$$I(\mathbf{1}) = \bigoplus_E r_E A_E, \quad (v_{A_E})_0 = r_E, \quad \dim A_E = D/f_E, \quad \theta_{A_E} = 1. \tag{4.3}$$

These are the formal-codegree and induction statements of [38, Theorems 2.5,2.13].

**Theorem 4.1** (Multiplicity-weighted low induction)**.** *For each $E$ let $v_E \in \mathbb{Z}_{\geq 0}^{6}$ satisfy*

$$(v_E)_0 = r_E, \qquad d \cdot v_E = D/f_E, \qquad \sum_E r_E v_E = B_{0,*}. \tag{4.4}$$

*A necessary condition for a pseudo-unitary categorification is that at least one such collection has a nonnegative integral Gram completion*

$$Q = B - LL^{\mathsf{T}} = \sum_{Z \not\subset I(\mathbf{1})} v_Z v_Z^{\mathsf{T}}, \qquad L = (v_E)_E, \tag{4.5}$$

*and satisfies the central-indicator necessary conditions of [13, 11]. In particular, $Q$ is entrywise nonnegative and positive semidefinite. The columns of $L$ occur once each in $LL^{\mathsf{T}}$; they are not repeated $r_E$ times.*

*For a noncommutative rank-six ring the low unit multiplicities are $1, 1, 2$. Equations* (4.4) *and* (4.5)*, with these multiplicities, give a finite exhaustive low-column search.*

*Proof.* The first two identities of (4.4) are (4.3). Restricting $I(\mathbf{1})$ proves the third. Separating the distinct simples occurring in $I(\mathbf{1})$ from the sum (4.2) proves (4.5). This also explains why $r_E$ weights the row-sum equation but not the number of copies of a column in $L$.

Finiteness follows from $(v_E)_i^2 \leq B_{ii}$ and $r_E(v_E)_i \leq B_{0i}$. Thus each entry belongs to an explicitly bounded integer interval. All dimension equalities are checked in the exact number field. The usual trace identities involve only restriction multiplicities and categorical dimensions, so their proofs and the finite relative-trace construction do not require commutativity of $R$ once the correct low data and Gram matrix are supplied. Finally a noncommutative semisimple rank-six algebra with a dimension character has type (3.5), giving the stated multiplicities. □

In the implementation the one-dimensional character subspace is computed before selecting a Frobenius–Perron embedding. The unique matrix-block eigenvalue $c$ of the Casimir gives $f_E = c/2$, and the identity

$$\det(tI - C_R) = (t - c)^4 \det(tI - C_R|_{\mathrm{linear}})$$

is checked. A positive gate from Proposition 3.1(iv) is required before a pseudo-unitary exclusion is promoted to a general one. The five higher-multiplicity noncommutative Haagerup-type tensors in the census are excluded in this way. The multiplicity-one Haagerup ring and the group ring of $S_3$ are retained positive controls.

## 4.2 Balanced finite-indicator systems

The principal positive-dimension test is a single relative-trace construction. We strengthen it without introducing a new case-specific obstruction. Throughout this section suppose all dimensions lie in $K = \mathbb{Q}$ or $K = \mathbb{Q}(\sqrt{q})$, with $q > 1$ squarefree in the latter case. The low matrix $L$ is one of the possibilities of Theorem 4.1.

### 4.2.1 The inherited finite system

For a simple $b_i$, the Ng–Schauenburg identity [32] is

$$\sum_Z (v_Z)_i \, \dim(Z)\, \theta_Z^k = D\nu_k(b_i). \tag{4.6}$$

The first indicator vanishes for nonunit simples, and the second is zero for non-self-dual simples and $\pm 1$ for self-dual simples. Higher indicators satisfy the bounds used in the finite-indicator theorem of [11]. We use exactly that theorem, including its permitted indicator choices and rational inequalities, rather than assuming unknown indicators have prescribed values.

For a chosen height $h$, take joint normalized relative traces of $(\theta, \dots, \theta^h)$ to $K$. The exponent prime support is controlled by the categorical Cauchy theorem [5]. Let $P$ be the prime divisors of the norm of $D$, and let $f$ be the conductor of $K$ (take $f = 1$ for $K = \mathbb{Q}$). The trace vectors are covered by roots whose orders divide

$$M_h = \prod_{p \in P} p^{\max\{1, v_p(f)\} + \lfloor \log_p h \rfloor}, \tag{4.7}$$

together with the zero vector. As proved in [11], a higher prime-power layer contributes zero to every one of the first $h$ traces. Thus this is not a bound imposed on the actual twist orders. Ramanujan and quadratic Gauss sums compute the finite trace vectors exactly. Extra vectors may be retained as a relaxation; they cannot produce a false exclusion.

An admissible high column has zero unit entry, a nonnegative integral tail $v$, and $Q - vv^{\mathsf{T}} \succeq 0$. Entrywise Gram bounds and the algebraic-integrality condition $D/\dim(v) \in \mathcal{O}_K$ further restrict it. We also require $\dim(v)$ to be a $d$-number: the center is braided, so its simple squared norms are $d$-numbers by [37, Theorem 1.8]; in the positive spherical structure these are the squares of the dimensions, and [37, Corollary 2.9] removes the square. In a quadratic field this is the exact condition that the square of the trace divided by the norm is integral. Denote the finite list by $\mathcal{V}$, and the finite joint trace list by $\mathcal{T}$. Variables $x_{v,\tau} \geq 0$ give the number of high simples with column $v$ and trace type $\tau$; the linear relaxation allows nonintegral values. Gram identities, (4.6), and the cited higher-indicator inequalities give an integer linear system

$$Ax = b, \qquad x \geq 0, \tag{4.8}$$

after adding slack variables and clearing denominators. The code reconstructs this system from the tensor, the low matrix, and the indicator choices.

### 4.2.2 Duality and Galois balance

Write $v_i^* = v_{i^*}$. Low simples are stable under duality, so the high part is stable as well. Duality preserves the ribbon twist: $\theta_{Z^*} = \theta_Z$, not $\theta_Z^{-1}$. Consequently the actual integer counts satisfy

$$x_{v,\tau} = x_{v^*,\tau} \qquad (v \in \mathcal{V},\ \tau \in \mathcal{T}). \tag{4.9}$$

If a partner column is absent from the list, its variable is zero.

For the quadratic case let $\sigma(\sqrt{q}) = -\sqrt{q}$ and define, for a positive central dimension $\beta \in K$,

$$\gamma(\beta) = \left| \frac{D}{\sigma(D)} \sigma(\beta) \right|, \tag{4.10}$$

where absolute value is at the fixed real embedding. Let $c_\beta$ be the number of *distinct* low central simples of dimension $\beta$. The low multiplicity $r_E$ in $I(\mathbf{1})$ is not included in this count.

**Lemma 4.2** (Dimension-count balance)**.** *Galois symmetry permutes the central simples with dimension $\beta$ onto those with dimension $\gamma(\beta)$. Hence their high counts satisfy*

$$\sum_{\dim(v)=\beta,\tau} x_{v,\tau} - \sum_{\dim(v)=\gamma(\beta),\tau} x_{v,\tau} = c_{\gamma(\beta)} - c_\beta. \tag{4.11}$$

*Moreover $\gamma^2(\beta) = \beta$.*

*Proof.* Extend $\sigma$ to a cyclotomic field of the center. Its normalized $S$-matrix is $s = S/D$, since the center has global dimension $D^2$. Signed Galois symmetry gives

$$\sigma(s_{0Z}) = \epsilon_Z s_{0,\widehat{\sigma}(Z)}, \qquad \epsilon_Z \in \{1, -1\}.$$

The entry $s_{0Z}$ equals $\dim(Z)/D$. All central dimensions are positive. Taking absolute values therefore gives $\dim \widehat{\sigma}(Z) = \gamma(\dim Z)$, proving the permutation statement. Subtract the known low counts to obtain (4.11). Finally the map $u \mapsto (D/\sigma D)\sigma(u)$ squares to the identity. Taking absolute values commutes with changing a real sign, so the positive version squares to the identity as well. See also the squared-dimension form of Galois symmetry in [30, Equation (2.4)]. □

The factor $D/\sigma(D)$ and the absolute value in (4.10) are both essential. Omitting either does not give a valid positive dimension permutation in general.

**Theorem 4.3** (Balanced finite-indicator criterion)**.** *Append* (4.9) *and, when quadratic,* (4.11) *to the finite system of [11]. If for every admissible low matrix and every allowed second-indicator choice the resulting system has an exact Farkas certificate*

$$A^{\mathsf{T}} y \geq 0, \qquad b^{\mathsf{T}} y < 0, \tag{4.12}$$

*then there is no pseudo-unitary categorification. Coupled with a valid general-to-positive reduction, this excludes general complex categorification.*

*At a fixed height and with the same admissible low and high columns, this criterion contains the former unbalanced finite-indicator test: every old separating functional extends by zero on the appended rows.*

*Proof.* An actual category supplies integral nonnegative counts satisfying the inherited system and both new balances, by duality and Lemma 4.2. But (4.12) would imply $0 > b^{\mathsf{T}} y = x^{\mathsf{T}} A^{\mathsf{T}} y \geq 0$. The last assertion follows by appending zero coordinates to $y$. □

This containment justifies using the balanced theorem as the common finite-indicator framework. It does not imply that every earlier non-indicator criterion is redundant. In particular, the cheap arithmetic filters and signed reductions remain useful.

### 4.2.3 An integral refinement and its scope

For a high-column list $v_1, \dots, v_s$, enumerate nonnegative integers $m_j$ with

$$Q = \sum_{j=1}^{s} m_j v_j v_j^{\mathsf{T}}. \tag{4.13}$$

This is finite because each nonzero $v_j$ has a positive diagonal entry. For each frame append $\sum_\tau x_{v_j,\tau} = m_j$ to the balanced system. Exhaustive infeasibility for every frame would be a further valid obstruction. We implement this refinement principally to diagnose the frontier. The feasible point in Section 7.2 survives it. We do not count this unproductive refinement as an independent successful exclusion criterion.

## 4.3 Cyclic-character compatibility for tensor powers

The preceding indicator system bounds different higher indicators separately. There is additional compatibility: several of them are traces of powers of a single finite-order linear operator. This gives extra linear equations inside the same exact-certificate framework.

**Lemma 4.4** (Block rotation)**.** *In a pivotal fusion category, let $E_X^{(n)}$ be cyclic rotation on* $\mathrm{Hom}(1, X^{\otimes n})$*. For every positive divisor $r$ of $n$,*

$$(E_X^{(n)})^n = \mathrm{id}, \qquad \mathrm{Tr}((E_X^{(n)})^r) = \nu_{n/r}(X^{\otimes r}) = \sum_i [X^{\otimes r} : X_i]\nu_{n/r}(X_i). \tag{4.14}$$

*Here the trace is the ordinary trace on the displayed Hom space.*

*Proof.* Pivotal strictification preserves the rotation operators. In the strict pivotal setting, equation (5.4) in [33] states that consecutive rotations of adjacent tensor blocks equal the rotation of their tensor product. Induction gives $(E_X^{(n)})^r = E_{X^{\otimes r}, X^{\otimes(n-r)}}$, exactly the identity used in the proof of [33, Theorem 5.1]. If $r \mid n$, group the factors into $n/r$ consecutive blocks of length $r$. This identifies the operator with one rotation on the blocks. Taking traces gives the first indicator equality; the final equality is additivity of higher indicators on a direct sum. For $r = n$ it is the identity operator, with trace $[X^{\otimes n} : 1]$. □

Fix the dimension field $K$, the low columns, and the second indicators as in Section 4.2. For a positive integer $n$, let $\mathcal{U}_{K,n}$ be the finite set of vectors

$$(P_K(\zeta^r))_{r|n,\ r<n}, \qquad \zeta^n = 1,$$

where $P_K$ is normalized relative trace from a common cyclotomic extension to $K$. This definition is independent of enlarging the common extension. The vectors are computed by the same Ramanujan and quadratic Gauss sums used for the center twists, but their orders now divide $n$ exactly. In particular no unbounded-order approximation occurs in this auxiliary set.

**Theorem 4.5** (Joint tensor-rotation criterion)**.** *In a pseudo-unitary categorification with simple dimensions in $K$, for each simple $X$ and each $3 \le n \le h$ there are nonnegative integers $y_{X,n,u}$, $u \in \mathcal{U}_{K,n}$, such that*

$$\sum_u y_{X,n,u} = [X^{\otimes n} : 1], \tag{4.15}$$

$$\sum_u y_{X,n,u} u_r = \sum_i [X^{\otimes r} : X_i] P_K(\nu_{n/r}(X_i)) \qquad (r \mid n,\ r < n). \tag{4.16}$$

*Together with the center trace identities, these are linear equations in the existing high-center counts and the new $y$ variables after multiplication by $D$ and expansion in a rational basis*

*of $K$. Infeasibility of the nonnegative real relaxation is an obstruction to pseudo-unitary categorification. Combined with any proved sphericalization or pseudo-unitarity reduction, it is a general categorification obstruction.*

*Proof.* The operator in Lemma 4.4 is diagonalizable over $\mathbb{C}$ and has $n$th roots of unity as eigenvalues. Group its eigenvalues, with multiplicity, by their joint relative-trace vectors. This gives nonnegative integer counts satisfying (4.15) and (4.16). Use (4.6) to express each indicator on the right. Its denominator is the known scalar $D$, while its numerator is the known low contribution plus a linear combination of the high-center variables. Since $P_K$ is $K$-linear, multiplying by $D$ gives linear equations with coefficients in $K$. Clearing rational denominators gives an integer system. An integer solution would also be a nonnegative real solution; an exact separating functional therefore proves the claimed obstruction. □

For clarity, let $\ell_i$ be the dimension of the known low part of $I(X_i)$, with $\ell_0 = D$, and let $(v, \tau)$ index a high variable. The equation appended for a divisor $r$ is exactly

$$D\sum_u y_{X,n,u}u_r - \sum_{v,\tau} x_{v,\tau}\Big(\sum_{i>0}[X^{\otimes r} : X_i]v_i\Big)\dim(v)\tau_{n/r} = \sum_i [X^{\otimes r} : X_i]\ell_i. \tag{4.17}$$

This formula specifies the implemented rows without a graphical normalization ambiguity. The unit indicator is one, and unit-free high columns have unit entry zero.

The criterion strengthens the existing system by adding equations and variables with nonnegative constraints; no previously valid inequality is dropped. Old dual certificates extend by zero. Its height-six application excludes $R^{(6)}_{2,55}$, the self-dual Klein-four quadratic ring with two noninvertibles of dimension $2+\sqrt{6}$. The complete ten low-column alternatives and every permitted second-indicator choice are checked. The two-Frobenius–Perron-dimension reduction supplies the general scope. No separate theorem declaring this particular ring impossible is used by the program.

The cyclic-four ring $R^{(6)}_{2,4}$ has the same dimension type but does not satisfy that exclusion certificate. Section 7.2 records an exact integer-count feasible point for the stronger system. Thus neither dimension-type identification nor a failed numerical optimization can replace full-tensor testing.

## 4.4 Consequences of a proved twist exponent

Suppose the actual twists of $Z(\mathcal{C})$ have order dividing a proved integer $N$. In this subsection $\mathcal{C}$ is pseudo-unitary, $D = \mathrm{FPdim}(\mathcal{C})$, and all indicators are actual indicators, before relative trace projection. The center formula makes $\nu_k(X)$ periodic in $k$ modulo $N$. It also gives

$$|\nu_k(X)| \le D^{-1}\sum_Z [F(Z) : X]d_Z = d_X. \tag{4.18}$$

The dimension identity on the right is $\mathrm{FPdim}\, I(X) = Dd_X$.

**Proposition 4.6** (Full rotation spectra and prime recovery)**.** *Let $h_n(X) = [X^{\otimes n} : \mathbf{1}]$.*

1. *If $n \mid N$, the entire character of rotation on $\mathrm{Hom}(\mathbf{1}, X^{\otimes n})$ is determined by the actual center variables. Its value at the identity is $h_n(X)$. For $0 < r < n$, put $d = \gcd(n, r)$ and $m = n/d$. Then*

$$t_r = \mathrm{Tr}((E^{(n)}_X)^r) = \sigma_{r/d}\Big(\nu_m(X^{\otimes d})\Big), \qquad c_e = \frac{1}{n}\sum_{r=0}^{n-1} t_r\zeta_n^{-er} \in \mathbb{Z}_{\ge 0} \quad (0 \le e < n). \tag{4.19}$$

*Here $\sigma_{r/d}$ acts on $\mathbb{Q}(\zeta_m)$; any unit lift modulo $N$ gives the same answer.*

*2. If $p \nmid N$ is prime, then*

$$\nu_p(X) \in \mathbb{Z}, \qquad \nu_p(X) \equiv h_p(X) \pmod{p}, \qquad |\nu_p(X)| \le d_X. \tag{4.20}$$

*Consequently, for an integer bound $B_X \ge d_X$ and a prime $p > 2B_X$, the residue $h_p(X) \bmod p$ allows at most one value of $\nu_p(X)$ in $[-B_X, B_X]$. If no such value exists, categorification is impossible. Values obtained from primes in the same residue class modulo $N$ must agree.*

*Proof.* Lemma 4.4 identifies $(E_X^{(n)})^d$ with rotation on $m$ blocks of length $d$. This operator has $m$th roots as eigenvalues. Raising it to the power $r/d$, which is coprime to $m$, applies the cyclotomic automorphism $\sigma_{r/d}$ to its trace. This proves the first formula. The reduction map from units modulo $N$ to units modulo $m \mid N$ is surjective, and its kernel fixes $\mathbb{Q}(\zeta_m)$, so the choice of lift is immaterial. Fourier inversion gives the actual eigenvalue multiplicities $c_e$, proving their integral positivity.

For a prime $p \nmid N$, the indicator is both a rotation trace in $\mathbb{Q}(\zeta_p)$ and a center trace in $\mathbb{Q}(\zeta_N)$. The center field statement follows from its canonical congruence representation [49, Theorem II]. These fields intersect in $\mathbb{Q}$, so it is a rational algebraic integer. A rational character of $C_p$ has equal multiplicity $b$ at each primitive $p$th root. If the multiplicity at one is $a$, then $h_p = a + (p-1)b$ and $\nu_p = a - b = h_p - pb$. This proves the congruence; (4.18) proves the bound. An interval of length $2B_X < p$ contains at most one integer in a specified residue class. Finally, periodicity modulo $N$ proves the compatibility between primes. All $h_p \bmod p$ are obtained from based-ring matrix powers modulo $p$; a maximum row sum of a fusion matrix supplies an integer Perron–Frobenius bound $B_X$. No indicators are guessed. □

The same argument shows why an averaged primitive-root bucket cannot always be treated as an arbitrary integer count: for $p \nmid N$ its total multiplicity is divisible by $p-1$. Nonnegative real relaxations lose this divisibility, but exact modular row certificates retain it.

**Proposition 4.7** (Coprime higher Gauss sums of the center)**.** *Write $\tau_n = \sum_Z d_Z^2 \theta_Z^n$. For every $n$ coprime to $N$,*

$$\tau_n = \frac{D^2}{\sigma_{n^{-1}}(D)}, \tag{4.21}$$

*where the inverse is modulo $N$.*

*Proof.* Use the canonical anomaly-one representation of the center, so that $\tau_1 = D$, $S_{0Z} = d_Z/D$, and $T_{ZZ} = \theta_Z$. The signed Galois permutation $\pi_n$ satisfies

$$\sigma_n(d_Z)^2 = \frac{\sigma_n(D)^2}{D^2} d^2_{\pi_n(Z)}, \qquad \theta_{\pi_n(Z)} = \theta_Z^{n^2}$$

by the Galois formulas of [49, Theorem II]. Hence

$$\sigma_n(\tau_n) = \frac{\sigma_n(D)^2}{D^2} \sum_Z d^2_{\pi_n(Z)} \theta_{\pi_n(Z)} = \frac{\sigma_n(D)^2}{D}.$$

Applying $\sigma_{n^{-1}}$ gives (4.21). These are exact linear equations in the actual restriction-and-twist counts. They do not require an independent choice of higher central charges. □

## 4.5 Character-aware central induction

The dimension-only induction matrix forgets an important part of the formal-codegree correspondence: each of its low columns has a specified *character*, not only a specified dimension. Retaining these characters adds exact modular reciprocity and a joint Galois-spectrum condition to the same finite-indicator system. No classification of modular data, numerical eigenvalue decision, or enumeration of center fusion rings is required.

### 4.5.1 The character attached to a low column

Let $\mathcal{C}$ be pseudo-unitary, let $R = K_0(\mathcal{C})$ be commutative of rank $n$, and write $d_i = \mathrm{FPdim}(X_i)$ and $D = \sum_i d_i^2$. For a character $\chi : R \to \mathbb{C}$, put

$$f_\chi = \sum_i \chi(X_i)\chi(X_i^*), \qquad \beta_\chi = D/f_\chi.$$

Use the unnormalized modular matrix $S$ of $Z(\mathcal{C})$, so that $S_{1W} = \dim W$. The correspondence preceding [38, Theorem 2.13] identifies the simple summand $Z_\chi$ of $I(1)$ by the equality of characters

$$\chi(FW) = \frac{S_{Z_\chi, W}}{\beta_\chi} \qquad (W \in \mathrm{Irr}\, Z(\mathcal{C})). \tag{4.22}$$

Indeed, pull back $\chi$ along $F$. The resulting character of the modular fusion ring is its unique normalized $S$-row, and [38, Theorem 2.13] identifies the row label, its dimension $\beta_\chi$, and its multiplicity one in $I(1)$. Thus this statement uses the character-level correspondence, not an arbitrary choice among central simples of equal dimension.

For an integral restriction vector $v = (v_i)$ write $d(v) = \sum_i v_i d_i$ and $\chi(v) = \sum_i v_i \chi(X_i)$. If $a_\chi$ is the restriction vector of $Z_\chi$, symmetry of $S$ gives

$$\boxed{\beta_\chi \chi(a_\psi) = \beta_\psi \psi(a_\chi)}. \tag{4.23}$$

The unit column is labelled by the Frobenius–Perron character. When formal codegrees repeat, every dimension-compatible assignment of the remaining distinct characters must be considered. Selecting one assignment without exhausting the others would not be an obstruction.

### 4.5.2 Joint Galois closure

The next elementary spectral observation is useful well beyond the present induction problem.

**Lemma 4.8** (Joint spectral closure)**.** *Let $B$ be a commutative fusion ring and $b_1, \dots, b_s \in B$. The multiset*

$$\{(\varphi(b_1), \dots, \varphi(b_s)) : \varphi \in \mathrm{Irr}(B \otimes \overline{\mathbb{Q}})\}$$

*is invariant under every automorphism of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$. Consequently a finite set containing all these tuples must contain every Galois conjugate of each tuple that actually occurs.*

*Proof.* The multiplication equations of $B$ have integral coefficients. Composing a character with a field automorphism gives another character, and the inverse automorphism makes this operation a permutation. Equivalently, the commuting integer multiplication matrices have a Galois-invariant joint spectrum. Restriction to the specified elements preserves the multiset, including repeated restricted tuples. □

For a nonzero prospective restriction vector $v$ define

$$\Lambda(v) = \left( \frac{\beta_\chi \chi(v)}{d(v)} \right)_{\chi \in \mathrm{Irr}\, R}. \tag{4.24}$$

If $v = FW$ for a central simple $W$, then (4.22) and symmetry give

$$\frac{\beta_\chi \chi(v)}{d(v)} = \frac{S_{W, Z_\chi}}{\dim W}.$$

These are the joint eigenvalues of the integer fusion matrices of $Z_\chi$ evaluated at the character labelled by $W$. In particular, Lemma 4.8 applies to the *whole tuple*. Checking its coordinates separately would be weaker.

Fix a candidate low matrix $L$. Let $\mathcal{H}_L$ be any finite set containing all possible high restriction columns for that low matrix; the Gram and algebraic-integrality construction in Section 3.1 and the finite-indicator implementation supplies such a set. Let $\mathcal{A}_L$ be its union with the low columns, and set

$$\mathcal{T}_L = \{\Lambda(v) : v \in \mathcal{A}_L\}.$$

Choose a finite Galois field $K$ containing all character values and all $d_i$. Every coordinate of every tuple belongs to $K$. A column $v$ is *forbidden* if $\sigma(\Lambda(v)) \notin \mathcal{T}_L$ for some $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. A forbidden low column rejects $L$; a forbidden high column must have multiplicity zero. Notice that (4.24) uses the fixed character order and $D/f_\chi$, so this test is independent of the permutation used to label equal-dimensional low columns.

**Theorem 4.9** (Character-aware finite-indicator obstruction)**.** *For every low matrix of a pseudo-unitary categorification with commutative Grothendieck ring:*

*(i) there is a dimension-compatible bijection from its columns to* $\mathrm{Irr}\, R$ *satisfying* (4.23)*;*

*(ii) none of its low columns is forbidden by the joint Galois test;*

*(iii) the finite-indicator system remains feasible after all variables supported on forbidden high columns are set to zero.*

*Therefore the absence of an assignment in (i), failure of (ii), or an exact Farkas obstruction to (iii), for every candidate low matrix and every permitted second-indicator choice, excludes a pseudo-unitary categorification.*

*Proof.* The formal-codegree correspondence proves (i). Every actual central restriction belongs to $\mathcal{A}_L$. Lemma 4.8, applied to the low central simples, proves (ii) and the asserted zero multiplicities. The finite-indicator variables group actual central simples by restriction and joint relative twist trace. Every variable with a forbidden restriction is therefore zero; the remaining actual multiplicities satisfy the inherited equations and inequalities. After slack variables are inserted these have the form $Ax = b$, $x \geq 0$. A rational vector $y$ with $A^{\mathsf{T}}y \geq 0$ and $b^{\mathsf{T}}y < 0$ is impossible, because $b^{\mathsf{T}}y = x^{\mathsf{T}}A^{\mathsf{T}}y \geq 0$. All ordinary low-matrix rejections remain applicable before these additional tests. □

This is a strengthening of the existing finite-indicator framework, not a competing list of special forbidden rings. The implementation uses exact splitting fields of degree at most sixteen generated by quadratic eigenvalues of one or two ring generators. This is an implementation eligibility bound, not a hypothesis of the theorem. Every proposed character is checked against all multiplication equations; the complete set has cardinal $n$. Field automorphisms are constructed by splitting the primitive-element polynomial over $K$ and checking that all of its distinct roots have been obtained. The verifier reconstructs all labellings, all possible high columns, the forbidden set, and every rational separating vector. It never calls the numerical optimizer used to discover a vector.

### 4.5.3 General categorification and the multiplicity-two completion

Two illustrative tensors give a compact reproducibility summary:

| Ring | Low matrices | Reciprocity survivors | General reduction |
|---|---|---|---|
| $R^{(6)}_{2,20}$ | 77 | 2 | two FP dimensions |
| $R^{(6)}_{2,41}$ | 84 | 2 | two exhaustive signed tables |

For $R^{(6)}_{2,20}$, both surviving matrices and both permitted second indicators have exact height-six obstructions. The two-FP-dimension Galois reduction of Proposition 3.1 makes this a general

exclusion. For $R^{(6)}_{2,41}$, one reciprocity survivor is already rejected by the inherited height-six equations. For the other, there are 104 possible high columns; joint Galois closure forbids 66, and the restricted height-six system has an exact separating vector.

For this latter tensor the complete oriented signed enumeration has exactly two tables. Both are sign gauges of the original tensor, with sign vectors $(1,-1,-1,1,1,1)$ and $(1,1,1,1,1,1)$ in census order. The ordinary spherical-character test excludes every character outside the FP Galois orbit by a zero dimension or unequal dimensions on a dual pair. That orbit is excluded by Theorem 4.9. The signed-sphericalization reduction already established in the preceding sections therefore excludes *general* categorification, without assuming the conjectural existence of pivotal structures. These statements are recomputed from the tensors in the certificate, including the completeness of the oriented enumeration.

Table 2 records 23 certified examples of this common character-aware mechanism. Their stored proof trees include the complete sphericalization or Galois reduction required for general nonexistence. In particular, the signed calculation for $R^{(6)}_{3,132}$ is detailed in Section 4.5.4. The certificate controls recompute these applications without a numerical optimizer. This illustrative collection is not a separate allocation of the entire census: the current sequential counts, eligibility outcomes and bounded-search diagnostics appear in Section 7.1 and its machine-readable ledger.

| Exact multiplicity | Example certificates |
|---:|---:|
| 2 | 2 |
| 3 | 2 |
| 4 | 6 |
| 5 | 3 |
| 6 | 10 |
| Total | 23 |

Table 2: Illustrative character-aware induction certificates for Theorem 4.9, generated from the supplied exact proof examples. These are examples of the mechanism, not a disjoint allocation of the current census.

The multiplicity-three applications are detailed in Section 4.5.4. The complete consecutive classification range now reaches multiplicity three by the stabilizer-rotation argument of Theorem 5.25. The counting bound eight is not presented as a categorification bound.

### 4.5.4 Modular equations retained inside the indicator system

We describe one further strengthening of Theorem 4.9. It uses the same high-column and relative-trace variables; it does not require enumerating an unknown center fusion ring. Its purpose here is also to specify exactly which stronger necessary conditions the former multiplicity-three residual satisfies before the stabilizer constraint is imposed.

Assume that all ring characters and positive dimensions lie in a real field $K$ of degree at most two, and let $P$ be the normalized relative trace from a common cyclotomic field to $K$. For a central simple $W$ write $v_W = FW$, $d_W = d(v_W)$, and

$$H_{aW} = S_{Z_a,W} = \beta_a \chi_a(v_W).$$

Here $Z_a$ ranges over the labelled low objects. Thus $H_{aW}$ denotes an entry of the unnormalized matrix $S$.

**Proposition 4.10** (Joint dimension, twist and low modular constraints)**.** *In addition to the preceding induction conditions, every pseudo-unitary categorification satisfies the following statements.*

*Every coordinate of $\Lambda(v_W)$ is an algebraic integer. Under a Galois automorphism $\sigma$, the pair consisting of a simple dimension and its low fusion-eigenvalue tuple is permuted according to*

$$(d_W, \Lambda(v_W)) \longmapsto \left( \left| \frac{D}{\sigma(D)} \sigma(d_W) \right|, \sigma(\Lambda(v_W)) \right). \tag{4.25}$$

*This permutation preserves every joint relative twist-trace type*

$$(P(\theta_W), P(\theta_W^2), \ldots, P(\theta_W^h)).$$

*Finally, for any two low labels $a, b$,*

$$\sum_W P(\theta_W) H_{aW} H_{bW} = D S_{Z_a, Z_b}. \tag{4.26}$$

*Thus the finite indicator system may impose equality of multiplicities within each joint orbit in* (4.25)*, separately for each trace type, and may impose all equations* (4.26)*. A missing orbit member forbids a candidate column; an exact infeasibility certificate for the enlarged system is a pseudo-unitary obstruction.*

*Proof.* The entries of $\Lambda(v_W)$ are eigenvalues of integer fusion matrices, so they are algebraic integers. Use the canonical normalized modular representation of the center, namely $S/D$ and $T$. Its Galois symmetry gives

$$\sigma(S_{0W}/D) = \epsilon_\sigma(W) S_{0,\widehat{\sigma W}}/D, \qquad \Lambda(v_{\widehat{\sigma W}}) = \sigma(\Lambda(v_W)),$$

where $\epsilon_\sigma(W) \in \{1, -1\}$; see [5, equations (2.12)–(2.13)]. Positivity of all simple Frobenius–Perron dimensions gives exactly (4.25), including its absolute value.

For this canonical center normalization, the Galois symmetry of $T$ is $\theta_{\widehat{\sigma W}} = \sigma^2(\theta_W)$, by [5, Theorem 2.9(b)]. A square of a Galois automorphism fixes $K$, since its degree is at most two. The normalized trace $P$ is invariant under automorphisms over $K$, and cyclotomic Galois groups are abelian. Consequently $P(\theta_{\widehat{\sigma W}}^j) = P(\theta_W^j)$ for every $j$. The simultaneous dimension-and-eigenvalue permutation therefore gives multiplicity balances within each trace type, including the low objects in the twist-one type. This statement concerns multiplicities of tuples; it does not assert that a restriction vector itself is Galois invariant.

The positive Gauss sum of the center is $D$. Its modular relation is

$$D S_{Z_a, Z_b} = \theta_{Z_a} \theta_{Z_b} \sum_W \theta_W S_{Z_a, W} S_{W, Z_b},$$

by [5, equation (2.6)]. Both low twists are one. All the other coefficients in this equation belong to $K$, by the character formula. Applying $P$ proves (4.26). The actual central simples satisfy the resulting rational equations and nonnegative multiplicity constraints. An exact Farkas obstruction therefore applies as in Theorem 4.9. □

The code reconstructs all finite Galois conjugates, checks algebraic integrality by integer characteristic polynomials, and adds coefficient rows over $\mathbb{Q}$. This is one refinement of the common induction system, not an independently allocated family of forbidden rings. In particular, the final general exclusion of $R_{3,132}^{(6)}$ already has a certificate using Theorem 4.9; no stronger criterion is credited when an inherited one suffices.

### 4.5.5 Exact multiplicity-three outcome and the higher scan

The general certificate for $R_{3,132}^{(6)}$ has the following finite structure. Its positive type is $(1, 1, d, d+1, d+1, d)$, with $d = 4 + \sqrt{21}$, and its global dimension is $168 + 36\sqrt{21}$. There are 172 low matrices. Modular reciprocity rejects 168; the other four have exact height-six indicator certificates after the joint-spectrum exclusions. The general sphericalization reduction exhausts sixteen oriented signed tables. Eight are sign gauges of the original table; for the other eight, all character branches are rejected by the checked norm or zero-dimension tests. These are certificate statistics reconstructed from the full tensor. They are not premises selected by its identifier.

Corollary 6.12 supplies the unitary model for $R_{3,218}^{(6)}$, and Theorem 5.25 excludes $R_{3,215}^{(6)}$. This completes multiplicity three. The earlier feasible induction and modular data for the latter ring remain regression controls: they demonstrate the strict need for the additional stabilizer-rotation condition, and are not categorical models.

The same framework applies at every tested multiplicity. In particular, non-positive spherical branches are treated by Section 3.13; a positive-dimension obstruction alone would not establish general nonexistence.

## 4.6 Ambient cyclotomic traces for quartic dimension fields

The relative-trace method does not require the dimension field to be quadratic. One may instead embed it into a cyclotomic field and retain all coefficients of the resulting normalized traces.

**Proposition 4.11** (A finite ambient trace domain)**.** *Let $L = \mathbb{Q}(\zeta_F)$ contain all simple dimensions of a pseudo-unitary fusion category, and let $S$ be the prime support supplied by the categorical Cauchy theorem. Suppose every prime dividing $F$ lies in $S$. For height $h$, put*

$$B = \prod_{p \in S} p^{\max\{v_p(F),1\}+\lfloor \log_p h \rfloor}.$$

*Every joint normalized relative-trace vector*

$$\left( \frac{\mathrm{Tr}_{L(\theta)/L}(\theta^j)}{[L(\theta) : L]} \right)_{j=1}^{h}$$

*of a root of unity whose order has prime support in $S$ is either zero, or occurs among roots of unity of order dividing $B$.*

*Proof.* If $p^a$ divides the order with $a > \max\{v_p(F), 1\} + \lfloor \log_p h \rfloor$, then for each $1 \le j \le h$ the $p$-primary part of $\theta^j$ has order strictly larger than $p^{\max\{v_p(F),1\}}$. Summing over the last degree-$p$ cyclotomic layer gives zero: its conjugates differ by the complete set of $p$th roots of unity. This layer fixes $L$ and the other primary components, so transitivity of trace gives zero in $L$. Otherwise the order divides $B$. Since $F \mid B$, all remaining traces can be computed inside $\mathbb{Q}(\zeta_B)$ by averaging the automorphisms $\zeta_B \mapsto \zeta_B^a$ with $a \equiv 1 \pmod F$. □

Use each pair consisting of an admissible high restriction column $v$ and a trace vector as a nonnegative count variable. The Gram equations remain integral. Apply normalized relative trace to the central indicator formula of [32, Theorem 4.1]; the low induction objects have twist one. Expanding in the power basis of $L$ gives rational linear equations for the first three moments. The usual second indicator choices are exhausted. When $3 \nmid F$ and $3 \notin S$, the third indicator lies both in $\mathbb{Q}(\zeta_3)$ and in a cyclotomic field of conductor coprime to three, hence is rational. Its cyclic multiplicities give the necessary description

$$\nu_3(Y) = a - b/2, \qquad a, b \ge 0, \qquad a + b = \dim \mathrm{Hom}(\mathbf{1}, Y^3).$$

The count $b$ includes both primitive third-root eigenspaces. For a four-dimensional cubic space with the anticommuting stabilizer of Proposition 5.21, impose in addition $\nu_3(Y) \le 1$. These conditions are linear after adding nonnegative slack variables. Allowing rational multiplicities enlarges the necessary system and is therefore legitimate for exclusion.

### 4.7 Full-character traces and low Verlinde compatibility

The full character field can contain useful information that disappears under projection to the Frobenius–Perron dimension field. We retain this information by applying the ambient trace construction of Proposition 4.11 to a cyclotomic field containing all characters. The following formula makes this extension inexpensive.

**Lemma 4.12** (Direct cyclotomic relative trace)**.** *Let $K = \mathbb{Q}(\zeta_F)$ and let $\theta = \zeta_m^b$ have exact order $m$. Put $d = \gcd(m, F)$ and $r = m/d$. If $\gcd(r, F) > 1$, the normalized relative trace of $\theta$ to $K$ is zero. Otherwise it is*

$$\frac{\mathrm{Tr}_{K(\theta)/K}(\theta)}{[K(\theta) : K]} = \frac{\mu(r)}{\varphi(r)}\,\zeta_d^{br^{-1}},$$

*where the inverse is taken modulo $d$, and the last factor is $1$ when $d = 1$.*

*Proof.* If a prime divides both $r$ and $F$, its exponent in $m$ exceeds its exponent in $F$. The last cyclotomic layer at this prime fixes $K$ and the other primary components; the corresponding primitive root sum is zero. Transitivity of normalized trace proves the first assertion. Otherwise $r$ is coprime to $F$, and hence to $d$. The Chinese remainder identity writes

$$\zeta_m^b = \zeta_d^{br^{-1}}\zeta_r^{bd^{-1}}.$$

The first factor lies in $K$. Averaging the second factor over the cyclotomic Galois group gives the primitive root sum $\mu(r)$ divided by $\varphi(r)$. Coprimality ensures that adjoining $K$ does not change this average. □

Apply the lemma separately to the powers $\theta^j$. With the finite bound $B$ of Proposition 4.11, this computes the complete joint trace domain directly in $K$, without constructing $\mathbb{Q}(\zeta_B)$. An independent implementation compares this formula with explicit Galois averaging in compositum fields: all 1056 exact comparisons, covering 176 conductor–order pairs, agree.

**Proposition 4.13** (Full-character moment compatibility)**.** *Suppose a commutative fusion ring has a pseudo-unitary categorification. Let $D$ be its global dimension and let $K = \mathbb{Q}(\zeta_F)$ contain its full character table. For every possible low induction frame, the nonnegative counts of high restriction columns and joint twist traces satisfy the following conditions:*

(i) *the induction Gram identities and the central indicator formulas;*

(ii) *for every simple $X$, every tested order $n$, and every proper divisor $r$ of $n$, the cyclic rotation character identity*

$$\mathrm{Tr}(E_{X,n}^r) = \nu_{n/r}(X^r), \qquad \dim \mathrm{Hom}(\mathbf{1}, X^n) = [X^n : \mathbf{1}];$$

(iii) *the low-pair modular equations and the integrality and Galois compatibility of the joint fusion eigenvalues;*

(iv) *equality of the multiplicities of paired types $(\lambda, \tau)$ and $(\sigma\lambda, \sigma^2\tau)$, where $\lambda$ is the joint low fusion eigenvalue vector and $\tau$ the joint normalized twist-trace vector.*

*Thus exact infeasibility of all these finite systems excludes a pseudo-unitary categorification.*

*Proof.* The Gram identities and the indicator formula are the established induction constraints; the latter are applied coefficientwise in $K$ using [32, Theorem 4.1]. The cyclic rotation operator has $n$th roots of unity as eigenvalues, and the divisor-power identity follows by grouping the tensor factors into blocks of length $r$; see [33]. Introduce nonnegative variables for these rotation eigenvalue multiplicities and apply normalized relative trace. The resulting equations are rational after expansion in a power basis.

For low central objects $A, B$, both twists are one, so the center identity $STS = DT^{-1}ST^{-1}$ gives the low-pair equations. Their joint eigenvalues are algebraic integers and occur with Galois-invariant multiplicities. Under a Galois automorphism the fusion eigenvalues transform by $\sigma$, whereas the canonical center twists transform by $\sigma^2$. Normalized relative trace commutes with these automorphisms, proving the paired-type equality. If an ambient automorphism requires a lift to a compositum, the resulting normalized trace is independent of that lift. The twist-one low objects contribute the known constant terms. Relaxing integer multiplicities to nonnegative real variables enlarges the set of possible solutions, so an exact separating functional proves the claimed necessary obstruction. □

There is a complementary refinement using low fusion products. Write $v_Z$ for the restriction column of a central simple $Z$ and $d(v) = \sum_i v_i \operatorname{FPdim}(b_i)$. For a low central simple $A_a$ associated to a character $\chi_a$ of formal codegree $f_a$, put

$$t_a(v) = \frac{D}{f_a}\frac{\chi_a(v)}{d(v)}.$$

Then $t_a(v_Z) = S_{A_a,Z}/d_Z$.

**Corollary 4.14** (Self-dual low-product constraints)**.** *For self-dual input, all low central simples are self-dual. The general word and restriction formulas in Section 4.8 therefore apply with $a^* = a$ and real $\lambda_a(Z) = t_a(v_Z)$. In particular the cubic moments are the low fusion coefficients, and summing the nonnegative high coefficients gives the restriction-product equations and their coordinatewise upper bounds.*

*Proof.* The input character values are real. Hence the corresponding low $S$-rows are real, and $S_{a^*,Z} = \overline{S_{a,Z}}$ together with invertibility of $S$ forces $a^* = a$. The assertion follows from the general formulas, retaining the same bounded high coefficient variables. □

The rotation multiplicities remain integers after grouping eigenvalues by joint normalized-trace tuples: each grouped count is a sum of eigenspace dimensions. The same applies to counts of central simples with a fixed restriction column and joint twist-trace tuple. The implementation may retain these integer variables and test congruences. Only rows with integer-count variables are used for a congruence witness. For the next two rings, nonnegative linear infeasibility already suffices: each stored ray $y$ satisfies $y^T A \geq 0$ and $y^T b < 0$ over the integers. No numerical optimizer is used by the final verifier.

**Theorem 4.15.** *The rings $R^{(6)}_{4,211}$ and $R^{(6)}_{4,243}$ have no complex fusion categorification.*

*Proof.* For $R^{(6)}_{4,211}$, the simple dimensions are $(1, 1, d, 2d+1, 2d+1, d)$, where $d = 2+\sqrt{6}$, and $D = 120 + 48\sqrt{6}$. Its full character field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, so take $F = 24$. At height three the Cauchy prime support $\{2, 3\}$ gives $B = 144$ and 53 joint trace types. The exhaustive low enumeration has 130 frames. Only two frames have relevant character assignments: each has two assignments and four second-indicator choices. All 16 systems are excluded by exact rays from Proposition 4.13. Here even the weaker equality of total joint-type Galois multiplicities suffices; the coupled trace refinement is not needed.

For $R^{(6)}_{4,243}$, the dimensions are $(1, 1, d, d+1, d+1, d)$ with $d = 7+\sqrt{57}$ and $D = 456+60\sqrt{57}$. The full character field is quadratic. The height-six systems use Proposition 4.14, including

restriction-product bounds. There are 210 low frames. Four relevant frames contribute respectively two, one, one, and two character assignments, each with eight second-indicator choices. The resulting 48 systems all have exact separating rays.

Finally, general scope is supplied independently by unrestricted signed sphericalization. For the two rings the complete signed enumerations have respectively 128 and 1584 tables, in 4 and 53 sign-gauge orbits. Exactly one orbit in each case is gauge isomorphic to the original tensor; the remaining character branches are rejected by respectively 8 and 176 exact arithmetic obstructions. The surviving nonzero dimension characters belong to the Frobenius–Perron Galois orbit. Thus any complex categorification would admit a pseudo-unitary Galois conjugate, contradicting the preceding exclusions. □

The full certificates and exact replay reports are retained in `research/round3/m4`. The verifiers `software/round3_m4.py` and `software/round3_verlinde.py` reconstruct the complete branch domains from the input tensors.

### 4.8 Joint restriction and word Gram realization

The cubic Verlinde, higher-word and integer-square conditions are different projections of one necessary integral Gram factorization. Retaining the restriction coordinates in that factorization makes it substantially stronger without introducing new categorical assumptions.

**Lemma 4.16** (Simple restriction support)**.** *Let $F : \mathcal{B} \to \mathcal{C}$ be a tensor functor between fusion categories. Suppose a proposed complete list of restriction columns for simples of $\mathcal{B}$ is available. Call a column low if its tensor-unit coordinate is positive, and let $v$ be the restriction column of a simple object $Z$. Suppose $\mathbf{1}_{\mathcal{B}}$ is the only simple object whose restriction column has positive tensor-unit coordinate and is componentwise bounded by the based-ring product $vv^*$. Then the tensor-unit coordinate of $vv^*$ is one. Moreover, there is at most one simple of $\mathcal{B}$ having restriction column $v$.*

*Proof.* Simplicity gives $[ZZ^* : \mathbf{1}_{\mathcal{B}}] = 1$. Every simple constituent of $ZZ^*$ has restriction componentwise bounded by $vv^*$. Only the tensor unit can contribute to the unit coordinate of $F(ZZ^*)$, so that coordinate is one. If distinct simples $Z, W$ had restriction $v$, then $[ZW^* : \mathbf{1}_{\mathcal{B}}] = 0$, whereas the unit coordinate of $F(ZW^*)$ would be positive. A nonunit low constituent would be required, contrary to the hypothesis. □

Let $\mathcal{C}$ be pseudo-unitary, put $D = \mathrm{FPdim}(\mathcal{C})$, and write $F : Z(\mathcal{C}) \to \mathcal{C}$ for the forgetful functor. Fix distinct low central simples $A_0 = \mathbf{1}, A_1, \dots, A_{\ell-1}$, their duality permutation $a \mapsto a^*$, and their restriction columns $u_a = [F(A_a)]$. We assume that their fusion eigenvalues

$$\lambda_a(Z) = \frac{S_{A_a,Z}}{S_{\mathbf{1},Z}}$$

are known from the restriction column $v_Z = [F(Z)]$. Here $S$ is normalized and $S_{\mathbf{1},Z} = d(v_Z)/D$, where $d(v) = \sum_i v_i \,\mathrm{FPdim}(X_i)$. For a commutative input ring and the character-labelled low simples of canonical induction, the existing character formula gives

$$\lambda_a(v) = \frac{D}{f_a}\frac{\chi_a(v)}{d(v)},$$

where $\chi_a$ has formal codegree $f_a$. The low simples are distinguished objects even when their restriction columns coincide. Fix a complete high restriction profile: for every admissible nonzero column $v$, its multiplicity is $m_v$.

For a word $P = A_{a_1} \cdots A_{a_k}$, set $\lambda_P(v) = \prod_j \lambda_{a_j}(v)$. The empty word is the unit. Dualizing a word dualizes each index; the order is immaterial in the commutative central Grothendieck ring.

**Theorem 4.17** (Joint restriction–word Gram condition)**.** *Choose finitely many words $P_1, \dots, P_s$ in the low simples. Their pairings and their low-simple coefficients are determined by the profile:*

$$H_{pq} = \sum_{Z \in \operatorname{Irr} Z(\mathcal{C})} \frac{d(v_Z)^2}{D^2} \overline{\lambda_{P_p}(v_Z)} \lambda_{P_q}(v_Z), \tag{4.27}$$

$$L_{pa} = \sum_{Z} \frac{d(v_Z)^2}{D^2} \overline{\lambda_a(v_Z)} \lambda_{P_p}(v_Z). \tag{4.28}$$

*These numbers are nonnegative integers. Let $r_p$ be the restriction of $P_p$ after its low constituents are removed, and put*

$$G = H - LL^{\mathsf{T}}, \qquad Q = \sum_v m_v v v^{\mathsf{T}}, \qquad B_{pi} = (r_p)_i.$$

*There are nonnegative integral vectors $x_Z = ([P_p : Z])_{p=1}^s$ for the high central simples such that*

$$\begin{pmatrix} Q & B^{\mathsf{T}} \\ B & G \end{pmatrix} = \sum_{Z \text{ high}} \begin{pmatrix} v_Z \\ x_Z \end{pmatrix} \begin{pmatrix} v_Z \\ x_Z \end{pmatrix}^{\mathsf{T}}. \tag{4.29}$$

*In particular $G$ and the joint block matrix are positive semidefinite, have nonnegative integral entries, and each $x_Z$ satisfies*

$$x_Z \in \operatorname{im}_{\mathbb{Q}} G, \qquad 0 \le (x_Z)_p \le \min \left\{ \lfloor \sqrt{G_{pp}} \rfloor, \min_{i:(v_Z)_i > 0} \left\lfloor \frac{(r_p)_i}{(v_Z)_i} \right\rfloor \right\}.$$

*Proof.* In a commutative based ring the matrices of a simple and its dual are transposes and commute. They are therefore normal and simultaneously diagonalizable, with $\lambda_{a^*}(Z) = \overline{\lambda_a(Z)}$. The Verlinde formula in the modular center, whose global dimension is $D^2$, gives

$$[P : \mathbf{1}] = \sum_Z \frac{d(v_Z)^2}{D^2} \lambda_P(v_Z).$$

Apply this identity to $P_p^* P_q$ and to $A_a^* P_p$. Rigidity identifies the resulting numbers with $\dim \operatorname{Hom}(P_p, P_q)$ and $[P_p : A_a]$, respectively. This proves (4.27) and (4.28), including their integral positivity. It also explains why complex conjugation cannot be omitted when the low simples are not self-dual.

Decompose each $P_p$ into its low and high constituents. The orthonormality of the simple basis for the Frobenius pairing gives $G_{pq} = \sum_{Z \text{ high}} [P_p : Z][P_q : Z]$. Applying the tensor functor $F$ to the same decomposition gives $r_p = \sum_{Z \text{ high}} [P_p : Z] v_Z$. These are the lower-right and mixed blocks of (4.29); its upper-left block is the fixed high induction profile. Thus the displayed matrix is an integral Gram matrix.

For a real matrix $X$, the images of $X$ and $XX^{\mathsf{T}}$ agree. Apply this to the matrix of coefficient vectors $x_Z$. Since all entries are rational, each $x_Z$ lies in the rational image of $G$. A diagonal entry of $G$ is a sum of their squared coordinates, giving the square-root bound. Each coordinate of $r_p$ is a sum of nonnegative integers $(x_Z)_p (v_Z)_i$, giving the restriction bound. □

Equivalently, writing $X$ for the matrix with columns $x_Z$, the lower and mixed blocks are

$$G = XX^{\mathsf{T}}, \qquad \sum_{Z \text{ high}} (x_Z)_p v_Z = r_p. \tag{4.30}$$

All terms in the theorem are computed from the input tensor, a complete profile, a character labelling and the chosen words. Thus a nonintegral pairing, a negative entry or an impossible finite Gram factorization rejects that specified profile. No hand choice of central fusion coefficients is part of the criterion.

### 4.8.1 Exact finite domains and conditional ellipsoids

The coordinate bounds already make the coefficient-vector search finite. A stronger bound exploits the whole Gram matrix, including its mixed block.

**Lemma 4.18** (Exact rank-one Gram domains)**.** *Let $M$ be a rational positive-semidefinite matrix, and choose a principal submatrix $M_{JJ}$ of size* $\operatorname{rank} M$ *that is invertible. For a rational vector $z$, the condition $M - zz^{\mathsf{T}} \succeq 0$ is equivalent to*

$$z = M_{\bullet J} M_{JJ}^{-1} z_J, \qquad z_J^{\mathsf{T}} M_{JJ}^{-1} z_J \le 1.$$

*Consequently every column in a Gram factorization belongs to this exact ellipsoid in the rational image of $M$.*

*Proof.* Positivity of $M - zz^{\mathsf{T}}$ forces $z$ to be orthogonal to $\ker M$, hence to lie in $\operatorname{im} M$. On that image, use the independent coordinates indexed by $J$. Congruence reduces the assertion to positive definiteness of $M_{JJ}$ and positivity of $M_{JJ} - z_J z_J^{\mathsf{T}}$. Conjugating by the positive square root of $M_{JJ}^{-1}$ gives $I - ww^{\mathsf{T}} \succeq 0$, equivalently $w^{\mathsf{T}} w \le 1$, proving the formula. All matrices required for checking the final formula are rational; the square root is only used in this proof. □

For the joint matrix in (4.29), fix the restriction column $v$. Choose independent principal coordinates $J$ of $Q$, and put

$$\begin{aligned} c_v &= B_{\bullet J} Q_{JJ}^{-1} v_J, \\ C &= G - B_{\bullet J} Q_{JJ}^{-1} B_{\bullet J}^{\mathsf{T}}, \\ \rho_v &= 1 - v_J^{\mathsf{T}} Q_{JJ}^{-1} v_J. \end{aligned} \tag{4.31}$$

The necessary image identities for $v$ and $B^{\mathsf{T}}$ are checked first. Block Gaussian congruence then turns the joint matrix into $\operatorname{diag}(Q, C)$ and its column $(v, x)$ into $(v, x - c_v)$. Lemma 4.18 therefore gives

$$x - c_v \in \operatorname{im}_{\mathbb{Q}} C, \qquad (x - c_v)_K^{\mathsf{T}} C_{KK}^{-1} (x - c_v)_K \le \rho_v,$$

where $K$ is any independent principal coordinate set for $C$. If $C = 0$, this simply forces $x = c_v$. A negative $\rho_v$ is impossible. This calculation uses the inverse of an explicit nonsingular rational matrix, not a numerical pseudoinverse.

To enumerate the domain, select independent coordinates of $x$ itself. The image condition expresses all other coordinates as rational affine functions of these, and the theorem bounds each free coordinate in a finite integer interval. During enumeration, interval bounds for the unassigned coordinates discard a branch only when some reconstructed coordinate cannot lie in its prescribed interval. In addition, any assigned prefix of independent coordinates satisfies its own principal ellipsoid inequality: it is obtained by minimizing over the remaining real coordinates. A prefix violating that exact rational inequality can also be discarded. At each leaf check integrality, every coordinate bound, the full ellipsoid inequality, and the componentwise inequality $zz^{\mathsf{T}} \le M$. Every admissible integer column follows a unique unpruned path, and every retained leaf satisfies all stated constraints. A node or time limit interrupts this enumeration and supplies no completeness claim.

Let $\mathcal{Q}_v$ be the resulting complete finite domain for restriction type $v$. Its zero vector is retained whenever allowed: a high central simple may occur in none of the selected words. Introduce nonnegative integer multiplicities $z_{v,x}$, for $x \in \mathcal{Q}_v$. The remaining necessary conditions are the integer linear equations

$$\sum_{x \in \mathcal{Q}_v} z_{v,x} = m_v, \qquad \sum_{v,x} z_{v,x} x x^{\mathsf{T}} = G, \qquad \sum_{v,x} z_{v,x} x_p v = r_p. \tag{4.32}$$

The upper-left block follows from the first equations and the fixed profile, so these enforce the complete joint Gram factorization. Exact rational separation or an exact integer obstruction rejects a profile. A solution supplies compatible coefficients for the selected words, and makes no categorical existence assertion.

The optimizer-free implementation is `software/round5_unified.py`. It uses the same theorem for quadratic, cubic or longer words, retains complex duality, and records incomplete domains separately. Exact unimodular changes of coordinates may shorten the ellipsoid enumeration without changing its integral points.

Categorical controls independently derive the centers of $\mathrm{Vec}_{C_2}$, $\mathrm{Vec}_{C_3}$ and $\mathrm{Rep}(S_3)$. For the last, finite centralizer character sums give its eight-simple center and exact Verlinde gives all 512 fusion coefficients. Every selected quadratic or cubic nonunit low word has a nonzero high part. Its actual integral assignments satisfy the joint Gram and coupled balancing systems; replacing all twists by one violates two product-balancing rows. The complex pointed control separately detects omission of the dual index. These optimizer-free controls accompany `verify_domains.py`, `pointed_center_control.py` and `s3_center_control.py` in `research/round5/unification/`.

### 4.8.2 Completing the action of the low simples

**Proposition 4.19** (Finite word-action completion)**.** *Expand an integral solution of* (4.32) *into individual proposed central simples. Let $V$ be their restriction matrix, including the low columns, and let $u(P)$ be the complete coefficient vector of each retained word. If these data come from a categorification, there exist nonnegative integral matrices $N_a$, with $(N_a)_{ZY} = [A_aY : Z]$, satisfying*

$$N_a u(P) = u(A_aP), \qquad N_a e_0 = e_a, \qquad N_{a^*} = N_a^{\mathsf{T}}, \qquad N_aN_b = N_bN_a,$$
$$VN_a = N_{F(A_a)}V,$$

*where the word equation is imposed whenever both words are retained. Each entry is bounded by the restriction product:*

$$0 \le (N_a)_{ZY} \le \min_{i:(v_Z)_i>0} \left\lfloor \frac{([F(A_a)]v_Y)_i}{(v_Z)_i} \right\rfloor.$$

*For every $k \ge 1$, one also has the unweighted spectral identity*

$$\mathrm{Tr}(N_a^k) = \sum_{Z \in \mathrm{Irr}\, Z(\mathcal{C})} \lambda_a(v_Z)^k. \tag{4.33}$$

*Equivalently, if the proposed center has rank $r$, the first $r$ trace identities specify the characteristic polynomial*

$$\det(tI - N_a) = \prod_Z (t - \lambda_a(v_Z)).$$

*If duality on the proposed simple objects has been selected, rigidity also gives $(N_a)_{ZY} = (N_a)_{Y^*Z^*}$. In particular every $N_a$ is normal, since it commutes with $N_{a^*} = N_a^{\mathsf{T}}$. Consequently completion of the selected low fusion actions is a finite exact problem for each integral word-coefficient assignment.*

*Proof.* Multiplication in the central Grothendieck ring gives the word equation and the unit column. Rigidity gives $N_{a^*} = N_a^{\mathsf{T}}$; braidedness gives commutativity. Applying the forgetful tensor functor to each product gives the restriction equation. Its terms are nonnegative, so the contribution of any single constituent is bounded in every positive coordinate of its nonzero restriction column. This proves the finite entry bound.

The regular fusion representation is diagonalized by the modular $S$-matrix. Its eigenvalues for multiplication by $A_a$ are $\lambda_a(v_Z)$, one for every central simple $Z$, counted with the multiplicities in the complete profile. Taking traces proves (4.33); no dimension-square weight occurs here. Newton's identities give the equivalent characteristic polynomial. The further entry identity is the Frobenius reciprocity equality $[A_aY : Z] = [A_aZ^* : Y^*]$. The entry bounds make the matrix domain finite. All displayed identities can therefore be checked exactly after exhaustive bounded enumeration. □

A contradiction excludes only the coefficient assignment being completed; to exclude its restriction profile, all integral coefficient assignments must be covered. Conversely, compatible low actions need not specify products of every pair of high simples. Even a full based ring and modular datum still require a categorical construction before an existence claim.

**Corollary 4.20** (Unit-column saturation)**.** *Let $\mathcal{A} \subset \mathrm{Mat}_r(\mathbb{Q})$ be the rational algebra generated by proposed known fusion actions and the identity. If they extend to a fusion ring on the proposed simple basis, then $\mathrm{ev}_0(M) = Me_0$ is injective on $\mathcal{A}$. Moreover,*

$$Me_0 \in \mathbb{Z}^r \Longrightarrow M \in \mathrm{Mat}_r(\mathbb{Z}), \qquad Me_0 \in \mathbb{Q}^r_{\geq 0} \Longrightarrow M \in \mathrm{Mat}_r(\mathbb{Q}_{\geq 0}).$$

*Both conditions admit finite exact tests before unknown fusion actions are reconstructed.*

*Proof.* Every $M$ in the generated algebra is multiplication by the element $z$ whose coefficient vector is $Me_0$. Thus $M = \sum_i z_i N_i$. This gives injectivity and both implications. Compute a rational basis of $\mathcal{A}$ and its unit-column image. Injectivity is a rank test. A basis of the saturated lattice $\mathrm{ev}_0(\mathcal{A}) \cap \mathbb{Z}^r$, obtained by exact integer normal forms, suffices to test integrality of its lifted matrices. The nonnegative cone of unit columns is rational polyhedral; its finitely many generating rays suffice to test entrywise positivity. Equivalently, each possible negative matrix entry gives a rational linear feasibility problem in that cone. □

### 4.8.3 The earlier moment conditions as projections

For $P = A_aA_b$, formula (4.28) is precisely $N^{A_c}_{A_a,A_b}$, with the factor $\lambda_{c^*}$ required in the complex case. The corresponding diagonal entry of $H$ is

$$\dim \mathrm{Hom}(A_aA_b, A_aA_b) = \sum_Z (N^Z_{A_a,A_b})^2. \tag{4.34}$$

In the earlier cubic notation this says

$$t_{abc} = \frac{1}{D^2} \sum_Z d(v_Z)^2 \lambda_a(v_Z)\lambda_b(v_Z)\lambda_c(v_Z), \qquad N^{A_c}_{A_a,A_b} = t_{abc^*}. \tag{4.35}$$

Writing $q_{ab,v} = \sum_{Z:\, v_Z = v} N^Z_{A_a,A_b}$ for high types, the restriction identity is

$$u_au_b = \delta_{a,b^*}u_0 + \sum_{c \neq 0} t_{abc^*}u_c + \sum_{v \text{ high}} q_{ab,v}v. \tag{4.36}$$

The off-diagonal entries couple different products, so retaining only these individual squared norms loses necessary constraints. For integers $0 \leq n \leq M$ and $k \geq 0$,

$$n^2 - (2k+1)n + k(k+1) = (n-k)(n-k-1) \geq 0, \qquad n^2 \leq Mn.$$

Summing these inequalities over a restriction type recovers the integer-square cuts. They are valid relaxations of (4.32), rather than separate categorical principles. The vectorized full Morita-action Gram factorization uses the same finite-domain lemma after its own categorical Gram identity has been established.

There is also a useful exact variance bound, omitting zero-multiplicity types, when the type totals $T_{pv} = \sum_{Z:\, v_Z = v} (x_Z)_p$ are known. Write $T_{pv} = m_v q_{pv} + r_{pv}$ with $0 \le r_{pv} < m_v$. Then

$$G_{pp} \ge \sum_v \bigl(m_v q_{pv}^2 + r_{pv}(2q_{pv} + 1)\bigr). \tag{4.37}$$

Indeed, among $m_v$ nonnegative integers with fixed sum, the smallest sum of squares occurs when their values differ by at most one: replacing $a, b$ with $a - 1, b + 1$ decreases the sum whenever $a \ge b + 2$. Subtracting the type averages also proves $G - T \operatorname{diag}(m_v^{-1}) T^{\mathsf{T}} \succeq 0$. If this matrix is zero, every coefficient vector in a restriction type equals that type's average; their integrality is then forced.

If an additional word has zero Schur complement against the previously selected words, its high coefficient vector is a forced rational linear combination of their vectors. Checking its nonnegative integrality therefore strengthens the finite domains without adding a free coordinate. This follows directly from the image identity in Lemma 4.18, applied after adjoining that word.

### 4.8.4 Coupling product coefficients to projected twists

The same coefficient vectors can be coupled to the finite twist-trace types, imposing balancing on individual products rather than only aggregate modular moments. Care with duality is essential here. Our normalized $S$ follows the modular-representation convention of [5, equation (2.3)]; its balancing identity is

$$D\theta_a \theta_b S_{a,b} = \sum_Z N^Z_{A_a^*, A_b} \theta_Z d(v_Z).$$

Thus, for low simples, whose twists are one, the product $A_a A_b$ has right-hand side $DS_{a^*,b}$, not $DS_{a,b}$ in general.

**Proposition 4.21** (Projected product balancing)**.** *Let $K$ contain the dimensions and the known low $S$-entries, and let $P$ be normalized relative trace from a common cyclotomic field to $K$. Refine the coefficient counts in* (4.32) *to $z_{v,x,\tau}$, where $\tau = (P(\theta_Z), \dots, P(\theta_Z^h))$ is a permitted joint trace type. If the selected words include $P_p = A_a A_b$, then*

$$\sum_{v,x,\tau} z_{v,x,\tau}\, x_p\, d(v) \tau_1 = DS_{a^*,b} - \sum_{c \text{ low}} L_{pc} d(u_c). \tag{4.38}$$

*These equations, the Gram and restriction equations, and every indicator or Galois-multiplicity equation must use the same count variables, with their respective marginals.*

*Proof.* Apply the balancing identity to $A_a^*, A_b$; the double dual has the same simple label as $A_a$. Separate the low constituents, whose twists are one. The other coefficients and the dimensions belong to $K$, so normalized relative trace replaces only $\theta_Z$ by $\tau_1$. Grouping the central simples by $(v, x, \tau)$ proves the formula. Summing over $\tau$ recovers the coefficient counts, while summing over $x$ recovers the original restriction/twist counts. Consequently all earlier necessary equations hold for these marginals. Using independent variables for the two systems would forget the compatibility asserted here. □

Equation (4.38) concerns a product of two specified simple objects with known twists. It does not attach a scalar twist to an arbitrary longer word. Nor is a normalized trace type silently treated as an actual root of unity.

**Lemma 4.22** (Actual twists from an extremal projected trace)**.** *Let $\tau_k$ be the normalized relative trace of $\theta_Z^k$ to a retained cyclotomic field. If $\tau_k^m = 1$, then $\theta_Z^k = \tau_k$ and $\theta_Z^{km} = 1$. Consequently, if every occupied projected twist type has such an extremal coordinate, all actual lifts of that projected assignment can be enumerated in a finite root-of-unity group.*

*Proof.* At the chosen complex embedding the normalized trace is the average of the conjugates of $\theta_Z^k$, all on the unit circle. Since $|\tau_k| = 1$, equality in the triangle inequality forces every conjugate to equal $\tau_k$, including $\theta_Z^k$ itself. The root bound follows. For several occupied types, take the least common multiple of their resulting exponents and check every power-trace coordinate of every root in that group. Integer compositions of each type's multiplicity enumerate all multisets of actual lifts. □

Before combining actual roots into projected trace types, retain their individual dimension capacities. In the pseudo-unitary setting, put

$$M_X(\lambda) = \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}} \sum_{\substack{W \in \operatorname{Irr} Z(\mathcal{D}) \\ \theta_W = \lambda}} d_W [F_{\mathcal{D}}(W) : X].$$

Corollary 4.31 and positivity give, for every central simple of restriction $v$ and twist $\lambda$,

$$d(v) v_X \le M_X(\lambda) \qquad (X \in \operatorname{Irr} \mathcal{D}). \tag{4.39}$$

A projected column/type pair is therefore impossible if none of its allowed actual roots satisfies all these inequalities. Coincident relative traces do not add the capacities available to a single object. More generally, the individual-root mass equations can be imposed as an exact integral allocation problem for the proposed columns. These are direct consequences of the same projection identity, independent of any choice of simultaneous word coefficients.

## 4.9 Lifted central subcategories and adjunction Gram matrices

Projection to a fusion subcategory constrains each central column and its twist. When the induction data additionally force an actual nondegenerate fusion subcategory of the center, its entire action can be retained.

**Proposition 4.23** (Central packets and restricted adjunction)**.** *Let $\mathcal{B} \subset Z(\mathcal{C})$ be a nondegenerate braided fusion subcategory and let $\mathcal{E} = \mathcal{B}'$ be its centralizer. For $b \in \operatorname{Irr} \mathcal{B}$ and $Y \in \operatorname{Irr} \mathcal{E}$, put $w_b = [F(b)]$ and $v_Y = [F(Y)]$. The complete central restriction profile is the disjoint union of the packets*

$$\{w_b v_Y : b \in \operatorname{Irr} \mathcal{B}\}, \qquad Y \in \operatorname{Irr} \mathcal{E}.$$

*Their twists are $\theta_b \theta_Y$. Let $T = F|_{\mathcal{E}}$, let $I$ be its right adjoint, and set $A = I(\mathbf{1})$. With restriction matrix $V_{iY} = [T(Y) : X_i]$ and fusion convention $(N_A)_{XY} = [Y \otimes A : X]$, one has*

$$V^{\mathsf{T}} V = N_A. \tag{4.40}$$

*Neither this identity nor its proof requires $T$ to be dominant.*

*Proof.* The nondegenerate centralizer decomposition gives a braided equivalence $\mathcal{B} \boxtimes \mathcal{E} \simeq Z(\mathcal{C})$; see [16]. Its simple objects are precisely the pairs $(b, Y)$. Tensoriality of $F$ gives their restriction $w_b v_Y$, and the ribbon product gives their twist. Coincident restriction columns do not identify the corresponding simple pairs.

Semisimplicity gives the right adjoint $I$. Rigidity, tensoriality of $T$ and adjunction yield, naturally in $X, Y \in \mathcal{E}$,

$$\begin{aligned} \operatorname{Hom}_{\mathcal{E}}(X, I(T(Y))) &\cong \operatorname{Hom}_{\mathcal{C}}(T(X), T(Y)) \\ &\cong \operatorname{Hom}_{\mathcal{C}}(T(Y^* \otimes X), \mathbf{1}) \\ &\cong \operatorname{Hom}_{\mathcal{E}}(Y^* \otimes X, A) \\ &\cong \operatorname{Hom}_{\mathcal{E}}(X, Y \otimes A). \end{aligned}$$

Thus $I(T(Y)) \cong Y \otimes A$. For simple $X, Y$, the dimension of the second space is the scalar product of their restriction columns, while the dimension of the last space is $(N_A)_{XY}$, proving (4.40). □

For a forced Ising subcategory with simples $1, g, \sigma$, a packet has columns $v, gv, \sigma v$ and twists $\lambda, -\lambda, z\lambda$, with the appropriate primitive sixteenth root $z$. In particular the rank of the center is divisible by three. All packet orientations and all inherited Ising choices must be covered. This factors the center; it does not assert a tensor-product decomposition of $\mathcal{C}$.

The algebra $A$ is the $\mathcal{E}$-part of canonical unit induction: for simple $Y \in \mathcal{E}$, $[A : Y] = \dim \mathrm{Hom}_{\mathcal{C}}(F(Y), \mathbf{1})$. Consequently, when the packet assignments force $A = \mathbf{1} \oplus c$, the full fusion matrix of $c$ is already known: $N_c = V^{\mathsf{T}}V - I$. Every other fusion matrix must commute with it, have its prescribed unit column, and satisfy $N_{Y^*} = N_Y^{\mathsf{T}}$. Since $\mathcal{E}$ is braided, these matrices are normal and mutually commuting. Together with the restriction identities, these give finite exact generator constraints before any modular-data completion. No converse identity for $VV^{\mathsf{T}}$ is asserted: it does not generally equal multiplication by $F(A)$.

**Corollary 4.24** (Grading separates the adjunction summands)**.** *Suppose $\mathcal{E}$ has a known group grading and $I(\mathbf{1}) = \mathbf{1} \oplus \bigoplus_a A_a$, where the nonunit simple summands $A_a$ have pairwise distinct grading degrees. Write $\gamma_Y$ for the degree of a simple $Y$. Then*

$$(N_{A_a})_{ZY} = (V^{\mathsf{T}}V - I)_{ZY}[\gamma_Z = \gamma_{A_a}\gamma_Y].$$

*Thus the individual fusion matrices are fixed by the restriction matrix and a proposed assignment of degrees to simples.*

*Proof.* The adjunction identity gives $V^{\mathsf{T}}V - I = \sum_a N_{A_a}$. Grading homogeneity permits a contribution in position $(Z, Y)$ only from the summand with degree $\gamma_Z\gamma_Y^{-1}$; by hypothesis there is at most one such summand. This proves the formula. □

Pointed monodromy supplies such gradings. For two summands of opposite $\{+1, -1\}$ charges, the two masks are respectively the equal-charge and opposite-charge entries. Assignments on repeated restriction columns can be enumerated by their charge counts, up to permutations of copies.

Full faithfulness of $F|_{\mathcal{B}}$ does not imply a tensor factorization of $\mathcal{C}$, even when $\mathcal{B}$ is semion. For example, take $\mathcal{C} = \mathrm{Vec}^{\omega}_{\mathbb{Z}/4}$ with $\omega(a, b, c) = i^{a\lfloor (b+c)/4 \rfloor}$ for representatives $0 \le a, b, c < 4$. The element 2 lifts centrally with half-braiding $\gamma(x) = e^{3\pi i x/4}$; its square lift is the unit and its self-braiding is $-i$. Thus its semion subcategory forgets fully faithfully, whereas a tensor factorization would identify the fusion group $\mathbb{Z}/4$ with $(\mathbb{Z}/2)^2$. The exact replay `central_semion_counterexample.py` checks all 256 cocycle pentagons, 16 half-braiding hexagons and four square-lift identities by arithmetic modulo eight.

For a pointed central subgroup $G$, put $q(g) = \theta_g$ and $b(g, k) = q(gk)/(q(g)q(k))$. Write $\epsilon_g(Z)$ for the scalar monodromy on $g \otimes Z$ when $Z$ is simple. Since $g$ is invertible, this scalar is nonzero.

**Lemma 4.25** (Monodromy charges of word constituents)**.** *For a low word $P = A_{a_1} \cdots A_{a_r}$ define $\epsilon_g(P) = \prod_j \epsilon_g(A_{a_j})$. Every simple constituent $Z$ of $P$ has $\epsilon_g(Z) = \epsilon_g(P)$ for every $g \in G$. Consequently the words on which a coefficient vector $x_Z$ is positive must all have the same charge character of $G$. Moreover,*

$$\epsilon_k(g \otimes Z) = b(k, g)\epsilon_k(Z).$$

*For the character-labelled induction lows in our $S$ convention,*

$$\epsilon_g(A_a) = \frac{DS_{g^*,A_a}}{d(A_a)} = \chi_a(F(g^*)) = \chi_a(F(g))^{-1}.$$

*For involutions, the last two values agree.*

*Proof.* The braiding hexagon expresses monodromy with a tensor product as the product of the monodromies with its factors, whenever these are scalar. Thus monodromy on $g \otimes P$ is the displayed scalar product. Naturality with respect to each simple inclusion $Z \to P$ gives the constituent assertion. Taking one factor to be an invertible $g$ gives the transport formula, since its monodromy with $k$ is $b(k, g)$. The $S$-matrix equality uses the same dual-index convention as Proposition 4.21; substituting the low character formula gives the final identities. □

In particular the pairings of differently charged words vanish, and no nonnegative Gram column may mix their supports. This gives a cheap exact sieve before word-action completion or twist coupling. A zero coefficient vector imposes no charge choice and remains admissible.

**Proposition 4.26** (Pointed stabilizer divisibility)**.** *Let* $\mathrm{rad}(b) = \{g : b(g, k) = 1 \text{ for all } k \in G\}$. *For a restriction column $v$, set $K_v = \{g \in G : [F(g)]v = v\}$ and let $m_v$ count the central simples with that column. Then*

$$\frac{|K_v|}{|K_v \cap \mathrm{rad}(b)|} \mid m_v.$$

*Proof.* If $g \otimes Z \cong Z$, the transport formula in Lemma 4.25 forces $b(k, g) = 1$ for every $k$, hence $g \in \mathrm{rad}(b)$. The group $K_v$ acts on the finite set of simples with restriction $v$. Every stabilizer $H$ is contained in $K_v \cap \mathrm{rad}(b)$, so each orbit size $|K_v|/|H|$ is a multiple of the displayed divisor. Summing orbit sizes proves the assertion. The forgetful functor need not be fully faithful on $G$. □

## 4.10 Modular-word closure and finite representation completion

Once a complete twist profile is fixed, the known low rows also determine part of the modular representation by exact linear algebra. This is a separate use of the same central data: no fusion coefficients are guessed. Write $S, T$ for the canonical normalized matrices of $Z(\mathcal{C})$, $N = \mathrm{ord}(T)$, and $C = S^2$ for charge conjugation. The canonical representation factors through $SL_2(\mathbb{Z}/N\mathbb{Z})$. For $a$ coprime to $N$, let $Q_a$ be the signed permutation defined by $\sigma_a(S) = SQ_a$, where $\sigma_a(\zeta_N) = \zeta_N^a$. The congruence and Galois formulas of [49, Theorem II] give, for $ab \equiv 1 \pmod N$,

$$ST^bS = T^{-a}SQ_aT^{-a}. \tag{4.41}$$

In particular, if all twist-one simples are among the known lows, then $Q_a$ preserves that set. Its signed action can be determined from the known low block whenever the matching signed columns are unique.

**Proposition 4.27** (Exact modular closure and aggregate Verlinde test)**.** *Suppose the action of $S$ is known on a subspace $W$ of the $C$-fixed space, and $T$ is known on the entire simple basis. The following operations are necessary and exact.*

1. *Known pairs $(v, Sv)$ satisfy every linear relation among their first coordinates. A zero first coordinate with nonzero proposed image is an obstruction.*
2. *If $u, v \in W$ satisfy $u = Tv$, then*

$$S(TSu) = T^{-1}Sv.$$

   *Both this pair and its reverse may be adjoined to the known action.*

3. *If $P, Q, R$ are nonnegative integral sums of central simples whose $S$-images are known, then*

$$\sum_j \frac{(SP)_j(SQ)_j(SR)_j}{S_{0j}} = [PQR : \mathbf{1}] \in \mathbb{Z}_{\geq 0}. \tag{4.42}$$

*Thus this test applies to an entire class of indistinguishable restriction and twist columns even before its individual rows are known.*

*Proof.* The first assertion is linearity. For the second, use the anomaly-one relation $STS = T^{-1}ST^{-1}$ and $u = Tv$:

$$STSu = T^{-1}ST^{-1}u = T^{-1}Sv.$$

Charge conjugation commutes with both $S$ and $T$, so the new vectors are still $C$-fixed. On this space $S^2 = 1$, which justifies reversing the pair. The starting low basis must therefore be verified self-dual; self-duality of all high simples is not assumed. Finally, the trilinear Verlinde formula gives (4.42) for simples. Additivity proves the statement for nonnegative sums, with no assumption on duality within a class. □

The process terminates after at most the central rank many increases of the known span. A constant rank is a statement about this closure process, not by itself a proof that the unknown matrix is unique. If its orthogonal complement is explicitly verified to be $T$-stable, its remaining representation can instead be enumerated in the finite quotient. For any proved exponent bound $N$, the Chinese remainder theorem gives

$$SL_2(\mathbb{Z}/N\mathbb{Z}) \cong \prod_{p^a \| N} SL_2(\mathbb{Z}/p^a\mathbb{Z}).$$

The representation factors through this quotient even when the actual twist order is a proper divisor of $N$. Enumerating each finite group, its conjugacy classes and its class-algebra multiplication determines all irreducible characters: the simultaneous primitive eigenspaces of the class algebra give the central characters, and character orthogonality determines their degrees. Tensor products then give all irreducibles of the finite quotient. Comparing their exact $T$-spectra with the remaining spectrum is a finite integral multiplicity problem. Applied to the entire simple basis, this gives the necessary condition that its unweighted twist-multiplicity vector be a nonnegative integral sum of irreducible $T$-spectra. Applied to a proved invariant complement, it gives the corresponding condition on that complement. The action of $-I$ must also agree with an allowable duality permutation of the original simple basis. On each twist eigenspace the multiplicity of its $+1$ eigenvalue is at least that of its $-1$ eigenvalue: a dual pair contributes one of each, and a self-dual simple contributes only $+1$. The difference is at least the number of known self-dual simples of that twist. These inequalities apply to the full representation, or after subtracting the known contribution when a complement is used.

If the remaining $T$-eigenvalues are distinct and their representation is unique, all intertwiners preserving $T$ are diagonal. Symmetry of $S$ fixes their relative squares; when these ratios are nonzero it leaves only finitely many sign choices, modulo common scalar. Each resulting matrix is then checked for the full modular relations, the prescribed known rows, and every Verlinde coefficient. Passing these checks supplies necessary fusion and modular data; it does not construct an associator or establish categorifiability.

## 4.11 Subcategory projection and exact twist support

Information about the center of a proper fusion subcategory imposes constraints on individual restriction columns of the ambient center. It does not require a braiding on the ambient fusion category, or an extension of a chosen subcategory braiding.

**Proposition 4.28.** *Let $\mathcal{D} \subset \mathcal{C}$ be a full fusion subcategory of a pivotal fusion category, with the inherited pivotal structure. Let $P(V)$ be the sum of all simple summands of $V$ belonging to $\mathcal{D}$. If $(V, c)$ is an object of $Z(\mathcal{C})$, then $P(V)$ has a canonical $\mathcal{D}$-half-braiding. If $V$ is central simple with twist $\lambda$, every simple constituent of $P(V)$ in $Z(\mathcal{D})$ has twist $\lambda$. Consequently, writing $F_{\mathcal{D}} : Z(\mathcal{D}) \to \mathcal{D}$ for the forgetful functor, the $\mathcal{D}$-part of a central restriction column belongs to the nonnegative integral span of the columns $[F_{\mathcal{D}}(W)]$ with $\theta_W = \lambda$.*

*Proof.* The complement of $\mathcal{D}$ is stable under tensoring on either side by objects of $\mathcal{D}$. Indeed, if a simple $S \notin \mathcal{D}$ and $X, T \in \mathcal{D}$ satisfy $\mathrm{Hom}(T, X \otimes S) \neq 0$, rigidity makes $S$ a constituent of $X^* \otimes T \in \mathcal{D}$, a contradiction. Hence $P$ is a projection of $\mathcal{D}$-bimodule categories. Put $W = P(V)$ and choose $i : W \to V$, $p : V \to W$ with $pi = 1_W$. For $X \in \mathcal{D}$ the maps

$$c_X^W = (1_X \otimes p)c_X(i \otimes 1_X)$$

are isomorphisms. Naturality and the half-braiding identity restrict to this summand, so they make $W$ an object of $Z(\mathcal{D})$.

It remains to check the twist, rather than merely the half-braiding. Suppress associators and use the Drinfeld isomorphism

$$u_V = (\mathrm{ev}_V \otimes 1_{V^{**}})(c_{V^*} \otimes 1_{V^{**}})(1_V \otimes \mathrm{coev}_{V^*}) : V \longrightarrow V^{**}.$$

The duality identities give

$$(1_{V^*} \otimes p^{**})\,\mathrm{coev}_{V^*} = (p^* \otimes 1_{W^{**}})\,\mathrm{coev}_{W^*},$$
$$\mathrm{ev}_V(p^* \otimes 1_V) = \mathrm{ev}_W(1_{W^*} \otimes p).$$

Using naturality of $c$ in its second argument between these two identities gives $p^{**}u_V i = u_W$. If $j$ denotes the pivotal isomorphism, then $p^{**}j_V i = j_W$. With the convention $\theta = u^{-1}j$, central simplicity gives $u_V = \lambda^{-1}j_V$, and therefore $u_W = \lambda^{-1}j_W$. Thus $\theta_W = \lambda 1_W$. Decomposing $W$ into central simples proves the integral-span assertion. The opposite twist convention gives the same conclusion when used in both centers. □

**Proposition 4.29** (Modular subcategory projection)**.** *Suppose $\mathcal{C}$ is pseudo-unitary and $\mathcal{D} \subset \mathcal{C}$ is a full fusion subcategory, both with their positive spherical structures. Let $\Pi : Z(\mathcal{C}) \to Z(\mathcal{D})$ be the additive central projection constructed above, and put*

$$B_{WZ} = \dim \mathrm{Hom}_{Z(\mathcal{D})}(W, \Pi Z).$$

*For the canonical normalized modular matrices of the two centers,*

$$S_{\mathcal{D}}B = BS_{\mathcal{C}}, \qquad T_{\mathcal{D}}B = BT_{\mathcal{C}}, \qquad B_{W,\mathbf{1}} = \delta_{W,\mathbf{1}}. \tag{4.43}$$

*In particular, each individual simple $W \in Z(\mathcal{D})$ has the mass and capacity bounds*

$$\sum_Z B_{WZ}d_Z = \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}}d_W, \qquad B_{WZ}d_Z \leq \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}}d_W. \tag{4.44}$$

*The second inequality concerns one ambient central simple and one subcategory central simple, even if several $W$ have the same twist and forgetful image.*

*Proof.* Regard $\mathcal{M} = \mathcal{C}$ as a $\mathcal{D}$–$\mathcal{C}$ bimodule by tensor multiplication. It is indecomposable, since its right $\mathcal{C}$ action alone generates every simple from the unit. The full center of this module category is a Lagrangian algebra

$$L \in Z(\mathcal{D}) \boxtimes Z(\mathcal{C})^{\mathrm{rev}};$$

see the module-category full-center construction [53, Section 6, Theorem 6.2] and the correspondence [54, Proposition 4.8]. We identify its multiplicities explicitly. For $W \in Z(\mathcal{D})$ and $Z \in Z(\mathcal{C})$, define bimodule endofunctors on $\mathcal{M}$ by

$$L_W(X) = F_{\mathcal{D}}(W) \otimes X, \qquad R_Z(X) = X \otimes F_{\mathcal{C}}(Z).$$

Their left and right module structures use the respective half-braidings. The defining full-center adjunction identifies the coefficient of $W \boxtimes Z^*$ with $\dim \mathrm{Nat}_{\mathcal{D}-\mathcal{C}}(L_W, R_Z)$. A right $\mathcal{C}$-module natural transformation is determined by its component at the unit, a map $f : F_{\mathcal{D}}(W) \to F_{\mathcal{C}}(Z)$. Its left $\mathcal{D}$-linearity says precisely

$$c^Z_X(f \otimes 1_X) = (1_X \otimes f)c^W_X \qquad (X \in \mathcal{D}).$$

Because its source belongs to $\mathcal{D}$, $f$ factors uniquely through $P(F_{\mathcal{C}}(Z))$. The displayed condition then says that this factorization is a morphism $W \to \Pi Z$ in $Z(\mathcal{D})$. Consequently

$$L \cong \bigoplus_{W,Z} B_{WZ}\, W \boxtimes Z^*.$$

This fixes the dual convention: when $\mathcal{D} = \mathcal{C}$, $B$ is the identity and $L$ is the usual diagonal Lagrangian algebra.

The class of a Lagrangian algebra is fixed by the canonical normalized $S$ and $T$. Indeed, the Morita description identifies it with induction of the unit in a center, where this is [38, Theorem 2.5, Proposition 2.8]; the positive spherical structures make the center equivalence ribbon. Let $C_{\mathcal{C}}$ be the duality permutation of the ambient center. The coefficient matrix of $L$ is $BC_{\mathcal{C}}$, and the reversed center has matrix $\overline{S_{\mathcal{C}}} = S_{\mathcal{C}}^{-1}$ and inverse twists. Thus invariance is

$$S_{\mathcal{D}} B C_{\mathcal{C}} S_{\mathcal{C}}^{-1} = B C_{\mathcal{C}}, \qquad T_{\mathcal{D}} B C_{\mathcal{C}} T_{\mathcal{C}}^{-1} = B C_{\mathcal{C}}.$$

Duality commutes with both matrices, so these equalities give the first two identities in (4.43). Projection sends the ambient central unit to the subcategory central unit, proving the last identity. Finally compare the ambient-unit column of $S_{\mathcal{D}} B = B S_{\mathcal{C}}$ and use $S_{\mathcal{D},W0} = d_W/D_{\mathcal{D}}$ and $S_{\mathcal{C},Z0} = d_Z/D_{\mathcal{C}}$. Positivity of all summands proves the individual capacity inequality. □

**Corollary 4.30** (Disjoint central projection rows)**.** *In the setting of Proposition 4.29, suppose every ambient central simple projects either to zero or to a single simple of $Z(\mathcal{D})$ with multiplicity one. Then, for a positive integer $r$,*

$$BB^{\mathsf{T}} = rI.$$

*Thus $B^{\mathsf{T}}/\sqrt{r}$ is an isometric intertwining embedding of the entire canonical modular representation of $Z(\mathcal{D})$ into that of $Z(\mathcal{C})$. Its $T$-spectrum can be subtracted before the finite-quotient representation test. Any signed charge-conjugation bound for the complement is obtained by subtracting this known subrepresentation's signed trace from the corresponding full-space bound; it need not be nonnegative.*

*Proof.* The hypothesis says each column of $B$ is zero or a standard basis vector. Hence $BB^{\mathsf{T}}$ is diagonal, with entries equal to the numbers of simples in the projection fibers. Each entry is positive by (4.44). The normalized modular representations are unitary, so the intertwining identities imply that $BB^{\mathsf{T}}$ commutes with $S_{\mathcal{D}}$ and $T_{\mathcal{D}}$. Comparing its $W, \mathbf{1}$ entries and using $S_{\mathcal{D},W\mathbf{1}} = d_W/D_{\mathcal{D}} > 0$ shows that all diagonal entries equal the same positive integer $r$. The embedding and the subtraction statements follow by taking adjoints. The orthogonal complement is a representation space, but is not required to have a simple basis permuted by charge conjugation. □

The disjoint-column hypothesis must be verified. The exact categorical control `research/round5/unification/subcategory_intertwiner_control.py` derives $S$, $T$ and $B$ from finite group character sums for $\mathrm{Vec}(C_2), \mathrm{Vec}(C_3) \subset \mathrm{Vec}(S_3)$. Both rectangular modular identities and all individual masses hold. The second projection has rank five although its target center has rank nine, illustrating why a general subcategory inclusion alone does not justify subtracting the entire target representation. Mutating one restriction multiplicity violates the checked $S$-identity.

For an Ising subcategory, the two twist-minus-one simples of its center are $\mathbf{1} \boxtimes g$ and $g \boxtimes \mathbf{1}$, each of dimension one. Hence every high ambient central simple whose restriction contains $g$ has dimension at most $D_{\mathcal{C}}/4$. This is stronger than the combined twist-minus-one mass $D_{\mathcal{C}}/2$. For $\mathcal{D} = \mathrm{Vec}$, the same proposition says that the multiplicity vector of $I_{\mathcal{C}}(\mathbf{1})$ is fixed by $S$ and $T$; thus the full modular representation contains a trivial constituent.

The projection also gives exact, twist-resolved dimension identities. Here and below spherical structures are inherited, so the centers carry their standard ribbon structures.

**Corollary 4.31.** *Let $\mathcal{C}$ be spherical and $\mathcal{D} \subset \mathcal{C}$ a full fusion subcategory. Put $D_{\mathcal{C}} = \dim(\mathcal{C})$ and $D_{\mathcal{D}} = \dim(\mathcal{D})$. For a simple $X \in \mathcal{D}$ and a root of unity $\lambda$,*

$$\sum_{\substack{Z \in \operatorname{Irr} Z(\mathcal{C}) \\ \theta_Z = \lambda}} \dim(Z)[F_{\mathcal{C}}(Z) : X] = \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}} \sum_{\substack{W \in \operatorname{Irr} Z(\mathcal{D}) \\ \theta_W = \lambda}} \dim(W)[F_{\mathcal{D}}(W) : X].$$

*An empty right-hand side is zero. In the pseudo-unitary case all weights are positive and dimensions can be replaced by Frobenius–Perron dimensions.*

*Proof.* Choose $N$ divisible by the orders of all twists in $Z(\mathcal{D})$. Proposition 4.28 shows that every central simple contributing to the left-hand side has twist in $\mu_N$. The indicator formula [32] gives, for $1 \le n < N$,

$$D_{\mathcal{C}} \nu_n(X) = \sum_Z \theta_Z^n \dim(Z)[F_{\mathcal{C}}(Z) : X].$$

The same formula holds in $\mathcal{D}$, and the indicators agree: their defining pivotal rotations act on the same spaces $\mathrm{Hom}(1, X^{\otimes n})$. For $n = 0$, the corresponding equality is $\dim I_{\mathcal{C}}(X) = D_{\mathcal{C}} \dim(X)$, and similarly in $\mathcal{D}$. Fourier inversion on $\mu_N$ proves the assertion. □

**Corollary 4.32.** *In the pseudo-unitary setting the following subcategory constraints hold.*

1. *For a Fibonacci subcategory with basis $(1, a)$, $a^2 = 1+a$, write $\varphi = \mathrm{FPdim}(a) = (1+\sqrt{5})/2$. A nonzero projected column $(r, s)$ has either twist $1$ and $r \ge s$, or $r = 0$ and twist $\zeta_5^2$ or $\zeta_5^{-2}$. For each of the latter twists, the $a$-mass in Corollary 4.31 is $D_{\mathcal{C}}\varphi/(\varphi+2)$; the twist-one $a$-mass is $D_{\mathcal{C}}\varphi^2/(\varphi+2)$.*
2. *For an Ising subcategory with basis $(1, g, \sigma)$, a nonzero projected column $(r, s, t)$ has one of the following forms: twist $1$ with $t = 0$ and $r \ge s$; twist $-1$ with $r = t = 0$; or one of four Ising twists $\{z, -z, z^{-1}, -z^{-1}\}$ with $r = s = 0$, where $z$ is an appropriate primitive sixteenth root. The $g$-masses at $1$ and $-1$ are each $D_{\mathcal{C}}/2$. The $\sigma$-mass at each of its four twists is $D_{\mathcal{C}}\sqrt{2}/4$.*

*In particular, a high column, whose unit coordinate is zero, cannot combine $g$ and $\sigma$ in its Ising projection, or contain either $a$, $g$ or $\sigma$ at twist one in the respective cases.*

*Proof.* The rank-two and Ising classifications [36, 16] provide nondegenerate braidings on these subcategories. For either such category $\mathcal{D}$, its center is $\mathcal{D} \boxtimes \mathcal{D}^{\mathrm{rev}}$ and the restriction of $(U, V)$ is $U \otimes V$. For Fibonacci the four columns are

$$(1, 0), \quad (0, 1), \quad (0, 1), \quad (1, 1),$$

with twists $1, z, z^{-1}, 1$. Positivity gives the usual Fibonacci modular $S$-matrix; balancing gives $z^2 + \varphi z + 1 = 0$, hence $z = \zeta_5^{\pm 2}$. The center dimensions are respectively $1, \varphi, \varphi, \varphi^2$. For Ising, the twist-one columns are $(1, 0, 0)$ twice and $(1, 1, 0)$ once; the twist-$-1$ columns are $(0, 1, 0)$ twice; and each of $z, -z, z^{-1}, -z^{-1}$ has the column $(0, 0, 1)$ once. Their dimensions are $1, 2, \sqrt{2}$ as determined by these columns. Taking their additive spans and applying the preceding corollary gives every assertion. All inherited Ising spherical choices must be retained. If the second indicator of $\sigma$ is fixed, the identity $z^2 + z^{-2} = \nu_2(\sigma)\sqrt{2}$ selects the appropriate set of four twists. □

**Computational use.** The criteria first restrict the allowed high-column/twist pairs and the twist-one low columns. They then add the exact mass identities to the integer linear equations. When using normalized relative twist traces, one projects the identities to the chosen field, combining coincident trace types; one must not confuse a projected trace with an actual twist. These are necessary conditions only. General exclusions still require the separately checked general-to-pseudo-unitary reduction.

### 4.12 Fibonacci projection over a biquadratic character field

The projection obstruction of Proposition 4.28 can be combined with full-character induction without computing in an unnecessarily large cyclotomic field. We record the exact reduction used here.

**Proposition 4.33** (Projected twist traces)**.** *Let $K \subseteq L = \mathbb{Q}(\zeta_F)$ be a Galois field containing all the characters of a commutative fusion ring, and suppose that* $\mathrm{Gal}(K/\mathbb{Q})$ *has exponent at most two. Let $P$ be the allowed prime support of the Frobenius–Schur exponent and $h \geq 2$. For a root of unity $\theta$, put*

$$T_K(\theta) = \left( \frac{\mathrm{Tr}_{K(\theta)/K}(\theta^j)}{[K(\theta) : K]} \right)_{1 \leq j \leq h}.$$

*Every such vector with root order supported on $P$ is either zero or is obtained from a root whose order divides*

$$B = \prod_{p \in P} p^{\max(1, v_p(F)) + \lfloor \log_p h \rfloor}.$$

*These vectors are computed exactly by first taking normalized trace to $L$ and then averaging over* $\mathrm{Gal}(L/K)$*. The induction Gram, rotation, and low-character equations remain necessary after this projection. Their Galois multiplicity equalities may be imposed separately for each projected twist vector.*

*Proof.* The tower identity for normalized traces gives the stated computation. The conductor bound follows from Lemma 4.12: if an order has a prime exponent larger than the displayed bound, every power $1 \leq j \leq h$ still has a nontrivial final cyclotomic layer over $L$, and its relative trace vanishes. Averaging once more preserves zero. Otherwise the order divides $B$.

All coefficients other than roots of unity in the induction and low-character equations lie in $K$. The normalized trace is $K$-linear, so applying it preserves these equations. The variables counting central simple objects and cyclic eigenspaces remain nonnegative; identifying equal projected vectors merely sums these variables and thus weakens the necessary system.

For Galois balance, a modular Galois automorphism acts on joint character eigenvalues by $\sigma$ and on the canonical twist by $\sigma^2$. The latter automorphism restricts to the identity on $K$, by the exponent-two hypothesis. Normalized trace to $K$ is invariant under this automorphism. Consequently the projected twist vector is unchanged along the corresponding permutation of central simples. This proves the separate multiplicity equalities. No integrality of the projected trace, or of a solution to the relaxed linear system, is asserted. □

For the ring $R_{6,8,3664}$, the Fibonacci basis object is $b_1$: $b_1^2 = 1 + b_1$. Its Frobenius–Perron character field and global dimension are

$$K = \mathbb{Q}(\sqrt{5}, \sqrt{6}), \qquad D = 300 + 132\sqrt{5} + 120\sqrt{6} + 54\sqrt{30}.$$

The formal codegree polynomials are

$$t^2 - 15t + 45, \qquad t^4 - 1200t^3 + 18000t^2 - 86400t + 129600,$$

each with multiplicity one. Thus one may use $F = 120$, $h = 3$, and $P = \{2, 3, 5\}$. The bound is $B = 720$, and there are exactly 43 distinct projected vectors. Arithmetic takes place in degree four, whereas the ambient cyclotomic field has degree 32.

By Corollary 4.32, each low column $c$ necessarily satisfies $c_1 \leq c_0$. For a high column, $c_0 = 0$; if $c_1 > 0$, its twist must be $\zeta_5^2$ or $\zeta_5^{-2}$. The two roots have the same normalized trace to the totally real field $K$, so this condition leaves one projected twist vector for such a column. Columns with $c_1 = 0$ retain the full necessary trace domain.

**Theorem 4.34.** *The fusion ring $R_{6,8,3664}$ has no categorification over $\mathbb{C}$.*

*Proof.* Unrestricted signed sphericalization gives 768 signed quotients and 24 sign orbits. One orbit is gauge equivalent to the original ring; the other orbits are eliminated by 36 exact dimension-character checks. The unique surviving nonzero dimension-character component is the Frobenius–Perron Galois orbit. The general-to-pseudo-unitary reduction therefore applies.

For the pseudo-unitary case, exhaustive low induction enumeration gives 13 frames. Four fail an ordinary moment inequality; three fail the Fibonacci twist-one projection inequality; four have no compatible low-character labelling. The remaining two frames each have one labelling and one allowed second-indicator tuple. Their necessary systems have, respectively, 708 equations in 2184 variables and 663 equations in 1792 variables. Both systems have certified rational separating vectors. The verifier reconstructs every field, frame, character assignment, indicator branch, and rational equation from the full input tensor, and checks $A^T y \geq 0$, $b^T y < 0$ exactly. The complete certificate is `research/round4/m8/R6_8_3664_certificate.json`, replayed by `software/round4_m8_subfield.py`. □

## 4.13 Ising twist masses at multiplicity eight

The two multiplicity-eight rings $R_{6,8,311}$ and $R_{6,8,1660}$ have formal codegrees with characteristic polynomial

$$(t-4)^2(t^4 - 576t^3 + 11520t^2 - 73728t + 147456).$$

They contain Ising basis subcategories, with labels $(g, \sigma) = (b_4, b_3)$ and $(b_1, b_4)$, respectively. Their different duality involutions remain part of the full tensor input; the common polynomial does not identify the rings.

Here the entire character field embeds in $\mathbb{Q}(\zeta_{24})$. A pseudo-unitary Ising category has twists $1, -1, \xi$, where $\xi$ is a primitive sixteenth root. Its center has four $\sigma$-restriction objects, with twists $\xi, -\xi, \xi^{-1}, -\xi^{-1}$. On taking normalized trace to $\mathbb{Q}(\zeta_{24})$, their height-three vectors become

$$(0, \eta, 0), \qquad (0, \eta^{-1}, 0), \qquad \eta = \xi^2,$$

with each vector occurring for two of the four objects. The sign $\epsilon = \nu_2(\sigma)$ determines the unordered pair:

$$\eta + \eta^{-1} = \epsilon\sqrt{2}.$$

Both possible signs are retained whenever permitted by the second indicator conditions.

Corollaries 4.32 and 4.31 now give the following simultaneous constraints. Every low column satisfies $c_\sigma = 0$ and $c_g \leq c_0$; moreover

$$\sum_{\text{low } Z} d_Z c_g(Z) = D/2.$$

A high column with $c_g > 0$ has twist $-1$ and $c_\sigma = 0$. A high column with $c_\sigma > 0$ has $c_g = 0$ and one of the two displayed projected vectors. For each of those vectors the exact dimension mass is $D\sqrt{2}/2$. These equations are imposed together with the complete induction Gram, full-character reciprocity, Galois balance, and cyclic rotation equations. They do not assume that the Ising braiding extends to the ambient category.

**Theorem 4.35.** *The fusion ring $R_{6,8,311}$ has no categorification over $\mathbb{C}$.*

*Proof.* The unrestricted signed reduction has 96 quotients and eight sign orbits. One orbit is gauge equivalent to the original ring, and 13 exact dimension-character checks reject all other possible categorical dimension components. The surviving component is the Frobenius–Perron Galois orbit, so it suffices to exclude a pseudo-unitary categorification.

The complete low induction domain has two frames. Each has two character labellings and four allowed second-indicator tuples. With $F = 24$ and height three, the complete order bound is 144 and there are 53 normalized trace vectors. The first frame gives systems of 387 equations in 802 variables; the second gives 1017 equations in 1896 variables. Each of the 16 branches has an exact rational separating vector. The direct certificate is `research/round4/m8/R6_8_311_certificate.json`; the full-tensor verifier is `software/round4_m8_ising.py`. □

For $R_{6,8,1660}$, the unrestricted signed reduction has 640 quotients in 24 sign orbits and 58 nongauge dimension-character rejections. It proves sphericality and a pseudo-unitary Galois reduction. Morita reconstruction over $1 + g$ returns the same based ring, so it supplies no transfer to the excluded non-self-dual partner. The structural reduction in Section 7.5 uses this conclusion.

## 4.14 Twist-resolved subcategory constraints at higher multiplicity

The projection in Proposition 4.28 can be used before enumerating a complete modular datum. In particular, it couples a restriction column to its twist; checking the column and the twist separately loses information. We record two explicit source tables which make this necessary condition fully effective.

**The half-$E_6$ subcategory.** Suppose that the based subring on $\{1, g, x\}$ satisfies $g^2 = 1$, $gx = xg = x$, and $x^2 = 1 + g + 2x$. Put $d = 1 + \sqrt{3}$ and $D_0 = 2 + d^2 = 6 + 2\sqrt{3}$. Hagge–Hong classify its four categorifications [47, Theorem 1 and Sections 4.7–4.8]. The relations imply that every tensor automorphism of the identity is trivial: its scalar on $x$ satisfies $\eta_x^2 = \eta_x$, and then $\eta_g = 1$. Hence a pivotal structure, when present, is unique. Two Galois conjugates have $\dim(x) = 1 - \sqrt{3} < 0$ and cannot be pseudo-unitary. The two remaining cases are complex conjugate. Their center restriction columns and twists are given by Hong–Rowell–Wang [48, Section 2.1]. Writing $\zeta = \exp(2\pi i/12)$, one table is

| $Z$ | $1$ | $Y$ | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $U$ | $V$ | $W$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $F(Z)$ | $1$ | $g$ | $x$ | $x$ | $x$ | $x$ | $x$ | $1+x$ | $g+x$ | $1+g+x$ |
| $\theta_Z$ | $1$ | $-1$ | $\zeta^{-3}$ | $\zeta^{5}$ | $\zeta^{5}$ | $\zeta^{2}$ | $\zeta^{-4}$ | $1$ | $-1$ | $1.$ |

The other table conjugates every twist. Complex conjugation of the entire putative pseudo-unitary category preserves its based ring and its positive dimensions, and exchanges these two subcategory choices. Thus exclusion for the displayed choice excludes both choices; this is a reduction of existence, not an omission of a categorical branch.

For a high induction column $c$, the unit coordinate is zero. Consequently, $c_g > 0$ forces $\theta = -1$ and $c_g \geq c_x$; if $c_g = 0 < c_x$, the twist must be one of the four displayed $X$-twists. A low column has twist one, and its coordinates must satisfy $0 \leq c_g \leq c_x \leq c_1$. These statements concern all structure constants, with the subring located by its multiplication table, rather than by census labels.

The twist masses give additional equalities. If the ambient category has FP dimension $D$, Corollary 4.31 implies

$$\sum_{Z:\,\theta_Z=\lambda} c_{x,Z}\,\mathrm{FPdim}(Z) = \frac{D}{D_0} M_\lambda,$$

where the successive values of $M_\lambda$ at $\lambda = \zeta^{-3}, \zeta^5, \zeta^2, \zeta^{-4}, -1, 1$ are

$$d, \quad 2d, \quad d, \quad d, \quad d+1, \quad 2d+3.$$

Known low columns are subtracted from the right-hand sides. The existence reduction by conjugating the entire category, given above, covers the conjugate source table as well. The implementation combines it with the full induction Gram equations, the integrality of joint character spectra, and equality of multiplicities along Galois orbits. Optional refinements add the low Verlinde coefficients and restriction products.

**Cyclic pointed subcategories of order three.** For a generator $g$ of a pointed $C_3$ subcategory, all associator classes are represented by $s \in \{0, 1, 2\}$. A standard cocycle representative is

$$\omega_s(a, b, c) = \exp\left(\frac{2\pi i s a}{3} \left\lfloor \frac{b+c}{3} \right\rfloor\right), \qquad 0 \le a, b, c < 3.$$

Solving the one-dimensional half-braiding equations gives, over the underlying object $g^a$, the three twists

$$\exp\left(2\pi i \left(\frac{sa^2}{9} + \frac{ab}{3}\right)\right), \qquad b \in \{0, 1, 2\}.$$

Thus a central object whose restriction contains $g$ or $g^2$ has $\theta^3 = \exp(2\pi i s/3)$. For the ambient field $\mathbb{Q}(\zeta_{24})$, the three height-three relative-trace types when $s = 0$ are $(\lambda, \lambda^2, 1)$, $\lambda^3 = 1$. For $s = 1, 2$, the three possible ninth roots have the same normalized relative-trace type $(0, 0, \exp(2\pi i s/3))$. All three associator classes must be retained. No triviality assumption on the pointed associator is used.

**Exact arithmetic and scope.** The supplementary implementations retain rational equations and integer separating rays. For the half-$E_6$ input with FP field $\mathbb{Q}(\sqrt{3}, \sqrt{5})$, normalized traces from $\mathbb{Q}(\zeta_{60})$ to the real FP field reduce the height-three twist domain from 131 to 40 types. A field pullback is accepted only after exact reconstruction of every coordinate. The Galois group of this real field has exponent two, so the square action on these relative twist traces is trivial; joint-spectrum multiplicity equalities can therefore be imposed separately for each relative-trace type. Additional negative-stabilizer rotation conditions are imposed only after exhaustive reconstruction of the relevant involution module. Surviving systems remain necessary relaxations and do not provide categorifications or modular data.

**Certified applications and the remaining multiplicity-five input.** For $R_{6,6,49}$ the complete low domain has 353 frames, of which only one remains after the ordinary and character-assignment checks. Its four assignments, three pointed associator classes, and two second-indicator choices give 24 exact separating rays. The established two-FP-value reduction gives unrestricted categorical scope. For $R_{6,6,1035}$ there are 12 low frames, of which two admit a character assignment. Each has two second-indicator choices; all four full cyclotomic projection systems have exact separating rays. The unrestricted signed reduction exhausts 448 signed tables in 14 sign orbits and reduces every possible categorification to the pseudo-unitary problem. These two exclusions complete the exact multiplicity-six stratum. The final half-$E_6$ certificate uses the full conductor-60 system; its validity does not depend on the exploratory real-field MILP infeasibility results or on additional Clifford rotation constraints.

The multiplicity-five input $R^{(6)}_{5,1204}$ requires integral cyclic-rotation multiplicities in addition to these linear constraints; its complete exclusion is proved in Theorem 4.39.

## 4.15 Integer-square linearization and central-invertible transport

The complex low-product and fourth-moment identities are special cases of Section 4.8. They retain dual indices even when the original ring is not self-dual. The following linear consequences are useful when a complete integral Gram decomposition is too large. For $P = A_a A_b$, put $q_{ab,v} = \sum_{F(Z)=v}[P : Z]$, let $x_v$ count that restriction type, and let $B_{ab,v}$ be the coordinatewise restriction upper bound from the unified theorem. Write $Q_{ab} = [P : P]$.

Introduce $z_{ab,v} = \sum_{F(Z)=v}(N^Z_{A_a,A_b})^2$ and corresponding square variables for the known low coefficients. The integer-square inequalities in Section 4.8 apply coefficientwise. Summing them over a restriction type gives

$$(2k+1)q_{ab,v} - k(k+1)x_v \le z_{ab,v} \le B_{ab,v}q_{ab,v}. \tag{4.45}$$

For a low simple, replace $x_v$ by 1 and $q_{ab,v}$ by its individual Verlinde multiplicity. Equation (4.34) becomes the sum of these square variables, together with the unit contribution $\delta_{a,b^*}$. Any finite selection of the lower inequalities is valid; the implementation supports both all $0 \le k < B$ and a stated cutoff. This gives an exact linear strengthening, with integrality retained in bounded discovery experiments. Failure of a mixed-integer optimizer to finish is recorded only as an incomplete search.

### 4.15.1 Transport by central invertibles

Suppose the low frame has exactly one column equal to $e_0$. A central invertible restricting to the unit must occur in $I(\mathbf{1})$, so the forgetful homomorphism on invertible objects has trivial kernel. Its image is therefore a subgroup $K$ of the based-ring invertible group, and each $g \in K$ has exactly one invertible lift. For any such lift $L$, tensoring by $L$ permutes all central simples and sends a restriction vector $v$ to the coefficient vector of $gv$. In particular, the total counts of these restriction types agree, including low objects.

**Lemma 4.36** (Indicator transport by a central invertible)**.** *Let $\mathcal{C}$ be spherical and let $b \in Z(\mathcal{C})$ be invertible of order $m$, with $\dim b = 1$. Write $bX = F(b)\otimes X$. For every $n \ge 1$, $\nu_n(X^*) = \nu_n(X)$; whenever $m \mid n$,*

$$\nu_n(bX) = \theta_b^n \nu_n(X).$$

*Proof.* For the adjoint $I$ of $F$, the projection formula gives $I(bX) \cong b \otimes I(X)$. For each central simple $Z$, write $\lambda_Z$ for its monodromy scalar with $b$. The hexagon and $b^m \cong \mathbf{1}$ imply $\lambda_Z^m = 1$, while balancing gives $\theta_{bZ} = \theta_b\theta_Z\lambda_Z$. Thus $\theta_{bZ}^n = \theta_b^n\theta_Z^n$ and $\dim(bZ) = \dim Z$. Apply the spherical center formula [32, Theorem 4.1]:

$$\nu_n(X) = \frac{1}{\dim \mathcal{C}} \sum_{Z \in \operatorname{Irr} Z(\mathcal{C})} [F(Z) : X] \dim Z\, \theta_Z^n.$$

Relabelling $Z$ by $Z^*$ proves the duality identity, since $\theta_{Z^*} = \theta_Z$ and $\dim Z^* = \dim Z$; see also [32, Remark 4.2]. No complex conjugation or unitarity is used. □

There is a stronger constraint when $g^2 = 1$ and $\nu_2(g) = -1$. The lift has order two and twist $\theta_L = \pm i$. Its monodromy with a simple $Z$ is a scalar in $\{1, -1\}$: monodromy with invertibles is multiplicative, and $L^2 = \mathbf{1}$. The balancing identity therefore gives

$$\theta_{LZ} = \pm i\theta_Z.$$

In particular $LZ \ncong Z$, so this permutation is a fixed-point-free involution. When the ambient cyclotomic field contains $i$, normalized relative traces transform by $(t_j)_j \mapsto ((\pm i)^j t_j)_j$. One can encode the resulting matching with nonnegative integral edge variables between allowed pairs

of restriction/trace types. An edge joining distinct types contributes one to each marginal; a loop in the *type* graph contributes two, because it still represents two distinct simples. This last convention is necessary when relative trace loses the underlying twist information. These matching equations are necessary only; for several lifts they do not impose all simultaneous group-action relations.

### 4.16 Bounded induction profiles and word moments

Induction multiplicity profiles give a finite necessary domain before searching for twists or fusion coefficients. A profile specifies the multiplicity of each restriction column.

Fix a low induction frame $L$ and let $v_1, \dots, v_t$ be the remaining admissible high restriction columns. If $m_j$ is the number of central simples with column $v_j$, then

$$\sum_{j=1}^{t} m_j v_j v_j^{\mathsf{T}} = G, \qquad m_j \in \mathbb{Z}_{\geq 0}, \qquad m_j \leq \min_{i:(v_j)_i>0} \left\lfloor \frac{G_{ii}}{(v_j)_i^2} \right\rfloor,$$

where $G$ is the induction Gram matrix after subtracting $LL^{\mathsf{T}}$. The joint character spectrum supplies additional linear equations equating Galois multiplicities. All these equations have integer coefficients.

**Proposition 4.37** (Exact bounded affine enumeration)**.** *For an integer system $Am = b$ with prescribed finite bounds $0 \leq m_j \leq u_j$, row reduction over $\mathbb{Q}$, followed by the bounded search below, enumerates every integral solution exactly once.*

*Proof.* Choose the nonpivot coordinates as free variables. Each pivot coordinate is an affine rational function of them. Clear its denominator separately. Enumerate the free coordinates in any fixed order within their given integer bounds. At a partial assignment, the unused coordinates give an explicit lower and upper bound for each pivot numerator: for a coefficient $a$ and remaining variable in $[0, u]$, its contribution belongs to $[\min(0, au), \max(0, au)]$. Discard a node only if one such interval misses the allowed interval for that numerator. At a leaf check divisibility by each cleared denominator and the original equations. Every integral solution follows a unique unpruned path; every retained leaf satisfies the original system and all bounds. A node or time budget interrupts this argument and is therefore recorded as incomplete, never as an exclusion. □

There is a useful inexpensive sieve after enumeration. Write $D$ for the global dimension of the original pseudo-unitary category and $d_Z$ for a central dimension. If $A_1, \dots, A_s$ are the nonunit low central simples, put $\lambda_a(Z) = S_{A_a,Z}/S_{\mathbf{1},Z}$, with the normalized modular matrix $S$. The character and restriction data determine these numbers. For every word $w = (a_1, \dots, a_k)$, simultaneous diagonalization of the fusion matrices gives

$$\dim \mathrm{Hom}(\mathbf{1}, A_{a_1} \otimes \cdots \otimes A_{a_k}) = \sum_{Z} \frac{d_Z^2}{D^2} \prod_{j=1}^{k} \lambda_{a_j}(Z) \in \mathbb{Z}_{\geq 0}. \tag{4.46}$$

No duality assumption on the $A_a$ is needed. In degree three these are precisely the low fusion multiplicities after the appropriate dual label is inserted. Their restrictions must leave a nonnegative remainder in every low product. Both tests are exact linear evaluations of a fixed profile. We subsequently check all nonunit low words through degree six.

### 4.17 An integral rotation obstruction at multiplicity five

The ring $R^{(6)}_{5,1204}$ has no nontrivial proper fusion subring. Its simple dimensions are

$$1, \quad 3+2\sqrt{3}, \quad 3+2\sqrt{3}, \quad 7+4\sqrt{3}, \quad 6+4\sqrt{3}, \quad 8+4\sqrt{3}.$$

We combine a forced central product with integrality of cyclic-rotation eigenspaces.

**Proposition 4.38.** *Every complex categorification of $R^{(6)}_{5,1204}$ is spherical and has a pseudo-unitary Galois conjugate. Its central induction matrix belongs, up to permutation of central simples, to an explicitly certified list of* 171 *restriction profiles. Their central ranks lie between* 22 *and* 51.

*Proof.* The unrestricted signed-sphericalization test of Section 3.4 enumerates 512 associative signed quotient tables, partitioned into 16 sign-gauge orbits. One orbit consists of gauges of the original fusion tensor. The exact dimension-character tests reject all 46 character components on the other orbit representatives. As in the sphericality argument of Proposition 3.27, a categorification must therefore admit a spherical structure. Among the character components of the original ring, the only one without a zero simple dimension is the Frobenius–Perron orbit. Conjugating to its positive embedding gives pseudo-unitarity. The positive global dimension is $336 + 192\sqrt{3}$.

We may consequently apply the central-induction criteria with positive dimensions. Exhaustive low-column enumeration gives 39 frames. Only two admit the required character assignments. Both assignments on the first frame are excluded by exact full-character twist equations. The second frame has three identical low restriction columns; its six assignments are precisely the permutations of those columns. Relabelling the corresponding low simples identifies these six necessary-data problems, so one representative suffices.

For this representative, write the restriction Gram equations, joint Galois multiplicity equations, and low cubic-word equations as an integer nonnegative system $An = b$. Here $n$ includes both high-column counts and low fusion multiplicities. The supplement records 165 exact bound certificates. Each upper bound $n_j \le B$ is proved by a separating functional for the augmented system

$$An = b, \qquad n_j - s = B + 1, \qquad n, s \ge 0;$$

its infeasibility and integrality exclude $n_j \ge B + 1$. Replacing $n_j$ by $-n_j$ gives lower bounds in the same way. Every bound is checked against the preceding certified system. Bounded affine-lattice enumeration produces exactly 39800 high restriction profiles. Requiring integral nonnegative low-word multiplicities through degree six and nonnegative restriction remainders leaves 1833 profiles.

For all unordered pairs of nonunit low simples, apply Theorem 4.17. The profiles determine 43 distinct joint restriction–word Gram matrices. The exact fixed-column Schur bounds of Lemma 4.18 enumerate every possible joint coefficient column. The resulting count, Gram, and restriction systems exclude 1568 profiles by exact separating functionals. The height-three full-character twist systems exclude another 91. Two additional joint count systems have inconsistent parity equations. Finally, adjoining all cubic words in the last three low simples gives an exact joint Gram obstruction for one further profile. Thus precisely

$$1833 - 1568 - 91 - 2 - 1 = 171$$

profiles remain in the certified necessary domain. None is discarded because a numerical optimizer reports infeasibility or exhausts its running-time budget.

The supplement reconstructs this complete finite domain and checks every separating and congruence certificate in exact arithmetic. □

**Theorem 4.39.** *The ring $R^{(6)}_{5,1204}$ has no complex fusion categorification.*

*Proof.* By Proposition 4.38, it suffices to exclude the 171 profiles in the pseudo-unitary setting. Write the simple basis as $(1, X, Y, Z, W, V)$ in the displayed dimension order. Every profile has

a unique central simple $U$ with $F(U) = X + Y$. All base simples are self-dual, so uniqueness gives $U \simeq U^*$. The tensor gives

$$F(U^2) = 2 \cdot 1 + 2X + 2Y + 2Z + 4W + 4V.$$

The central unit occurs exactly once. The low restriction columns are

$$(1,0,0,0,0,0), \quad (1,2,2,4,4,4), \quad (1,2,2,5,4,5),$$

and three copies of $(1,1,1,1,2,2)$. After removing the central unit, the remaining unit coordinate is one, so precisely one further low simple occurs. Coordinatewise bounds force its column to be $(1,1,1,1,2,2)$. The remaining high restriction is

$$X + Y + Z + 2W + 2V.$$

Its value under $c \mapsto c_W - c_Z$ is one. In every retained profile, all active high columns except the pure $W$ column have nonpositive value under this functional. Thus a central simple restricting to $W$ is necessary. Exactly 72 profiles lack this column, leaving 99.

For each remaining profile, apply Proposition 4.13 through order six in $K = \mathbb{Q}(\zeta_{24})$. The global dimension $336 + 192\sqrt{3}$ has norm 2304, so the Cauchy prime support is $\{2,3\}$. Proposition 4.11 gives the complete finite image of joint normalized traces, represented at height six by roots of orders dividing 288 together with the zero tuple. This bounds trace types, not actual twist orders. On the columns singled out by Proposition 3.20, the actual twist order divides 12; only these columns are restricted to twelfth-root tuples. All five nonunit second indicators are $+1$, as forced by odd self-dual multiplicities in the tensor squares.

The necessary equations use the induction Gram identities, indicator formulas, every divisor moment of cyclic rotations of orders three through six, low-pair modular equations and joint Galois balance. For a rotation $E$, group its eigenvalues by the tuple of normalized relative traces of all powers occurring in these equations. The variable for a tuple is the sum of the corresponding eigenspace dimensions, hence a nonnegative integer. Normalizing traces changes coefficients, not multiplicities. Consequently the exact equations $Ax = b$, $x \geq 0$, retain integral rotation-count variables even if high-simple counts are relaxed to real numbers.

The supplement supplies a short integral-bound certificate for each profile. To prove $x_j \leq B$ for an integral rotation count, adjoin the opposite branch $x_j \geq B + 1$, encoded by a nonnegative slack, and check an integer ray $y$ with $y^T A \geq 0$ and $y^T b < 0$. A lower bound is proved by excluding $x_j \leq B - 1$. Subsequent steps may use preceding certified bounds. The 99 profiles require 112 such bounds: 86 use one and 13 use two. A final exact ray excludes each resulting system, giving 211 rays altogether. All rounded variables are rotation counts. The checker reconstructs the equations and the full $72 + 99 = 171$ partition, then verifies every dot product with integers; numerical optimization is disabled. Thus all necessary profiles are impossible. □

# 5 Module categories and Morita constraints

These criteria use module categories forced by algebra objects and complement the central-induction system. They do not impose a pivotal structure on the original category. Field-automorphism and tensor-sign reductions are kept with the arithmetic reductions in Section 3.

## 5.1 An involution-module obstruction

**Theorem 5.1** (Antiinvariant module representation)**.** *Let $R$ be a commutative fusion ring and let $g \neq 1$ be an invertible basis element with $g^2 = 1$. Suppose the set $J = \{x : gx = x\}$ is*

*nonempty. If $R$ has a complex fusion categorification, there are integer matrices $T_i \in M_{|J|}(\mathbb{Z})$, indexed by all basis elements, satisfying*

$$T_0 = I, \qquad T_i T_j = \sum_k N_{ij}^k T_k, \qquad T_{i^*} = T_i^{\mathsf{T}}, \tag{5.1}$$

$$|(T_i)_{xy}| \le N_{ix}^y, \qquad (T_i)_{xy} \equiv N_{ix}^y \pmod 2 \qquad (x, y \in J). \tag{5.2}$$

*Failure of this finite system is a general categorification obstruction.*

*Proof.* Let $\mathcal{C}$ be a hypothetical categorification. The additive category generated by a fixed simple $x$ is a rank-one module category over the pointed subcategory generated by $g$. Choose isomorphisms $g \otimes x \simeq x$. The scalar module-associativity equation says that the pointed associator three-cocycle is a coboundary. Thus that subcategory is tensor equivalent to $\mathrm{Vec}_{C_2}$, and $A = 1 \oplus g$ admits the regular connected separable algebra structure. This is a use of module associativity, not an assumption that $g$ has a central lift. Standard separable-algebra module theory is as in [16, Chapter 7].

Let $\mathcal{M} = \mathcal{C}_A$ be the category of right modules. Its free and forgetful functors $I_A, F_A$ satisfy $F_A I_A(X) = X \oplus (X \otimes g)$. Adjunction shows that a two-element $g$-orbit gives one simple module, whose restriction is the sum of the two simples in the orbit. A fixed simple $x$ gives two nonisomorphic simples $M_x^+, M_x^-$, each restricting to $x$. Indeed, $\dim \mathrm{End}(I_A(x)) = 2$, so the squared multiplicities in its semisimple decomposition sum to two. The mixed Gram entries exclude constituents from other orbits. Every simple module occurs in a free module because $A$ is separable.

The algebra automorphism $\alpha$ acting as $+1$ on 1 and $-1$ on $g$ induces an involution $Q$ of $\mathcal{M}$ by twisting the right action. It fixes the module attached to a two-element orbit and interchanges $M_x^+$ and $M_x^-$. The latter assertion follows directly because the nonzero action map $x \otimes g \to x$ changes sign, whereas an endomorphism of the simple underlying object is scalar. This twisting functor commutes with the left action of $\mathcal{C}$.

Consequently the $(-1)$-eigenspace of $Q$ on $K_0(\mathcal{M}) \otimes \mathbb{Q}$ is a representation of $R$, with basis $e_x = [M_x^+] - [M_x^-]$, $x \in J$. Between two fixed pairs the action matrix has a block $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$. Forgetting modules gives $a + b = N_{ix}^y$, while its action on the antiinvariant basis has coefficient $a - b$. Nonnegativity and integrality of $a, b$ give (5.2). The module associator gives the multiplication identity in (5.1). Finally module Frobenius reciprocity makes the action of $i^*$ the transpose of the action of $i$. All $e_x$ have the same squared length two, so the restricted matrices obey the same transpose identity. No spherical or unitary structure has been used. □

**Proposition 5.2** (Exact generator enumeration)**.** *The conditions of Theorem 5.1 can be decided by a finite integer search. If a selected set of basis elements generates $R \otimes \mathbb{Q}$ as an algebra, it suffices to enumerate their matrices, reconstruct the other matrices by rational polynomial expressions, and check* (5.1)–(5.2)*.*

*Proof.* Every entry has an explicit finite interval and parity. Choose monomials $w_0 = 1, w_1, \dots, w_{r-1}$ forming a rational basis of the rank-$r$ algebra, and let $V$ have as its rows their coefficients in the based-ring basis. The expression for $b_i$ is row $i$ of $V^{-1}$ applied to the monomials. Thus any representation is uniquely reconstructed from the selected generator matrices. Testing all the ring relations, transpose identities, and entry bounds after reconstruction accepts every possible solution and nothing outside the stated necessary system. □

The direct enumeration implementation treats $|J| \le 2$ and chooses one or two generators when such a set is found. Applying it to the census gives nine complete exclusion certificates, including $R_{3,79}^{(6)}$ and $R_{3,216}^{(6)}$. The respective generator searches contain 4608 and 1536 assignments. These are automatic applications of one theorem; their indices are not inputs to the test. The spectral refinement in Section 5.2 also treats larger fixed-point sets and supplies two further exclusions with $|J| = 4$.

## 5.2 Spectral pruning of the antiinvariant module

The matrix system in Theorem 5.1 is useful beyond the two-dimensional antiinvariant space. Its entry bounds impose an additional spectral restriction that makes larger fixed-point sets accessible.

**Proposition 5.3** (Perron-factor exclusion)**.** *Retain the notation of Theorem 5.1, and let $i = i^*$. Suppose the nonnegative regular fusion matrix $N_i$ is irreducible. Then no $T_i$ can have an eigenvalue whose minimal polynomial contains* $\mathrm{FPdim}(i)$ *as a root. Moreover, if for every $x \in J$ there are $a, y \in J$ with $N_{ax}^y$ odd, then $T_g = I$.*

*Proof.* Write $B_i = (N_{ix}^y)_{x,y\in J}$. Since $J$ is a proper subset of the basis, the strict Perron–Frobenius inequality for a proper principal submatrix of an irreducible nonnegative matrix gives

$$\rho(T_i) \le \rho(|T_i|) \le \rho(B_i) < \rho(N_i) = \mathrm{FPdim}(i).$$

The characteristic polynomial of the integer matrix $T_i$ is rational; thus an irreducible factor containing $\mathrm{FPdim}(i)$ cannot divide it. The bounds on $T_g$ make it diagonal with entries $\pm 1$. For $a \in J$, the identity $ga = a$ gives $T_g T_a = T_a$. A diagonal entry $-1$ would therefore force the corresponding row of every such $T_a$ to vanish, contradicting the indicated odd entry. □

If $i$ generates $R \otimes \mathbb{Q}$, express each basis element as a rational polynomial $p_a(i)$. The preceding proposition removes the Perron factor from the possible minimal polynomial of $T_i$; when $T_g = I$, every remaining factor must also divide $p_g(t) - 1$. Since $T_i \equiv B_i \pmod 2$, its characteristic polynomial must reduce to that of $B_i$ modulo two. Enumerate all products of the remaining irreducible factors of total degree $|J|$, allowing repetitions. Each product fixes both $\mathrm{Tr}(T_i)$ and $\mathrm{Tr}(T_i^2) = \sum_{x,y}(T_i)_{xy}^2$. These equalities reduce the finite entry search substantially. Reconstruction by the $p_a$ and verification of the entire system (5.1)–(5.2) finish the test. The search is exhaustive because every possible module representation has one of the enumerated rational characteristic polynomials.

**Proposition 5.4.** *The rings $R_{8,2945}^{(6)}$ and $R_{8,3071}^{(6)}$ have no categorification as complex fusion categories.*

*Proof.* In both rings the involution is $g = b_1$ and $J = \{b_2, b_3, b_4, b_5\}$; all basis elements are self-dual. All tensor identities used below are checked automatically from the census tensors in the supplementary certificates.

For $R_{8,2945}^{(6)}$, the element $b_3$ generates the rational algebra, its regular matrix is irreducible, and

$$\det(tI - N_3) = t(t-2)(t^2 - 23t - 1)(t^2 + t - 1).$$

The Perron factor is $t^2 - 23t - 1$. The parity criterion gives $T_g = I$, and the character with $b_3$-value zero has $g$-value $-1$, so this character cannot occur. Thus only $t - 2$ and $t^2 + t - 1$ remain. The fixed block has characteristic polynomial $t^4 + t^2 + 1$ modulo two. Of the three degree-four products of the two remaining factors, only $(t^2 + t - 1)^2$ has this reduction. Hence

$$\mathrm{Tr}(T_3) = -2, \qquad \mathrm{Tr}(T_3^2) = 6, \qquad T_3^2 + T_3 - I = 0.$$

In the ordered basis $J$, the required odd entries of $T_3$ are the last two diagonal entries and the symmetric pairs $(1,4), (2,3)$. Their squared contributions already sum to six. Consequently

$$T_3 = \begin{pmatrix} 0 & 0 & 0 & \varepsilon \\ 0 & 0 & \delta & 0 \\ 0 & \delta & -1 & 0 \\ \varepsilon & 0 & 0 & -1 \end{pmatrix}, \qquad \varepsilon, \delta \in \{1, -1\}.$$

Rational reconstruction, reduced modulo $t^2+t-1$, gives $T_2 = -3T_3 - 4I$. Its first diagonal entry is $-4$, contradicting the bound $|(T_2)_{11}| \le N_{b_2b_2}^{b_2} = 2$.

For $R_{8,3071}^{(6)}$, the regular matrix $N_3$ is irreducible and

$$\det(tI - N_3) = t^3(t+2)(t^2 - 26t + 24).$$

The last factor is the Perron factor. After removing it, the symmetric matrix $T_3$ has eigenvalues in $\{0, -2\}$, so $T_3^2 = -2T_3$. Every diagonal entry of its fixed block is six, and therefore every diagonal entry $d$ of $T_3$ is even. The diagonal of the matrix identity gives

$$d^2 + \sum_{y \neq x} (T_3)_{xy}^2 = -2d.$$

It follows that $d \in [-2, 0] \cap 2\mathbb{Z} = \{-2, 0\}$, and either choice forces all off-diagonal entries in that row to vanish. This contradicts $N_{b_3b_2}^{b_3} = 7$, which requires the corresponding off-diagonal entry to be odd. □

The exact generator search checks respectively four and zero trace-and-norm-compatible matrices for these two records. The second search uses $b_5$ as a rational algebra generator; the displayed direct argument uses $b_3$ to make the contradiction shorter. Both certificates are general categorification obstructions, with no pivotal assumption.

### 5.3 Descent to a rank-three near-group subring

**Theorem 5.5** (Rank-three subring obstruction)**.** *Suppose the Grothendieck ring of a complex fusion category contains three distinct basis elements $1, g, x$ satisfying*

$$g^2 = 1, \qquad gx = xg = x, \qquad x^2 = 1 + g + mx.$$

*Then $m \le 2$.*

*Proof.* The three basis elements are closed under products and duals, so their simple objects generate a fusion subcategory with Grothendieck ring $K(1, 0, m, 0)$. By [38, Theorem 4.1], categorifiability of this ring implies $m \le 2$. The cited theorem uses the Galois-pseudounitary reduction for this near-group family and does not impose a pivotality hypothesis on the ambient category. □

In $R_{4,162}^{(6)}$, the basis elements $(b_0, b_3, b_4)$ have precisely these products with $m = 4$. The exact certificate checks all coefficients in their nine products, including the vanishing coefficients outside this subring. Theorem 5.5 therefore excludes this ring from general complex categorification. This automatic scan formed part of the campaign through multiplicity eight and is also included in the new multiplicity-nine pipeline.

### 5.4 Small-dimension Morita obstructions

The next criteria use categorical Morita equivalence rather than additional indicator-height searches. They apply to general complex fusion categories. In particular, no chosen pivotal structure enters their hypotheses.

### 5.4.1 A cyclic-index bound and dimension thirty

For group-theoretical data $(G, \omega, H, \psi)$, the simple objects of $\mathcal{C}(G, \omega, H, \psi)$ are indexed by a double coset $HgH$ and an irreducible projective representation $\pi$ of $S_g = H \cap gHg^{-1}$. Their Frobenius–Perron dimensions are

$$[H : S_g] \dim \pi. \tag{5.3}$$

The objects supported on $H$ form a fusion subcategory equivalent to $\mathrm{Rep}(H)$. One can see this after replacing the data by an equivalent adapted-cocycle presentation with $\psi = 1$ and $\omega|_{G\times G\times H} = 1$; see [40, Proposition 2.5]. These descriptions also follow from [21].

**Lemma 5.6** (Cyclic-index bound)**.** *Suppose a finite group $G$ has a cyclic normal subgroup $B$ of index at most two. Every simple object of every group-theoretical category $\mathcal{C}(G, \omega, H, \psi)$ has Frobenius–Perron dimension at most two.*

*Proof.* Put $K = H \cap B$. Since every subgroup of a cyclic group is characteristic, $K$ is normal in $G$. Thus $K \subseteq S_g \subseteq H$ and $[H : K] \leq 2$. If $[H : S_g] = 2$, then $S_g = K$ is cyclic. Every irreducible projective representation of a cyclic group over $\mathbb{C}$ has dimension one, so (5.3) is two.

If $S_g = H$, it suffices to bound the dimensions of projective representations of $H$. In the twisted group algebra of $H$, the subalgebra on $K$ is commutative and semisimple: its cyclic generator satisfies $u^{|K|} = c \neq 0$, a polynomial with distinct roots. The remaining coset, when present, permutes its weight spaces in orbits of size at most two. In an irreducible module there is one such orbit. For an orbit of size one, a coset generator squares to a nonzero scalar and hence has a one-dimensional invariant subspace. For an orbit of size two, a line in one weight space and its translate span an invariant two-dimensional subspace. Irreducibility therefore gives dimension at most two. This argument includes arbitrary projective factor sets. □

**Proposition 5.7** (Integral dimension thirty)**.** *Every simple object of an integral fusion category of Frobenius–Perron dimension* 30 *has dimension one or two.*

*Proof.* The category is group-theoretical by [17, Theorem 9.2]. Its presenting group has order thirty. Every group of order thirty is solvable: the possible numbers of Sylow five- and three-subgroups are respectively $1, 6$ and $1, 10$; if neither is one, the corresponding nonidentity elements already number $24 + 20 > 29$. A normal Sylow subgroup and the solvable quotient prove the claim. Hall's theorem now supplies a subgroup of order fifteen. It is cyclic by the Sylow theorems and normal because it has index two. Lemma 5.6 applies. □

The adjective *integral* is essential: the statement does not claim the same bound for nonintegral categories of total dimension thirty. The certificate checks a positive integral dimension character, not just an integer value of the total dimension.

### 5.4.2 A Morita dichotomy in dimension sixty

**Lemma 5.8.** *A fusion category of Frobenius–Perron dimension sixty is either solvable or categorically Morita equivalent to $\mathrm{Vec}^{\omega}_{A_5}$ for some three-cocycle $\omega$. If it is integral and solvable, has exactly two invertible simples, and satisfies $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$, every simple has dimension at most four.*

*Proof.* By [17, Theorem 9.16], the category is weakly group-theoretical. It is therefore Morita equivalent to a nilpotent category of the same dimension, obtained from vector spaces by successive extensions by finite groups whose orders multiply to sixty. If all these groups are solvable, refining their normal series into cyclic prime-order factors shows that the category is solvable. Solvability is invariant under categorical Morita equivalence; see [17, Section 4.2].

Otherwise an extension group has nonsolvable order dividing sixty. All proper divisors involve at most two primes, apart from thirty, which was handled above. Burnside's theorem and the fact that the unique nonabelian simple group of order sixty is $A_5$ imply that this group is $A_5$ itself. Its order uses the entire dimension. The nilpotent category is consequently pointed, of the form $\mathrm{Vec}^{\omega}_{A_5}$.

For the second assertion, a nontrivial last step of a solvable construction is a cyclic prime-order extension or equivariantization. An extension would give a nontrivial grading, contrary to $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$. In an equivariantization $\mathcal{C} = \mathcal{D}^{C_p}$, the canonical $\mathrm{Rep}(C_p)$ is a fusion subcategory. Exactly two invertible simples force $p = 2$, and $\mathrm{FPdim}(\mathcal{D}) = 30$.

The simples of a $C_2$-equivariantization are indexed by orbits of simples of $\mathcal{D}$ and projective representations of their stabilizers. Those stabilizers are trivial or cyclic of order two, so the projective representations have dimension one. A simple dimension in $\mathcal{C}$ is therefore either $d$ or $2d$, with $d$ a simple dimension in $\mathcal{D}$. Since $\mathcal{C}$ is integral, $d$ is rational; as an algebraic integer it is an integer. Thus $\mathcal{D}$ is integral. Proposition 5.7 gives $d \leq 2$, proving the bound four. □

### 5.4.3 Cocycle-independent involution returns

For a normalized three-cocycle on a group of order two, its value on the triple of nonidentity elements is unchanged by a normalized coboundary. In particular, triviality of its restriction class implies that this value is one.

**Theorem 5.9** (Involution-return indicator)**.** *Let $\mathcal{C}(G, \omega, H, \psi)$ be group-theoretical, let $g \in G$, and put $S = H \cap gHg^{-1}$. Suppose every element in*

$$T_g = \{r \in gH : r^2 \in S\}$$

*is an involution conjugate in $G$ to an element of $H$. For the simple object $M(g, \pi)$ associated to an irreducible projective representation $\pi$ of $S$, one has*

$$\nu_2(M(g, \pi)) = \frac{|T_g| \dim \pi}{|S|}. \tag{5.4}$$

*In particular, if $T_g \neq \varnothing$, these simples are self-dual. This conclusion holds for every admissible $\omega, \psi$.*

*Proof.* Use the equivalent adapted presentation supplied by [40, Proposition 2.5]; its cocycle restricts to one on $H^3$. If $r$ is an involution conjugate into $H$, the restriction class on $\langle r \rangle$ is trivial. The preceding coboundary observation gives $\omega(r, r, r) = 1$ in this adapted presentation, independently of any remaining values of the cocycle.

The second-indicator specialization of [40, Theorem 4.2, equation (4.6)] is

$$\nu_2(M(g, \pi)) = \frac{1}{|S|} \sum_{r \in T_g} \pi_{-2}(r) \chi_\pi(r^{-2}).$$

For an involution, the scalar $\pi_{-2}(r)$ in that formula is $\omega(r, r, r)^{-1} = 1$, whereas $\chi_\pi(r^{-2}) = \chi_\pi(1) = \dim \pi$. This proves (5.4). Group-theoretical categories have their positive spherical structures; a simple with nonzero second indicator is self-dual. Self-duality is preserved under tensor equivalence, so the conclusion is independent of the presentation. □

**Corollary 5.10** (Dihedral supports in $A_5$)**.** *In every category $\mathcal{C}(A_5, \omega, D_{2q}, \psi)$ with $q \in \{3, 5\}$, every simple of dimension $q$ has second indicator one. In particular all such simples are self-dual.*

*Proof.* The dihedral subgroup has order $2q$. Its ordinary irreducible representations have dimensions one or two, so a simple of dimension $q > 2$ is not supported on it. Formula (5.3) then forces stabilizer order two and a one-dimensional projective representation. For these double cosets the following exact group data hold:

| $\lvert H\rvert$ | double-coset sizes | $\lvert S\rvert$ | $\lvert T_g\rvert$ when $\lvert S\rvert = 2$ |
|---|---|---|---|
| 6 | $6, 18, 36$ | 2 | 2 |
| 10 | $10, 50$ | 2 | 2 |

Every element in $T_g$ is an involution. All fifteen involutions in $A_5$ are conjugate, and $H$ contains an involution. Thus Theorem 5.9 gives $\nu_2 = 2/2 = 1$.

These group assertions are certified independently from the sixty even permutations of five letters. The verifier generates every order-$2q$ subgroup using an element of order $q$ and an involution, checks that they form one conjugacy class, and enumerates every right coset and every square-return element. It does not test only a selected coset or an assumed cocycle. The coset lists are included in each certificate. □

**Theorem 5.11** (Tensor-checkable small-dimension obstruction)**.** *Let $R$ be an integral fusion ring. Each of the following conditions excludes a general complex categorification.*

(a) $\mathrm{FPdim}(R) = 30$ *and some basis dimension exceeds two.*

(b) $\mathrm{FPdim}(R) = 60$*, the adjoint based subring is $R$, and exactly two basis elements are invertible. Some basis dimension exceeds four. For a prime $q \in \{3, 5\}$ the complete set of based-subring dimensions is $\{1, 2, 2q, 60\}$, and a basis element of dimension $q$ is not self-dual.*

*Proof.* Part (a) is Proposition 5.7. For (b), Lemma 5.8 excludes solvability and gives a presentation $\mathcal{C}(A_5, \omega, H, \psi)$. The canonical $\mathrm{Rep}(H)$ subcategory forces $|H| \in \{1, 2, 2q, 60\}$. If $|H| \leq 2$, every simple dimension is at most two, contradicting the stated dimension exceeding four. If $H = A_5$, the category is $\mathrm{Rep}(A_5)$ and has only one invertible simple, because $A_5$ is perfect. Thus $H$ is an order-$2q$ subgroup, necessarily a conjugate of $D_{2q}$. Corollary 5.10 contradicts the non-self-dual dimension-$q$ basis element. □

### 5.4.4 Uniform census application

The implementation supplies the same tensor test at every multiplicity. It first checks a positive integral dimension character; it then computes the adjoint closure, all based subrings, duality, and the number of invertibles. Illustrative applications are

| ring | sorted dimension type | branch | $q$ |
|---|---|---|---|
| $R^{(6)}_{2,51}$ | $(1, 1, 2, 2, 5, 5)$ | (b) | 5 |
| $R^{(6)}_{3,15}$ | $(1, 1, 2, 3, 3, 6)$ | (b) | 3 |
| $R^{(6)}_{4,19}$ | $(1, 1, 1, 1, 1, 5)$ | (a) | – |

The program finds these applications from the tensor, without selecting labels in advance. The current census scan and its exact dimension checks are recorded by `software/morita_controls.py`. The criterion also excludes $R^{(6)}_{2,25}$ and $R^{(6)}_{2,29}$. Overlap with other criteria is counted only once in the sequential exclusion table.

## 5.5 Two-sided module Gram constraints

Let $\mathcal{C}$ be a fusion category, let $A$ be a connected separable algebra, and let $\mathcal{M} = \mathcal{C}_A$. Write $M_i$ for the action matrix of a simple $X_i \in \mathcal{C}$, and $\mathcal{D} = \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{M})$ for the dual category, with a

fixed choice of opposite tensor convention. All dimension statements here use Frobenius–Perron dimensions, not a positive pivotal structure.

**Proposition 5.12** (Two-sided module Gram identities)**.** *For every simple $m \in \mathcal{M}$, let $B_m$ be the matrix of the evaluation functor $\mathcal{D} \to \mathcal{M}$, $Y \mapsto Y(m)$. Then*

$$B_m B_m^{\mathsf{T}} = \sum_i (M_i)_{mm} M_i. \tag{5.5}$$

*The tensor unit column of $B_m$ is $e_m$. If $d_n$ are the positive module dimensions in any fixed normalization, each column $v$ belonging to a simple $Y \in \mathcal{D}$ satisfies*

$$\sum_n v_n d_n = d_m \operatorname{FPdim}(Y). \tag{5.6}$$

*Proof.* The internal endomorphism algebra of $m$ in $\mathcal{C}$ is $A_m = \bigoplus_i (M_i)_{mm} X_i$, up to the harmless replacement of every index by its dual; these multiplicities agree by module reciprocity. The module category is equivalently $\mathcal{C}_{A_m}$, sending $m$ to the free rank-one module. The dual category is the category of $A_m$-bimodules. Evaluation becomes the functor forgetting the left action, whose adjoint is free left multiplication by $A_m$. Their composite on $\mathcal{M}$ is therefore the action of $A_m$. Adjunction gives (5.5). Evaluation of the tensor unit is $m$, giving the specified column. The commuting action of the dual category on the positive module dimension vector has eigenvalue $\operatorname{FPdim}(Y)$; this gives (5.6). These assertions are standard consequences of separable-algebra Morita theory; see [16, Chapter 7]. □

### 5.5.1 The split-involution input and a finite system

Suppose $g^2 = 1$ is an invertible object fixing a simple object of $\mathcal{C}$. The module-associativity argument in the main manuscript supplies the algebra $A = 1 \oplus g$. Free two-element $g$-orbits give one simple of $\mathcal{M}$; fixed simples give two. The sign automorphism of $A$ interchanges the two actions on every fixed simple. Thus the action matrices $M_i$ can be reconstructed from the full tensor and an anti-invariant representation satisfying the parity, transpose and entry bounds of Theorem 5.1.

Normalize the dimension of the free module $A$ to one. An orbit module has dimension $\operatorname{FPdim}(X)$ and a split fixed module has dimension $\operatorname{FPdim}(X)/2$. These module dimensions need not be algebraic integers. At $m = A$, the Gram matrix in (5.5) is $I + M_g$. Since $M_g$ is an involutive permutation, its nonnegative integral Gram columns are forced: a fixed point has two unit columns, and a transposed pair has one column supported on that pair. Hence (5.6) determines the multiset of simple dimensions of $\mathcal{D}$. One of the unit-dimension columns is its tensor unit.

For another simple $m$, remove its known tensor-unit column. For every remaining prescribed dual dimension $e$, enumerate nonzero integer vectors $v \geq 0$ with

$$v_i^2 \leq Q_{ii}, \quad v_i v_j \leq Q_{ij}, \quad \sum_i v_i d_i = e d_m, \qquad Q = \sum_i (M_i)_{mm} M_i - e_m e_m^{\mathsf{T}}.$$

The integer multiplicities of these columns must add their outer products to $Q$ and respect the prescribed number of columns of each dual dimension. Relaxing the multiplicities to nonnegative rationals gives a necessary rational linear system. An exact Farkas vector proves infeasibility. A general exclusion requires infeasibility for every admissible anti-invariant representation; selecting only one is invalid.

The executable implementation treats at most two fixed simples and simple-dimension fields of degree at most four. It enumerates all admissible generator matrices and reconstructs the complete representation. The vector search uses exact field coordinates and explicit integer

bounds. A domain-budget limit or an ineligible fixed-point count gives no conclusion. Negative checking, when a certificate is proposed, reconstructs the full domain and checks only exact arithmetic.

The implementation in `software/module_dual_gram.py` is a uniform necessary-condition test. It does not assign categorical existence when the relaxed system is feasible. Its successful certificates, if any, must enter the same exact-checking and disjoint allocation process as every other obstruction.

### 5.6 Coupling all Morita action matrices

The two-sided module test can be strengthened by retaining the complete associativity pairing, rather than its restrictions to one module simple at a time. The strengthening uses Frobenius–Perron positivity only to recognize module categories; it makes no pivotal or spherical assumption.

**Theorem 5.13** (Full Morita action Gram identity)**.** *Let ${}_{\mathcal{C}}\mathcal{M}_{\mathcal{D}}$ be an invertible bimodule category, with simple objects $m_0, \dots, m_{s-1}$. Write*

$$L_i(a,b) = [X_i \otimes m_a : m_b], \qquad R_j(a,b) = [m_a \otimes Y_j : m_b]$$

*for its left and right action matrices. Then*

$$\sum_j R_j(a,d)R_j(b,c) = \sum_i L_i(b,a)L_i(c,d) \qquad (0 \le a,b,c,d < s). \tag{5.7}$$

*Consequently the matrix*

$$G_{(a,d),(b,c)} = \sum_i L_i(b,a)L_i(c,d)$$

*has a nonnegative integral Gram factorization whose columns are the vectorized matrices $R_j$. Each such column belongs to $\operatorname{im}_{\mathbb{Q}} G$, and its coordinate indexed by $(a,d)$ is at most $\lfloor\sqrt{G_{(a,d),(a,d)}}\rfloor$. The corresponding matrices commute with every $L_i$, include the identity, and are permuted by transposition.*

*Proof.* Use duals in the Morita bicategory. The multiplicity of $Y_j$ in $m_b^* m_c$ is $R_j(b,c)$, by Frobenius reciprocity. Thus the multiplicity of $m_d$ in $m_a(m_b^* m_c)$ is the left-hand side of (5.7). The multiplicity of $X_i$ in $m_a m_b^*$ is $L_i(b,a)$, so associativity identifies the same multiplicity with its right-hand side. If $W$ is the matrix of vectorized right actions, this is $G = WW^{\mathsf{T}}$. Hence $\operatorname{im}_{\mathbb{R}} W = \operatorname{im}_{\mathbb{R}} G$; since all matrices are rational, each column of $W$ belongs to the stated rational image. A diagonal entry of $G$ is the sum of the squares of the corresponding entries of $W$, which proves the bound. The commuting left and right actions, the tensor unit of $\mathcal{D}$, and $R_{j^*} = R_j^{\mathsf{T}}$ prove the remaining assertions. □

Suppose now that $A = \mathbf{1} \oplus g$ is the split involution algebra and $m_0 = A$ in $\mathcal{C}_A$. The existence argument for this algebra and the exhaustive module-action construction are those of the preceding module criterion. Since the internal end of $m_0$ is $A$, setting $c = d = 0$ in (5.7) gives

$$\sum_j R_j(\,\cdot\,,0)R_j(\,\cdot\,,0)^{\mathsf{T}} = I + L_g. \tag{5.8}$$

Here $L_g$ is an involutive permutation matrix. Its right-hand side has a unique nonnegative integral Gram factorization with nonzero columns, up to permutation of columns: a fixed point contributes two copies of its unit vector, and a transposition contributes the sum of its two unit vectors. Indeed, zero off-diagonal entries prevent columns from meeting different orbits. On a fixed point the squared entries sum to two; on a transposition the diagonal entries

and the off-diagonal entry are all one. These observations prove the assertion orbit by orbit. Transposition supplies the same finite list for the possible first rows of each right action matrix.

The implementation intersects the rational image of $G$ with these prescribed first rows and columns. Choosing free coordinates among the actual matrix entries, the diagonal bounds in Theorem 5.13 give a finite integer box. Every candidate is checked for nonnegativity, the entrywise necessary bounds $vv^{\mathsf{T}} \leq G$, and commutation with the left actions. If the candidates are $v_1, \ldots, v_t$, the unknown multiplicities must satisfy

$$\sum_{h=1}^{t} z_h v_h v_h^{\mathsf{T}} = G - \operatorname{vec}(I)\operatorname{vec}(I)^{\mathsf{T}}, \qquad z_h \in \mathbb{Z}_{\geq 0},$$

together with the first-column multiplicities from (5.8), after removing the unit action. An exact Farkas certificate for the relaxation $z_h \geq 0$ excludes that module-action branch. As before, exclusion of the fusion ring requires exhausting every eligible branch. The program `software/full_morita_gram.py` reconstructs the finite domains and checks separating vectors using integer arithmetic.

**Proposition 5.14** (The involution dual of the first residual)**.** *If $\mathcal{C}$ categorifies $R^{(6)}_{3,215}$ and $g$ is its nontrivial invertible simple, then $\mathbf{1} \oplus g$ admits a connected separable algebra structure and*

$$K_0({}_A\mathcal{C}_A) \cong K_0(\mathcal{C}) \quad \text{as based rings}, \qquad A = \mathbf{1} \oplus g.$$

*This assertion does not identify the two fusion categories up to tensor equivalence.*

*Proof.* The involution fixes two simples, so the previously established split involution argument applies. Exhaustive reconstruction gives sixteen module-action branches. Two violate the rank bound $\operatorname{rank} G \leq \operatorname{rank} K_0({}_A\mathcal{C}_A)$, and twelve have exact Farkas certificates for the full Gram system. Each of the remaining two has Gram rank six and six dual simples. Therefore its six right action matrices are linearly independent; in particular they cannot repeat. Exhaustive selection from the finite candidate lists gives exactly one Gram frame for each branch, up to permuting its columns. Multiplying those six matrices reconstructs a nonnegative integral fusion tensor, and a recorded based permutation identifies all its coefficients with $R^{(6)}_{3,215}$.

The complete sixteen-branch record is `research/new_frontier/frontier_full_morita.json`. The script `research/new_frontier/verify_frontier_full_morita.py` reconstructs the module actions and candidate domains, checks all rank and Farkas obstructions, proves uniqueness of each surviving frame by finite exact enumeration, and verifies every matrix product and the based-ring identification. Verification does not call an optimizer. □

The two surviving frames are based bimodule data, not associators or categorifications. In particular the proposition does not supply an algebra $\mathbf{1} \oplus X$ for a noninvertible simple $X$.

## 5.7 Integral multiplicities in the Morita Gram identity

The full Morita action Gram identity of Theorem 5.13 has an elementary integral refinement. The unknown attached to a possible right-action matrix counts simple objects of the dual category; it is an integer, even when distinct simple objects have the same action matrix. Retaining this integrality detects obstructions invisible in the rational nonnegative relaxation.

**Proposition 5.15** (Morita Gram congruence obstruction)**.** *Fix one of the exhaustively enumerated module actions for a forced split involution algebra $\mathbf{1} \oplus g$. Let $Ax = b$ be its full Morita Gram system, including the prescribed first-column multiplicities, with all possible right-action matrices retained. If there exist $q > 1$ and an integral row vector $y$ such that*

$$yA \equiv 0 \pmod{q}, \qquad yb \not\equiv 0 \pmod{q},$$

*then this module action cannot occur in a complex fusion category.*

*Proof.* Every categorification would give a vector $x$ of nonnegative integral multiplicities satisfying $Ax = b$. Multiplication by $y$ would give $yb = yAx \equiv 0 \pmod q$, a contradiction. All entries of $A$ and $b$ come from the integral fusion and module actions. The argument therefore requires no pivotal or spherical structure and no positivity assumption on categorical dimensions. □

For discovery, Gaussian elimination over $\mathbb{F}_2$ produces the row vector $y$. Verification only multiplies integer vectors and checks the two congruences. Repeated equality rows may be removed by a deterministic rule. The completeness of the right-action candidate list is exactly the finite Gram enumeration already used in Section 5.6; no new search cutoff is introduced.

**Proposition 5.16.** *The rings* `R6_5_638` *and* `R6_7_1225` *admit no categorification by a fusion category over* $\mathbb{C}$.

*Proof.* The unrestricted signed reduction is verified separately for each full fusion tensor. There are respectively 11 800 and 46 856 signed tables, partitioned into 418 and 1 560 sign orbits. Each has one original gauge orbit. Every other orbit is excluded by its exact character obstructions; within the original orbit the only admissible nonzero dimension characters lie in the Frobenius–Perron Galois orbit. It therefore suffices to exclude pseudo-unitary categorification.

In each case the involution-module enumeration has sixteen branches. Two fail the full Morita rank bound. Twelve fail Proposition 5.15, with $q = 2$. Both remaining branches force negative stabilizer commutators on the two fixed simple objects, indexed 4 and 5 in the stored tensor. Their cubic rotation spaces have common dimension $n = 4$ for `R6_5_638` and $n = 6$ for `R6_7_1225`. The higher-dimensional Clifford rotation criterion of Section 5.10 applies. The exponent prime supports are respectively $\{2, 5, 13\}$ and $\{2, 5, 29\}$, so the third indicators belong respectively to

$$\{-2, 1\} \qquad \text{and} \qquad \{-3, 0\}.$$

The actual separating certificates already hold on the weaker intervals $[-2, 1]$ and $[-3, 3/2]$; they do not rely on selecting a discrete indicator value.

For either tensor the complete low-induction enumeration gives 91 frames, of which 31 have a nonempty character-labelling set. The 50 labellings and four second-indicator choices yield 200 exact height-six systems. Every system has a verified integral separating functional. The two dimension fields are $\mathbb{Q}(\sqrt{65})$ and $\mathbb{Q}(\sqrt{145})$; their low-column counts are respectively $(217, 225, 225, 61, 61)$ and $(469, 481, 481, 127, 127)$. All possible twist orders are covered by the inherited relative-trace construction. Thus the two remaining module branches also fail. □

The exact verifier is `software/round3_integral_morita.py`, with certificate kind `general_integral_morita_rotation`. The full finite domains, congruence vectors, separating vectors, and optimizer-free replay reports are stored in `research/round3/higher/`.

## 5.8 An order-six Morita–grading obstruction

The dimension reconstruction above can force a dual category whose based ring has a short exhaustive parametrization. This supplies two further exclusions without assuming a pivotal structure on the original category.

**Theorem 5.17.** *Let $\mathcal{C}$ be a fusion category, and let $g$ be an invertible involution fixing a simple object. Set $A = \mathbf{1} \oplus g$. Suppose the simple Frobenius–Perron dimensions of $\mathcal{D} = {}_A\mathcal{C}_A$ are*

$$\underbrace{1, \dots, 1}_{6}, \qquad \underbrace{d, \dots, d}_{6}, \qquad d = 2 + \sqrt{5}.$$

*Then $\mathcal{C}$ has a faithful $C_3$ grading. In particular, a fusion ring with trivial universal grading and the stated forced dual dimension profile cannot be categorified. This excludes $R^{(6)}_{5,100}$ and $R^{(6)}_{5,408}$ over $\mathbb{C}$.*

*Proof.* The algebra $A$ exists by the fixed-object argument of Theorem 5.1. We first exhaust the possible based rings of $\mathcal{D}$. Its invertible objects form a group $G$ of order six. The multiplicity of an invertible $h$ in $\rho\rho^*$, for a noninvertible simple $\rho$, is one precisely when $h\rho \cong \rho$. The identity $d^2 = 1 + 4d$ and irrationality of $d$ therefore imply that the stabilizer of $\rho$ is trivial. Thus the noninvertibles are exactly $h\rho$, $h \in G$. The right action is free for the same reason, so

$$\rho h = \phi(h)\rho, \qquad \rho^2 = u + \sum_{h\in G} a_h h\rho, \qquad \sum_{h\in G} a_h = 4, \tag{5.9}$$

where $\phi \in \mathrm{Aut}(G)$, $u \in G$, and $a_h \in \mathbb{Z}_{\geq 0}$. Associativity gives $\phi^2 = \mathrm{Ad}(u)$ and $\phi(u) = u$; comparison of $(\rho^2)h$ with $\rho(\rho h)$ also makes the coefficient function invariant under

$$k \longmapsto \phi^2(h)k\phi(h)^{-1}.$$

Only $C_6$ and $S_3$ can occur as $G$.

*The cyclic case.* The automorphism $\phi$ is either the identity or inversion. Inversion would make $(a_h)$ constant on cosets of the square subgroup of order three, contradicting $\sum a_h = 4$. Hence $\phi$ is the identity. Use additive labels $0, \dots, 5$ for $C_6$. Duality gives $\rho^* = (-u)\rho$ and

$$u \in \mathbb{Z}/6\mathbb{Z}, \qquad a_k \geq 0, \qquad \sum_{k=0}^{5} a_k = 4, \qquad a_k = a_{u-k}. \tag{5.10}$$

There are exactly 60 pairs $(u, a)$ satisfying these conditions. Each supplies the entire candidate tensor by

$$h \cdot k = h + k, \qquad h \cdot (k\rho) = (h + k)\rho, \qquad (h\rho)(k\rho) = h + k + u + \sum_j a_j(h + k + j)\rho.$$

An exact exhaustive calculation partitions the 60 pairs as follows:

| Number | Exact consequence |
|---|---|
| 42 | A formal-codegree norm obstruction |
| 6 | A closed based subring isomorphic to $R^{(6)}_{2,20}$ |
| 12 | A faithful $C_3$ grading with neutral invertible involution |

The first row is a general obstruction here: by [41, Lemma 3.1], a category with these two distinct Frobenius–Perron dimensions is Galois conjugate to a pseudo-unitary category with the same based ring. In that category $\mathrm{FPdim}(\mathcal{D})/f$ must be an algebraic integer for every formal codegree $f$. The cyclic candidate ring is commutative, so the Casimir characteristic roots used in the check are its formal codegrees, with no representation dimension correction. Its global dimension is

$$D = 6 + 6d^2 = 60 + 24\sqrt{5}, \qquad p_D(t) = t^2 - 120t + 720.$$

For each of the 42 cases, the certificate gives the exact minimal polynomial $q$ of a formal codegree and a prime $p$ for which

$$\frac{v_p(|p_D(0)|)}{\deg p_D} < \frac{v_p(|q(0)|)}{\deg q}. \tag{5.11}$$

This contradicts integrality of $D/f$: take the norm in any common number field containing $D$ and $f$. The second row uses the previously established general exclusion for $R_{2,20}^{(6)}$, whose original exact certificate is replayed, not just its allocated status. For each of the remaining 12 pairs the grading is explicitly

$$\deg(h) = h \pmod 3, \qquad \deg(h\rho) = h + 2u \pmod 3. \tag{5.12}$$

Every nonzero structure constant respects this degree assignment. The unique nonidentity invertible involution, namely $3 \in C_6$, has degree zero.

*The nonabelian case.* Every automorphism of $S_3$ is inner. Replacing $\rho$ by an invertible multiple makes $\phi$ the identity. Then $u$ is central, so $u = 1$, and $a$ is constant on conjugacy classes. Its values $(a_e, a_t, a_c)$ on the identity, transpositions and three-cycles satisfy

$$a_e + 3a_t + 2a_c = 4.$$

The complete domain and its obstructions are

| $(a_e, a_t, a_c)$ | Obstruction |
|---|---|
| $(4, 0, 0)$ | The rank-two subring $\rho^2 = 1 + 4\rho$ |
| $(2, 0, 1)$ | The closed subring $R_{2,20}^{(6)}$ |
| $(0, 0, 2)$ | The closed subring $R_{2,21}^{(6)}$ |
| $(1, 1, 0)$ | A one-dimensional formal-codegree norm obstruction |

The first row is excluded by [36]. In the next two rows the subring is generated by $C_3$ and $\rho$; its based isomorphism and the previously established general certificate are checked exactly. For the last row, extend the sign representation of $S_3$ by assigning $\rho$ a root $\lambda$ of $\lambda^2 + 2\lambda - 1 = 0$. Equation (5.9) verifies this as a one-dimensional representation of the full ring. Its genuine formal codegree is

$$f = \sum_{h \in S_3} 1 + \sum_{h \in S_3} \lambda^2 = 6(1 + \lambda^2) = 24 \pm 12\sqrt{2}, \qquad q(t) = t^2 - 48t + 288.$$

Here $\rho^* = \rho$. Since $v_2(720) = 4$ and $v_2(288) = 5$, condition (5.11) holds. The two-dimension reduction again makes this a general categorical obstruction. In particular, this step does not identify an unweighted Casimir eigenvalue from a higher-dimensional irreducible representation with a formal codegree. All nonabelian possibilities have now been excluded.

*Returning through Morita duality.* Consequently $\mathcal{D}$ has one of the faithful gradings (5.12). Regard $A$ as the distinguished simple right module in $\mathcal{M} = \mathcal{C}_A$, and let $B = \underline{\mathrm{End}}_{\mathcal{D}}(A)$. Double Morita duality identifies $\mathcal{C}$ with ${}_B\mathcal{D}_B$, up to the opposite tensor convention; see [16, Chapter 7]. For a simple $A$-bimodule $Y$, its multiplicity in $B$ is

$$\dim \mathrm{Hom}_{\mathcal{M}}(A \otimes_A Y, A) = \dim \mathrm{Hom}_{\mathcal{M}}(Y, A).$$

There are exactly two simple bimodules restricting to the right module $A$: the regular bimodule and its twist by the nontrivial character of $C_2$. Each has multiplicity one. This can also be read directly from the two unit columns at $A$ in the Gram matrix $I + M_g$ of Proposition 5.12; off-diagonal entries in that row vanish. All other simple bimodules have zero multiplicity. Thus $B = \mathbf{1} \oplus b$, where $b$ is a nontrivial invertible involution. In the cyclic case $b$ is necessarily $3 \in C_6$, hence $B$ lies in the neutral component.

A connected separable algebra in the neutral component of a faithful group grading transfers that grading to its bimodule category. Indeed, the left and right algebra actions preserve each homogeneous support, and tensor product over the algebra multiplies the degrees. Every

component is nonzero: for a nonzero homogeneous object $Y$, the free bimodule $B \otimes Y \otimes B$ is nonzero and has the same degree. It follows that $\mathcal{C}$ is faithfully $C_3$-graded.

Finally, for $R^{(6)}_{5,100}$ the involution is basis label 4, the fixed labels are $3, 5$, and the exhaustive anti-invariant generator search has 16 tuples. For $R^{(6)}_{5,408}$ these data are respectively 1, $4, 5$, and 8. Each search has exactly one admissible representation, in which the involution acts as $+I$ on the two anti-invariant coordinates. The Gram reconstruction forces exactly the dimension profile in the statement. In both original tensors the multiplicative closure of the supports of all $XX^*$ is the whole basis. Thus the adjoint subcategory would be all of $\mathcal{C}$, which rules out any nontrivial faithful grading. This proves the two claimed exclusions. □

**Exact finite certificate.** The implementation is `software/new_morita_order6.py`, with audit output in `research/model_continuation/morita_order6_certificates.json`. The input certificate stores the involution, fixed labels, generator search size, every admissible anti-invariant representation, and the reconstructed dual dimensions in exact field coordinates. Verification reconstructs the exhaustive representation domain, rather than trusting a proposed list of branches. The finite dual routine enumerates every composition of 4 into six nonnegative parts, applies exactly (5.10), and checks that there are 60 cases. It constructs each full tensor, factors its Casimir polynomial exactly, and checks the recorded norm witness, the full grading identities, or the closed-subring support and unit-preserving based isomorphism. The four $S_3$ class triples are independently exhausted. The inherited certificates for $R^{(6)}_{2,20}$ and $R^{(6)}_{2,21}$ are replayed before they are used. Thus the audit output is a reproducible record of a complete finite necessary-condition calculation, with no unresolved branch silently discarded.

A separate replay, `research/model_continuation/replay_morita_order6.py`, checks equality with the saved finite audit and rejects mutations of the involution, branch list, generator count, representation matrices, dual dimension counts, and input basis labels. Its checks use explicit exceptions and remain active under `python -O`.

### 5.9 Forced pointed associators and the multiplicity-four frontier

The next obstruction uses the associator of a pointed subcategory to restrict central twists. It is a tensor criterion and applies to an entire family.

**Proposition 5.18.** *Let $m \geq 0$, and suppose a fusion category has invertible objects $1, t, t^2, t^3$, with $t$ of order four, and two self-dual noninvertible objects $X, Y$ satisfying*

$$tX = Y = Xt, \qquad tY = X = Yt,$$
$$X^2 = Y^2 = 1 + t^2 + m(X + Y), \qquad XY = t + t^3 + m(X + Y).$$

*Then the pointed associator has class 2 in $H^3(C_4, \mathbb{C}^\times) \cong \mathbb{Z}/4\mathbb{Z}$. In a pseudo-unitary realization, every simple central summand of $I(t)$ or $I(t^3)$ has twist satisfying $\theta^4 = -1$.*

*Proof.* Put $H = \langle t^2 \rangle$. Because $t^2$ fixes a simple object, the associator restricts trivially to $H$: the associativity scalars of the action on that object give a two-cochain trivializing the restriction. Thus its class in $H^3(C_4, \mathbb{C}^\times)$ is either 0 or 2.

Suppose first that the class is zero. The regular pointed algebra $A = 1 \oplus t \oplus t^2 \oplus t^3$ exists. Its right-module category has three simple objects. The invertible orbit contributes $A$; the orbit $\{X, Y\}$ contributes two simples $M_+, M_-$, each with underlying object $X \oplus Y$. Indeed its stabilizer is $C_2$, whose projective irreducible representations are the two ordinary one-dimensional representations, because $H^2(C_2, \mathbb{C}^\times) = 0$. This is the usual orbit–stabilizer description of modules over a regular pointed algebra (equivalently, the classification of modules over its action groupoid algebra).

Let $d = \mathrm{FPdim}(X) = \mathrm{FPdim}(Y)$, so $d^2 = 2 + 2md$ and $d = m + \sqrt{m^2+2}$. Normalize module dimensions by $\mathrm{FPdim}(A) = 4$; their vector is $(1, d/2, d/2)$. Every invertible fixes the simple module $A$ and permutes $M_+, M_-$. On their difference line its action is a character $\chi : C_4 \to \{\pm 1\}$, hence $\chi(t^2) = 1$. The action matrix of $X$ is symmetric, since $X$ is self-dual. Its first row and column are $(0, 1, 1)$ by free–forgetful adjunction. The dimension eigenvector therefore forces its remaining block to be

$$\begin{pmatrix} a & 2m-a \\ 2m-a & a \end{pmatrix} \quad (a \in \mathbb{Z}).$$

Thus its eigenvalue on the difference line is the integer $\lambda = 2a - 2m$. Applying the fusion relation for $X^2$ on this line gives

$$\lambda^2 = 2 + m(1 + \chi(t))\lambda.$$

For $\chi(t) = -1$ this says $\lambda^2 = 2$; for $\chi(t) = 1$ it says $(\lambda - m)^2 = m^2 + 2$. Both are impossible for integers: $m^2 + 2$ lies strictly between consecutive squares for $m \geq 1$, and equals 2 for $m = 0$. Hence class 0 is impossible.

For completeness use representatives $a, b, c \in \{0, 1, 2, 3\}$ and standard cyclic cocycles

$$\omega_k(a, b, c) = \exp\left(\frac{2\pi i k}{4} a \left\lfloor \frac{b+c}{4} \right\rfloor\right).$$

Their restrictions have $\omega_k(2, 2, 2) = (-1)^k$, confirming that the restriction kernel is $\{0, 2\}$. The canonical pseudo-unitary indicator formula gives $\nu_4(t) = i^k$ and $\nu_4(t^3) = i^{9k}$; for $k = 2$ both equal $-1$. Reversing the cocycle convention inverts these values and has no effect on this conclusion.

The center indicator formula reads

$$\mathrm{FPdim}(\mathcal{C})\nu_4(t) = \sum_{Z \subset I(t)} [Z : I(t)] \, \mathrm{FPdim}(Z)\theta_Z^4.$$

The positive coefficients on the right sum to $\mathrm{FPdim}(I(t)) = \mathrm{FPdim}(\mathcal{C})$. Equality in the triangle inequality forces every $\theta_Z^4 = -1$. The same argument applies to $t^3$; compare [41, Proposition 3.8]. □

**Lemma 5.19.** *Under the hypotheses of Proposition 5.18, $m$ is even. For the algebra $1 \oplus t^2$, its anti-invariant module action on the two fixed-object coordinates is, up to sign gauge,*

$$T_t = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T_{t^2} = -I, \qquad T_X = T_Y = 0.$$

*In particular both noninvertibles have the anticommuting stabilizer used in Proposition 5.21.*

*Proof.* The involution-module construction gives $T_t = \left(\begin{smallmatrix} 0 & \epsilon \\ \delta & 0 \end{smallmatrix}\right)$, where $\epsilon, \delta \in \{\pm 1\}$, and $T_{t^2} = \epsilon\delta I$. Since $X, Y$ are self-dual, $T_X$ and $T_Y$ are symmetric integer matrices; they commute with $T_t$ and satisfy $T_Y = T_t T_X$. If $\epsilon\delta = 1$, commutation makes the diagonal entries of $T_X$ equal. Its eigenvalue $\lambda$ on the $+1$ eigenspace of $T_t$ is therefore an integer. But the relation for $X^2$ gives $\lambda^2 = 2 + 2m\lambda$, which is impossible as above. Consequently $\epsilon\delta = -1$. A symmetric matrix commuting with this skew-symmetric $T_t$ is scalar, so $T_X = aI$. The symmetry of $T_Y = aT_t$ forces $a = 0$. Finally every entry of $T_X$ is congruent modulo two to the corresponding coefficient of multiplication by $X$ on $\{X, Y\}$, namely $m$. Hence $m$ is even. Changing the sign of one fixed-object coordinate gives the displayed sign gauge. □

**Theorem 5.20.** *The fusion ring $R_{4,6}^{(6)}$ has no categorification over $\mathbb{C}$.*

*Proof.* Its full tensor has the family in Proposition 5.18 with $m = 4$ and $(t, t^2, t^3, X, Y) = (b_1, b_5, b_2, b_3, b_4)$. All simple Frobenius–Perron dimensions belong to $\{1, 4 + 3\sqrt{2}\}$. The two-dimension Galois-pseudo-unitarity theorem therefore reduces general existence to pseudo-unitary existence, without a pivotality assumption on the original category.

Lemma 5.19 gives exactly one anti-invariant module representation for the split involution algebra $1 \oplus t^2$, after sign gauge. The finite module checker independently reconstructs it. The action of $t^2$ is $-I$ on both fixed-object coordinates. The cubic spaces of $X$ and $Y$ each have dimension four; Proposition 5.21 therefore forces

$$-2 \le \operatorname{Re} \nu_3(X), \operatorname{Re} \nu_3(Y) \le 1.$$

The exact central-induction calculation has ten low frames and four second-indicator assignments per frame. All low summands have twist one. Every high restriction column containing $t$ or $t^3$ is restricted by Proposition 5.18 to $\theta^4 = -1$. In the normalized relative-trace domain over $\mathbb{Q}(\sqrt{2})$, this means its fourth coordinate is exactly $(-1, 0)$. Keeping only those trace types is necessary; actual roots with $\theta^4 = -1$ have precisely that coordinate under every relative Galois embedding. The order-four trace system includes all root orders with the allowed prime support $\{2, 3\}$, including arbitrarily large orders through their zero trace type.

Each of the resulting forty rational nonnegative systems has a recorded integral separating vector. Verification reconstructs the low frames, the unique module branch, the entire joint trace domain, the four second-indicator assignments, and the forced-twist variable deletion. For every system $Az = b$, its vector $y$ satisfies $A^T y \ge 0$ and $b^T y < 0$. Therefore no feasible central induction data, and hence no pseudo-unitary categorification, exists. This contradicts the general-to-pseudo-unitary reduction. □

**Certificate and implementation.** The tensor-based implementation is `software/round3_smallest.py`. Its verifier never calls an optimizer; the discovery routine merely proposes exact rays. The certificate, proof audit, all-census applicability scan, positive controls and optimized replay are retained in `research/round3/smallest/`.

## 5.10 A stabilizer obstruction to scalar pivotal rotation

**Proposition 5.21** (Anticommuting stabilizers and third rotation)**.** *Let $\mathcal{C}$ be a spherical fusion category, let $g$ be an invertible of order two for which $A = \mathbf{1} \oplus g$ is an algebra, and let $Y$ be a simple object with $gY \cong Y \cong Yg$. Suppose that left multiplication by $g$ exchanges the two right $A$-module structures on $Y$. Put $m = \dim \operatorname{Hom}(\mathbf{1}, Y^{\otimes 3})$. Then $m$ is even. If $m > 0$, the pivotal third-rotation operator on this space is not scalar. More precisely,*

$$\nu_3(Y) = -(a + b\zeta_3 + c\zeta_3^2), \qquad a, b, c \in \mathbb{Z}_{\ge 0}, \quad a + b + c = m/2.$$

*If 3 is coprime to the Frobenius–Schur exponent, this becomes*

$$\nu_3(Y) \in \{-m/2 + 3b : 0 \le b \le \lfloor m/4 \rfloor\}.$$

*In particular, if $m = 2$ under this coprimality hypothesis, then*

$$\nu_3(Y) = -1.$$

*Proof.* Use strict monoidal notation and an associative identification $q : g^2 \xrightarrow{\sim} \mathbf{1}$ supplied by the algebra $A$. Choose isomorphisms $l : gY \to Y$ and $r : Yg \to Y$, normalized by

$$l(\mathrm{id}_g \otimes l) = q \otimes \mathrm{id}_Y, \qquad r(r \otimes \mathrm{id}_g) = \mathrm{id}_Y \otimes q.$$

The two normalized right module structures are $r$ and $-r$. Transporting the right action on $gY$ through $l$ gives

$$r' = l(\mathrm{id}_g \otimes r)(l^{-1} \otimes \mathrm{id}_g).$$

The hypothesis says $r' = -r$, or equivalently

$$r(l \otimes \mathrm{id}_g) = -l(\mathrm{id}_g \otimes r). \tag{5.13}$$

This also shows directly how the hypothesis is read from the anti-invariant module matrices: it is $T_g[Y, Y] = -1$.

Define an invertible endomorphism of $Y^2$ by

$$a = (r \otimes l)(\mathrm{id}_Y \otimes q^{-1} \otimes \mathrm{id}_Y),$$

and put $P = a \otimes \mathrm{id}_Y$, $Q = \mathrm{id}_Y \otimes a$ on $Y^3$. Expanding either composite inserts two copies of $q^{-1}$, with common intermediate tensor word $YggYggY$. The endpoint maps agree. On the middle $gYg$, the two composites use the two sides of (5.13). Consequently

$$PQ = -QP.$$

The restrictions to $V = \mathrm{Hom}(\mathbf{1}, Y^3)$ remain invertible. Taking determinants shows that $\dim V$ is even.

Let $E$ be pivotal cyclic rotation moving the last $Y$ to the first position. The canonical cyclic map

$$\mathrm{Hom}(\mathbf{1}, B \otimes Y) \longrightarrow \mathrm{Hom}(\mathbf{1}, Y \otimes B)$$

is natural in the entire object $B$. Applying this naturality to $B = Y^2$ and $a \in \mathrm{End}(B)$ gives the exact identity

$$EP = QE.$$

The construction and its naturality are those of [33, Definition 3.1, Lemma 3.4 and Theorem 5.1]. No braiding or self-duality sign is used. If $E$ were scalar on nonzero $V$, then $P = Q$, contradicting $PQ = -QP$ and invertibility.

The module-action normalizations also give $a^2 = I$: after two pair insertions, the first endpoint applies $r(r \otimes \mathrm{id}_g)$, the second applies $l(\mathrm{id}_g \otimes l)$, and the two resulting copies of $q$ cancel the inserted copies of $q^{-1}$ by associativity. Set $A_j = E^j P E^{-j}$ for $j \in \mathbb{Z}/3\mathbb{Z}$. These are involutions, and the anticommutation of $A_0 = P$ and $A_1 = Q$, together with cyclic conjugation, makes them pairwise anticommuting. They therefore give a representation of the complex Clifford algebra $\mathrm{Cl}_3 \cong M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$. In that algebra the element

$$U = \frac{-I + A_0A_1 + A_1A_2 + A_2A_0}{2}$$

satisfies $U^3 = I$, $UA_jU^{-1} = A_{j+1}$, and has trace $-1$ on each two-dimensional simple module. These identities also follow by direct multiplication using $A_i^2 = I$ and $A_iA_j = -A_jA_i$. The eight-dimensional universal algebra calculation is reproduced with rational coefficients by `research/round2_audit/clifford_rotation.py`; it verifies $U^3 = I$ and $UA_j = A_{j+1}U$. Its scalar coefficient is $-1/2$, while the three bivectors are traceless on either two-dimensional simple, giving trace $-1$ without a matrix-choice assumption. Thus $S = U^{-1}E$ commutes with the Clifford action and satisfies $S^3 = I$. On its multiplicity spaces, of total dimension $m/2$, let $a, b, c$ be the multiplicities of $1, \zeta_3, \zeta_3^2$ for $S$. Taking the trace of $E = US$ proves the asserted formula. Reversing the cyclic convention exchanges $b$ and $c$, so the same value set applies to the standard indicator convention.

On the other hand, the central-twist formula puts $\nu_3(Y)$ in $\mathbb{Q}(\zeta_N)$, where $N$ is the Frobenius–Schur exponent. If $3 \nmid N$, then $\mathbb{Q}(\zeta_N) \cap \mathbb{Q}(\zeta_3) = \mathbb{Q}$. Hence $b = c$, and $a + 2b = m/2$ gives the rational-value list. □

*Remark* 5.22 (No self-duality requirement). The stabilizer-rotation proposition applies equally to a non-self-dual simple $Y$. Its hypotheses and proof use isomorphisms $gY \cong Y \cong Yg$, not an isomorphism $Y \cong Y^*$. In particular, rigidity gives

$$m = \dim \mathrm{Hom}(\mathbf{1}, Y^{\otimes 3}) = N_{Y,Y}^{Y^*};$$

one must not replace the last index by $Y$ unless $Y$ is self-dual. The pair-insertion operators act on $Y^3$ itself, and naturality of pivotal rotation applies to $Y^2$ and its endomorphism $a$ without dualizing $Y$. Thus the same Clifford multiplicity formula applies to each member of a dual pair separately.

**Corollary 5.23** (Projected third-indicator spectrum)**.** *Under the hypotheses of Proposition 5.21,*

$$-m/2 \le \mathrm{Re}\,\nu_3(Y) \le m/4, \qquad \mathrm{Re}\,\nu_3(Y) \in \{m/4 - 3a/2 : a = 0, 1, \ldots, m/2\}.$$

*Let $K$ be any number field satisfying $K \cap \mathbb{Q}(\zeta_3) = \mathbb{Q}$. The normalized relative trace of $\nu_3(Y)$ to $K$ equals its real part and therefore belongs to the displayed finite set. This includes every real number field and the complex cyclotomic field $\mathbb{Q}(\zeta_8)$. If instead $K$ contains $\zeta_3$, the unprojected possibilities are the full set*

$$\{-(a + b\zeta_3 + c\zeta_3^2) : a, b, c \in \mathbb{Z}_{\ge 0},\ a + b + c = m/2\}.$$

*Proof.* The preceding proposition gives $\nu_3(Y) = -(a + b\zeta_3 + c\zeta_3^2)$ with $a + b + c = m/2$. Its real part is $-a + (b + c)/2 = m/4 - 3a/2$, which gives the stated set and bounds. Under the intersection hypothesis, $K(\zeta_3)/K$ is quadratic and its nontrivial automorphism exchanges $\zeta_3$ and $\zeta_3^2$. Consequently its normalized relative trace sends both roots to $-1/2$, and sends the indicator to $-a + (b + c)/2$. Transitivity gives the same value when the trace is taken from any larger common field. The quadratic field $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ does not embed into $\mathbb{Q}(\zeta_8)$: the latter's three quadratic subfields are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$. This proves the stated complex-field example. When $K$ contains $\zeta_3$, normalized trace fixes the entire indicator, so the original Clifford multiplicity set must be retained. □

**Corollary 5.24.** *Every pseudo-unitary categorification of $R_{3,215}^{(6)}$, in the basis of dimensions $(1, 1, d, d, d - 1, d + 1)$, must satisfy*

$$\nu_3(b_4) = \nu_3(b_5) = -1.$$

*Proof.* The exact full Morita-action calculation for $A = 1 + g$ leaves two branches, both with $T_g = -I$ on the two fixed simples $b_4, b_5$. They satisfy $N_{44}^4 = N_{55}^5 = 2$, and both are self-dual. Their third-rotation spaces therefore have dimension two. The global dimension is $136 + 32\sqrt{17}$, whose norm is $1088 = 2^6 \cdot 17$. The Cauchy theorem [5, Theorem 3.9] excludes the prime 3 from the Frobenius–Schur exponent. Apply Proposition 5.21. □

## 5.11 Exact certificate and exhaustive coverage

The obstruction is implemented without a distinguished census label in `software/stabilizer_rotation.py`. Its input is the multiplication tensor. The certificate first reconstructs the complete list of anti-invariant actions for the forced algebra $\mathbf{1} \oplus g$. For the multiplicity-three frontier this list has sixteen entries. Twelve are rejected by exact separating functionals for the full Morita-action Gram system, and two by its rank bound. In each of the remaining two entries, left multiplication by $g$ exchanges both pairs of module structures on the fixed simples $b_4, b_5$. Equivalently, the diagonal entries of $T_g$ on their anti-invariant lines are both

$-1$. Their cubic invariant spaces have dimension two. The pair-insertion theorem therefore forces $\nu_3(b_4) = \nu_3(b_5) = -1$, since the Cauchy prime support is $\{2, 17\}$.

The central calculation is exhaustive; it does not assume the rank-twenty-two modular datum retained elsewhere as a relaxation witness. All ninety-one low induction frames are reconstructed from the tensor and its formal codegrees. Thirty-one frames admit character labellings; there are fifty labellings in total and four possible second-indicator assignments. The certificate contains an exact Farkas vector for each of these two hundred systems. Each system includes the induction Gram equations, relative twist traces through order three, cyclic-rotation character constraints, joint Galois compatibility of dimensions and eigenvalues, and the low-row modular relations. It additionally imposes the two forced third indicators. If $Ax = b$ is such a system, its recorded integral vector $y$ is checked to satisfy

$$A^{\mathsf{T}} y \geq 0, \qquad b^{\mathsf{T}} y < 0.$$

Every feasible central decomposition supplies $x \geq 0$, so these inequalities exclude every branch. The variables allow all roots of unity whose prime support is $\{2, 17\}$ via their finite joint relative-trace types; no finite upper bound on the actual exponent or central rank is imposed. For clarity, the order-three relative-trace domain has exactly ten joint patterns. Write a root's order as $2^a 17^b$. If $b \geq 2$ or $a \geq 3$, its normalized traces at powers $1, 2, 3$ all vanish. Otherwise $a \in \{0, 1, 2\}$ and $b \in \{0, 1\}$ give the remaining nine distinct patterns, with the two quadratic signs when 17 divides the order. For primitive seventeenth roots the first trace is $(-1 \pm \sqrt{17})/16$; the quadratic character of 2 is $+1$ and that of 3 is $-1$. This elementary enumeration explicitly includes arbitrarily large exponent orders through the all-zero pattern. Integral high-column decompositions are not needed for this final exclusion.

The general-scope wrapper is `software/galois_pseudo_reduction.py`. It reconstructs the full unrestricted signed-sphericalization enumeration, including all 1480 signed quotient tables and all sign orbits. Every nongauge character component is rejected by its exact arithmetic certificate, and the only nonzero dimension-character orbit of the original ring is checked directly to be its Frobenius–Perron orbit. Consequently every hypothetical complex categorification has a pseudo-unitary Galois conjugate to which the preceding obstruction applies. No positivity normalization is imposed during the signed enumeration. The combined certificate is stored in `certificates/stabilizer_rotation_215.json`; the public replay entry point is `stabilizer_rotation.verify_general(R,c)`.

**Theorem 5.25** (Resolution of the multiplicity-three frontier)**.** *The based ring $R^{(6)}_{3,215}$ has no complex fusion categorification.*

*Proof.* The unrestricted signed-sphericalization calculation of Proposition 3.27, independently reconstructed by the general certificate above, shows that any hypothetical categorification has a pseudo-unitary Galois conjugate. Work with that conjugate.

The complete involution-module domain has sixteen branches. The full Morita-action Gram identities exclude fourteen, and either remaining branch satisfies the hypotheses of Corollary 5.24. Consequently both $\nu_3(b_4)$ and $\nu_3(b_5)$ equal $-1$.

The exhaustive central-induction calculation in Section 5.11 excludes those two equalities for every permitted low frame, character assignment and second-indicator choice. Each of its 200 integral separating vectors is checked against the reconstructed rational system; no numerical infeasibility output is used as a certificate. Thus no pseudo-unitary categorification exists, contradicting the reduction. □

This proof uses no identification of a hypothetical Drinfeld center with a known braided category. A cyclic-eight modular datum satisfies the central-induction conditions, but the local stabilizer and rotation constraints prevent it, and every other central-induction branch, from arising here.

### 5.12 Further stabilizer exclusions at multiplicity four

**Corollary 5.26.** *The ring $R^{(6)}_{4,416}$ has no complex fusion categorification.*

*Proof.* The same local theorem excludes `R6_4_416` without changing its hypotheses to fit the input. The unique relevant invertible involution is $b_1$, with fixed simples $b_4, b_5$. Both exhaustive anti-invariant module actions have negative stabilizer sign on both fixed objects. Each cubic invariant space has dimension four. The Clifford-rotation trace therefore lies in $\{-2, 1\}$, since the Cauchy prime support is $\{2, 5\}$. In particular its upper bound is one.

The quadratic central-induction calculation reconstructs all 895 low frames. After character reciprocity and joint Galois pruning, two relevant character labellings remain. All sixteen second-indicator choices for each labelling are excluded by exact Farkas rays, using the interval bound $-2 \le \nu_3(b_4), \nu_3(b_5) \le 1$. Thus only 32 separating vectors are needed, and no discrete guess for the third indicators is required.

For general scope, the unrestricted signed enumeration has 2376 quotient tables in 83 sign orbits. All nongauge branches are rejected by 222 exact character obstructions. Of the original nonzero dimension-character components, the non-Frobenius–Perron component has global dimension polynomial $x^2 - 32x + 128$ and fails the norm condition at prime 5, required by the Frobenius–Perron global dimension polynomial $x^2 - 800x + 3200$. Consequently the same general-to-Galois-pseudo-unitary reduction applies. The combined general-scope certificate is `certificates/stabilizer_rotation_416.json`; it has been replayed under optimized Python with all numerical optimizer entry points disabled.

□

**The ring $R^{(6)}_{4,174}$.** The four invertible basis elements have Frobenius–Perron dimension 1; the remaining two have dimension $d = 4 + 3\sqrt{2}$, with $d^2 = 2 + 8d$. Thus the global dimension is $72 + 48\sqrt{2}$, whose norm is 576. The involution-module calculation has a single antiinvariant branch, and its stabilizer commutator is negative on both noninvertible simples. Each cubic invariant space has dimension four. Since the global-dimension prime support contains 3, the exact relative traces of their third indicators are allowed to be $-2, -1/2$, or 1. The computation first uses the weaker interval $[-2, 1]$.

At trace height six, there are ten low induction frames. Eight character labellings survive the earlier exact conditions; their full second-indicator assignments give 128 systems. Every system admits an exact separating ray, already with the interval constraint. Hence no pseudo-unitary categorification exists. The two distinct Frobenius–Perron dimensions supply the previously established general-to-Galois-pseudo-unitary reduction, so the exclusion holds for arbitrary complex fusion categories. The composite certificate uses the general higher-stabilizer verifier, with the reduction payload `two_fp_values`; all matrix and branch data are reconstructed from the input tensor.

**The ring $R^{(6)}_{4,427}$.** Write $d = 7 + 5\sqrt{2}$, so that $d^2 = 1 + 14d$. The Frobenius–Perron dimensions are $(1, 1, d, d, d-1, d+1)$ and the global dimension is $400 + 280\sqrt{2}$, of norm 3200. There is one antiinvariant involution-module branch. Its stabilizer commutator is negative on the last two simples, whose cubic invariant spaces both have dimension four. The prime 3 does not divide the global-dimension norm; the Clifford criterion therefore restricts both third indicators to $\{-2, 1\}$.

The trace-height-six computation enumerates 315 low induction frames. The surviving character assignments comprise eight labellings and 32 second-indicator systems. Exact separating rays exclude every system even using the weaker interval $[-2, 1]$. This proves pseudo-unitary nonexistence. The independent unrestricted signed-quotient reduction has 2608 tables, forming 87 sign-gauge orbits, with 239 nongauge character rejections. Its exact character certificates

reduce every categorification to the Galois-pseudo-unitary case. Combining the two verified components gives general nonexistence through the same `general_stabilizer_rotation_higher` verifier.

**Theorem 5.27** (Quartic stabilizer-trace exclusions)**.** *The rings $R_{4,258}^{(6)}$, $R_{4,325}^{(6)}$, $R_{8,570}^{(6)}$ and $R_{8,2203}^{(6)}$ have no complex categorification.*

*Proof.* The tensor-only anti-invariant module enumeration has one possible representation for each ring. The order-two invertible has mixed stabilizer sign $-1$ on the self-dual simple $b_5$ in the two multiplicity-four rings, and on $b_4$ in the two multiplicity-eight rings. Each of these cubic invariant spaces has dimension four. Thus the preceding bound applies in every pseudo-unitary categorification.

For $R_{4,258}^{(6)}$, set $u = 2\cos(2\pi/20)$. The exact dimension-field embedding gives

$$D = 12u^3 + 20u^2 - 20u - 20, \qquad p_D(t) = t^4 - 120t^3 + 2400t^2 - 16000t + 32000.$$

The Cauchy prime support is $\{2, 5\}$. At height three, Proposition 4.11 gives $B = 40$ and 31 distinct trace vectors, including zero. The complete low-induction search has three frames and four second-indicator choices per frame. Every one of these twelve systems has an exact separating ray.

For $R_{4,325}^{(6)}$, set $u = 2\cos(2\pi/16)$. Here

$$D = 8u^3 + 16u^2, \qquad p_D(t) = t^4 - 128t^3 + 2560t^2 - 16384t + 32768.$$

The prime support is $\{2\}$, the height-three order bound is 32, and there are 25 trace vectors. The two low frames and four second-indicator choices per frame give eight exact separating rays. For both $R_{8,570}^{(6)}$ and $R_{8,2203}^{(6)}$, take $u = 2\cos(2\pi/20)$. The shared dimension-field data are

$$D = 160u^3 + 300u^2 - 220u - 400, \qquad p_D(t) = t^4 - 1400t^3 + 44000t^2 - 400000t + 800000.$$

Again $S = \{2, 5\}$, $B = 40$, and the trace domain has 31 vectors. Each ring has seven low frames. There is one second-indicator choice for $R_{8,570}^{(6)}$ and four for $R_{8,2203}^{(6)}$, giving respectively seven and twenty-eight exact separating rays. These computations establish pseudo-unitary nonexistence in all four cases.

The general certificates also reconstruct all signed sphericalization quotients and their character obstructions. For $R_{4,258}^{(6)}$ there are 32 quotients in one sign-gauge orbit. For $R_{4,325}^{(6)}$ there are 96 quotients in four orbits; the three nongauge orbits require nine character rejections. The two multiplicity-eight rings give respectively 96 quotients in six orbits with ten nongauge character rejections, and 1472 quotients in 46 orbits with 111 such rejections. The surviving orbit in each case is gauge isomorphic to the original ring, and the nonpositive original characters are excluded by the recorded exact tests. Consequently any complex categorification would have a Galois conjugate with the positive spherical dimensions just excluded.

The canonical implementation is `software/cyclotomic_stabilizer_trace.py`. Its `verify_general` entry point checks both the sphericalization reduction and the complete pseudo-unitary certificate. It reconstructs the module domain, field embedding, trace domain, low frames, high columns and second-indicator alternatives from the input tensor; no census identifier enters a mathematical test. Every retained high column satisfies the exact integral dimension quotient and central simple $d$-number conditions. All separating inequalities are checked with integers. The certificate files are `certificates/cyclotomic_stabilizer_R6_4_258.json`, `certificates/cyclotomic_stabilizer_R6_4_325.json`, `certificates/cyclotomic_stabilizer_R6_8_570.json` and `certificates/cyclotomic_stabilizer_R6_8_2203.json`. □

The same ambient-field computation applies to the other quartic inputs recorded in `research/round2_literature/quartic/`. Its exploratory modular refinement keeps global Galois multiplicity balance; it does not copy the quadratic assertion that Galois squares preserve every relative twist-trace type. A surviving linear system is retained as a scoped necessary relaxation and does not supply a category.

# 6 Categorical models and exact tensor reconstruction

Positive records are established by tensor matches to constructions, not by lack of an obstruction. We use Deligne products of the standard rank-two and rank-three unitary categories, the $TT_3$ category of [14, Definition 2.5], and the positive quadratic constructions of [30, Proposition 5.1, Corollary 5.8]. We use the latter results only for existence, not as a negative classification oracle. Every displayed tensor is reconstructed from its defining rules and compared under unit-preserving permutations with the census.

## 6.1 A central fermion and a cyclic-four quadratic ring

Let $\mathcal{D}$ be the unitary near-group category $\mathrm{Ad}(E_6)$ with basis $1, h, X$ and rules

$$h^2 = 1, \qquad hX = Xh = X, \qquad X^2 = 1 + h + 2X. \tag{6.1}$$

Its noninvertible dimension is $d = 1 + \sqrt{3}$, and $D = \dim \mathcal{D} = 4 + 2d$.

**Lemma 6.1** (Central fermion)**.** *There is an invertible $z \in Z(\mathcal{D})$ with $F(z) = h$, $z^2 = 1$, and $\theta_z = -1$. The full subcategory generated by $z$ is the unitary symmetric category* sVec.

*Proof.* The formal codegrees are $D$, $D' = 6 - 2\sqrt{3}$, and 2. Thus $I(1) = 1 \oplus A \oplus B$, with dimensions $1, 1 + d, 2 + d$. Since $FI(1) = 3 \cdot 1 + h + 2X$, comparison of the unit multiplicities and of the coefficients of $1, d$ gives

$$F(A) = 1 + X, \qquad F(B) = 1 + h + X.$$

All three low twists are one.

The induction Gram matrix in the basis $(1, h, X)$ is

$$\begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 8 \end{pmatrix}.$$

Only $B$ among the low objects occurs in $I(h)$, once. The remaining endomorphism dimension is $3 - 1 = 2$, so the remainder consists of two distinct simples, both with multiplicity one. Its restriction is $FI(h) - F(B) = 2h + X$. Consequently these two objects are $z, W$ with $F(z) = h$ and $F(W) = h + X$. The object $z$ has dimension one and is invertible. Its square restricts to the unit; the only central simple restricting to the unit is the unit itself, as the decomposition of $I(1)$ shows. Therefore $z^2 = 1$.

The pointed associator class on $\langle h \rangle$ is trivial: the rank-one module category on $X$ supplies a two-cochain whose coboundary is the restricted associator. In the positive spherical structure, $\nu_2(h) = 1$. Equation (4.6) at $k = 2$ implies that every twist in $I(h)$ squares to one, because all dimensions are positive and their total is $D$. At $k = 1$ it gives

$$0 = (2 + d) + \theta_z + (1 + d)\theta_W.$$

The known positive summand has exactly the total dimension of the other two. Equality in the triangle inequality forces $\theta_z = \theta_W = -1$. Finally an order-two invertible in a unitary ribbon category with twist $-1$ has self-braiding $-1$ in its canonical positive structure. Its generated pointed braided category is sVec. □

**Proposition 6.2** (Fermionic zesting)**.** *The ring $R^{(6)}_{2,2}$ has a unitary categorification. More generally, if a unitary fusion category $\mathcal{D}$ has a central fermion $z$ of order two, the rules*

$$(U,a)\star(V,b)=(U\otimes V\otimes F(z)^{ab},a+b),\qquad a,b\in\mathbb{Z}/2, \tag{6.2}$$

*on $\mathcal{D}\boxtimes \mathrm{Vec}_{\mathbb{Z}/2}$ admit a unitary associator.*

*Proof.* Use the central two-cocycle $\lambda(a,b)=z^{ab}$ and normalized associativity scalar $\tau(a,b,c)=i^{abc}$, where the exponents use representatives $0,1$. The only central interchange obstruction is $c_{\lambda(a,b),\lambda(c,d)}=(-1)^{abcd}$. A direct calculation gives

$$\begin{aligned} bcd+a(b\oplus c)d+abc-(a\oplus b)cd-ab(c\oplus d)\\ \equiv 2abcd \pmod 4, \end{aligned}$$

where $\oplus$ is addition modulo two. Thus the coboundary of $\tau$ is precisely that interchange obstruction. The associative-zesting construction of [9, Definition 3.2, Proposition 3.4], in the relative-center formulation also recalled in [3, Definition 2.1], therefore supplies the pentagon and rigidity for (6.2). All structural maps used are unitary, including the scalar $\tau$, so the resulting fusion category is unitary. This use of a known construction does not claim to have independently solved its individual $F$-symbol equations.

Apply Lemma 6.1. Let $g=(1,1)$, identify $g^2=(h,0)$, and write $Y=(X,0)$ and $X_1=(X,1)$. Then the invertibles form $C_4$,

$$gY=X_1,\quad gX_1=Y,\quad Y^2=X_1^2=1+g^2+2Y,\quad X_1Y=g+g^3+2X_1.$$

These rules reconstruct the full tensor that is based-isomorphic to $R^{(6)}_{2,2}$. The supplement checks that equality and all sixteen scalar coboundary identities exactly. □

The proposition is a positive construction, not an assertion that this fusion category or its ring has not appeared in another presentation. It is particularly useful here because a ring that survived the first obstruction pass is thereby removed from the residual list for the correct reason: existence rather than an attempted exclusion.

## 6.2 Finite-group and group-theoretical models

For $G=(C_3\times C_3)\rtimes C_4$, with the order-four rotation $(x,y)\mapsto(-y,x)$, the character ring is reconstructed from four linear characters and two induced characters. Exact character inner products give all fusion coefficients and identify $R^{(6)}_{2,11}$.

We also compute the untwisted group-theoretical categories $\mathcal{C}(G,1,H,1)$ for

$$(G,H)=(A_4,C_2),\quad (A_5,D_{10}),\quad (A_5,S_3).$$

Here $D_{10}$ has order ten. The finite-group construction is standard; we recall its explicit tensor calculation to distinguish it from a mere comparison of dimension types. Realize the category as $H$-bimodules in $\mathrm{Vec}_G$. Its simples are supported on double cosets $HgH$ and indexed by irreducible representations of $H\cap gHg^{-1}$. The tensor product is relative tensor product over the group algebra of $H$. Equivalently it is convolution of the corresponding equivariant vector bundles. The character of a convolution fiber is computed by summing over the middle $H$-orbits and taking invariants under the stabilizer; averaging the finite stabilizer action gives its multiplicities. This is a finite exact character-inner-product calculation. The code lists the groups as permutations, computes the double cosets and stabilizers, and uses their exact character values. It then checks the resulting complete fusion tensors.

The three outputs are respectively $R^{(6)}_{2,36}$, $R^{(6)}_{2,105}$, and $R^{(6)}_{3,141}$. The last model is also discussed in [2, Section 6.5]. Finite groups, ordinary unitary representations, and relative tensor product over a finite-dimensional $C^*$-algebra give unitary fusion categories in these untwisted cases.

### 6.3 ADE modules, cyclic Izumi models, and a finite group

The following constructions add five realizations at multiplicity two and one at multiplicity three. Each constructor builds its fusion tensor without reading the census, then searches for a full based-ring match.

**Proposition 6.3** (ADE modules and their zesting)**.** *The rings $R^{(6)}_{2,75}$, $R^{(6)}_{2,14}$, $R^{(6)}_{2,128}$ and $R^{(6)}_{2,140}$ admit unitary categorifications. They are, respectively, the $E_6$ module category over $SU(2)_{10}$, its fermionic grading zest, the $D_6$ module category over $SU(2)_8$, and the local $D_{10}$ module category over $SU(2)_{16}$.*

*Proof.* Use the unitary $SU(2)_k$ category with simples $V_0, \dots, V_k$ and fusion coefficients

$$N_{ij}^{l} = 1 \quad \Longleftrightarrow \quad |i-j| \le l \le \min(i+j, 2k-i-j), \quad i+j+l \equiv 0 \pmod 2.$$

The $E_6$ algebra is $V_0 \oplus V_6$ at level ten. The $D$-type algebra is $V_0 \oplus V_k$ at levels divisible by four. Their existence, separability, module categories and the ADE module graphs are the Kirillov–Ostrik construction [28, Theorem 6.1 and Section 7]. These algebras and their module tensor products have the standard unitary structures.

For $E_6$, label the graph vertices $0, \dots, 5$ and its edges

$$01, \quad 12, \quad 23, \quad 34, \quad 25,$$

with unit vertex zero. Multiplication by the free image of $V_1$ is the adjacency matrix $B$. The rows $(B^j)_{0,*}$, $0 \le j \le 5$, form an invertible matrix $V$. Centrality of the free image of $V_1$ and the cyclicity of the unit vector imply that all multiplication matrices are uniquely determined by $N_i = \sum_j (V^{-1})_{ij} B^j$. The exact constructor evaluates these matrices, checks nonnegative integral entries and all tensor axioms, and matches $R^{(6)}_{2,75}$.

The parity of the original $SU(2)$ labels descends to the module category; in these coordinates it is $(0,1,0,1,0,1)$. The free image of $V_{10}$ is the degree-zero invertible at vertex four. The canonical central free functor sends the order-two fermion $V_{10}$ to a central fermion $g$ in the module category. Hence associative zesting may multiply the product of two odd objects by $g$. Its obstruction is the fermionic sign $(-1)^{abcd}$, trivialized by the scalar cochain $i^{abc}$, exactly as in Proposition 6.2. All sixteen cochain identities are checked. This constructs a unitary fusion category and its tensor is $R^{(6)}_{2,14}$. The construction uses the actual grading and central fermion, not just a ring automorphism.

For the $D$ cases put $m = k/2$. The orbit modules are $O_j$, $0 \le j < m$, and the fixed object $V_m$ splits as $U_+ \oplus U_-$. The free image of $V_j$ is $O_{\min(j,k-j)}$ unless $j = m$. Products of two orbit modules are obtained by applying this induction rule to the displayed $SU(2)$ coefficients. For an orbit module times $U_+ \oplus U_-$ use the same rule; the difference $U_+ - U_-$ has eigenvalue zero for odd $j$ and $(-1)^{j/2}$ for even $j$. At the two levels used here, $k \equiv 0 \pmod 8$, the resolved objects are self-dual. The fixed-point resolution formulas of [28, Section 7, $D_{2m}$ fusion table] give the remaining products. Equivalently, in the local sector of this congruence class,

$$U_\pm^2 = U_\pm + O_0 + O_4 + \cdots + O_{m-4}, \qquad U_+ U_- = O_2 + O_6 + \cdots + O_{m-2}.$$

Together with Frobenius reciprocity these formulas are the constructor's complete specification. At $k = 8$ use all six modules; at $k = 16$ use the local modules $O_0, O_2, O_4, O_6, U_+, U_-$. Their tensors match $R^{(6)}_{2,128}$ and $R^{(6)}_{2,140}$ respectively. □

**Proposition 6.4** (The Izumi cyclic-five model)**.** *The ring $R^{(6)}_{2,143}$ is the Grothendieck ring of the commutative even part of the unitary Izumi $C_5$ subfactor.*

*Proof.* The existence of the cyclic-five Izumi subfactor and the relevant even part are established in the literature summarized in [23, Section 6.6]. Its commutative fusion rules are those in [18, Theorem 7]. To specify the full tensor used for the match, write $1, a, b_1, b_2, c_1, c_2$, identify indices $i$ and $-i$ modulo five, and put $b_0 = c_0 = 0$, $\Sigma = a + b_1 + b_2 + c_1 + c_2$. The products are

$$a^2 = 1 + \Sigma, \qquad ab_i = \Sigma - b_i, \qquad ac_i = \Sigma + c_i, \qquad b_i c_j = \Sigma,$$
$$b_i b_j = \delta_{ij} 1 + (1 - \delta_{ij})a + \Sigma - a - b_{i+j} - b_{i-j},$$
$$c_i c_j = \delta_{ij} 1 + (1 + \delta_{ij})a + \Sigma - a + c_{i+j} + c_{i-j}.$$

All products commute and all simples are self-dual. The independent constructor specializes these formulas at five and verifies all 216 coefficients of the matched tensor. Existence is the cited subfactor construction, not a consequence of the based-ring axioms alone. □

**Proposition 6.5** (An exact finite-group model)**.** *The ring* $R^{(6)}_{3,47}$ *is categorified by* $\mathrm{Rep}(GL(3, \mathbb{F}_2)) \simeq \mathrm{Rep}(PSL(2, 7))$.

*Proof.* Generate the group as all invertible $3 \times 3$ binary matrices, with actual matrix multiplication. It has 168 elements and six conjugacy classes of sizes $1, 21, 56, 42, 24, 24$, ordered by element orders $1, 2, 3, 4, 7, 7$. Put $a = (-1 + \sqrt{-7})/2$ and $b = \overline{a}$. The candidate character table is

| | 1 | 2 | 3 | 4 | 7 | 7 |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 3 | 3 | −1 | 0 | 1 | $a$ | $b$ |
| $\overline{3}$ | 3 | −1 | 0 | 1 | $b$ | $a$ |
| 6 | 6 | 2 | 0 | 0 | −1 | −1 |
| 7 | 7 | −1 | 1 | −1 | 0 | 0 |
| 8 | 8 | 0 | −1 | 0 | 1 | 1 |

.

We verify its representation-theoretic meaning, not just orthogonality. The program computes the multiplication constants of the six class sums in $\mathbb{C}[GL(3, \mathbb{F}_2)]$. For each row $\chi$ it checks that $|C|\chi(C)/\chi(1)$ is a unital character of this class algebra. The six characters are distinct and therefore exhaust its primitive central characters. By semisimplicity they are the normalized central characters of the actual irreducible representations. Row orthogonality and positivity of the displayed degrees fix the normalization, so the rows are the actual irreducible characters. Taking their product inner products gives nonnegative integer structure constants, evaluated exactly in $\mathbb{Q}(\sqrt{-7})$. The resulting full tensor is based-isomorphic to $R^{(6)}_{3,47}$. A finite-group representation category has its usual unitary realization. □

## 6.4 Categorical structures, extensions, and central quotients

The following determinations use categorical constructions, followed by exact identification of their full fusion tensors. A based-ring isomorphism transports both existence and a proved obstruction to a specified categorical structure. It does not turn spherical existence into pseudo-unitary existence. All identifications in this section are recomputed by `software/literature_models.py`; no census label is an input to the construction.

### 6.4.1 Schopieray's spherical model

Schopieray constructs the rank-six modular category $\mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$ in [43, Example 1, equations (1)–(2)]. Its displayed fusion matrices, in the order $(X_0, \ldots, X_5)$, give our tensor $R^{(6)}_{2,124}$ after the permutation

$$(0, 1, 5, 4, 2, 3).$$

Here, and in the electronic catalogue, a permutation $p$ means $N_{ij}^{\text{census}\,k} = N_{p(i),p(j)}^{\text{source}\;\;p(k)}$. There can be more than one such permutation; the catalogue stores the lexicographically first one. The six matrices are independently transcribed in the constructor and all 216 coefficients are compared. Schopieray's Theorem 1 proves that this based ring admits no pseudo-unitary categorification. Its exact existence record is therefore

| | general | pivotal | spherical | pseudo-unitary | unitary |
|---|---|---|---|---|---|
| $R_{2,124}^{(6)}$ | yes | yes | yes | no | no. |

The last entry follows because a unitary fusion category is pseudo-unitary. This is a ring-level nonexistence theorem, not only the failure of a particular spherical structure to have positive dimensions. It does not contribute to the count of *general* exclusions.

### 6.4.2 A nilpotent extension and an Izumi even part

Write $H_8$ for the Kac–Paljutkin Hopf algebra of dimension eight. Choi–Lu–Sun give extensions of $\mathrm{Rep}(H_8)$ with one additional simple object in [7, equation (1.4), Section 7.1 and Tables 3–4]. With basis $(1, a, b, ab, N, D)$ and $a^2 = b^2 = 1$, their fusion rules are

$$\begin{aligned} gN = N, \quad gD = D, \quad N^2 = \sum_{g \in C_2^2} g, \\ ND = DN = 2D, \qquad D^2 = \sum_{g \in C_2^2} g + 2N. \end{aligned} \tag{6.3}$$

The relation $gN = N$ is meant for each $g \in C_2^2$, as is $gD = D$. The source provides existence; it is not inferred from associativity of these integer rules.

For a separate categorical justification of unitarizability, the successive adjoint subcategories determined by (6.3) have bases

$$\{1, a, b, ab, N, D\} \supset \{1, a, b, ab, N\} \supset \{1, a, b, ab\} \supset \{1\}.$$

Thus every categorification with these rules is nilpotent. A nilpotent fusion category is weakly group-theoretical and is unitarizable by [19, Theorem 5.20]. The program recomputes this adjoint series by support closure. The source tensor matches $R_{2,69}^{(6)}$ with the identity permutation. This proves unitary *existence*; neither nilpotency alone nor a tensor-axiom check is used to assert existence.

The dual even part of the Izumi $C_4$ subfactor is a unitary fusion category. Its multiplication table in [26, p. 27], in the source order $(1, \kappa, \beta\kappa, \chi, \sigma, \beta)$, matches $R_{2,98}^{(6)}$ after the permutation

$$(0, 5, 1, 2, 4, 3).$$

The constructor recovers every entry of that table before making the comparison. The equality is with $R_{2,98}^{(6)}$, not with the dimension-matching tensor $R_{2,41}^{(6)}$. Equality of dimension lists would not establish this identification.

### 6.4.3 A free central-boson quotient

A *central boson* here means an invertible object $\widetilde{g}$ in the unitary Drinfeld center $Z(\mathcal{D})$, generating a symmetric subcategory equivalent to $\mathrm{Rep}(C_2)$, whose underlying object $g$ in $\mathcal{D}$ is nontrivial. This includes the half-braiding and the Tannakian structure. Centrality of the basis element $[g]$ in $K_0(\mathcal{D})$ is not a substitute for this hypothesis.

**Proposition 6.6** (Free central-boson quotient)**.** *Let $\mathcal{D}$ be a unitary fusion category and let $\widetilde{g}$ be a central boson as above. Suppose tensoring with $g$ has no fixed point on* $\mathrm{Irr}(\mathcal{D})$*. Then the right modules over the central algebra $A = 1 \oplus g$ form a unitary fusion category $\mathcal{D}_A$. Its simples are indexed by the two-element $g$-orbits. For orbit representatives $i, j, k$ its fusion coefficients are*

$$\widetilde{N}_{[i],[j]}^{[k]} = N_{ij}^{k} + N_{ij}^{gk}. \tag{6.4}$$

*Proof.* The algebra $A$ is the image of the regular function algebra of $C_2$ under the central Tannakian embedding. It is connected and separable, and is commutative as a central algebra. The half-braiding makes right $A$-modules into a tensor category with product $\otimes_A$; see [16, Theorem 8.23.3]. Use the standard dagger Frobenius structure on the regular algebra. Work first with its unitary modules. Every algebraic module is a summand of a free module by separability; an idempotent in the finite-dimensional $C^*$-endomorphism algebra of a free unitary module is conjugate to an orthogonal projection. Thus these unitary modules represent the whole algebraic module category up to equivalence. Relative tensor products are the images of the self-adjoint separability projections. The unitary associativity maps of $\mathcal{D}$ descend to unitary maps between their images, and the module morphism spaces inherit the adjoint operation. Thus $\mathcal{D}_A$ is a unitary fusion category.

The free-module functor $F(i) = i \otimes A$ is a tensor functor. By adjunction,

$$\mathrm{Hom}_A(F(i), F(j)) \cong \mathrm{Hom}_{\mathcal{D}}(i, j \oplus jg).$$

For simple $i$, freeness of the $g$-action makes the endomorphism algebra of $F(i)$ one-dimensional, so $F(i)$ is simple. The same formula shows $F(i) \cong F(j)$ precisely when $j = i$ or $j = gi$. Every simple module is a summand of a free module, since $A$ is separable, and hence is one of these simple free modules. Finally

$$F(i) \otimes_A F(j) \cong F(i \otimes j).$$

Collecting the two constituents $k, gk$ in each orbit gives (6.4). It also proves independence of the chosen representatives. □

Apply the proposition to

$$\mathcal{D} = \mathrm{Ising} \boxtimes PSU(2)_6, \qquad g = (\psi, V_6).$$

In the principal unitary braidings, $\psi$ and $V_6$ are order-two fermions. Their product has self-braiding $+1$ and generates a Tannakian $\mathrm{Rep}(C_2)$, with central lift supplied by the product braiding. The simple labels of $PSU(2)_6$ are $V_0, V_2, V_4, V_6$; tensoring by $V_6$ exchanges $V_0, V_6$ and $V_2, V_4$. Therefore the diagonal action on the twelve simples of $\mathcal{D}$ is free. Formula (6.4) gives a rank-six unitary category, whose tensor is $R_{2,94}^{(6)}$. The checker reconstructs the parent tensor, checks all twelve action labels, computes the six orbits and verifies the displayed formula and its independence of representatives. Its orbit order is specified in `data/diagonal_fermion_quotient.json`; the final census permutation is $(0, 3, 1, 2, 4, 5)$.

The construction uses *all* right $A$-modules with the tensor product supplied by the central lift. It does not identify them with the local modules of a braided algebra and does not assert that the quotient inherits the ambient braiding. In particular, no transparency hypothesis has been silently added or inferred from the fusion ring.

## 6.5 A diagonal fermion with a fixed point

The free-action formula of Section 6.4 does not apply to a fixed simple object. Its two resolutions can have different duality depending on the local rotation data. We give a calculation which resolves the fixed point and determines every fusion coefficient.

We use positive spherical structures in unitary categories. If $f^2 = 1$, $x^* = x$, and $t : f \otimes x \longrightarrow x$ is an isomorphism, denote by $t^{\mathsf{T}} : x \otimes f \longrightarrow x$ its pivotal transpose, using the specified self-dual identifications of $f$ and $x$. Equivalently it is obtained from $t$ by inserting a coevaluation for $x$ on the right and evaluating the left pair. Associators are suppressed in this notation. Normalize $t$ so that the action of two copies of $f$ is the identity.

**Lemma 6.7** (Duality of a resolved fixed point)**.** *Let $(f, c)$ be a central boson of order two, and let $V$ be a self-dual simple fixed by $f$. Suppose $t : fV \to V$ is a normalized action and*

$$t^{\mathsf{T}} c_{f,V} = \delta t, \qquad \delta \in \{1, -1\}. \tag{6.5}$$

*For the central algebra $A = 1 \oplus f$, the two simple left $A$-modules with underlying object $V$ have actions $t$ and $-t$. Their duality is $V_\epsilon^* = V_{\delta\epsilon}$, where $\epsilon = \pm 1$.*

*Proof.* The two actions satisfy the module equation because $t$ is normalized. They are simple and inequivalent since $V$ is simple and a scalar intertwiner cannot change the sign of its nonzero action. Their sum is the free module on $V$: the commuting algebra of that free module is the two-dimensional semisimple algebra of its stabilizer.

The dual left action is obtained by transposing the given action and then moving $f$ from the left to the right of $V^*$ with its specified half-braiding. This is exactly the composite in (6.5). The normalization $f^2 = 1$ makes the inverse action equal to the action, and the symmetric Frobenius structure of $A$ supplies the ordinary pivotal transpose. Thus its sign is $\delta\epsilon$. Alternatively, on the one-dimensional ordinary vacuum channel of $V \otimes V$, the projector for the relative tensor product of signs $\epsilon, \eta$ has scalar $(1 + \epsilon\eta\delta)/2$. The unit occurs precisely when $\eta = \delta\epsilon$, giving the same duality rule. □

**Theorem 6.8** (Equal-phase fermionic product)**.** *Let $\mathcal{E}_m$ and $\mathcal{E}_n$ be unitary fusion categories with respective self-dual bases $1, f, x$ and $1, f', y$ and rules*

$$f^2 = 1, \quad fx = xf = x, \quad x^2 = 1 + f + mx, \qquad (f')^2 = 1, \quad f'y = yf' = y, \quad y^2 = 1 + f' + ny.$$

*Assume specified central fermions $f, f'$ and normalized actions $t, s$ satisfy*

$$t^{\mathsf{T}} c_{f,x} = -it, \qquad s^{\mathsf{T}} c_{f',y} = -is. \tag{6.6}$$

*The categories have specified positive pivotal identifications used in these equations. Then $m, n$ are even, and the category of modules over the central algebra*

$$A = 1 \oplus (f \boxtimes f') \quad \text{in} \quad \mathcal{E}_m \boxtimes \mathcal{E}_n$$

*is a unitary fusion category of rank six. With basis $1, g, a, b, u, v$, its duality fixes $1, g, a, b$ and interchanges $u, v$. Its complete multiplication is determined by commutativity and*

$$g^2 = 1, \quad ga = a, \quad gb = b, \quad gu = v, \quad gv = u, \tag{6.7}$$

$$a^2 = 1 + g + ma, \quad b^2 = 1 + g + nb, \quad ab = u + v, \tag{6.8}$$

$$au = av = b + \tfrac{m}{2}(u + v), \qquad bu = bv = a + \tfrac{n}{2}(u + v), \tag{6.9}$$

$$\begin{aligned} u^2 = v^2 = g + \tfrac{m}{2}a + \tfrac{n}{2}b + \tfrac{mn}{4}(u + v), \\ uv = 1 + \tfrac{m}{2}a + \tfrac{n}{2}b + \tfrac{mn}{4}(u + v). \end{aligned} \tag{6.10}$$

*If $d_m = \mathrm{FPdim}(x)$ and $d_n = \mathrm{FPdim}(y)$, its dimensions are $1, 1, d_m, d_n, d_m d_n/2, d_m d_n/2$.*

*Proof.* The product of the specified fermions is a central boson: its self-braiding is $(-1)(-1) = 1$. Its central regular algebra $A$ is connected and separable. Modules over this algebra have relative tensor product, with tensor unit $A$, and form a unitary fusion category. This uses all

modules over a *central* algebra; no braiding on the quotient is asserted. Write $Q$ for the tensor free-module functor.

There are four free orbits and one fixed point in the nine simples of the Deligne product. The corresponding modules are

$$1 = Q(1,1), \quad g = Q(f,1), \quad a = Q(x,1), \quad b = Q(1,y), \qquad Q(x,y) = u \oplus v.$$

On the fixed point $V = x \boxtimes y$, use the action $t \boxtimes s$. Its square is one. The transpose and half-braiding in a Deligne product factor, so (6.6) gives

$$(t \boxtimes s)^{\mathsf{T}} c_{f\boxtimes f',V} = (-i)^2 (t \boxtimes s) = -(t \boxtimes s).$$

Lemma 6.7 proves $u^* = v$. The other four objects are self-dual because they are tensor free modules on self-dual objects. The module dimension equals the underlying dimension divided by $\dim A = 2$, giving the asserted dimensions and rank.

Tensoring a resolved fixed point by $g$ changes its action sign. Indeed, to form the relative product with $Q(f,1)$ the diagonal algebra line is moved past $(f,1)$; the scalar is $c_{f,f} = -1$. Identifying $(f,1)V$ with $V$ by $t \boxtimes \mathrm{id}$ leaves no other scalar, by the normalized square of $t$. Hence $gu = v$, $gv = u$. Taking duals gives $ug = v$, $vg = u$, so $g$ is central on these objects. The free-module tensor identity gives $g^2 = 1$, $ga = ag = a$, $gb = bg = b$ and (6.8).

Since $g$ fixes $a$, the two products $au, av$ are equal, and likewise $ua = va$. Their respective sums, computed by the tensor functor $Q$, both equal $2b + m(u+v)$. This proves the first half of (6.9) on both sides and also forces $m$ even. The second half and the evenness of $n$ follow similarly. Thus all products involving $a, b, g$ commute. Since $v = gu$ and $g$ is central, $u$ and $v$ commute as well.

Now $u^2 = v^2$ and $(u^2)^* = v^2 = u^2$. The coefficient of 1 in $u^2$ is zero and that of $g$ is one. Frobenius reciprocity and (6.9) give coefficients $m/2, n/2$ on $a, b$. Self-duality of $u^2$ makes the coefficients of $u, v$ equal. Put $e = d_m d_n/2$ and call the common coefficient $k$. From $d_m^2 = 2 + m d_m$ and $d_n^2 = 2 + n d_n$ one obtains

$$e^2 = 1 + \tfrac{m}{2} d_m + \tfrac{n}{2} d_n + \tfrac{mn}{2} e.$$

Dimension comparison gives $k = mn/4$ and proves the square formula. Multiplication by $g$ then gives the formula for $uv$. □

**Corollary 6.9** (Three tensor identifications)**.** *The cases $(m,n) = (0,0), (0,2), (2,2)$ of Theorem 6.8 have unitary realizations and identify, respectively, the census rings*

$$R^{(6)}_{1,3}, \qquad R^{(6)}_{2,14}, \qquad R^{(6)}_{2,40}.$$

*In particular, $R^{(6)}_{2,40}$ is the diagonal-fermion quotient of two* identical *unitary half-$E_6$ categories.*

*Proof.* For $m = 0$ use the Ising category with positive dimension $\sqrt{2}$ and Frobenius–Schur sign $+1$. In the conventions of [1, Section 2.1, equations (9)–(13)], choose $A = e^{3\pi i/8}$; then $d = -A^2 - A^{-2} = \sqrt{2}$ and $A^4 = -i$. The displayed fermion half-braiding and pivotal normalizations give (6.6).

For $m = 2$ use unitary half-$E_6$, the category with (6.1). The local data in [1, Section 7.1, equation (235), Section 7.2, equation (240), Appendix D.1, equation (467)] give the normalized action square one, pivotal rotation scalar one, and the half-braiding scalar $-i$ on the noninvertible object. They therefore give exactly the second input needed in (6.6). Categorical existence and unitarity are those of the Ising and half-$E_6$ constructions, not inferences from these few scalar identities alone.

The code reconstructs (6.7)–(6.10) for the three pairs, checks every fusion-ring axiom, and matches all 216 coefficients under unit-fixing permutations. For $(2,2)$ the census basis is $(1, u, v, g, a, b)$, giving the stated identification. □

The two half-$E_6$ factors must not be replaced without comment by a category and its complex conjugate. Conjugation changes $-i$ to $+i$; the product of the rotation phases is then $+1$, and Lemma 6.7 makes the two resolutions self-dual. The relative-product description of the 3311 category in [27, Section 5.1] uses the opposite/conjugate factors. Our non-self-dual resolution is therefore not a dimension-only reidentification of that model. The construction uses central boson condensation with a fixed point, not the earlier free-orbit formula.

### 6.6 The exceptional $G_2$ level-four model

The cyclic-four tensor $R^{(6)}_{2,4}$ has a unitary realization. We identify it with the exceptional quantum subgroup $\mathcal{E}_4(G_2)$ associated with the conformal embedding

$$G_{2,4} \ \subset \ \mathrm{Spin}(14)_1. \tag{6.11}$$

The embedding, vacuum algebra, annular matrix and branching matrix used below are given in [8, Section 4.5, pp. 29–32]. We use existence of this conformal-embedding algebra, not existence inferred merely from a commuting modular-invariant matrix.

Let $\mathcal{B}$ be the unitary WZW category of $G_2$ at level four, with the long fundamental weight first, as in [8]. The vacuum algebra is

$$A = V_{(0,0)} \oplus V_{(0,3)}, \qquad \dim A = 6 + 2\sqrt{6}.$$

It is the connected commutative separable algebra, with its unitary structure, supplied by (6.11). Thus $\mathcal{B}_A$, the category of all right modules with tensor product over $A$, is an ordinary unitary fusion category. Its subcategory of local modules is the level-one category of Spin(14). This distinction between all modules and local modules matters: the former has six simples, the latter only four.

**Proposition 6.10.** *The Grothendieck ring of $\mathcal{B}_A$ is based-isomorphic to $R^{(6)}_{2,4}$. In particular this ring has a unitary categorification.*

*Proof.* In the source ordering $(1, X, Y, h, g, g^{-1})$, the basic annular matrix and branching matrix are

$$N_X = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad E = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

The source weights, indexing the rows of $E$, are

$$(0,0), (0,1), (0,2), (1,0), (0,3), (1,1), (0,4), (1,2), (2,0).$$

The free-module image of $V_{(0,1)}$ is the simple $X$, by the second row of $E$, so its annular action really is multiplication by $X$. The free-module functor is central, hence $X$ commutes with every object. The local simples are $1, h, g, g^{-1}$, corresponding to the vacuum, vector, and two spinor representations. The level-one fusion group of $D_7$ is $P/Q \simeq C_4$; equivalently its vector is the square of either spinor. Thus $g^2 = h$ and $g^4 = 1$. These four objects remain distinct and invertible in $\mathcal{B}_A$, since the local subcategory is a full tensor subcategory.

Reading the displayed matrix gives

$$X^2 = 1 + h + 2X + 2Y, \qquad hX = X, \qquad gX = g^{-1}X = Y.$$

The image $X$ is self-dual, since $V_{(0,1)}$ is self-dual. Moreover $Y = gX$, so $Y^* = Xg^{-1} = Y$. The category is tensor generated by the central object $X$ and the cyclic local group, so its ring is commutative. Associativity now determines all remaining products:

$$\begin{aligned} g^4 = 1, \quad gX = Y, \quad gY = X, \\ X^2 = Y^2 = 1 + g^2 + 2X + 2Y, \qquad XY = g + g^3 + 2X + 2Y. \end{aligned} \tag{6.12}$$

These are the required full fusion rules, with census order $(1, g, g^{-1}, X, Y, h)$; in source coordinates the permutation is $(0, 4, 5, 1, 2, 3)$. The resulting 216 coefficients, including the nontrivial duality $g^* = g^{-1}$ and the self-duality of $X, Y$, agree with $R^{(6)}_{2,4}$. □

The mathematical inputs to this identification and its arithmetic checks are separated. Existence and unitarity come from the conformal embedding. The source annular and branching matrices determine the module action; the cyclic local group then determines its full fusion ring. Independently, the constructor calculates all nine $G_{2,4}$ fusion matrices by Weyl alternants in $\mathbb{Q}(\zeta_{24})$, and verifies

$$N^{\mathcal{B}}_i E = E\Big(\sum_a E_{ia} N^{\mathcal{B}_A}_a\Big), \qquad EE^{\mathsf{T}} = I + N^{\mathcal{B}}_{(0,3)}.$$

The latter is free-module adjunction for the stated vacuum algebra. All entries, all unit-fixing identifications with the census, and the cyclic group action are exact. A dimension match alone would not prove the proposition: it would fail to distinguish the cyclic-four ring from the Klein-four rings with the same dimension list.

The dimensions are $(1, 1, 1, 1, 2 + \sqrt{6}, 2 + \sqrt{6})$ and $\mathrm{FPdim}(\mathcal{B}_A) = 24 + 8\sqrt{6}$. The published construction and its exact tensor identification provide a positive control for the finite necessary systems in Section 7.2.

## 6.7 Inversion equivariantization of an odd near-group category

The positive case at multiplicity three is an existing unitary model, not a proposed solution of new pentagon equations. We use the coherent order-two action constructed in [25, Section 13.3]. The following formulation separates that categorical input from the reconstruction of the entire fusion tensor.

**Proposition 6.11** (An inversion-equivariantization constructor)**.** *Let $n$ be odd, and let $\mathcal{N}$ be a unitary near-group category with invertibles $\alpha_j$, $j \in \mathbb{Z}/n\mathbb{Z}$, and one further simple $\rho$, with*

$$\alpha_j \rho = \rho \alpha_j = \rho, \qquad \rho^2 = \bigoplus_j \alpha_j \oplus n\rho.$$

*Assume a coherent unitary $C_2$-action fixes $\rho$, takes $\alpha_j$ to $\alpha_{-j}$, fixes the unit channel of $\rho^2$, and acts on a chosen basis $(T_j)_j$ of $\mathrm{Hom}(\rho, \rho^2)$ by $T_j \mapsto T_{-j}$. Assume moreover that the action on the pointed part has the standard inversion equivariantization $\mathrm{Rep}(\widehat{C}_n \rtimes C_2)$. Then $\mathcal{N}^{C_2}$ is a unitary fusion category of rank $(n + 7)/2$. Its simples are*

$$1, g, x_+, x_-, v_1, \ldots, v_{(n-1)/2}.$$

*Writing $v_0 = 1 + g$ and $v_j = v_{-j}$, with indices modulo $n$, its products are determined by*

$$g^2 = 1, \qquad gx_+ = x_-, \qquad gx_- = x_+, \qquad gv_i = v_i.$$

$$
\begin{aligned}
v_i v_j &= v_{i+j} + v_{i-j}, \\
v_i x_\pm = x_\pm v_i &= x_+ + x_-, \\
x_+^2 = x_-^2 &= 1 + \sum_i v_i + \frac{n+1}{2} x_+ + \frac{n-1}{2} x_-, \\
x_+ x_- = x_- x_+ &= g + \sum_i v_i + \frac{n-1}{2} x_+ + \frac{n+1}{2} x_-.
\end{aligned}
$$

*All its simples are self-dual. The dimensions are* $1, 1, d, d, 2, \ldots, 2$*, where* $d > 0$ *satisfies* $d^2 = n + nd$*.*

*Proof.* For an action on a semisimple category, equivariant simple objects are obtained from orbits of simple objects and irreducible projective representations of their stabilizers. In the present explicitly given action, the fixed simples $1, \rho$ have their indicated equivariant structures and each admits the two sign choices. Denote these by $1, g$ and $x_+, x_-$. Each nontrivial inversion orbit has trivial stabilizer and yields a single $v_i$ with underlying object $\alpha_i \oplus \alpha_{-i}$. This proves the rank and dimension statements. Tensoring with $g$ changes the sign of an equivariant structure and therefore exchanges $x_+, x_-$ while fixing every $v_i$.

The pointed equivariantization hypothesis gives the displayed products of the $v_i$. For $v_i\rho$, the two summands of the underlying object are exchanged by the action. Its $+1$ and $-1$ eigenspaces are both one-dimensional, proving $v_i x_\pm = x_+ + x_-$, and likewise for products in the opposite order.

The unit channel in $\rho^2$ contributes 1 to $x_+^2$. The other invertible channels are paired and contribute the $v_i$. In the $\rho$-multiplicity space, inversion fixes exactly $T_0$ and pairs the other $n-1$ basis vectors. Its eigenspaces have dimensions $(n+1)/2$ and $(n-1)/2$. This proves the formula for $x_+^2$. Multiplying by $g$ proves the mixed product, and multiplying twice proves $x_-^2 = x_+^2$. The unit coefficients in these formulas show self-duality. A coherent unitary action has a unitary equivariantization; thus the construction uses actual associativity data, not merely an automorphism of a based ring. □

**Corollary 6.12.** *The ring* $R^{(6)}_{3,218}$ *admits a unitary categorification.*

*Proof.* Izumi's $C_5 + 5$ category has the inversion symmetry of [25, Example 13.8]. Section 13.3 constructs an actual order-two automorphism commuting with $\rho$, not just an outer automorphism class. Its action on the isometries is $S_j \mapsto S_{-j}$ and $T_j \mapsto T_{-j}$. Theorems 13.17–13.18 of that source establish precisely the hypotheses and product decompositions used above, including the pointed representation subcategory.

The source scalar symmetries can also be checked directly: with $\zeta = \exp(2\pi i/5)$, the quadratic phases are $a(j) = \zeta^{2j^2}$ and the nonzero $b$-phases have exponents $(4, 1, 1, 4)$ at $j = 1, 2, 3, 4$, each divided by $\sqrt{5}$. Both lists are inversion invariant. The dimension used here is derived from the fusion equation,

$$
d = \frac{5 + 3\sqrt{5}}{2}, \qquad d^2 = 5 + 5d, \qquad \dim(\mathcal{N}^{C_2}) = 45 + 15\sqrt{5}.
$$

Thus no independently copied numerical dimension enters the constructor. The source supplies existence and the coherent action; the finite checks validate the specified permutations and scalar symmetries.

In the proposition's order $(1, g, x_+, x_-, v_1, v_2)$, the census order is $(1, g, x_-, x_+, v_1, v_2)$. The function `inversion_rules(5)` constructs every product and the recorded permutation $(0, 1, 3, 2, 4, 5)$ identifies all 216 structure constants with $R^{(6)}_{3,218}$. This is a based-ring identification, not a match of Frobenius–Perron dimensions alone. □

The generic formula is not a claim that the categorical input exists for every odd $n$. The positive registry supplies only the cited $n = 5$ input. This constructor and its full based isomorphism are included in Table 3 and in the electronic model catalogue.

## 6.8 Higher subfactor and affine-group models

### 6.8.1 The six-object Asaeda–Haagerup category

The unitary category $\mathcal{AH}_1$ has the six-object fusion table in [24, Table 1]. In its source order $(1, \psi, \chi, \sigma, \zeta, \tau)$, the permutation $(0, 1, 3, 2, 5, 4)$ identifies all 216 structure constants with $R^{(6)}_{6,1764}$. The supplement encodes the source table separately from the census and checks this complete equality. Existence comes from the published subfactor category, not from the arithmetic of that table. This is a known unitary model. Its global dimension coinciding with that of another tensor, in particular $R^{(6)}_{3,215}$, proves neither a based-ring isomorphism nor a Morita equivalence.

### 6.8.2 A regular-orbit representation-ring constructor

**Proposition 6.13.** *Let $V$ be a nontrivial finite abelian group and suppose that $H$ acts freely and transitively on $\widehat{V} \setminus \{1\}$. Put $h = |H| = |V| - 1$. The irreducibles of $G = V \rtimes H$ consist of the inflations of* $\mathrm{Irr}(H)$ *and one additional self-dual representation $\rho$ of dimension $h$. Their products satisfy*

$$\sigma\rho = (\dim \sigma)\rho, \qquad \rho^2 = \sum_{\sigma \in \mathrm{Irr}(H)} (\dim \sigma)\sigma + (h-1)\rho. \tag{6.13}$$

*Consequently these rules, together with* $\mathrm{Rep}(H)$*, define a unitary categorical model, whenever the stated group action is supplied.*

*Proof.* The irreducibles of a semidirect product by an abelian normal subgroup are obtained from $H$-orbits in $\widehat{V}$ and representations of their stabilizers. The trivial orbit gives the inflated representations of $H$. The single nontrivial orbit has trivial stabilizer and gives $\rho = \mathrm{Ind}_V^G \lambda$, for any nontrivial $\lambda$. Inversion preserves this orbit, so $\rho$ is self-dual. Restriction of an inflated $\sigma$ to $V$ is $(\dim \sigma)1$, and induction gives the first equation. Its reciprocity then gives the multiplicity $\dim \sigma$ of each inflation in $\rho^2$. If the remaining multiplicity of $\rho$ is $a$, comparing dimensions yields $h^2 = \sum_\sigma (\dim \sigma)^2 + ah = h + ah$, so $a = h - 1$. The representation category of a finite group is unitary. □

Take $V = \mathbb{F}_3^2$ and let $H$ be generated by

$$I = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The identities $I^2 = J^2 = -1$ and $IJ = -JI$ identify $H$ with $Q_8$. Its orbit on the eight nonzero vectors is free and transitive; the contragredient action has the same property. Thus Proposition 6.13 gives a six-object category of dimension 72, with simple dimensions $(1, 1, 1, 1, 2, 8)$ and multiplicity seven. It realizes $R^{(6)}_{7,572}$.

The constructor generates the actual 72-element semidirect product, checks its multiplication and action, and computes the four linear characters, the inflated two-dimensional quaternion character, and the induced eight-dimensional character. It verifies orthogonality, completeness by the sum of squares, and all tensor coefficients by character inner products. The resulting tensor is then compared to the census. This test is an application of the proposition to explicit group data, not a rule accepting all abstract rings with these dimensions.

## 6.9 The complete catalogue and group-theoretical constructions

The positive part of the classification is recorded as explicitly as the negative part. Table 3 contains one row for every one of the 57 positive rings, including all twenty-one rings of multiplicity one. The electronic catalogue records every constructor used, not merely one representative: there are 64 constructor records, with an explicit unit-fixing basis permutation from each output to its census representative. These permutations are verified entry by entry. The names in the table are presentation labels, not hypotheses in a categorification test. The symbols U and S distinguish unitary existence from spherical existence with proved nonexistence of a pseudo-unitary categorification. The single S entry is not counted among the unitary models.

### 6.9.1 Small constructions and conventions

Here $PSU(2)_k$ means the even subcategory of the principal unitary $SU(2)_k$ category, and $D_9$ has order eighteen. The indicated $C_2$ action on $C_3 \times C_3$ is inversion. In $C_3 \rtimes C_4$, a generator of $C_4$ acts by inversion on $C_3$; the action therefore factors through $C_2$. The positive Ising-type rings in the table are denoted $\mathcal{I}(G, r)$, where $|G| = 4$, a subgroup $H$ of order two is fixed, and $r \in \{0, 1\}$. Their basis consists of $G$ and $\sigma_0, \sigma_1$. An element of $G/H \cong C_2$ translates the subscripts, and

$$\sigma_i \sigma_j = \sum_{a \in G:\, \bar{a} = i+j+r} a.$$

The unitary realizations are the Ising zestings constructed in [29, Section 4.3]; the metaplectic category $SO(5)_2$ and its listed zest are treated there as well. The program reconstructs these formulas, ordinary finite-group character products, the Haagerup tensor, and the $SU(2)$ fusion rules before matching the resulting tensors. Thus the multiplicity-one positives do not depend on a cardinality-complement argument.

Table 3: All positively identified rings in the census. U means a unitary realization; S means a spherical realization and no pseudo-unitary categorification (hence no unitary one). The numbers of types U and S are given in Table 1. Here $R_{m,j} = R^{(6)}_{m,j}$. Every constructor and its based isomorphism are recorded electronically.

| Ring | Model or construction | Type | Source |
|---|---|---|---|
| $R_{1,1}$ | $\mathcal{I}(C_4, 1)$: cyclic, non-self-dual Ising zest | U | [29] |
| $R_{1,3}$ | $\mathcal{I}(C_4, 0)$: cyclic, self-dual Ising zest | U | [29] |
| $R_{1,4}$ | $\mathrm{Rep}(C_3 \rtimes C_4)$ | U | [29] |
| $R_{1,6}$ | $\mathrm{Vec}_{C_2} \boxtimes \mathrm{Vec}_{C_3} \simeq \mathrm{Vec}_{C_6}$ | U | [29] |
| $R_{1,7}$ | Tambara–Yamagami category for $C_5$ | U | [29] |
| $R_{1,9}$ | $\mathcal{I}(C_2^2, 1)$: non-self-dual Ising zest | U | [29] |
| $R_{1,10}$ | Zested $SO(5)_2$ category | U | [29] |
| $R_{1,11}$ | $\mathrm{Fib} \boxtimes \mathrm{Vec}_{C_3}$ | U | [29] |
| $R_{1,14}$ | $\mathrm{Vec}_{S_3}$ | U | [29] |
| $R_{1,15}$ | Haagerup category $\mathcal{H}_3$ | U | [23] |
| $R_{1,20}$ | $\mathrm{Vec}_{C_2} \boxtimes$Ising; also $R(0, 0)$ | U | [29, 30] |
| $R_{1,21}$ | $\mathrm{Vec}_{C_2} \boxtimes \mathrm{Rep}(S_3)$; also $R(0, 1)$ | U | [29, 30] |
| $R_{1,23}$ | $\mathrm{Vec}_{C_2} \boxtimes PSU(2)_5$ | U | [29] |
| $R_{1,24}$ | $PSU(2)_{10}$ (the other 3311 even-part ring) | U | [29, 27] |
| $R_{1,26}$ | $SO(5)_2$ | U | [29] |
| $R_{1,30}$ | $\mathrm{Rep}(D_9)$, with $\lvert D_9 \rvert = 18$ | U | [29] |

Continued on the next page

Table 3 continued

| Ring | Model or construction | Type | Source |
|---|---|---|---|
| $R_{1,32}$ | $\mathrm{Rep}((C_3 \times C_3) \rtimes C_2)$, inversion action | U | [29] |
| $R_{1,34}$ | $PSU(2)_{11}$ | U | [29] |
| $R_{1,36}$ | $\mathrm{Fib} \boxtimes \mathrm{Ising}$ | U | [29] |
| $R_{1,37}$ | $\mathrm{Fib} \boxtimes \mathrm{Rep}(S_3)$ | U | [29] |
| $R_{1,38}$ | $\mathrm{Fib} \boxtimes PSU(2)_5$ | U | [29] |
| $R_{2,2}$ | Fermionic $C_2$-zest of $\mathrm{Ad}(E_6) \boxtimes \mathrm{Vec}_{C_2}$ | U | [9] |
| $R_{2,4}$ | $\mathcal{E}_4(G_2)$: modules for $G_{2,4} \subset \mathrm{Spin}(14)_1$ | U | [8] |
| $R_{2,11}$ | $\mathrm{Rep}((C_3 \times C_3) \rtimes C_4)$ | U | [16] |
| $R_{2,14}$ | Fermionic grading zest of the $E_6$ category | U | [9] |
| $R_{2,34}$ | $TT_3$ | U | [14] |
| $R_{2,36}$ | $\mathcal{C}(A_4, 1, C_2, 1)$ | U | [2] |
| $R_{2,40}$ | Diagonal-fermion quotient of two identical half-$E_6$ categories | U | [1] |
| $R_{2,49}$ | Quadratic category $R'(2)$ | U | [30] |
| $R_{2,53}$ | $\mathrm{Vec}_{C_2} \boxtimes \mathrm{Ad}(E_6)$; also $R(0,2)$ | U | [30] |
| $R_{2,69}$ | $C_2$-extension of $\mathrm{Rep}(H_8)$ | U | [7] |
| $R_{2,71}$ | $\mathcal{C}(S_4, \omega, C_2^2, \psi)$ of Theorem 6.16 | U | [21] |
| $R_{2,75}$ | $E_6$ modules over $SU(2)_{10}$ | U | [28] |
| $R_{2,94}$ | Diagonal-fermion quotient of $\mathrm{Ising} \boxtimes PSU(2)_6$ | U | [16] |
| $R_{2,97}$ | Dual Goodman–de la Harpe–Jones 3311 even part | U | [27] |
| $R_{2,98}$ | Dual even part of the Izumi $C_4$ subfactor | U | [26] |
| $R_{2,105}$ | $\mathcal{C}(A_5, 1, D_{10}, 1)$ | U | [2] |
| $R_{2,124}$ | Schopieray: $\mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$ | S | [43] |
| $R_{2,127}$ | $\mathrm{Fib} \boxtimes \mathrm{Ad}(E_6)$ | U | [28] |
| $R_{2,128}$ | $D_6$ modules over $SU(2)_8$ | U | [28] |
| $R_{2,140}$ | Local $D_{10}$ modules over $SU(2)_{16}$ | U | [28] |
| $R_{2,143}$ | Commutative even part of the Izumi $C_5$ subfactor | U | [18] |
| $R_{2,144}$ | Principal $G_2$ level-three category | U | [42] |
| $R_{3,47}$ | $\mathrm{Rep}(GL(3, \mathbb{F}_2)) \simeq \mathrm{Rep}(PSL(2,7))$ | U | [2] |
| $R_{3,141}$ | $\mathcal{C}(A_5, 1, S_3, 1)$ | U | [2] |
| $R_{3,218}$ | Inversion equivariantization of Izumi's $C_5 + 5$ near-group category | U | [25] |
| $R_{4,22}$ | Neutral modules for $(B_2)_7 \subset (D_7)_1$ | U | §6.14 |
| $R_{4,91}$ | Neutral modules for $(C_3)_4 \subset (B_{10})_1$ | U | §6.14 |
| $R_{4,249}$ | Sign-swap equivariantization of the equal half-$E_6$ quotient | U | §6.12 |
| $R_{4,322}$ | Diagonal-fermion quotient of half-$E_6 \boxtimes PSU(2)_6$ | U | §6.11 |
| $R_{4,629}$ | Neutral modules for $(B_2)_{12} \subset (E_8)_1$ | U | §6.14 |
| $R_{5,17}$ | Izumi's $C_5 + 5$ near-group category | U | [25] |
| $R_{6,345}$ | $\mathcal{C}_{4,8,4}$, with ring $R(4,6)$ | U | [30] |
| $R_{6,1712}$ | Relative Fibonacci center $Z_{\mathrm{Fib}}(TT_3)$ | U | §6.13 |
| $R_{6,1764}$ | Asaeda–Haagerup category $\mathcal{AH}_1$ | U | [24] |
| $R_{7,572}$ | $\mathrm{Rep}((C_3 \times C_3) \rtimes Q_8)$ | U | [16] |
| $R_{7,2274}$ | Extended Haagerup category $\mathcal{EH}_1$ | U | [4, 22, 6] |

### 6.9.2 Exact Weyl-character reconstruction

We use the principal unitary categories $\mathcal{C}(\mathfrak{g}, k)$ at positive integral level. Their construction and the Weyl-alcove and character formulas are reviewed in [42]; these are categorical inputs, not consequences of a numerically fitted modular matrix. Normalize long roots to have squared length two, let $W$ be the Weyl group, and put $\rho = \sum_i \lambda_i$. For dominant weights in the level-$k$ alcove define

$$C_{\lambda\mu} = \frac{\sum_{w\in W} \det(w) \exp\bigl(-2\pi i (w(\lambda+\rho), \mu+\rho)/(k+h^\vee)\bigr)}{\sum_{w\in W} \det(w) \exp\bigl(-2\pi i (w\rho, \mu+\rho)/(k+h^\vee)\bigr)}. \tag{6.14}$$

These are the fusion characters. Their products determine the fusion coefficients by

$$C_{\lambda\mu} C_{\nu\mu} = \sum_\tau N_{\lambda\nu}^\tau C_{\tau\mu}.$$

All denominators and the required character-matrix rank are checked in the cyclotomic number field. Exact Gaussian elimination determines the coefficients, which are then checked to be nonnegative integers and to satisfy every character equation, including columns not used to solve the linear system. This also works for the root-lattice subcategory of type $A_2$, using a full-rank submatrix of its restricted characters.

For $G_2$ at level three, the short fundamental weight is first, and the input data are

$$((\lambda_i, \lambda_j)) = \begin{pmatrix} 2/3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad h^\vee = 4, \qquad \Lambda_3 = \{(0,0), (1,0), (2,0), (3,0), (0,1), (1,1)\}.$$

The calculation takes place in $\mathbb{Q}(\zeta_{21})$, and produces a rank-six tensor of multiplicity two.

**Proposition 6.14.** *The tensor obtained from* (6.14) *for* $(\mathfrak{g}, k) = (G_2, 3)$ *is based-isomorphic to* $R^{(6)}_{2,144}$. *In particular this ring admits a unitary modular categorification.*

*Proof.* The Weyl construction supplies the category and the displayed character formula supplies its fusion rules. The exact calculation just described is implemented independently of the census in `software/quantum_rank2.py`. The resulting 216 coefficients and a basis permutation are recorded in the model catalogue; the checker recomputes them and verifies their equality to the census tensor. This is a full-tensor identification, not an identification by dimension type. □

### 6.9.3 The dual 3311 even part

The dual even-part identification is

$$R^{(6)}_{2,97} \cong K_0\bigl((A_{11}^{\mathrm{even}})^*_{E_6^{\mathrm{even}}}\bigr). \tag{6.15}$$

The category on the right is the dual even part of the Goodman–de la Harpe–Jones 3311 subfactor and is unitary. Its complete multiplication table is Figure 2 of [27, Section 5.1]. The independent constructor uses the basis $(1, B, C, C', D, g)$ of that figure; in particular

$$C^2 = 1 + g + 2C, \quad (C')^2 = 1 + g + 2C', \quad CC' = B + D, \quad gC = C, \quad gC' = C', \quad gB = D.$$

The other products are reconstructed from the same published table and all tensor axioms and the based isomorphism in (6.15) are checked. The other 3311 even-part ring is already $R^{(6)}_{1,24}$. Neither identification implies that another ring with the same Frobenius–Perron dimensions is categorifiable; in particular it does not settle the non-self-dual ring $R^{(6)}_{2,40}$.

### 6.9.4 A three-double-coset realization criterion

For finite $G$, a subgroup $H$, and unit-modulus normalized cochains $\omega \in Z^3(G,\mathbb{C}^\times)$ and $\psi \in C^2(H,\mathbb{C}^\times)$ with $\omega|_H = \delta\psi$, write $\mathcal{C}(G,\omega,H,\psi)$ for the category of $\psi$-twisted $H$-bimodules in $\mathrm{Vec}_G^\omega$, with the conventions of [21]. For $u \in G$ put $H^u = H \cap uHu^{-1}$. Simple objects over $HuH$ correspond to irreducible projective representations of $H^u$ with factor

$$\psi^u(a,b) = \psi(a,b)\psi(u^{-1}b^{-1}u, u^{-1}a^{-1}u) \cdot \frac{\omega(a,b,u)\omega(a,bu,u^{-1}b^{-1}u)}{\omega(abu,u^{-1}b^{-1}u,u^{-1}a^{-1}u)}. \tag{6.16}$$

Such a simple has Frobenius–Perron dimension $[H:H^u]\dim\pi$. These statements, and support of tensor products on products of double cosets, are [21, Theorems 4.1 and 5.1]. We call a factor on an abelian group nondegenerate when its twisted group algebra has one irreducible representation; equivalently its commutator bicharacter has trivial radical.

**Theorem 6.15** (Three-double-coset realization)**.** *Suppose $H$ is abelian of order $h = s^2 > 1$ and*

$$G = H \sqcup tH \sqcup HuH, \qquad N_G(H) = H \sqcup tH, \qquad H^u = \{1\}.$$

*If the factor $\psi^t$ in* (6.16) *is nondegenerate, then $\mathcal{C}(G,\omega,H,\psi)$ is a unitary categorification of the commutative fusion ring with basis $\widehat{H} \sqcup \{X,Y\}$ and rules*

$$\chi X = X\chi = X, \qquad \chi Y = Y\chi = Y, \qquad XY = YX = sY,$$
$$X^2 = \sum_{\chi\in\widehat{H}} \chi, \qquad Y^2 = \sum_{\chi\in\widehat{H}} \chi + sX + (h-2)Y. \tag{6.17}$$

*Its dimension type is $(1^h, s, h)$.*

*Proof.* The component over $H$ is $\mathrm{Rep}(H)$, so it consists of the $h$ invertibles indexed by $\widehat{H}$ with their ordinary group multiplication. Nondegeneracy of $\psi^t$ gives one simple $X$ of dimension $s$ over $tH$. The trivial stabilizer of $HuH$ gives one simple $Y$ of dimension $h$. Hence these are all the simple objects.

The normalizer is a subgroup with quotient $N_G(H)/H$ of order two. Consequently $tH$ is closed under inversion. Its complement $HuH$ in $G$ is also closed under inversion and is stable under multiplication on either side by $N_G(H)$. Since $X$ and $Y$ are unique on their supports, both are self-dual and all the invertibles fix them on both sides. Support forces $X^2$ into $\mathrm{Rep}(H)$. Frobenius reciprocity gives $[X^2:\chi] = [\chi X : X] = 1$, proving the first square formula. Similarly, $XY$ and $YX$ are supported on $HuH$, and dimensions give $XY = YX = sY$. Reciprocity gives $[Y^2:\chi] = 1$ and $[Y^2:X] = s$. The remaining coefficient is therefore determined by $h^2 = h + s^2 + h[Y^2:Y]$, namely $[Y^2:Y] = h-2$.

Finally $\mathrm{Vec}_G^\omega$ is unitary for unit-modulus $\omega$. The admissible twisted regular algebra of $H$, with its normalized regular inner product, is a connected separable $C^*$-algebra object. Its bimodule category is unitary. This is the standard unitary form of the group-theoretical construction, and proves existence rather than merely associativity of the displayed ring. □

### 6.9.5 An explicit twisted $S_4$ model

The following cochain certificate makes the realization criterion applicable to a residual that is not realized by the corresponding untwisted construction.

Realize $G = PGL(2,\mathbb{F}_3) \cong S_4$ by its action on $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0),(0,1),(1,1),(1,2)\}$. In each projective class choose the matrix $s(g)$ whose first nonzero entry in row-major order is

one. Define

$$s(g)s(k) = (-I)^{\beta(g,k)}s(gk), \qquad \alpha(g) = \begin{cases} 0, & \det s(g) = 1, \\ 1, & \det s(g) = 2, \end{cases} \qquad \omega(g,k,l) = (-1)^{\alpha(g)\beta(k,l)}.$$

Here $\beta$ is a normalized $\mathbb{F}_2$-valued 2-cocycle from the central extension $GL(2,\mathbb{F}_3)$, and $\alpha$ is a homomorphism. The cup-product identity therefore proves that $\omega$ is a normalized 3-cocycle. Take $H = \{1, a, b, c\}$ represented, respectively, by

$$I, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Its action is the subgroup generated by the two disjoint transpositions on the displayed ordered projective line, so $H \cong C_2^2$. Put $\psi(x,y) = i^{q(x,y)}$, with $q(1,x) = q(x,1) = 0$ and

| $q$ | $a$ | $b$ | $c$ |
|---|---|---|---|
| $a$ | 0 | 2 | 0 |
| $b$ | 3 | 0 | 1 |
| $c$ | 1 | 0 | 1 |

in $\mathbb{Z}/4\mathbb{Z}$. (6.18)

For every $x, y, z \in H$ the admissibility identity is

$$q(y,z) + q(x,yz) - q(xy,z) - q(x,y) \equiv 2\alpha(x)\beta(y,z) \pmod 4.$$

It follows either by substitution in the nine-entry table or by the complete $4^3$-triple check in the supplement.

**Theorem 6.16.** *For the cochains just defined, $\mathcal{C}(S_4, \omega, C_2^2, \psi)$ is a unitary categorification of $R_{2,71}^{(6)}$.*

*Proof.* Choose

$$t = \left[\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\right], \qquad u = \left[\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}\right].$$

Matrix multiplication over $\mathbb{F}_3$ gives exactly three double cosets, of sizes $4, 4, 16$, represented by $1, t, u$. The first two form $N_G(H)$, and $H^u$ is trivial. Substitution in (6.16) gives the following exponent matrix for $\psi^t$, in the order $(1, a, b, c)$:

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 3 & 0 & 1 \\ 0 & 2 & 3 & 1 \end{pmatrix} \pmod 4.$$

Its commutator bicharacter has trivial radical: each nonidentity row has an entry different from the corresponding transposed entry. Thus its twisted group algebra is $M_2(\mathbb{C})$, and Theorem 6.15 applies with $h = 4$, $s = 2$.

The constructor regenerates the twenty-four projective matrices, their faithful permutation action, the subgroup, all double cosets, and every stabilizer factor. It checks the admissibility identity and, in addition to the cup-product proof, all $24^4$ cocycle identities. It then uses (6.17) to reconstruct the full tensor and checks its based isomorphism with $R_{2,71}^{(6)}$. The complete data are in `data/twisted_s4_certificate.json`. Thus no inference from Frobenius–Perron dimensions alone is used. □

The three additions in this section are positive determinations, not new exclusions. In particular, searching only untwisted group-theoretical models would miss the last construction. All allocated negative certificates remain applications of the unified obstruction framework stated earlier in the paper.

### 6.10 Exceptional and near-group source identifications

Two source tensors must be included independently of any obstruction search. The principal even part $\mathcal{EH}_1$ of the extended Haagerup subfactor is unitary by its subfactor construction [4]. Its fusion-category structure and rank-six fusion rules are described in [22, Section 2.1]; its six fusion matrices are explicitly displayed in [6, Section 5.1]. In the latter source order, the permutation $(0,1,3,4,5,2)$ gives $R^{(6)}_{7,2274}$. The largest coefficient is seven. The constructor transcribes the published matrices and checks all 216 coefficients, not merely the dimension field or formal codegrees.

The unitary $C_5+5$ near-group category in [25, Example 13.8] has basis $\alpha_0,\dots,\alpha_4,\rho$ and products

$$\alpha_i\alpha_j=\alpha_{i+j},\qquad \alpha_i\rho=\rho\alpha_i=\rho,\qquad \rho^2=\sum_{i=0}^{4}\alpha_i+5\rho.$$

Its tensor identifies $R^{(6)}_{5,17}$ with permutation $(0,1,4,3,2,5)$. The positive root of the dimension equation is $(5+3\sqrt{5})/2$. This is the near-group input itself, distinct from the inversion equivariantization of Corollary 6.12. Both source identifications are independent of the census and both are matched under all unit-preserving permutations.

### 6.11 A mixed diagonal-fermion quotient

Let $\mathcal{E}$ be the unitary half-$E_6$ category of (6.1), and let $\mathcal{P}=PSU(2)_6$. The object $h$ of $\mathcal{E}$ has the central fermion of Lemma 6.1. In the standard unitary $SU(2)_6$ category, $V_6$ is an order-two fermion; restricting its half-braiding gives a central fermion in $\mathcal{P}$. Thus $(h,V_6)$ is a central boson in $\mathcal{E}\boxtimes\mathcal{P}$.

**Proposition 6.17.** *The category of modules over $1\oplus(h,V_6)$ in $\mathcal{E}\boxtimes\mathcal{P}$ is a unitary fusion category with Grothendieck ring $R^{(6)}_{4,322}$.*

*Proof.* The second factor has basis $(V_0,V_2,V_4,V_6)$, with $V_6$ interchanging $V_0,V_6$ and $V_2,V_4$. The diagonal action is therefore free, even on objects whose first component is $X$. Proposition 6.6 applies without a fixed-point choice. In the product order $(1,h,X)\times(V_0,V_2,V_4,V_6)$, its permutation of the twelve simples is

$$(7,6,5,4,3,2,1,0,11,10,9,8).$$

Take orbit representatives $(0,1,2,3,8,9)$. The quotient formula of Proposition 6.6, applied to the independently constructed half-$E_6$ and $SU(2)_6$ tensors, gives the census tensor after permutation $(0,3,1,2,4,5)$. The implementation checks every product and its independence of the chosen representative. Existence and unitarity come from the central separable algebra; the tensor comparison identifies its Grothendieck ring. □

### 6.12 A sign-twisted swap on a fermionic product

The next construction has a categorical hypothesis. Associativity of the integer tensor for a formal parameter does not provide its input category.

**Theorem 6.18** (Sign-swap equivariantization)**.** *Let $\mathcal{E}$ be a unitary fusion category with simples $1,f,x$, a specified central fermion $f$, and rules*

$$f^2=1,\quad fx=xf=x,\quad x^2=1+f+mx.$$

*Suppose $d = \mathrm{FPdim}(x)$ is irrational. There is a unitary fusion category with invertible group $G = C_2 \times C_2$ and two self-dual simples $X, Y$, fixed by all invertibles, with*

$$\begin{aligned} X^2 &= \sum_{g\in G} g + mX + 2Y, \\ XY = YX &= 2X + 2mY, \\ Y^2 &= \sum_{g\in G} g + 2mX + m^2 Y. \end{aligned} \tag{6.19}$$

*Their dimensions are $2d$ and $d^2$. The Ising input $m = 0$ gives $R^{(6)}_{2,69}$, and the half-$E_6$ input $m = 2$ gives $R^{(6)}_{4,249}$.*

*Proof.* In $\mathcal{E}\boxtimes\mathcal{E}$ put $b = f\boxtimes f$ and $A = 1\oplus b$. This is the regular algebra of a central Tannakian subcategory. The factor-swap tensor functor $S$ preserves the central lift of $b$. Equip $A$ with the algebra isomorphism $\phi : S(A) \to A$ that is $+1$ on 1 and $-1$ on $b$. It satisfies $\phi S(\phi) = \mathrm{id}$. Transport of module actions through $S$ and $\phi$ defines a coherent unitary $C_2$ action $T$ on $\mathcal{Q} = (\mathcal{E}\boxtimes\mathcal{E})_A$. The category we use is the ordinary equivariantization $\mathcal{Q}^{C_2}$.

The free $b$-orbits give four simples of $\mathcal{Q}$: the unit, $g = (f\boxtimes 1)A$, and $a = (x\boxtimes 1)A$, $a' = (1\boxtimes x)A$. The fixed object $x\boxtimes x$ gives two simples $u_+, u_-$, each of dimension $d^2/2$. To see the action explicitly, choose the right $f$-action $t : x\otimes f \to x$ with square one, using the module associativity constraint. Such a normalization exists because $1\oplus f$ is the regular separable algebra and its two actions on $x$ differ by sign. The right $b$-actions on the fixed object are $\pm(t\boxtimes t)$. Swap preserves this tensor, while $\phi$ changes its sign. Hence $T$ interchanges $u_+, u_-$; it also interchanges $a, a'$ and fixes $1, g$.

An orbit of size two has trivial stabilizer and yields one simple in the equivariantization; a fixed simple has two equivariant structures, since projective representations of $C_2$ over $\mathbb{C}$ can be linearized. Thus there are four invertibles, one simple $X$ over $a\oplus a'$ and one simple $Y$ over $u_+\oplus u_-$. Their dimensions are $2d, d^2$. Their supports are invariant under duality, so $X, Y$ are self-dual. They are fixed by all invertibles: tensoring preserves their distinct dimensions and there is only one simple of either dimension.

We check the group law on the four invertibles; dimensions alone would not distinguish $C_4$ from $C_2\times C_2$. Work in the pointed part $\mathrm{sVec}\boxtimes\mathrm{sVec}$, with trivial associator and the specified self-braidings $-1$. Let $e_0, e_1$ be the two homogeneous components of $g = (f, 1)A$, in degrees $(f, 1), (1, f)$. An isomorphism $g\otimes_A g \to A$ has the normalized coefficients

$$\mu_{00} = 1, \qquad \mu_{01} = b, \qquad \mu_{10} = -b, \qquad \mu_{11} = -1.$$

The minus sign in balancing is forced by the half-braiding of $b$ with $(f, 1)$; it is not a choice of a ring sign. Give $g$ the equivariance $T(e_0) \mapsto ie_1$, $T(e_1) \mapsto -ie_0$. Its square is the identity. On each of the four pairs $(e_r, e_s)$, the displayed $\mu$ intertwines this tensor-square equivariance with $\phi$, which acts on $A$ as $(1, -1)$. Therefore this lift of $g$ squares to the tensor unit. Together with the independent sign representation of $C_2$, it generates $C_2\times C_2$. The six scalar equalities (two for coherence, four for multiplication) are also checked exactly over $\mathbb{Q}(i)$.

Frobenius reciprocity now puts each invertible once in $X^2, Y^2$ and none in $XY$. Write $d^2 = 2 + md$. Comparing coefficients of $1, d$ in $(2d)^2$, $(2d)d^2$ and $d^4$ forces respectively the coefficients in (6.19); their uniqueness uses the irrationality of $d$. For example $X^2 = \sum_g g + aX + cY$ gives $8 + 4md = 4 + 2c + (2a + cm)d$, so $c = 2$ and $a = m$. For $XY = aX + cY$, comparison gives $c = 2m, a = 2$; for $Y^2 = \sum_g g + aX + cY$, it gives $c = m^2, a = 2m$. All operations used preserve unitarity.

The Ising and half-$E_6$ inputs and their central fermions are supplied by standard Ising theory and Lemma 6.1 (see also [1]). Their tensors, constructed from (6.19), match the census with permutations $(0, 1, 2, 3, 5, 4)$ and the identity. These two categorical inputs, not arbitrary values of $m$, are admitted by the model registry. □

### 6.13 Reconstruction from a relative center

For a full fusion subcategory $\mathcal{D} \subset \mathcal{C}$, the relative center $Z_{\mathcal{D}}(\mathcal{C})$ consists of objects of $\mathcal{C}$ equipped with a coherent half-braiding against objects of $\mathcal{D}$. It is not the Müger centralizer in a braided category. It is a fusion category, is Morita equivalent to $\mathcal{D} \boxtimes \mathcal{C}^{\mathrm{rev}}$, and has dimension $\mathrm{FPdim}(\mathcal{D})\,\mathrm{FPdim}(\mathcal{C})$ [20, Remark 2.4]. For unitary input, its realization as the dual of the natural unitary module category gives a unitary fusion category.

**Proposition 6.19** (Forced relative-center reconstruction)**.** *Suppose $\mathcal{D} = \langle a \rangle \subset \mathcal{C}$ is a Fibonacci subcategory, so $a^2 = 1 + a$. In the simple basis of $\mathcal{C}$, let $L_a, R_a$ be the matrices of left and right multiplication by $a$, and put*

$$B = I + L_a R_a, \qquad L = (e_0, e_a, e_a, e_0 + e_a), \qquad Q = B - LL^{\mathsf{T}}.$$

*Starting from $Q$, whenever a diagonal entry is one, its column is a forced nonnegative integral Gram column. Subtract its outer product and repeat. If this process reaches zero, it determines the entire restriction matrix $F = (L, H)$ of $Z_{\mathcal{D}}(\mathcal{C})$.*

*Assume additionally that the high columns are pairwise distinct, distinct from the low columns, and individually invariant under duality in $\mathcal{C}$. Then the relative center is self-dual. Impose the known* $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ *multiplication on its first four simples and the ring-homomorphism equations for $F$. If these linear equations in the symmetric nonunit structure constants have a unique solution, that solution is its complete Grothendieck tensor.*

*Proof.* For the forgetful functor and its adjoint, relative central induction has underlying object $FI(V) = V \oplus aVa$; equivalently it is the finite semisimple coend $\bigoplus_{U \in \mathrm{Irr}\,\mathcal{D}} U \otimes V \otimes U^*$. Adjunction gives the Gram identity $FF^{\mathsf{T}} = B$. The full inclusion $\mathcal{D} \subset \mathcal{C}$ induces a fully faithful tensor inclusion $Z(\mathcal{D}) \subset Z_{\mathcal{D}}(\mathcal{C})$: the half-braiding condition and all morphism spaces between objects of $\mathcal{D}$ are unchanged. The standard nondegenerate Fibonacci braiding identifies $Z(\mathcal{D})$ with $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, whose four restriction columns are exactly $L$.

If $Q_{ii} = 1$, exactly one remaining simple has restriction coefficient one in row $i$, and all the others have coefficient zero. Its full restriction column is therefore $Q_{\bullet i}$. Subtracting its outer product is mandatory. At zero residual no further nonzero restriction column is possible. This proves the asserted uniqueness without a numerical factorization or a conjectural central rank.

The low four simples are self-dual. The additional hypotheses force every high simple to be self-dual, because duality permutes restriction columns and no other simple has the same column. A based ring all of whose simples are self-dual is commutative. Frobenius reciprocity makes its nonunit coefficients symmetric in all three indices. Its forgetful functor is a ring homomorphism and the low multiplication is already known. Hence its actual tensor solves the stated linear system. Uniqueness determines that tensor; categorical existence was supplied before solving the system. If a pivot or uniqueness test fails, the recipe is inconclusive, not an exclusion or a claimed construction from arbitrary ring data. □

**Corollary 6.20.** *For the unitary category $TT_3$ and either of its Fibonacci generators, $Z_{\mathrm{Fib}}(TT_3)$ has Grothendieck ring $R^{(6)}_{6,1712}$.*

*Proof.* Use the source basis $(1, f^{(2)}, \rho, \sigma, \bar{\sigma}, \mu)$ of [14, Definition 2.5] and $a = \rho$. Its dimension vector is $(1, \varphi^3, \varphi, \varphi^2, \varphi^2, \varphi)$, where $\varphi = (1 + \sqrt{5})/2$. The source provides the unitary category and its complete multiplication table. The Gram procedure of Proposition 6.19 gives

$$F = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Indeed, after removal of the low columns the row of $\sigma$ is a diagonal-one pivot; after its forced subtraction the row of $f^{(2)}$ is another such pivot. Their removal leaves zero. Denote the columns by $(1, p, q, pq, X, Y)$. Then $p^2 = 1 + p$, $q^2 = 1 + q$, and the complete remaining rules are

$$\begin{aligned} pX = qX &= Y, & pY = qY &= X + Y, \\ (pq)X &= X + Y, & (pq)Y &= X + 2Y, \\ X^2 &= 1 + pq + 2X + 2Y, \\ XY &= p + q + pq + 2X + 4Y, \\ Y^2 &= 1 + p + q + 2pq + 4X + 6Y. \end{aligned}$$

For detail on uniqueness, there are $\binom{7}{3} = 35$ symmetric nonunit unknowns. The low products and all 36 restriction-product identities give 312 scalar linear equations of exact rank 35. The verifier reconstructs this integer system and its unique nonnegative integral solution, and independently checks every based-ring axiom. The displayed rules also permit direct substitution into every identity.

The full tensor matches $R^{(6)}_{6,1712}$ under $(0, 1, 2, 4, 5, 3)$. Repeating the calculation with $a = \mu$ gives a based-isomorphic result. Applying the same recipe to $\mathcal{C} = \mathcal{D} = \mathrm{Fib}$ recovers $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ as an independent positive control. Thus neither the rank-six answer nor its census identification is an input to the reconstruction. □

### 6.14 Three neutral components of conformal extensions

The same construction realizes three further census rings. We write $\mathcal{B}(\mathfrak{g}, k)$ for the principal unitary WZW category and use finite Dynkin labels for its simple objects.

**Theorem 6.21.** *The rings $R^{(6)}_{4,22}$, $R^{(6)}_{4,91}$ and $R^{(6)}_{4,629}$ admit unitary categorifications. They are the neutral components of the module categories for the conformal embeddings*

$$(B_2)_7 \subset (D_7)_1, \qquad (C_3)_4 \subset (B_{10})_1, \qquad (B_2)_{12} \subset (E_8)_1,$$

*respectively.*

**Common construction and identification principle.** A unitary conformal extension supplies an actual connected commutative separable algebra $A$ in its source category $\mathcal{B}$. Consequently $\mathcal{E} = \mathcal{B}_A$ is a unitary fusion category, $F : \mathcal{B} \to \mathcal{E}$, $U \mapsto U \otimes A$, is a central tensor functor, and the local modules are the level-one target category $\mathcal{L}$. These facts use the conformal algebra, not just a modular invariant [54, Sections 6.1–6.2]. In each case $A$ belongs to the even part of the source parity grading. The two homogeneous parts of any module are submodules. Thus simples have definite parity, relative tensor product respects parity, and odd free modules make the grading faithful. Its neutral component $\mathcal{C} = \mathcal{E}_0$ is therefore an actual unitary fusion category.

We identify its full based ring from free induction and local modules. Two standard tests resolve the few possible tensor completions. First, [54, Corollary 3.30] and [16, Proposition 8.23.11] give

$$Z(\mathcal{E}) \simeq \mathcal{B} \boxtimes \mathcal{L}^{\mathrm{rev}}, \qquad Z(\mathcal{C}) \simeq \big(\mathrm{Rep}(C_2)' \subset Z(\mathcal{E})\big)_{\mathrm{Rep}(C_2)}, \tag{6.20}$$

where $\mathrm{Rep}(C_2)$ is the canonical grading subcategory. If a commutative candidate ring has formal codegree $f$, the actual known center must contain a simple of dimension $\dim(\mathcal{C})/f$ [38, Theorem 2.13]. Second, indicators can be computed directly from the actual center and its forgetful functor [32]. These tests concern the constructed category and do not presume that archived modular data have been categorified.

**The $B_2$ level-seven component.** The source [8, Section 3.5, Figure 11(a)] gives

$$A = (0,0) \oplus (1,6) \oplus (2,2) \oplus (5,0), \qquad \mathcal{L} = \mathcal{B}(D_7, 1).$$

Parity is the second Dynkin coordinate modulo two. The six even vertices, in source order $(1, 9, 10, 2, 5, 6)$, will be denoted $(1, x, y, g, e, f)$. Here $e = F(1,0)$ and $g = F(7,0)$ are simple, hence central in the fusion ring, and $e$ is self-dual. The published vector-action graph gives

$$E = N_e = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 & 2 & 2 \end{pmatrix}.$$

The local category has fusion group $C_4$; an odd local spinor $z$ satisfies $z^2 = g$ [55, Theorem 7.1]. The odd simple $s = F(0,1)$ is self-dual and central, and $zs$ is one of $x, y$. Consequently $(zs)^* = gzs$: duality on this pair is the action of $g$. Dimensions force $g$ to fix $e, f$ and either fix or interchange $x, y$. The first row of $E^2$ gives $e^2 = 1 + g + 2f$; the matrix $(E^2 - I - N_g)/2$ is integral only for the interchange. Thus $x^* = y$.

Put $G = N_g$, $H = (E^2 - I - G)/2$ and $X = N_x$. The equation $ef = x + y + 2e + 2f$ and rigidity imply

$$X + GX = EH - 2E - 2H, \qquad X^{\mathsf{T}} = GX.$$

Also $X$ commutes with $E, G, H$, and its unit row is the basis vector of $x$. These are linear equations over $\mathbb{Q}$ in the 36 entries of $X$. Their coefficient matrix has rank 36. The unique solution is nonnegative integral and, with $N_y = GX$ and $N_f = H$, gives exactly all 216 coefficients of $R^{(6)}_{4,22}$. The accompanying checker extracts the graph directly from the source PDF and reconstructs this linear system, its rank, and the full tensor comparison.

**The $B_2$ level-twelve component.** Here [8, Section 3.6, Figure 13] gives

$$\begin{aligned} A = {} & (0,0) \oplus (0,6) \oplus (1,8) \oplus (3,2) \oplus (3,8) \\ & \oplus 2(4,4) \oplus (6,6) \oplus (7,2) \oplus (12,0). \end{aligned}$$

The target category $\mathcal{L}$ is Vec, so $A$ is Lagrangian and $Z(\mathcal{E}) \simeq \mathcal{B}(B_2, 12)$. The even graph component has source vertices $(1, 3, 7, 8, 4, 6)$. The central simple $x = F(1,0)$ has multiplication matrix of characteristic polynomial $t^2(t^4 - 5t^3 + 10t - 5)$. Every eigenvalue has multiplicity at most two. A noncommutative matrix summand $M_n(\mathbb{C})$ of the complexified fusion ring would contribute at least $n^2 \geq 4$ to the multiplicity of an eigenvalue of this central element. The ring is therefore commutative.

The only duality permutations fixing $1, x$ and commuting with $N_x$ are the identity and the interchange of vertices $7, 8$. In the latter case, Frobenius reciprocity and associativity with $x$ force coefficients $1/2$ and $3/2$, which are impossible. For identity duality the same linear constraints leave one parameter $t$. In the indicated basis $(1, x, u, v, w, z)$,

$$N_u = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 4-t & t-1 & 1 & 2 \\ 0 & 1 & t-1 & 3-t & 2 & 2 \\ 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 1 & 2 & 2 & 2 & 3 \end{pmatrix}.$$

All other products are determined and associative; integrality and nonnegativity give $t \in \{1,2,3\}$. The middle value is exactly $R^{(6)}_{4,629}$. The outer values have formal codegrees $3, 15$.

The unique nontrivial invertible $J = (12, 0)$ in $\mathcal{B}(B_2, 12)$ is a boson. Thus (6.20) identifies the actual center as $(\mathcal{B}(B_2, 12)_{\mathrm{ad}})_{\langle J\rangle}$. Among the 49 even weights, the involution $(a, b) \mapsto (12 - a - b, b)$ has 21 ordinary orbits and seven fixed points. Each ordinary orbit gives one simple of the original Weyl dimension; each fixed point gives two simples of half that dimension. This determines all 35 simple dimensions of the center. With $D = \dim(A)/2$, exact Weyl arithmetic in $\mathbb{Q}(\zeta_{60})$ shows that neither $D/3$ nor $D/15$ occurs. The formal-codegree test excludes both outer completions, proving the required tensor identification.

**The $C_3$ level-four component.** The adjoint conformal branching theorem [56, Theorem 3.9 and Proposition 4.1] gives the three local Ising modules $1, \psi, \sigma$ with underlying source objects

$$\begin{aligned} U(1) = A &= 000 \oplus 210 \oplus 004 \oplus 121, \\ U(\psi) &= 200 \oplus 030 \oplus 022 \oplus 301, \\ U(\sigma) &= 2(111). \end{aligned}$$

Their dimensions are $1, 1, \sqrt{2}$ and their twists are $1, -1, \zeta_{16}^5$. The source parity is $a + c$ modulo two; there are 19 even weights. For free modules the exact induction Gram matrix is

$$G_{\lambda\mu} = \langle F(\lambda), F(\mu)\rangle = \sum_{\alpha} [A : \alpha] N^{\mu}_{\lambda\alpha}.$$

The three local columns are known from branching, while $\tau = F(010)$ has norm one and is orthogonal to them. Subtracting these four columns leaves $2vv^{\mathsf{T}}$ with $v \geq 0$ integral and $v_{002} = 1$. The row of norm two must be a sum of two distinct unit vectors. Rank one forces every other residual row to be proportional to it. Thus there are exactly two further simples $x, y$, with identical restriction columns, and $F(002) = x + y$. Dominance of free induction rules out further simples.

The free image has rank five in the six-dimensional complexified fusion ring. It is central, so the ring is commutative: a noncommutative matrix block would make the difference between algebra and center dimensions at least three. The objects $1, \psi, \sigma, \tau$ are self-dual, leaving only the alternatives $x^* = y$ and self-duality of both $x, y$. Source fusion, the induction matrix, the Ising products, Frobenius reciprocity and associativity give a bounded exact completion problem. Its four or five free integer parameters lie between zero and four, as follows from coefficient nonnegativity. In the non-self-dual case there is one tensor, equal in all 216 entries to $R^{(6)}_{4,91}$. There are three self-dual possibilities, with respectively

$$x^2 = y^2 = 1 + 2y + 2\tau, \qquad 1 + x + y + 2\tau, \qquad 1 + 2x + 2\tau.$$

To exclude the middle possibility, note that a self-dual simple $X$ has a symmetric self-duality on $X \otimes X$. Indeed, if the pivotal transpose of $f : X \to X^*$ is $\epsilon f$, then the transpose of $f \otimes f$, under the reversed dual-tensor identification, is $\epsilon^2(f \otimes f)$. Every self-dual constituent of multiplicity one therefore has second indicator $+1$. The middle rule would give $\nu_2(x) = \nu_2(y) = 1$. On the other hand, the known center $Z(\mathcal{E}) \simeq \mathcal{B}(C_3, 4) \boxtimes \mathcal{L}^{\mathrm{rev}}$ and the displayed branching give

$$\nu_2(F(\mu)) = \frac{1}{\dim \mathcal{E}} \sum_{\lambda,a,\nu} \theta_\lambda^2 \theta_a^{-2} d_\lambda d_a [U(a) : \nu] N^{\mu}_{\nu\lambda}.$$

Exact evaluation in $\mathbb{Q}(\zeta_{32})$ gives $\nu_2(F(002)) = 0$, contradicting $\nu_2(x + y) = 2$.

For the outer two completions the formal codegrees include $8 \pm 4\sqrt{2}$. The boson $J = 004$ belongs to $A$, so $F(J) = 1$; the canonical grading subcategory in the known center is generated by $(J, 1)$. Formula (6.20) gives

$$Z(\mathcal{C}) \simeq (\mathcal{B}(C_3, 4)_{\mathrm{ad}})_{\langle J\rangle} \boxtimes \mathcal{L}^{\mathrm{rev}}.$$

The action $J(a,b,c) = (b, a, 4-a-b-c)$ has eight ordinary orbits and three fixed weights among the even weights. Fixed-point splitting gives rank 14 for the first factor and rank 42 for the center. Its dimension list contains neither $\dim(\mathcal{C})/(8+4\sqrt{2})$ nor $\dim(\mathcal{C})/(8-4\sqrt{2})$, as checked exactly by the Weyl formula. The formal-codegree test excludes both outer completions. Only the non-self-dual target remains, completing the proof of Theorem 6.21.

The supplement reconstructs the branching from the abelian ideals of $C_3$, certifies the source fusion coefficients by exact Weyl-character identities, and checks every finite completion and dimension comparison. All three arguments establish categorical existence and full tensor identifications; they do not classify associators on these rings.

# 7 Exact certificates and the remaining rings

## 7.1 Automated allocation and verification

The computation has three separate layers: the frozen source census, a discovery pipeline, and an exact categorification-certificate checker. Discovery is permitted to time out or to use numerical linear programming to propose a functional. Only the checker can validate a negative record. The final ledger contains one of four disjoint reporting classes: generally excluded, unitarily realized, spherically realized with no pseudo-unitary categorification, or unresolved. Five separate existence columns record the categorical structures, with no implication from spherical to pseudo-unitary existence. The pipeline rejects any intersection of its positive and negative sets.

Tables 4, 5 and 6 give a sequential view: a tensor removed at one stage is not counted as a new removal at any later stage. The order places inexpensive structural and arithmetic obstructions before larger finite branch searches. Effectiveness means the number of *additional* tensors decided at that point, rather than the number that a criterion could reject when run in isolation. The table is a transparent operational order, not a comparison of logical strength or a hardware-independent timing ranking. Ineligible hypotheses, an unfinished search and a feasible necessary relaxation all leave a ring undecided unless another accepted certificate or model resolves it.

The allocation in Table 7 is disjoint. It records the certificate family stored for each exclusion, while Tables 4, 5 and 6 make the sequential accounting explicit. Neither table asserts mathematical independence of the criteria. The signed and positive-dimension records call the same low-induction and finite-indicator framework internally. When an older unbalanced relative-trace certificate suffices, it is interpreted as a certificate for Theorem 4.3 with zero coefficients on the added balance equations. Likewise every inherited Farkas vector remains valid after the Galois-forbidden columns of Theorem 4.9 are removed: the inequalities on retained columns are unchanged. The stronger framework therefore retains the old exact witnesses rather than treating their special cases as unrelated obstructions.

Every certificate is verified from the original ordered tensor. The checker reconstructs number fields, verifies separating elements and factorizations, recomputes low-column and signed-table possibilities, and checks all integer inequalities. In particular it does not assume that a stored list of branches is exhaustive. For Farkas records it checks (4.12) with Python integers. During certificate verification the floating-point linear-programming entry point is replaced by a function that raises an error if called.

The source revision and tensor hashes fix the counting input. No census is regenerated for the categorification analysis. The positive check reconstructs the listed products, the zesting tensor and scalar coboundary, finite-group character rings, and group-theoretical convolution tensors, including the twisted $S_4$ cochains and their projective stabilizer factors. Exact Weyl characters reconstruct the $G_2$ model. Every positive entry, including every multiplicity-one entry, is matched by an explicit unit-fixing basis permutation; the complete 64-record

| Criterion | $m = 1$ | $m = 2$ | $m = 3$ | $m = 4$ | Total removed | Seconds |
|---|---|---|---|---|---|---|
| Input | 39 | 154 | 384 | 872 | — | — |
| Rank-two subring | 0/39 | 4/150 | 4/380 | 49/823 | 631 | 0.9 |
| $d$-number | 10/29 | 87/63 | 312/68 | 687/136 | 14753 | 11.6 |
| Zero spectrum | 2/27 | 0/63 | 1/67 | 0/136 | 4 | 0.6 |
| Lower-rank subring | 0/27 | 0/63 | 6/61 | 16/120 | 144 | 0.4 |
| Involution module | 4/23 | 3/60 | 19/42 | 13/107 | 233 | 52.0 |
| Dimension quotient | 0/23 | 1/59 | 2/40 | 2/105 | 20 | 6.5 |
| Cyclotomicity | 0/23 | 0/59 | 2/38 | 1/104 | 14 | 2.2 |
| Morita descent | 0/23 | 2/57 | 1/37 | 1/103 | 4 | 0.6 |
| Field automorphisms | 0/23 | 1/56 | 0/37 | 0/103 | 2 | 5.8 |
| Weighted induction | 0/23 | 1/55 | 1/36 | 1/102 | 7 | 5.5 |
| Sphericalization/induction | 2/21 | 33/22 | 32/4 | 85/17 | 693 | 857.6 |
| Dimension order | 0/21 | 0/22 | 0/4 | 0/17 | 1 | 184.1 |
| Character induction | 0/21 | 0/22 | 0/4 | 0/17 | 4 | 98.7 |
| Tensor rotation | 0/21 | 0/22 | 0/4 | 0/17 | 0 | 81.8 |
| Modular induction | 0/21 | 0/22 | 0/4 | 0/17 | 0 | 65.2 |
| Deferred exact checks | 0/21 | 0/22 | 0/4 | 0/17 | 1 | 157.2 |
| Added: rank-three subring | 0/21 | 0/22 | 0/4 | 1/16 | 1 | — |
| Added: spectral module | 0/21 | 0/22 | 0/4 | 0/16 | 2 | — |
| Added: order-six Morita | 0/21 | 0/22 | 0/4 | 0/16 | 2 | — |
| Added: character induction | 0/21 | 0/22 | 0/4 | 2/14 | 2 | — |
| Added: stabilizer rotation | 0/21 | 0/22 | 1/3 | 5/9 | 8 | — |
| Added: ramified dimension field | 0/21 | 0/22 | 0/3 | 0/9 | 4 | — |
| Added: biquadratic norm and trace | 0/21 | 0/22 | 0/3 | 0/9 | 4 | — |
| Added: formal-codegree trace ideal | 0/21 | 0/22 | 0/3 | 0/9 | 2 | — |
| Added: forced cyclic associator | 0/21 | 0/22 | 0/3 | 1/8 | 1 | — |
| Added: local signed spectra | 0/21 | 0/22 | 0/3 | 0/8 | 4 | — |
| Added: noncommutative signed spectra | 0/21 | 0/22 | 0/3 | 0/8 | 3 | — |
| Added: integral Morita constraints | 0/21 | 0/22 | 0/3 | 0/8 | 2 | — |
| Added: ambient character rotations | 0/21 | 0/22 | 0/3 | 2/6 | 2 | — |
| Added: subcategory center projection | 0/21 | 0/22 | 0/3 | 0/6 | 4 | — |
| Added: center rotation integrality | 0/21 | 0/22 | 0/3 | 0/6 | 1 | — |
| Constructed models | 21 | 22 | 3 | 5 | — | — |
| Unresolved | 0 | 0 | 0 | 1 | — | — |

Table 4: Historical executed ordered sieve and verified continuation, multiplicities one to four. Criterion cells are newly excluded / remaining; the total column covers the original multiplicities one through eight. Seconds are original pass wall times with 4 or 8 workers. Continuation dashes indicate that no comparable pass wall time is assigned. Timeouts remove nothing. The multiplicity-nine campaign is recorded separately.

| Criterion | $m = 5$ | $m = 6$ | $m = 7$ | $m = 8$ |
|---|---|---|---|---|
| Input | 1582 | 2768 | 3814 | 6999 |
| Rank-two subring | 81/1501 | 121/2647 | 113/3701 | 259/6740 |
| $d$-number | 1356/145 | 2442/205 | 3473/228 | 6386/354 |
| Zero spectrum | 0/145 | 1/204 | 0/228 | 0/354 |
| Lower-rank subring | 25/120 | 28/176 | 22/206 | 47/307 |
| Involution module | 49/71 | 23/153 | 78/128 | 44/263 |
| Dimension quotient | 6/65 | 2/151 | 4/124 | 3/260 |
| Cyclotomicity | 1/64 | 3/148 | 0/124 | 7/253 |
| Morita descent | 0/64 | 0/148 | 0/124 | 0/253 |
| Field automorphisms | 0/64 | 1/147 | 0/124 | 0/253 |
| Weighted induction | 1/63 | 1/146 | 1/123 | 1/252 |
| Sphericalization/induction | 58/5 | 140/6 | 115/8 | 228/24 |
| Dimension order | 0/5 | 0/6 | 0/8 | 1/23 |
| Character induction | 0/5 | 0/6 | 0/8 | 4/19 |
| Tensor rotation | 0/5 | 0/6 | 0/8 | 0/19 |
| Modular induction | 0/5 | 0/6 | 0/8 | 0/19 |
| Deferred exact checks | 0/5 | 0/6 | 1/7 | 0/19 |
| Added: rank-three subring | 0/5 | 0/6 | 0/7 | 0/19 |
| Added: spectral module | 0/5 | 0/6 | 0/7 | 2/17 |
| Added: order-six Morita | 2/3 | 0/6 | 0/7 | 0/17 |
| Added: character induction | 0/3 | 0/6 | 0/7 | 0/17 |
| Added: stabilizer rotation | 0/3 | 0/6 | 0/7 | 2/15 |
| Added: ramified dimension field | 0/3 | 0/6 | 2/5 | 2/13 |
| Added: biquadratic norm and trace | 0/3 | 0/6 | 1/4 | 3/10 |
| Added: formal-codegree trace ideal | 0/3 | 0/6 | 0/4 | 2/8 |
| Added: forced cyclic associator | 0/3 | 0/6 | 0/4 | 0/8 |
| Added: local signed spectra | 0/3 | 0/6 | 0/4 | 4/4 |
| Added: noncommutative signed spectra | 0/3 | 1/5 | 1/3 | 1/3 |
| Added: integral Morita constraints | 1/2 | 0/5 | 1/2 | 0/3 |
| Added: ambient character rotations | 0/2 | 0/5 | 0/2 | 0/3 |
| Added: subcategory center projection | 0/2 | 2/3 | 0/2 | 2/1 |
| Added: center rotation integrality | 1/1 | 0/3 | 0/2 | 0/1 |
| Constructed models | 1 | 3 | 2 | 0 |
| Unresolved | 0 | 0 | 0 | 1 |

Table 5: The same ordered sieve, multiplicities five to eight. Sphericalization/induction checks retained proof trees for $m \leq 7$ and discovers proofs at $m = 8$. Trees can use nested advanced filters; unfinished retained checks appear in the deferred row. Original advanced passes target $m \in \{3, 8\}$. Continuation rows apply additional complete certificates to the original survivors. This practical order is not claimed optimal.

| Criterion | Excluded / remaining | Worker seconds | Errors | Budgets |
|---|---|---|---|---|
| Input | 0 / 8809 | — | — | — |
| Rank-two subring | 211 / 8598 | 3.7 | 0 | 0 |
| $d$-number | 8277 / 321 | 26.7 | 0 | 0 |
| Zero spectrum | 0 / 321 | 0.4 | 0 | 0 |
| Lower-rank subring | 49 / 272 | 3.8 | 0 | 0 |
| Involution module | 134 / 138 | 203.5 | 0 | 1 |
| Dimension quotient | 4 / 134 | 4.3 | 0 | 0 |
| Cyclotomicity | 2 / 132 | 1.1 | 0 | 0 |
| Morita descent | 0 / 132 | 42.0 | 0 | 1 |
| Field automorphisms | 0 / 132 | 2.0 | 0 | 0 |
| Weighted induction | 1 / 131 | 1.3 | 0 | 0 |
| Sphericalization/induction trees | 109 / 22 | 3172.1 | 0 | 21 |
| Early exact arithmetic and Morita continuations | 11 / 11 | 67.6 | 0 | 0 |
| Dimension order | 0 / 11 | 796.0 | 0 | 10 |
| Character induction | 0 / 11 | 329.2 | 0 | 7 |
| Tensor rotation | 0 / 11 | 330.5 | 0 | 7 |
| Modular induction | 0 / 11 | 140.5 | 0 | 2 |
| Dual module Gram continuation | 0 / 11 | 9.1 | 0 | 0 |
| Advanced: trace ideal | 0 / 11 | 2.5 | 0 | 0 |
| Advanced: ramified field | 0 / 11 | 2.7 | 0 | 0 |
| Advanced: ramified signed | 0 / 11 | 3.0 | 0 | 0 |
| Advanced: biquadratic | 0 / 11 | 2.3 | 0 | 0 |
| Advanced: inert spectrum | 0 / 11 | 2.4 | 0 | 0 |
| Advanced: noncommutative spectrum | 0 / 11 | 2.6 | 0 | 0 |
| Advanced: cyclotomic extended | 0 / 11 | 5.1 | 0 | 0 |
| Advanced: full morita | 0 / 11 | 123.2 | 0 | 0 |
| Advanced: c4 associator | 0 / 11 | 2.7 | 0 | 0 |
| Advanced: gauss family | 0 / 11 | 93.9 | 0 | 1 |
| Completion: general exclusions | 11 / 0 | 390.5 | 0 | 0 |
| Constructed models | 0 | — | — | — |
| Unresolved | 0 | — | — | — |

Table 6: Separately executed exact-multiplicity-nine sieve and advanced continuation. Worker seconds sum the recorded per-tensor elapsed times and are not pass wall times. Errors and exhausted discovery budgets remove no rings. Every accepted exclusion has an independently replayed exact certificate. Earlier campaign timings and scope are unchanged.

| First retained certificate | Rings |
|---|---|
| Formal-codegree $d$-number obstruction | 23030 |
| Rank-two fusion subring | 842 |
| Complete oriented signed sphericalization | 467 |
| Involution-module representation obstruction | 352 |
| Commutative signed sphericalization | 192 |
| Complete lower-rank subring descent | 187 |
| Further general-to-positive certificates | 77 |
| Direct general-to-positive reduction | 70 |
| Subcategory dimension divisibility | 24 |
| Spectral involution-module obstruction | 19 |
| Cyclotomicity | 16 |
| Formal-codegree trace-ideal obstruction | 10 |
| Multiplicity-weighted noncommutative induction | 8 |
| Rank-three near-group subring descent | 7 |
| Biquadratic norm and trace obstruction | 5 |
| Stabilizer and cyclotomic traces | 4 |
| Solvability descent and involution-return indicators | 4 |
| Zero-spectrum obstruction | 4 |
| Stabilizer and averaged third rotation | 3 |
| Local signed-spectrum saturation | 3 |
| Noncommutative signed-spectrum saturation | 3 |
| Ramification and Casimir trace | 3 |
| Integral Morita and Clifford rotation | 2 |
| Signed involution fields and indicators | 2 |
| Cyclic-quartic trace and complementary codegrees | 2 |
| Diagonal associativity and trace saturation | 2 |
| Order-six Morita dual and grading obstruction | 2 |
| Noncommuting dimension-field automorphisms | 2 |
| Signed reduction with the common induction test | 2 |
| Cyclic associator and stabilizer rotation | 1 |
| Full Morita-action Gram obstruction | 1 |
| Full-character cyclotomic rotation | 1 |
| Center rotation integrality | 1 |
| Cyclic order-three projection | 1 |
| Fibonacci subfield and rotation | 1 |
| Half-$E_6$ projection | 1 |
| Ising projection and rotation | 1 |
| Low Verlinde and restriction products | 1 |
| Stabilizer and third-rotation obstruction | 1 |
| Signed dimension fields and different ideals | 1 |
| Free involution and indicators | 1 |
| Two-adic saturation and quadratic Gauss sums | 1 |
| Relative center and Galois spin orbits | 1 |
| Indicator normalization and signed spectra | 1 |
| Ramification and signed reduction | 1 |
| Different ideal and signed saturation | 1 |
| Trace ideal and signed saturation | 1 |
| Total | 25362 |

Table 7: Disjoint allocation of general exclusions. Nested criteria are not counted again.

constructor catalogue is checked for omissions. Regression tests compare the native integer routines with straightforward exact implementations, and deliberately corrupted certificates must be rejected.

This is an exact computer-assisted verification, not a formalization in a proof assistant. The categorical input theorems, their hypotheses, and the mathematical completeness argument remain part of the written proof. A checksum proves file integrity, not a mathematical theorem.

*Proof of Theorem 1.2.* Each negative record is an application of Theorems 5.1 and 3.3, or of Proposition 3.1, Corollary 3.2, or of an exhaustive signed/positive reduction followed by Theorem 4.1 or Theorem 4.3 and the cited rank-four or rank-five tests, with the tensor-sign reduction of Lemma 3.4 and the extension of Theorem 4.5 where recorded. The stronger character-aware system of Theorem 4.9, dimension-order test of Proposition 3.25, and signed refinement of Theorem 3.26 are used where recorded. The spectral refinement of Proposition 5.3, rank-three subring descent, Theorems 5.17 and 5.13, and Proposition 3.8 supply further certificates. The certificates also use the ramification criterion of Proposition 3.12, Theorem 3.9, the signed saturation corollary, and Proposition 3.11. The higher stabilizer and ambient cyclotomic refinements use Corollary 5.23 and Theorem 5.27, with a complete general-to-pseudo-unitary reduction whenever their inner certificate has only pseudo-unitary scope. Theorem 5.25 supplies the final general multiplicity-three exclusion: its sphericalization reduction, full Morita-action calculation, local rotation lemma and complete exact indicator separation are all required. The exact checker validates all hypotheses and exhausts the required finite alternatives. Theorem 4.39 adds the forced-product and integral rotation obstruction at multiplicity five. The signed-field, saturation, indicator and spin-orbit arguments of Section 7.6 exclude all eleven final multiplicity-nine cases. Thus no excluded tensor has a complex categorification. Section 6 constructs and tensor-matches all claimed positive examples, with the additional models in Sections 6.4 and 6.6, the higher models of Section 6.8, the conformal models of Theorem 6.21, and the complete catalogue in Section 6.9. In particular the spherical existence and pseudo-unitary nonexistence for $R^{(6)}_{2,124}$ use the full-tensor identification and the two cited results of Schopieray. No complement argument is used to assign the positive records. The census and disjoint ledger establish the counts in Table 1. The other records are left undecided, with no assertion of nonexistence or realizability. □

## 7.2 Reproduction

The full verifier checks the source tensor import, positive models and allocated negative certificates. The completed rank-six campaign covers all 25 362 exclusions by combining 25 355 passes from the initial full invocation with complete, unchanged Python-backend replays of its seven failed certificates. The two post-loop count and registry checks pass separately. This is not a single successful full invocation. Backend-dependent ordering of exact balance equations accounts for the selected replay requirement; the verification guide specifies the source-bound evidence and the supplied orchestration command, whose end-to-end execution remains untested.

The conformal models have the exact replay

```
python -O software/conformal_models.py --verify
```

Normal and optimized continuation runs keep all correctness checks active. Controls include positive categories and corrupted certificates that must be rejected. The verification guide distinguishes the scope of every checker. These procedures check categorification evidence; the cited census supplies the counting input.

## 7.3 The remaining frontier

Through multiplicity eight, the remaining tensors are precisely $R^{(6)}_{4,155}$ and $R^{(6)}_{8,1660}$. Every categorification of each is spherical and has a pseudo-unitary Galois conjugate, by the arguments below and the retained signed-sphericalization certificates. These reductions permit positive dimensions in the subsequent necessary conditions without assuming pivotality in the original existence question. The generalized near-group and Ising subcategories give smaller structural problems.

The current exact records are under `research/frontier/`, with one directory per tensor. Necessary induction profiles and relative-center rings are not assigned categorical existence. Only the three conformal constructions of Section 6.14 enter the positive catalogue.

## 7.4 The generalized near-group ring $R'(4)$

Write the basis of $R^{(6)}_{4,155}$ as $1, h, g, hg, X, Y$, where the invertibles form $C_2^2$, $h$ fixes $X, Y$, and $g$ exchanges them. The ring belongs to the family

$$Y = X^*, \qquad XY = 1 + h + m(X + Y), \qquad X^2 = g + hg + m(X + Y), \tag{7.1}$$

with $m = 4$. The following uniform argument eliminates signed-table search for this family.

**Proposition 7.1.** *Every complex fusion category with rules* (7.1)*, for $m \geq 0$, is spherical and has a pseudo-unitary Galois conjugate.*

*Proof.* In the sphericalization choose the dimension-one lifts $k_+$ of the invertibles, and choose $X_+$ and $Y_+ = g_+X_+$. Their common nonzero dimension $\delta$ forces $h_+X_+ = X_+$, $h_+Y_+ = Y_+$ and $X_+^* = Y_+$. The invertible part of the self-dual product $X_+Y_+$ is $1_+ + h_+$, by reciprocity; hence

$$X_+Y_+ = 1_+ + h_+ + s(X_+ + Y_+) + t(X_- + Y_-), \qquad s + t = m.$$

Thus $\delta^2 = 2 + r\delta$ with $r = 2(s - t)$; changing both lifts if necessary gives $0 \leq r \leq 2m$. The squared-norm global dimension and the Frobenius–Perron dimension are

$$d_r = 8 + r^2 \pm r\sqrt{r^2 + 8}, \qquad D_m = 8 + 4m^2 + 2m\sqrt{4m^2 + 8}.$$

The ratio $d_r/D_m$ is an algebraic integer [15, Proposition 8.22]. For nonzero even $r$, $r^2 + 8$ is not a square: $(v - r)(v + r) = 8$ would force $r = 1$. The same holds for $2m$. When $m > 0$, the compositum admits an automorphism negating both quadratic roots (or only the denominator root when $r = 0$). Multiplying the ratio by its image gives the positive rational algebraic integer

$$\frac{r^2 + 8}{4m^2 + 8} \leq 1.$$

It equals one, so $r = 2m$ and $t = 0$. For $m = 0$ this holds directly. The lift $g_+$ commutes with $X_+, Y_+$, since both orders have the same underlying simple and nonzero dimension. Therefore $X_+^2 = Y_+^2 = g_+X_+Y_+$, and all products of the chosen lifts close among them. They form a full spherical fusion subcategory projecting tensor-equivalently onto the original category. Its noninvertible dimension is $m \pm \sqrt{m^2 + 2}$; conjugating to the positive root gives pseudo-unitarity. □

For $m = 4$ the dimensions are

$$d = 4 + 3\sqrt{2}, \qquad D = 72 + 48\sqrt{2}.$$

There is no nontrivial faithful grading: the adjoint subring is the whole ring. The pointed subrings are its only proper nontrivial fusion subrings. Thus neither a grading nor rank-three subring descent resolves this case.

The retained central-induction certificate gives a finite necessary domain of six profiles, reduced below to the single center rank 52. For clarity, its categorical reductions are as follows. Pointed subcategory projection for all three involutions leaves one low frame and indicators $(\nu_2(X), \nu_2(Y), \nu_2(h), \nu_2(g), \nu_2(hg)) = (0, 0, 1, -1, -1)$. The $1 + h$ module action and stabilizer-rotation criterion constrain the third indicators. Exact Gram, Galois, word and balancing equations then force central lifts of $h, g, hg$. The lift of $h$ is fermionic: if it were bosonic, de-equivariantization would split the fixed object $X$ into dimensions $d/2$, whose norm is $-1/2$, contrary to integrality. A lift of $g$ generates a semion subcategory of the center, which splits off from the center.

In its modular centralizer the restricted adjoint algebra is $1 + A_1 + A_2$. For its restriction matrix $V$, adjunction gives $V^{\mathsf{T}}V = I + N_{A_1} + N_{A_2}$. The two low simples have opposite monodromy charges against the fermion. Consequently their multiplication matrices are the same-charge and opposite-charge blocks of $V^{\mathsf{T}}V - I$. Exhausting charge counts, duality, pointed actions, restriction products and ribbon balancing leaves the six profiles, each with a forced pair of low fusion matrices.

**Proposition 7.2** (Fermion traces and odd Gauss sums)**.** *The centralizer profiles* $275, 277, 279, 298, 306$ *cannot occur.*

*Proof.* Work in the positive spherical representative and normalize the central semion twist to $i$. Write

$$d = 4 + 3\sqrt{2}, \quad D = \dim \mathcal{C} = 72 + 48\sqrt{2}, \quad b = 2 + 4d = D/4 = 3\sqrt{2}d, \quad u = \zeta_8^{-1}.$$

Split the center as $Z(\mathcal{C}) \simeq \text{Semion} \boxtimes \mathcal{B}$; let $f \in \mathcal{B}$ be its fermion. The packet and charge enumeration, using Proposition 4.23 and Corollary 4.24, gives the following data. Exponents in the last column denote multiplicities; signs denote monodromy with $f$.

| profile | $r_+$ | $r_-$ | $r_+ - r_-$ | odd dimensions |
|---|---|---|---|---|
| 275 | 16 | 14 | 2 | $d^4, (2d)^8, b^2$ |
| 277 | 16 | 11 | 5 | $d^4, (2d)^4, 4d, b^2$ |
| 279 | 16 | 8 | 8 | $d^4, (4d)^2, b^2$ |
| 296 | 16 | 10 | 6 | $d^2, (2d)^4, (3d)^2, b^2$ |
| 298 | 16 | 7 | 9 | $d^2, (3d)^2, 4d, b^2$ |
| 306 | 16 | 6 | 10 | $(3d)^4, b^2$ |

All packet orientations and charge counts are reconstructed by `verify_center_cuts.py`. In every row the two $b$ objects have restrictions $1 + h + 2X + 2Y$ and $g + gh + 2X + 2Y$ and twists $1, -i$. Each is $f$-fixed, being unique in its restriction type.

Verlinde diagonalization gives $\#\,\mathrm{Fix}(f) = \mathrm{Tr}(N_f) = r_+ - r_-$. Balancing implies that a fixed simple is odd, immediately excluding 298 and 306. Even $f$-pairs have opposite twists and cancel in the Gauss sum. Odd pairs have equal twists, as do dual simples. Since $\tau^+(\mathcal{B}) = D/(1 + i)$ and the $b$ pair contributes $b^2(1 - i)$, the other odd contributions, divided by $d^2$, sum to $-24u$.

For 275, the $b$ pair accounts for all fixed points. The four $d$ objects restrict to $X, X, Y, Y$; the free $f$-pairs and duality force one common twist $\alpha$. The eight $2d$ objects form four pairs of twists $\beta_j$. Thus $\alpha + 2\sum_j \beta_j = -6u$, making the root of unity $\alpha$ divisible by 2 in the ring of algebraic integers, a contradiction. For 279, duality gives the four $d$ twists as $\alpha, \alpha, \beta, \beta$. If $\gamma, \delta$ are the $4d$ twists,

$$\alpha + \beta + 8(\gamma + \delta) = -12u.$$

The algebraic integer $(\alpha+\beta)/4$ has all conjugates of absolute value at most $1/2$, so it vanishes. Then $\gamma+\delta=-3u/2$ is not integral, again a contradiction.

For 277, the $b$ pair and the unique $4d$ simple supply three of five fixed points. Fixed points among the restrictions $X, X, Y, Y$ occur in multiples of four, since duality commutes with $f$. Hence these four are free with common twist $\alpha$; the $2d$ objects have two fixed twists $\beta, \gamma$ and one free-pair twist $\delta$. Write $\epsilon$ for the $4d$ twist. The Gauss equation is

$$\alpha+\beta+\gamma+2\delta+4\epsilon=-6u. \tag{7.2}$$

Use Galois symmetry in the full center, whose anomaly is trivial: $\theta_{\widehat{\sigma}Z}=\theta_Z^{a^2}$ [49, Theorem II]. Rational pointed eigenvalues preserve both monodromy charges, and the signed Galois permutation commutes with $N_f$, preserving fixedness. Pure dimensions $kd$ are preserved: when $\sigma(\sqrt{2})=-\sqrt{2}$, $D/\sigma(D)=1+4d$ and $(1+4d)\sigma(d)=-d$; the Galois dimension sign gives positive dimension $kd$. Consequently $\alpha, \delta, \epsilon$ are fixed by every square power and have orders dividing 24. The pair $\beta, \gamma$ has square-power orbit of size at most two. Cauchy's prime support is $\{2,3\}$, so their orders divide 48: the square subgroup modulo $2^a3^c$ has order $2^{\max(a-3,0)}3^{\max(c-1,0)}$. If a 16-part occurs, the nontrivial square sends the root to its negative (also at order 48), so $\gamma=-\beta$. Their first and third sums vanish, allowing replacement by an opposite pair of 24th roots for the following equations.

The spherical center formula [32, Theorem 4.1], grouping $Z\in\mathcal{B}$ with $gZ$, gives

$$\nu_3(X)=1+\frac{u}{6}S_3, \qquad S_3=\alpha^3+\beta^3+\gamma^3+2\delta^3+4\epsilon^3.$$

Indeed, their numerator is $d_Z\theta_Z^3([F(Z):X]-i[F(Z):Y])$; even $f$-pairs cancel, the $b$ terms contribute $4b=D$, and $2d/D=1/(6\sqrt{2})$. Thus $|\nu_3(X)-1|\le 3/2$. Proposition 5.21 gives the complete four-dimensional rotation domain $\{-2,-2\zeta_3,-2\zeta_3^2,1,\zeta_3,\zeta_3^2\}$. Every value except 1 is at distance at least $\sqrt{3}$ from 1; hence $S_3=0$.

Finally, the same verifier tests the $24^3$ choices of $\alpha, \delta, \epsilon$ in $\mathbb{Z}[t]/(t^8-t^4+1)$, looking up all unordered $\beta, \gamma$ sums. Equation (7.2) has 30 solutions and none has $S_3=0$. This finite integer calculation completes the exclusion. □

Only profile 296 remains, of center rank 52. This is a necessary condition, not a categorical construction. The semion factorization applies to the center.

**Proposition 7.3.** *If profile* 296 *is categorifiable, normalize the central semion to twist* $i$ *and put* $\mathcal{B}=\langle g\rangle'\subset Z(\mathcal{C})$. *Set* $d=4+3\sqrt{2}$, $b=2+4d$, $u=\zeta_8^{-1}$ *and* $w=\zeta_3$. *The twist multiset of* $\mathcal{B}$ *is forced as follows:*

| *h-charge* | *dimension* | *twists* |
|---|---|---|
| + | 1 | $1,-1$ |
| + | $d$ | $1,1,-1,-1$ |
| + | $2d$ | $w,-w,w^2,-w^2$ |
| + | $1+2d$ | $i,i,-i,-i$ |
| + | $1+4d$ | $1,-1$ |
| − | $d$ | $u,u$ |
| − | $2d$ | $uw,uw,uw^2,uw^2$ |
| − | $3d$ | $-u,-u$ |
| − | $b$ | $1,-i$ |

*Consequently* $Z(\mathcal{C})$ *has rank* 52 *and* $\mathrm{FSexp}(\mathcal{C})=24$.

*Proof.* The fermion trace is $16-10=6$. The six odd simples with unique charged restriction columns exhaust the fixed points. Gauss summation gives $\alpha+9\beta+4(\gamma+\delta)=-12u$ for

the remaining odd twists. The algebraic norm bound gives $\beta = -\alpha$; then $|2\sigma(\alpha/u) - 3| \le 2$ for every Galois conjugate forces $\alpha = u$ and $\{\gamma, \delta\} = \{uw, uw^2\}$. Low balancing leaves the even twists $a, -a, a, -a$ at dimension $d$ and $c, -c, e, -e$ at dimension $2d$. Write $t = a^2$, $p = c^2$, $q = e^2$, $F = 8 + 2t + 4p + 4q$. For the self-dual low objects $A, L$ of dimensions $1 + 4d, b$, the known fusion matrices give exactly

$$\nu_2(A) = \frac{|F|^2 + 36}{72}, \qquad \nu_2(L) = 1 - \frac{\operatorname{Im} F}{6}.$$

Since $|\operatorname{Im} F| \le 10$ and $\operatorname{Re} F \ge -2$, these indicators, each equal to $\pm 1$, force $F = 6$. Thus $t + 2(p + q) = -1$; the norm bound on the algebraic integer $(t + 1)/2$ gives either $t = 1$, $\{p, q\} = \{w, w^2\}$, or $t = -1$, $q = -p$. The fourth rotation has dimension 34 and half-turn trace $-2$, so $\nu_4(X) \in 2\mathbb{Z} + 2i\mathbb{Z}$. But

$$\nu_4(X) = 4 + \frac{1 + t^2 + 2p^2 + 2q^2}{3\sqrt{2}}$$

lies within $\sqrt{2}$ of 4, and hence equals 4. The second branch would require $2 + 4p^2 = 0$, an impossibility. This proves the table. The center spectrum is this spectrum together with its multiple by $i$. The accompanying exact verifier checks the moment identities and full low balancing; no root-order cutoff is assumed. This remains a necessary condition, not a realization or exclusion of profile 296. □

*Remark* 7.4 (What is determined). The profile and twist arguments above supply the inputs for Section 8, which determines the complete candidate $S$ matrix and fusion rules. Categorical existence remains unknown: even complete necessary modular data do not supply associators or the required algebra structure.

**Proposition 7.5** (Duality and indecomposability)**.** *In profile* 296*, the four simples of dimension* $1 + 2d$ *form two dual pairs. At this stage four residual duality assignments remain, giving respectively* $10, 12,$ *or* 14 *self-dual simples in* $\mathcal{B}$*. Section 8 subsequently forces exactly ten. The modular category* $\mathcal{B}$ *is prime, and its rank-*16 *fermion centralizer is a nonsplit supermodular category.*

*Proof.* The two dimension-$1 + 2d$ constituents of $I(1)$ correspond to the characters $\phi_\pm$ with $\phi_\pm(X) = \pm i\sqrt{2}$ and $\phi_\pm(Y) = \mp i\sqrt{2}$. For either constituent $M$ and every center simple $Z$, ribbon duality and the low-character formula give

$$S_{M^*,Z} = S_{M,Z^*} = d_M \phi_M(F(Z)^*).$$

Thus duality exchanges these two character rows and hence the two constituents, since $S$ is invertible. Removing their common semion factor gives the dual pair of twist $-i$ in $\mathcal{B}$; their fermion translates give the pair of twist $i$. All other dualities are forced except on the two odd $2d$ pairs, of twists $uw$ and $uw^2$. On each pair duality is either the identity or the transposition. The ten already self-dual simples therefore acquire zero, two, or four more.

A nontrivial modular factorization of rank 26 would have a rank-two factor. Its nonunit simple has dimension $x$ satisfying $x^2 = 1 + rx$, $r \in \mathbb{Z}_{\ge 0}$. If $x$ is irrational, its $\mathbb{Q}(\sqrt{2})$-norm is $-1$, whereas the available norms are

| $x$ | $1$ | $d$ | $2d$ | $3d$ | $1 + 2d$ | $1 + 4d$ | $b$ |
|---|---|---|---|---|---|---|---|
| $N(x)$ | $1$ | $-2$ | $-8$ | $-18$ | $9$ | $1$ | $36.$ |

If $x$ is rational, the same equation forces $x = 1, r = 0$. But $\mathcal{B}$ has only one nonidentity invertible, its fermion, so its rank-two pointed subcategory is degenerate. This proves primeness. Finally, a splitting $\langle f \rangle' \simeq \mathrm{sVec} \boxtimes \mathcal{A}$ would give a modular rank-eight subcategory $\mathcal{A} \subset \mathcal{B}$. Müger decomposition [57] would then force $8 \mid 26$, a contradiction. □

**Proposition 7.6** (Near-group dimension comparison)**.** *If the Frobenius–Perron dimension of the center of a near-group categorification of group order $n$ and multiplicity $m$ is a positive rational multiple of $D^2$, then*

$$(n, m) = (8, 16) \quad or \quad (32, 32).$$

*This is a necessary dimension comparison, not a construction or a Morita equivalence.*

*Proof.* Put $\rho^2 = n + m\rho$ and $\Delta = n + \rho^2$. Rational dimension is impossible here. For the quadratic field trace and norm,

$$\frac{\mathrm{Tr}(\Delta^2)}{\sqrt{N(\Delta^2)}} = \frac{m^2}{n} + 2, \qquad \frac{\mathrm{Tr}(D^2)}{\sqrt{N(D^2)}} = 34.$$

These quotients are unchanged by positive rational scaling, so $m^2 = 32n$, or $m = 8k, n = 2k^2$ for a positive integer $k$. Pass to a pseudo-unitary Galois representative using the two-dimension reduction in Proposition 3.1(iv). The formal-codegree argument of [41, Proposition 3.3] gives $n \mid m$: the relevant induced simple has dimension $1 + (m/n)\rho$, whose restriction forces $m/n$ to be an integer. This use of a Galois representative does not assume unitarity or pseudo-unitarity of the original category. Hence $k \mid 4$. The case $(n, m) = (2, 8)$ is excluded by [41, Theorem 1.1], leaving the two displayed possibilities. □

**Proposition 7.7** (No braided Morita representative)**.** *A categorification of $R^{(6)}_{4,155}$ is not Morita equivalent to any braided fusion category.*

*Proof.* Work with a pseudo-unitary Galois representative. Since the adjoint subcategory is the whole category, the forgetful map on central invertibles is injective. The central-induction certificate gives all four central lifts, with twists $1, -1, i, -i$, up to complex conjugation. Thus the center has no nonidentity invertible of twist one.

If a braided category $\mathcal{M}$ were Morita equivalent, it would also be pseudo-unitary. The equivalence of centers preserves their canonical positive ribbon structures. Every central invertible forgetting to the unit of $\mathcal{M}$ has twist one, so this forgetful map is injective. The image $a$ of the central semion has its canonical lift supplied by the braiding of $\mathcal{M}$; uniqueness forces that lift to be the semion itself. Hence $\langle a \rangle$ is nondegenerate, and braided decomposition gives $\mathcal{M} \simeq \mathrm{Semion} \boxtimes \mathcal{M}_0$ [57]. Its center then contains the nonidentity twist-one invertible $(a, a)$ from $Z(\mathrm{Semion}) \simeq \mathrm{Semion} \boxtimes \mathrm{Semion}^{\mathrm{rev}}$, a contradiction. Galois conjugation preserves braidings and Morita contexts, so the conclusion applies to the original category. □

## 7.5 The Ising subcategory at multiplicity eight

The ring $R^{(6)}_{8,1660}$ is self-dual. In the basis $1, g, X, gX, \sigma, \sigma X$ its products are determined by

$$g^2 = 1, \quad g\sigma = \sigma, \quad \sigma^2 = 1 + g, \qquad X^2 = 1 + (3(1 + g) + 4\sigma)X,$$

with commuting generators. Put $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$. Then $\mathrm{FPdim}\, X = d$ and $\mathrm{FPdim}\, \mathcal{C} = 4(1 + d^2)$. The retained unrestricted sphericalization certificate reduces every categorification to its pseudo-unitary Galois conjugate, without assuming a pivotal structure. The signed reduction and actual-root certificate in `research/frontier/R6_8_1660/` also force second indicators $(1, 1, 1, -1, 1)$ on the nonunit basis objects. The Ising subcategory gives the following small structural problem.

**Lemma 7.8.** *For a fusion subcategory $\mathcal{D} \subset \mathcal{C}$, the category $Z(\mathcal{D})$ has a central lift, with reverse braiding, to $Z(Z_{\mathcal{D}}(\mathcal{C}))$. In particular its Grothendieck ring is central in that of the relative center.*

*Proof.* Use strict notation and write a relative-central object as $(X, \gamma)$, where $\gamma_V : VX \to XV$ for $V \in \mathcal{D}$. For $b = (Y, \beta) \in Z(\mathcal{D})$, define its commutation with $(X, \gamma)$ by $\gamma_Y : YX \to XY$. To check that this is a relative-central morphism, we must verify

$$(\gamma_Y \otimes 1_V)(1_Y \otimes \gamma_V)(\beta_V \otimes 1_X) = (1_X \otimes \beta_V)(\gamma_V \otimes 1_Y)(1_V \otimes \gamma_Y).$$

The two half-braiding identities rewrite this as naturality of $\gamma$ at the morphism $\beta_V : VY \to YV$. Naturality in relative-central objects and tensor compatibility follow from their defining identities. On $Z(\mathcal{D})$ the resulting braiding is the reverse of its usual braiding. □

**Proposition 7.9** (Relative-center reconstruction)**.** *Any categorification $\mathcal{C}$ of $R^{(6)}_{8,1660}$ has a rank-eighteen relative center $\mathcal{E} = Z_{\mathcal{I}}(\mathcal{C})$, where $\mathcal{I} = \langle g, \sigma \rangle$ is Ising. Write $B = K_0(Z(\mathcal{I})) \cong K_0(\mathcal{I}) \otimes K_0(\mathcal{I})$ and $e = g \otimes g$. Its Grothendieck ring has basis $b, bt$ for the nine basis elements $b$ of $B$, with*

$$tb = bt, \qquad t^2 = L + At, \qquad L \in \{1, e\}, \qquad A \in B_{\geq 0}, \qquad A = LA.$$

*The forgetful map on $B$ is $F(a \otimes b) = ab$, and*

$$F(t) = X, \qquad F(A) = 3(1 + g) + 4\sigma, \qquad t^* = Lt.$$

*Proof.* The complement $\mathcal{M} = \langle X, gX, \sigma X \rangle$ is an $\mathcal{I}$-bimodule category. The functor $Y \mapsto YX$ is a left module equivalence from $\mathcal{I}$ onto $\mathcal{M}$: it takes the three simples bijectively to the three simples, or equivalently $\underline{\mathrm{End}}_{\mathcal{I}}(X) = 1$. Transport the right action to this regular left module. It becomes a tensor autoequivalence of $\mathcal{I}^{\mathrm{rev}}$, since it fixes all simple classes. Every tensor autoequivalence of an Ising category is isomorphic to the identity [57, Remark B.6(i)]. Thus $\mathcal{M}$ is the regular bimodule, with 1 corresponding to $X$.

Passing to bimodule centers gives a regular $Z(\mathcal{I})$-module $Z_{\mathcal{I}}(\mathcal{M})$ [20, Definition 2.1 and Remark 2.2]. Consequently its simples are $bt$, and $\mathcal{E}$ has eighteen simples. The lemma gives commutativity. The part of $t^2$ supported on $Z(\mathcal{I})$ forgets to 1, and hence is exactly 1 or $e$. Frobenius reciprocity gives $t^* = Lt$. All simples of $B$ are self-dual; taking duals of $t^2 = L + At$ therefore gives $A = LA$. The remaining statements follow by applying the forgetful functor. □

**Proposition 7.10** (An arithmetic relative-center domain)**.** *Using the previously proved reduction to a pseudo-unitary categorification, the above relative-center rings reduce to* 58 *labelled alternatives.*

*Proof.* The relative center inherits positive spherical dimensions from $\mathcal{C}$. Its dimension and dimension field are

$$D = 16(1 + d^2) = 576 + 384\sqrt{2} + 320\sqrt{3} + 224\sqrt{6}, \qquad K = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

Its dimension has minimal polynomial

$$z^4 - 2304z^3 + 184320z^2 - 4718592z + 37748736,$$

whose constant term is $2^{22}3^2$.

The Ising character rows are $(1, 1, \sqrt{2})$, $(1, 1, -\sqrt{2})$ and $(1, -1, 0)$, with formal codegrees $4, 4, 2$. Thus for a character $\chi$ of $B$, let $f_B$ be the product of these codegrees and write $a = \chi(A) = u + v\sqrt{2}$. If $\chi(L) = -1$, the identity $A = LA$ forces $a = 0$; its two extensions have formal codegree $2f_B$. If $\chi(L) = 1$, their formal codegrees are

$$f_{\pm} = \frac{f_B}{2}\left(4 + a^2 \pm a\sqrt{a^2 + 4}\right).$$

Every $D/f_\pm$ is an algebraic integer and every $f_\pm$ belongs to $K$ [38, Corollaries 2.14–2.15].

First, $f_+f_- = f_B^2(a^2+4)$ implies that

$$n(u,v) = (u^2+2v^2+4)^2 - 8u^2v^2$$

has no prime divisor other than 2 and 3. Second, the primitive minimal polynomial of $D/f_\pm$ must be monic. Finally, if $a \neq 0$, field membership implies $\sqrt{a^2+4} \in K$. A square root in $K = \mathbb{Q}(\sqrt{2}) \oplus \sqrt{3}\mathbb{Q}(\sqrt{2})$ of an element of $\mathbb{Q}(\sqrt{2})$ lies in one of those two summands. Taking its quadratic norm shows that $n(u,v)$ must be a square in $\mathbb{Q}$, hence an integer square.

The finite domain is particularly small. If $z$ is the coefficient of $\sigma \otimes \sigma$ in $A$, then $0 \leq z \leq 3$. The coefficients of $1 \otimes 1, g \otimes g$ sum to $3 - z$, those of $1 \otimes g, g \otimes 1$ also sum to $3 - z$, and the coefficients of the four $\sigma$-supported simples sum to four. This gives 1050 possible $A$ for $L = 1$ and six satisfying $A = eA$ for $L = e$. The exact checker produces

$$1056 \xrightarrow{\text{prime norms}} 158 \xrightarrow{D/f \text{ integral}} 130 \xrightarrow{\text{dimension field}} 58.$$

□

The following categorical conditions reduce this arithmetic domain further.

**Proposition 7.11** (Uniform indicator and central-lift constraints)**.** *Every pseudo-unitary categorification of $R_{8,1660}^{(6)}$ has*

$$\nu_2(g) = \nu_2(X) = \nu_2(gX) = \nu_2(\sigma X) = 1, \qquad \nu_2(\sigma) = -1,$$

*and its center contains a unique simple restricting to $g$ alone.*

*Proof.* We describe the finite certificate and the twist bound making it complete. For a central restriction column $v = (v_0, \ldots, v_5)$ put

$$a(v) = v_0 + v_1 + v_2 + v_3 + 2v_4 - 2v_5, \qquad b(v) = v_0 + v_1 - v_2 - v_3 + v_4 + v_5.$$

The saturated dimension-field characters of conductors 24 and 12 give $a(v) \neq 0 \Rightarrow \theta_v^{24} = 1$ and $b(v) \neq 0 \Rightarrow \theta_v^{12} = 1$, by Proposition 3.20. A high column containing $\sigma$ has primitive order-16 twist and $v_1 = 0$, by the Ising projection. Both displayed forms then vanish, forcing $v_2 + v_3 = 4v_4$ and $v_5 = 3v_4$. The individual root-mass capacity gives $v_4^2/2 \leq 1$, hence $v_4 = 1$. If both forms vanish and $v_4 = 0$, then $v_5 = 2v_1$ and $v_2 + v_3 = 3v_1$; a nonzero high column has $v_1 > 0$ and twist $-1$. Thus every actual twist has order dividing 48.

The first three indicators are already $+1$ by odd multiplicities in squares. Exhaust all four choices for the other two signs, all low frames and all character assignments. There are two low frames, each with two assignments, hence 16 global systems. The equations use all 48 actual roots, induction Gram identities, joint Galois balance, Ising root masses, low-product restriction constraints and cyclic rotation moments. Thirteen exact integer separating vectors exclude thirteen systems. Every one of the other three has the displayed indicator pattern. This conclusion precedes any specialization to individual high profiles.

For the central lift, bounded affine enumeration of the Gram, character, Galois and cubic-word equations gives 371 profiles in each assignment of the first frame and 2011 in each assignment of the second. Every profile contains exactly one copy of the high vector $(0,1,0,0,0,0)$. All domains are reconstructed by the retained exact profile checker; the uniform multiplicity is checked on all 4764 raw profiles. A simple with this restriction has dimension one and is the required central invertible. The sources, separating vectors and independent replay are indexed in `research/frontier/R6_8_1660/`. □

**Lemma 7.12** (Second indicator with a central factor)**.** *Let $\mathcal{A}$ be pivotal, let $b$ be self-dual in $Z(\mathcal{A})$ with simple underlying object, and let $t$ be self-dual and simple in $\mathcal{A}$. If $bt$ is simple, then $\nu_2(bt) = \nu_2(b)\nu_2(t)$.*

*Proof.* Use half-braidings $c_{b,x} : bx \to xb$. Choose a central self-duality $f_b : b \to b^*$ and a self-duality $f_t : t \to t^*$, and put $f = c_{b^*,t^*}(f_b \otimes f_t) : bt \to (bt)^*$. Dual compatibility gives $(c_{b^*,t^*})^* = c_{b^{**},t^{**}}$. Writing $j$ for the pivotal structure, naturality and centrality of $f_b$ give

$$\begin{aligned} f^* j_{bt} &= (f_t^* j_t \otimes f_b^* j_b) c_{b,t} \\ &= \nu_2(b)\nu_2(t)(f_t \otimes f_b) c_{b,t} = \nu_2(b)\nu_2(t) f. \end{aligned}$$

Since $bt$ is simple, this transpose scalar is its second indicator. No half-braiding for $t$ is needed. □

**Theorem 7.13** (A single relative-center ring)**.** *Up to exchanging the Ising factors, the relative center of a categorification of $R_{8,1660}^{(6)}$ must have*

$$L = e = (g, g), \qquad A = 2(\sigma, 1) + 2(\sigma, g) + 3(\sigma, \sigma).$$

*Proof.* The forgetful functor $F : \mathcal{E} \to \mathcal{C}$ is faithful and pivotal. It preserves second indicators for a self-dual simple whose image is simple: it preserves a nonzero self-duality and its transpose scalar, both computed in one-dimensional Hom spaces. If $L = 1$, apply Lemma 7.12 to $b = (\sigma, 1)$ and $t$. Their images, and that of $bt$, are $\sigma, X, \sigma X$. Proposition 7.11 gives their signs $-1, 1, 1$, contradicting the lemma. Thus $L = e$. Of the 58 arithmetic alternatives, only three have $L = e$: the stated coefficient, its factor exchange, and the all-ones coefficient $A = \sum_{b \in \mathrm{Irr}\, Z(\mathcal{I})} b$.

To exclude the last choice, realize $\mathcal{E}$ as the dual of $\mathcal{C} \boxtimes \mathcal{I}^{\mathrm{op}}$ acting on $\mathcal{C}$ by left and right tensor multiplication. Compatibility of a module endofunctor with the right action is exactly its relative half-braiding. Morita invariance therefore gives

$$|\operatorname{Inv} Z(\mathcal{E})| = 4 |\operatorname{Inv} Z(\mathcal{C})|.$$

All four invertibles of $\mathcal{E}$ belong to $B$ and have canonical central lifts. The forgetful map on central invertibles is surjective; its kernel is the group of tensor automorphisms of the identity. For the all-ones coefficient, $t^2$ contains $t$ and every $bt$. A tensor automorphism's scalars therefore satisfy $\lambda_t^2 = \lambda_t = 1$ and $\lambda_{bt} = \lambda_b = 1$. The kernel is trivial, so $|\operatorname{Inv} Z(\mathcal{E})| = 4$. This contradicts the nontrivial central invertible of Proposition 7.11. □

The last theorem is a necessary reduction, not a categorification of the remaining ring. Its universal grading is $C_2$: the first Ising spin and $t$ have odd degree. The neutral component has nine simples. Writing $p = (g, 1)$, $q = (1, g)$, $s = (1, \sigma)$ and $Y = (\sigma, 1)t$, its additional rules are

$$pY = Y, \qquad Y^* = qY, \qquad Y^2 = q(1 + p) + \big(4(1 + q) + 6s\big)Y,$$

together with $p^2 = q^2 = 1$, $qs = s$ and $s^2 = 1 + q$.

### 7.5.1 Two exact fermionic cuts on the remaining profiles

Write $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$ and $D = 4(1 + d^2)$. The actual-twist certificate leaves 1764 labelled restriction profiles, in branches of sizes $168, 798, 798$. The following reduction uses the already forced central fermion and second indicators. It makes no unitarity assumption and does not resolve the ring.

**Proposition 7.14** (Fermion profile reduction)**.** *The necessary restriction-profile domain for $R_{8,1660}^{(6)}$ reduces from* 1764 *to* 1727*. Exactly* 25 *profiles fail the fermion orbit, dimension, and rank conditions below, and* 12 *further profiles fail an integral second-indicator congruence. The two excluded sets are disjoint.*

*Proof.* Work in the retained pseudo-unitary Galois form. Let $\mathcal{M} = Z(\mathcal{C})$, and let $h$ be its unique nontrivial invertible, with $F(h) = g$ and $\theta_h = -1$. For a simple $Z$, write $\epsilon_Z \in \{1, -1\}$ for its monodromy with $h$. Balancing gives

$$\theta_{hZ} = -\epsilon_Z \theta_Z.$$

Thus the even sector consists of free $h$-pairs. Write its number of pairs as $p$, and write the odd sector as $s$ free pairs and $t$ fixed simples. The Verlinde diagonalization of $N_h$ gives

$$t = \mathrm{Tr}(N_h) = \sum_Z \epsilon_Z = 2p - 2s - t, \qquad p = s + t.$$

The two sectors each have dimension $D^2/2$.

The six low simples are the summands of $I(1)$ indexed by the six ring characters. Their monodromies are the characters' values on $g$. The first four are even, and their restrictions, in order $(1, g, X, gX, \sigma, \sigma X)$, are

$$(1,0,0,0,0,0), \quad (1,0,3,3,0,4), \quad (1,0,1,1,0,2), \quad (1,0,2,2,0,2).$$

They give four forced even pairs, whose high partners have the $g$-translated restrictions and twist $-1$. The two other low simples are odd. In low frame zero their distinct restrictions force an $h$-transposition. In low frame one the computation conservatively allows either a transposition or two fixed simples.

Remove the four forced high partners. Tensoring with $h$ exchanges the $X$ and $gX$ coordinates of each remaining restriction vector, so asymmetric vector types must occur in pairs with equal multiplicities. All remaining high vectors have zero $g$ coordinate. Let $E, O, T$ count the remaining even pairs, odd pairs, and odd fixed simples, and let $N$ be the number of remaining high simples. Then

$$(O, T) = \begin{cases} (N - 3E - 3,\ 4E + 6 - N), & \text{odd low transposition,} \\ (N - 3E - 2,\ 4E + 4 - N), & \text{two fixed odd low simples.} \end{cases}$$

The even dimension condition becomes

$$\sum_{\text{remaining even pairs}} \frac{d_Z^2}{d^2} = 180 + 120\sqrt{2}.$$

Here high vectors omit the unit coordinate. For a vector $(0, b, c, 0, a)$, the summand is $(b + c + a\sqrt{2})^2$. Every $\sigma$-containing type has $b + c = 4$, $\sigma$ coefficient 1, and $\sigma X$ coefficient 3; its summand is $36 + 24\sqrt{2}$.

There is also a pivotal parity condition. A self-duality of $Z$ with indicator $\eta$ induces a nondegenerate form with the same transpose sign on $F(Z)$. An odd multiplicity of a base simple of indicator $\delta$ forces $\eta = \delta$. Hence a type with odd $\sigma$ multiplicity and an odd positive-indicator multiplicity cannot be self-dual. Its total multiplicity is even. Its number of even $h$-pairs is also even: duality preserves twists, and therefore must pair distinct even $h$-orbits, whose two twists are opposite. This does not assert that $h$-fixed objects are self-dual.

A bounded integer calculation of these dimension and rank equations, with these orbit and parity conditions, excludes $3, 11, 11$ profiles in the three branches. A separate recursion over the three orbit counts independently verifies every rejected low-action branch.

For the second cut, put $\tau_n(\mathcal{M}) = \sum_Z d_Z^2 \theta_Z^n$. The modular indicator formula [32, Theorem 7.5] implies

$$\sum_{Z \in \mathrm{Irr}\,\mathcal{M}} \nu_2(Z) d_Z = \frac{\tau_2(\mathcal{M})\tau_{-2}(\mathcal{M})}{\dim \mathcal{M}}.$$

Indeed, multiply $\nu_2(Z) = (\dim \mathcal{M})^{-1} \sum_{A,B} N_{AB}^Z d_A d_B (\theta_A/\theta_B)^2$ by $d_Z$ and sum. The center identity [32, Theorem 4.1] gives $\tau_2(Z(\mathcal{C})) = D \sum_X d_X \nu_2(X)$. Dimensions and second indicators are real, and twists are roots of unity, so $\tau_{-2} = \overline{\tau_2} = \tau_2$. As $\dim \mathcal{M} = D^2$,

$$\sum_Z \nu_2(Z) d_Z = \big(2 - \sqrt{2} + (2 + \sqrt{2})d\big)^2.$$

The four even low restrictions are unique, so their simples are self-dual; the odd unit multiplicity forces indicator 1, as for their $h$-partners. The two odd low objects also have indicator 1. This is immediate in frame zero. In frame one their alternative would be a dual pair: after removing the four even pairs, the displayed identity would then require an additional term $(8\sqrt{3} + 4\sqrt{6})d$. Every remaining possible nonzero indicator contribution divided by $d$ belongs to $\mathbb{Q}(\sqrt{2})$; the $\sigma$-containing types have indicator zero. This rules out that alternative.

After removing all these forced contributions, the identity reads

$$\sum_{\text{remaining high } Z} \nu_2(Z) \frac{d_Z}{d} = 36 + 24\sqrt{2}.$$

For a restriction type of multiplicity $n_v$, its aggregate indicator $s_v$ satisfies $s_v \equiv n_v \pmod 2$, since nonself-dual objects occur in dual pairs. Also $\nu_2(hZ) = \nu_2(Z)$ by Lemma 4.36, applied to the braided lift of $h$ in $Z(\mathcal{M})$, whose twist is $-1$. Thus asymmetric $h$-paired vector types have equal aggregate indicators. For each type $(0, b, c, 0, a)$, let $q_v = b + c$ if it is $h$-invariant, and $q_v = 2(b + c)$ for an asymmetric pair. The rational coefficient is $\sum_v q_v s_v$. For each low-one labelling, profiles $1032, \dots, 1037$ force the integers $2q_v$ to be divisible by 8 and $\sum_v q_v n_v \equiv 0 \pmod 8$. The rational coefficient is therefore 0 modulo 8, whereas 36 is 4 modulo 8. These six profiles in each labelling are excluded.

The verifier `verify_fermion_profiles.py` in `research/frontier/R6_8_1660/fermion_profiles/` reconstructs the entire input domain from its hash-bound source files, checks all field identities by integer arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and records both disjoint excluded sets in `fermion_profiles_certificate.json`. It uses no optimizer and retains both low-one actions. The remaining count is $165 + 781 + 781 = 1727$. □

### 7.5.2 An involution and higher-indicator constraints

Put $T = \sigma X$, $S = sY$, and $\mathcal{A} = \mathcal{E}_{\mathrm{ad}}$. All categories below carry their canonical positive spherical structures.

**Proposition 7.15** (A tensor involution)**.** *There is a coherent tensor action of $C_2$ on $\mathcal{C}$ whose generator $\alpha$ exchanges $X, gX$ and fixes the other four simple classes. Consequently $\nu_n(X) = \nu_n(gX)$ for every $n \geq 1$; their entire $n$th pivotal rotation representations are isomorphic.*

*Proof.* The nine simples of $\mathcal{A}$ are $1, p, q, e, s, ps, Y, qY, S$, and the relative-center forgetful functor has fibers

$$\{1, e\} \mapsto 1, \quad \{p, q\} \mapsto g, \quad \{s, ps\} \mapsto \sigma, \quad \{Y, qY\} \mapsto T, \quad S \mapsto X \oplus gX.$$

It is therefore dominant, and $\mathrm{FPdim}\,\mathcal{A} = 2\,\mathrm{FPdim}\,\mathcal{C}$. The canonical central lift of $e = pq$ is a boson: its self-braiding is the product of the two Ising fermions' self-braidings. It generates a central $\mathrm{Rep}(C_2)$, with regular algebra $R = 1 \oplus e$. Moreover its half-braiding becomes trivial under $F$. Indeed, the explicit central lift above commutes with $(V, \gamma)$ by evaluating $\gamma$ at the underlying Ising object of $e$, which is isomorphic to 1; naturality and the unit identity give

ordinary unit commutation. Thus $F(R)$ is the split central algebra $1 \oplus 1$. Choosing either augmentation $F(R) \to 1$, base change defines a tensor functor

$$\overline{F} : \mathcal{A}_R \longrightarrow \mathcal{C}, \qquad M \longmapsto F(M) \otimes_{F(R)} 1.$$

It factors $F$ through the free-module functor and is dominant. De-equivariantization gives $\operatorname{FPdim} \mathcal{A}_R = \operatorname{FPdim} \mathcal{A}/2 = \operatorname{FPdim} \mathcal{C}$, so $\overline{F}$ is an equivalence. Equivariantization and de-equivariantization therefore identify $\mathcal{A}$ with $\mathcal{C}^{C_2}$ [57, Section 4.2, especially Theorem 4.18 and Proposition 4.26]. The underlying support of a simple equivariant object is a single group orbit; the displayed fibers force the stated action on simples. A tensor autoequivalence preserves the unique positive spherical structure and hence intertwines all pivotal rotations [33]. □

**Lemma 7.16** (Parity within a restriction fiber)**.** *In every central restriction type with equal* 1 *and* $g$ *multiplicities, the number of charge-even simples is even, and their twist multiset is invariant under* $\theta \mapsto -\theta$*. This applies to every remaining high type, including those asymmetric in* $X, gX$*.*

*Proof.* The braided autoequivalence $\widehat{\alpha} = Z(\alpha)$ preserves twists and fixes $h$, since $h$ is the unique central lift of $g$. The permutation $\beta(Z) = h \otimes \widehat{\alpha}(Z)$ of simple classes satisfies $\beta^2 = 1$, preserves monodromy charge $\epsilon_Z$, and has $\theta_{\beta(Z)} = -\epsilon_Z \theta_Z$. Its restriction vector is

$$(a_1, a_g, a_X, a_{gX}, a_\sigma, a_T) \longmapsto (a_g, a_1, a_X, a_{gX}, a_\sigma, a_T).$$

Thus it preserves each stated fiber and acts freely on its even part, pairing opposite twists. No assertion that tensoring with $h$ is a tensor autoequivalence is needed. □

**Proposition 7.17** (Odd rotation multiplicities)**.** *For every odd* $n$*, the cyclic rotation representation on* $\operatorname{Hom}_{\mathcal{C}}(1, T^n)$ *is twice that on* $\operatorname{Hom}_{\mathcal{E}}(1, Y^n)$*. In particular all its eigenvalue multiplicities are even, and, writing* $\omega = \zeta_3$*,*

$$\nu_3(T) \in \{2, 2\omega, 2\omega^2, -4, -4\omega, -4\omega^2\}.$$

*Proof.* For a pseudo-unitary category $\mathcal{D}$, the center formula gives

$$W_n(\mathcal{D}) := \sum_V d_V \nu_n(V) = \frac{\tau_n(Z(\mathcal{D}))}{\dim \mathcal{D}} \quad \text{[32, Theorem 4.1]}.$$

Hence $W_n$ is Morita invariant and multiplicative under Deligne products. Use the established Morita equivalence $\mathcal{E} \sim \mathcal{C} \boxtimes \mathcal{I}^{\mathrm{op}}$. For odd $n$, the Ising factor contributes $W_n = 1$; the odd component of $\mathcal{E}$ has zero indicators, and among the six $B$-simples in $\mathcal{A}$ only 1 contributes. Since $Y^* = qY$, dual invariance [32, Remark 4.2] yields

$$2\sqrt{2}\nu_n(Y) + 2\nu_n(S) = \nu_n(X) + \nu_n(gX) + \sqrt{2}\nu_n(T).$$

Indicators belong to $\mathbb{Q}(\zeta_n)$, which does not contain $\sqrt{2}$ for odd $n$. Comparing coefficients proves $\nu_n(T) = 2\nu_n(Y)$ and $\nu_n(X) + \nu_n(gX) = 2\nu_n(S)$.

To obtain all rotation powers, let $r$ be odd. Since $Y^* = eY$, duality gives $[Y^r : e] = [Y^r : 1]$. These are precisely the two simples in $\mathcal{A}$ whose forgetful image contains 1, so $\dim \operatorname{Hom}_{\mathcal{C}}(1, T^r) = 2 \dim \operatorname{Hom}_{\mathcal{E}}(1, Y^r)$. Expanding $Y^r$ in the nine simple classes and using the preceding indicator identities also gives $\nu_m^{\mathcal{C}}(T^r) = 2\nu_m^{\mathcal{E}}(Y^r)$ for every odd $m$. For the $k$th power of $n$th rotation put $r = \gcd(n, k)$ and $m = n/r$. Grouping $r$ adjacent tensor factors identifies rotation through $r$ places with $m$th rotation on the object's $r$th power; raising to $k/r$ applies the cyclotomic automorphism $\zeta_m \mapsto \zeta_m^{k/r}$ to its trace. The preceding equality therefore doubles every character value. The identity character value is covered by the dimension equality.

Finally the central fermion fixes $T$. Its self-braiding $-1$ gives the anticommuting left and right stabilizers of Proposition 5.21. Since $\dim \operatorname{Hom}(1, T^3) = 8$, that proposition gives $\nu_3(T) = -(a + b\omega + c\omega^2)$ with $a+b+c = 4$. The actual rotation multiplicities are $(4-a, 4-b, 4-c)$; their evenness makes $a, b, c$ even and leaves the six displayed values. □

### 7.5.3 Homogeneous-packet Vafa equations

We combine Proposition 7.15 and Lemma 7.16 with a Vafa congruence using only known induction matrices.

**Proposition 7.18.** *The* 1727 *restriction profiles reduce further to* 1670*. There are* 18 *exclusions from the involution and* 39 *additional exclusions from Anderson–Moore–Vafa equations and a sixteenth-indicator field test. Neither a braiding on* $\mathcal{C}$ *nor a braiding on its relative center is assumed.*

*Proof.* On simple labels of $Z(\mathcal{C})$ consider $\beta(Z) = h \otimes Z(\alpha)(Z)$. It preserves the monodromy sign $\epsilon_Z$ with $h$, and

$$\theta_{\beta(Z)} = -\epsilon_Z \theta_Z.$$

Its restriction vector exchanges the unit and $g$ coordinates and fixes the other four coordinates. Consequently, on every restriction fiber whose unit and $g$ multiplicities are both zero, its even sector is paired with opposite twists. The cardinality of this even sector is even. After the four forced low/high pairs are removed, all remaining high fibers have this form. In the previous orbit calculation this adds the condition that the number of even $h$-pairs joining two asymmetric restriction types is even. The other low-one alternatives remain allowed.

Here is the Vafa specialization used for the second cut. In any ribbon fusion category with twists $\theta_i = \zeta_N^{k_i}$, the Anderson–Moore–Vafa equation [16, Theorem 8.18.1] is

$$(\theta_i\theta_j\theta_m\theta_r)^{N_{ijm}^r} = \prod_p \theta_p^{N_{ij}^p N_{pm}^r + N_{im}^p N_{jp}^r + N_{jm}^p N_{ip}^r}.$$

Let $U = \bigoplus_i n_i i$ have one common twist $\zeta_N^{\kappa}$ on its constituents. Put $P = N_U$ and $a_j = [U^{\otimes j}]$. Multiplying the equations with weights $n_i n_j n_m$ gives the useful identity

$$(a_3)_r(3\kappa + k_r) \equiv 3\sum_p (a_2)_p P_{rp} k_p \pmod{N}. \tag{V}$$

This identity only requires the fusion matrix of the homogeneous-twist object $U$, rather than all simple fusion matrices.

Apply it in $Z(\mathcal{C})$, whose actual twist exponent divides 48. Let $R$ be its restriction matrix and $A = I(1)$. The projection formula $A \otimes Z \cong I(F(Z))$ gives $N_A = R^{\mathsf{T}}R$. Split $A = A_+ \oplus A_-$ into its four even and two odd low simples. Every constituent still has twist one. The fusion matrices of $A_+$ and $A_-$ are precisely the equal-charge and opposite-charge blocks of $R^{\mathsf{T}}R$, respectively. Thus they are known after an assignment of the fermion charges; no general center fusion reconstruction is involved.

In the base order $(1, g, X, gX, \sigma, \sigma X)$ write

$$u_+ = F(A_+) = (4, 0, 6, 6, 0, 8), \qquad u_- = F(A_-) = (2, 2, 6, 6, 0, 8),$$

and $H_+ = \sum_{Z \text{ even}} v_Z v_Z^{\mathsf{T}}$, where $v_Z = F(Z)$. For a remaining high simple $Z$, use $U = A_+$ if $Z$ is even and $U = A_-$ if $Z$ is odd. The coefficient of $Z$ in $U^3$ is

$$(a_3)_Z = v_Z^{\mathsf{T}} H_+ u_-.$$

For the even case, replacing $u_+$ by $u_-$ is legitimate: $H_+(u_+ - u_-) = (8, -8, 0, 0, 0, 0)$, which pairs to zero with every remaining high vector.

The four forced even low/high pairs contribute

$$H_+ u_- = (144, 144, 648, 648, 0, 848) + \text{contributions of remaining even pairs.}$$

For a sigma-free high vector $w = (0, 0, b, c, 0, a)$ put $x = b + c$. An even $h$-pair supported on $w, gw$ contributes, modulo 3,

$$(0, 0, 2xa, 2xa, 0, a^2).$$

The sigma-containing types have $a = 3$ and contribute zero modulo 3. The already imposed even dimension balance has square-root coefficient 120, and hence $\sum 2xa \equiv 0 \pmod 3$ over remaining even pairs. Put

$$T = \sum_{\text{remaining even pairs}} a^2 \pmod 3.$$

It follows that

$$(a_3)_Z \equiv (v_Z)_{\sigma X}(T - 1) \pmod 3.$$

Reducing (V) modulo 3 therefore gives the spin constraint

$$(v_Z)_{\sigma X}(T - 1)k_Z \equiv 0 \pmod 3. \tag{S}$$

Suppose a profile contains no sigma-free high type with positive sigma-X multiplicity divisible by 3. If $T \neq 1$, (S) says that every sigma-free high simple contributing to $I(\sigma X)$ has $3 \mid k_Z$, so its twist order divides 16. The sigma-containing high types already have twist order dividing 16; low objects and their four forced high partners have twists 1 and $-1$. Thus all simples in $I(\sigma X)$ would have sixteenth power of their twist equal to one. The center indicator formula would give

$$\nu_{16}(\sigma X) = \dim(\sigma X) = 4 + 3\sqrt{2} + 2\sqrt{3} + 2\sqrt{6}.$$

This is impossible: the left side is the trace of an order-sixteen cyclic rotation and lies in $\mathbb{Q}(\zeta_{16})$. The right side does not. For example, the automorphism $\zeta_{48} \mapsto \zeta_{48}^{17}$ fixes $\zeta_{16}$ and $\sqrt{2}$ but negates $\sqrt{3}$. Therefore such a profile must have $T = 1 \pmod 3$.

The exact bounded orbit calculation imposes the asymmetric-even parity condition and this single residue $T$. It excludes 18 profiles by parity and a further 39 by the spin/field condition, leaving branch sizes $164, 753, 753$. The portable verifier `verify_spin_profiles.py` first replays the input 1727-profile certificate and checks its surviving-index hashes. It uses only integer arithmetic and independently rechecks every rejected charge branch by a second recursion over the individual orbit counts. The small receipt `spin_profiles_certificate.json` records all excluded indices and the final count 1670. □

**Corollary 7.19.** *No categorification of $R^{(6)}_{8,1660}$ admits a braiding.*

*Proof.* Its Ising subcategory would be nondegenerate, so braided decomposition would give $\mathcal{C} \simeq \mathcal{I} \boxtimes \mathcal{D}$ with $\operatorname{rank} \mathcal{D} = 2$ [57, Corollary B.12 and Theorem 3.13]. Write $z$ for the nonunit simple of $\mathcal{D}$; then $z^2 = 1 + nz$. The dimension multiset forces $\operatorname{FPdim} z = d$, so $X$ corresponds to $1 \boxtimes z$ or $g \boxtimes z$. Its square cannot contain $\sigma \boxtimes z$, contrary to $N^{\sigma X}_{XX} = 4$. □

## 7.6 Completion at exact multiplicity nine

The stratum contains 8 809 rank-six based rings whose largest structure coefficient is exactly nine: 8 808 commutative rings and one noncommutative ring. None admits a complex fusion categorification. The initial exact sieve excludes 8 798 rings. The following arguments exclude its eleven residuals, without assuming pivotality or braiding. The only remaining general existence questions through the census bound are $R^{(6)}_{4,155}$ and $R^{(6)}_{8,1660}$.

The original and advanced campaign records retain their source-bound attempts, including unfinished searches. They supply no exclusion beyond their accepted certificates. The completion is a separate, disjoint set of eleven general certificates with normal and optimized exact replays. The certificates are applied to the fixed census tensors and identifiers.

### 7.6.1 Proof of the remaining eleven exclusions

**Theorem 7.20.** *None of the eleven rings*

$$R^{(6)}_{9,529},\ R^{(6)}_{9,1060},\ R^{(6)}_{9,2455},\ R^{(6)}_{9,3748},\ R^{(6)}_{9,7018},\ R^{(6)}_{9,7741},\ R^{(6)}_{9,7742},\ R^{(6)}_{9,8099},\ R^{(6)}_{9,8101},\ R^{(6)}_{9,8200},\ R^{(6)}_{9,8785}$$

*admits a fusion categorification over an algebraically closed field of characteristic zero. Consequently all* 8 809 *exact-multiplicity-nine rings are excluded, and none contributes a positive model.*

We prove the remaining cases by three common reductions: arithmetic of signed dimension fields, a small Casimir-deficit argument, and central induction. All finite domains below are derived from these reductions. The complete tensors, integer witnesses and independently executable verifiers are in `research/exclusions/m9_completion/`. No arbitrary conductor cutoff or unfinished associator search is used.

**Signed necessary data and two useful refinements.** For a self-dual input ring, sphericalization of an arbitrary hypothetical category gives the commutative signed algebra of Lemma 3.7. Indeed, the two lifts of a self-dual simple have opposite nonzero dimensions, whereas duality preserves dimension, so both lifts are self-dual. Its coefficients satisfy

$$t_{ijk} = t_{\pi(i,j,k)}, \qquad |t_{ijk}| \le N_{ijk}, \qquad t_{ijk} \equiv N_{ijk} \pmod 2.$$

Write $K$ for its dimension-character field, $\Delta$ for its dimension codegree and $T_0 = \sum N_{ijk}^2$. The symmetric integral regular matrices give a totally real semisimple algebra. Cyclotomicity makes $K$ abelian. Every original or signed codegree $f$ belongs to $K$, and $\Delta/f \in \mathcal{O}_K$: apply central induction to sphericalization, whose corresponding codegrees and global dimension are $2f$ and $2\Delta$. The six signed codegrees exceed one, have reciprocal sum one, are $d$-numbers, and have total at most $T_0$. The conjugates of $\Delta$ contribute $\mathrm{Tr}_K(\Delta)$ to this total, even when $\Delta$ itself generates a proper subfield of $K$. These statements are the earlier signed-field and trace-ideal criteria, using [38, Section 5.1 and Theorem 2.13] and [37, Theorem 1.2 and Corollary 1.4].

**Lemma 7.21** (Indicator normalization)**.** *For the self-dual signed algebra above, the lifts can be chosen so that*

$$N_{jji} - t_{jji} \equiv 0 \pmod 4 \qquad \text{for all } i, j.$$

*No further independent sign normalization is implicit in this choice.*

*Proof.* The Fuchs–Runkel–Schweigert theorem, in the pivotal formulation [60, Theorem 3.9], states that a self-dual simple of indicator $-1$ occurs with even multiplicity in $Y^* \otimes Y$; neither braiding nor simplicity of $Y$ is required. The two sphericalization lifts differ by the sign of their pivotal map. The transpose operator $\kappa \mapsto \kappa^* j$ on the one-dimensional self-duality space therefore changes sign, so their indicators are opposite. Choose the indicator-$+1$ lift of each simple. The negative lift of $i$ has multiplicity $(N_{jji} - t_{jji})/2$ in the square of the chosen lift of $j$, and that multiplicity is even. Apply the theorem inside sphericalization, without requiring a pivotal structure on the original category. □

**Lemma 7.22** (Diagonal associativity and saturation)**.** *Every self-dual signed table as above satisfies*

$$\sum_k |t_{iik} t_{jjk}| \ \ge\ \sum_k t_{ijk}^2. \tag{†}$$

*If $|t| = N$ and every nonunit $N_{iik}$ is positive, a sign gauge makes all repeated-index triples positive. Only the ten triples with three distinct nonunit indices then remain to be tested.*

*Proof.* The coefficient of 1 in $(i^2)(j^2) = (ij)^2$ gives $\sum_k t_{iik}t_{jjk} = \sum_k t_{ijk}^2$, proving the inequality. In the saturated case the unsigned original table gives the same sum with each summand replaced by its maximum $N_{iik}N_{jjk}$. Equality of the sums forces equality termwise. Thus for fixed nonunit $k$ the signs of $t_{iik}$ are independent of $i$. Gauge the basis element $k$ by this common sign. The transformed $t_{iik}$ is positive, since the gauge on $i$ occurs twice. This includes $i = k$. □

**Three small-deficit cases.** For $R^{(6)}_{9,7741}$ and $R^{(6)}_{9,7742}$ the trace-ideal lower bound is 6705, and the original Casimir traces are 6813 and 6805. For an unordered nonunit triple $q$ of weight $w_q$ (its number of ordered permutations), enumerate

$$a_q \in \{N_q, N_q - 2, \ldots\} \cap \mathbb{Z}_{\geq 0}, \qquad \sum_q w_q(N_q^2 - a_q^2) \leq 108 \quad \text{or} \quad 100.$$

These are respectively 389 and 241 magnitude profiles. Every profile except $a = N$ violates (†) for at least one pair $i, j$. This test is independent of all coefficient signs. Both tensors have $N_{iik} > 0$, so Lemma 7.22 leaves $2^{10}$ sign choices; checking the complete associativity identities leaves only the positive table. The resulting fusion-closed lifts give a spherical section. The previously established complete spherical obstructions for these two tensors therefore exclude arbitrary categorifications. Their exact spherical proof trees are replayed as part of the new certificate.

For $R^{(6)}_{9,8200}$, put $E = \mathbb{Q}(\sqrt{401})$. The original FP codegree is $D = 10\sqrt{401}(20 + \sqrt{401})$ and $T_0 = 8040$. Since $D \in 2\mathcal{O}_E$ and $\Delta/D$ is integral, $\Delta \in 2\mathcal{O}_K$. The characteristic polynomial $\mathrm{Norm}_{K/\mathbb{Q}}(X - \Delta)$ divides the signed Casimir polynomial and reduces to $X^{[K:\mathbb{Q}]}$ modulo two. The original Casimir polynomial reduces to $X^2(X + 1)^4$; hence $[K : \mathbb{Q}] \leq 2$, and $E \subset K$ gives $K = E$. The unique $\mathbb{F}_2$ character is $(1, 0, 0, 1, 0, 0)$. As two splits in $E$, reduction of every dimension embedding has this character, so $\Delta = \sum_i d_i^2 \equiv 2 \pmod{4\mathcal{O}_E}$. Its trace is divisible by four. Combining this with the trace ideal $4010\mathbb{Z}$ gives $\mathrm{Tr}_E(\Delta) = 8020$. The resulting deficit at most 20 is smaller than the least possible unsaturated loss 28. Saturation and the same $2^{10}$ sign test give a spherical section. Only the FP dimension orbit can occur; the quadratic Gauss obstruction, Proposition 3.11, then requires at least 3200 root contributions where only 770 are available. This excludes $R^{(6)}_{9,8200}$.

**The two cyclic-quartic cases.** For $i = 8099, 8101$ the original codegree polynomial is $FG_i$, where

$$\begin{aligned} F &= X^2 - 4225X + 21125, \\ G_{8099} &= X^4 - 300X^3 + 5250X^2 - 22500X + 28125, \\ G_{8101} &= X^4 - 300X^3 + 26250X^2 - 562500X + 703125. \end{aligned}$$

Each quartic is irreducible and generates $E = \mathbb{Q}(r)$, $r^4 - r^3 - 4r^2 + 4r + 1 = 0$. The integral basis is $1, r, r^2, r^3$, of discriminant 1125. The automorphism $r \mapsto -r^3 - r^2 + 3r + 2$ has order four, and $1 - 6r + 2r^3$ squares to five. Thus $E$ is cyclic quartic with unique quadratic subfield $\mathbb{Q}(\sqrt{5})$. Field containment gives $K = E$. Intersecting the six original codegree ideals gives the column lattices

$$L_{8099} = 65 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -8 & -13 & 15 & 0 \\ -7 & -4 & 0 & 15 \end{pmatrix}, \qquad L_{8101} = 65 \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 5 & 0 & 0 \\ -12 & -5 & 15 & 0 \\ 1 & -5 & 0 & 15 \end{pmatrix} \subset L_{8099}.$$

Consequently $\Delta = 65\theta$, where $\theta = a + br + cr^2 + dr^3$ satisfies

$$c \equiv -8a - 13b, \quad d \equiv -7a - 4b \pmod{15}, \qquad t := \mathrm{Tr}(\theta) = 4a + b + 9c + d \in \{15, 30, 45, 60\}.$$

Here $T_0 = 4525$ and the trace ideal is $975\mathbb{Z}$. Every conjugate of every trace-dual basis element lies between $-2$ and 2 (checked by exact Sturm counts), so total positivity gives $|a|, |b|, |c|, |d| < 2t$. This is an exhaustive finite box. For $P_\theta = X^4 + c_1X^3 + c_2X^2 + c_3X + c_4$, require $c_4^j \mid c_j^4$ for $1 \le j \le 4$, the $d$-number conditions. They remain necessary if $P_\theta$ is a power of a smaller minimal polynomial. The two remaining signed codegrees $f, g$ form a rational pair or a conjugate pair in $\mathbb{Q}(\sqrt{5})$. In lowest terms put

$$\frac{A}{B} = 1 + \frac{c_3}{65c_4} = \frac{1}{f} + \frac{1}{g} > 0.$$

Then $f + g = Ak$, $fg = Bk$, with $k \ge 1$ and $Ak \le 4525 - 65t$. Both roots of $X^2 - AkX + Bk$ must exceed one. Its discriminant is a square with integral roots, or five times a square with $Bk \mid (Ak)^2$. The exhaustive counts are

| $t$ | 15 | 30 | 45 | 60 |
|---|---|---|---|---|
| $\theta$ totally positive | 0 | 6 | 28 | 69 |
| $\theta$ passing $d$-number test | 0 | 6 | 12 | 21 |
| admissible complementary pairs | 0 | 0 | 0 | 0. |

The verifier reconstructs the maximal order, all codegree embeddings and ideal intersections by exact arithmetic; rational root intervals certify all positivity decisions. This proves both general exclusions.

**A common higher-field reduction for 7018 and 8785.** For these two rings, respectively set

$$(c, T_0, B) = (195, 4405, 4), \qquad (65, 4325, 13).$$

The original codegrees generate $E = \mathbb{Q}(\sqrt{5})$ and their ideal intersection is $c\sqrt{5}\mathcal{O}_E$. The full algebra modulo two is $\mathbb{F}_4^3$: the Frobenius matrix squares to the identity and its fixed space has dimension three. Therefore the signed order is etale at two. Every completion of $K$ at two is unramified of degree two, and its Frobenius restricts nontrivially to $E$. We have

$$[K : \mathbb{Q}] \in \{2, 4, 6\}, \quad \Delta \in c\sqrt{5}\mathcal{O}_K \cap \mathfrak{D}_K, \quad \mathrm{Tr}_K(\Delta) \le T_0.$$

The different condition is Theorem 3.23: orthogonality gives $\sum_i d_i \otimes d_i = (\Delta \otimes 1)e_K$; contracting against the inverse different shows $\Delta\mathfrak{D}_K^{-1} \subset \mathcal{O}_K$.

A cyclic quartic $K$ is impossible because its order-two Frobenius would act trivially on its unique quadratic subfield. In a biquadratic field write $K = E\mathbb{Q}(\sqrt{q})$, with $q > 1$ odd squarefree, $5 \nmid q$ and $q \equiv 1 \pmod 4$. The discriminants are coprime. Writing

$$\Delta = c\sqrt{5}(a + b\sqrt{5})/2, \qquad a, b \in \mathcal{O}_{\mathbb{Q}(\sqrt{q})},$$

shows that $b$ is totally positive with trace at most $B$. If $p \mid q$ but $p \nmid c$, the different puts $b$ in the ramified prime, so $p \mid \mathrm{Tr}(b)$. Thus $q/\gcd(q, c) \mid \mathrm{Tr}(b) \le B$. For $c = 195$ the only possible prime factors of $q$ are $3, 13$; for $c = 65$ they are $3, 7, 11, 13$, with the product of those other than 13 at most 13. The congruence $q \equiv 1 \pmod 4$ leaves only $q = 13$.

In this remaining field put

$$\Delta = \frac{c\sqrt{5}}{4}(x + y\sqrt{5} + z\sqrt{13} + w\sqrt{65}).$$

The maximal-order conditions are that $x, y, z, w$ have the same parity and $x - y - z + w \equiv 0 \pmod 4$. Positivity gives the complete bounds

$$1 \le y \le B, \quad x^2 < 5y^2, \quad 13z^2 < 5y^2, \quad 13w^2 < y^2.$$

Apply the degree-four $d$-number conditions and the complementary-pair test above, now allowing quadratic radicands $5, 13, 65$. There are seven $d$-number candidates for 7018, with no pair. For 8785 there are thirty-five; the only two surviving pairs give

$$P_\Delta = (X^2 - 1625X + 21125)^2, \qquad P_{f,g} = X^2 - 143X + 169.$$

Their product reduces to $(X^2 + X + 1)^3$, contradicting the required original Casimir reduction $(X^2 + X + 1)(X + 1)^4$.

For degree six, $K = EF$ with $F$ cyclic cubic and two completely split in $F$. Its conductor has the complete form $9^\epsilon \prod p$, where $\epsilon \in \{0, 1\}$ and the distinct primes satisfy $p \equiv 1 \pmod 3$. An order-three character is trivial on the pro-$p$ kernel for $p \neq 3$ and factors through modulus nine at three, which proves this form. The different exponents are $d_p = 2$ at tame primes and $d_3 = 4$ at the block nine, by the conductor–discriminant formula [59, Chapter V, Theorem 3.27]. Again $b \in \mathcal{O}_F$ is totally positive with trace at most $B$. The different forces $b \in \mathfrak{p}^{k_p}$, where $k_p = \max(0, d_p - 3v_p(c))$. The local codifferent definition gives

$$\mathrm{Tr}(\mathfrak{p}^k) = p^{\lfloor (k+d_p)/3 \rfloor}\mathbb{Z}_p, \qquad \mathrm{Norm}(b) \geq \prod_{p \mid f} p^{k_p}.$$

Indeed, containment of that trace ideal in $p^m\mathbb{Z}_p$ is equivalent to $p^{-m}\mathfrak{p}^k \subset \mathfrak{p}^{-d_p}$. For 7018 these trace divisibilities leave only conductors $9, 13, 117$; the first two make two inert. At 117 the trace of $b$ is three and its norm at least three, contradicting AM–GM's upper bound one. For 8785 the complete list is $7, 9, 13, 91, 117$: simultaneous blocks $7, 9$ would require trace divisible by 63, and every other tame prime is at least 19. The single blocks make two inert. At 91 the trace is seven and norm at least $49 > (7/3)^3$; at 117 the trace is nine and norm at least $81 > 27$. Thus both rings force $K = E$.

**The quadratic spectra and the last signed search.** For 7018 write

$$\Delta = (975y + 195x\sqrt{5})/2, \qquad 1 \leq y \leq 4, \quad x \equiv y \pmod 2, \quad x^2 < 5y^2, \quad 5y^2 - x^2 \mid 20y^2.$$

There are sixteen possibilities. Because the etale two-adic algebra is $\mathbb{F}_4^3$, every rational character component has even degree. The four complementary codegrees therefore partition into two $E$-conjugate pairs; rational pairs are repeated integers. For a pair $f, f' = (S \pm b\sqrt{5})/2$, set $P = ff'$ and require

$$P \mid \mathrm{Norm}(\Delta), \quad S^2 - 4P = 5b^2, \quad S \equiv b \pmod 2, \quad S > 2, \quad P > S - 1,$$

together with $P \mid S^2$ when $b \neq 0$, and integrality of $\Delta/f, \Delta/f'$. The two pairs must have total trace at most $4405 - 975y$ and reciprocal sum $1 - \mathrm{Tr}_E(1/\Delta)$. Only $(x, y) = (\pm 1, 1)$ remain, both with full polynomial

$$(X^2 - 975X + 190125)(X^2 - 975X + 1125)(X^2 - 975X + 7605).$$

Its reduction $(X^2 + X + 1)^3$ contradicts the original $(X^2 + X + 1)(X + 1)^4$, excluding 7018.

For 8785 the identical pair-enumeration procedure starts with $\Delta = (325y + 65x\sqrt{5})/2$, $1 \leq y \leq 13$. It leaves four spectra up to simultaneous quadratic conjugation. Three have $\Delta = D = (4225 + 1885\sqrt{5})/2$ or its conjugate, so [15, Proposition 8.23] reduces them to the pseudo-unitary obstruction below. The exceptional spectrum consists of the three pairs

$$\frac{1625 \pm 325\sqrt{5}}{2}, \qquad \frac{450 \pm 200\sqrt{5}}{2}, \qquad \frac{1170 \pm 520\sqrt{5}}{2}.$$

Its signed Casimir trace is 3245, hence its deficit is 1080. Lemma 7.21 leaves the explicit domain

$$t_q \in \{-N_q, -N_q+2, \dots, N_q\}, \qquad \sum_q w_q(N_q^2 - t_q^2) = 1080,$$

$$t_q \equiv N_q \pmod 4 \quad \text{if } q \text{ repeats an index.}$$

There are 35 nonunit triples. The independent verifier generates all entries of all commutators of the five multiplication matrices, giving 55 distinct polynomial equations. Depth-first enumeration evaluates each equation when its last variable is assigned and prunes only when the nonnegative accumulated deficit exceeds 1080. It visits 2 247 571 340 nodes and finds no table. All domains, equations, native integer bounds and successful termination are regenerated and checked; this is a complete finite proof, not a search deadline.

**Three rings with an involution.** We first give their exact pseudo-unitary certificates in a common form. For a commutative category let $G = \sum_i N_i N_{i^*}$ be the central induction Gram matrix. The six summands of $I(1)$ have restriction columns $v_j$ with unit coordinate one and dimension $D/f_j$. Full-character reciprocity is

$$(D/f_i)\chi_i(v_j) = (D/f_j)\chi_j(v_i).$$

For fixed low columns, the high columns and their relative twist moments satisfy Proposition 4.13. Its exact nonnegative system $Ax = b$ is impossible whenever an integer vector $y$ has $y^tA \ge 0$ and $y^tb < 0$.

For $529, 2455$ a common complete low-column prune is as follows. Let $u$ be the unit column, $M = G - uu^t > 0$, fix one other low column $v$, and put $w = G_{\bullet 0} - u - v$. The four remaining columns sum to $w$, so Cauchy's inequality gives $M - vv^t - ww^t/4 \succeq 0$. Writing $A = qM^{-1}$ integral with $q > 0$, this is exactly

$$q - v^tAv \ge 0, \quad 4q - w^tAw \ge 0, \quad (q - v^tAv)(4q - w^tAw) \ge (v^tAw)^2.$$

The five column domains shrink from $(817, 833, 833, 217, 217)$ to $(10, 9, 9, 10, 10)$. Exhausting their Cartesian product modulo permutations of equal-codegree columns gives 497 frames; ordinary Gram/moment constraints reject 406, and character reciprocity leaves six frames for 529 and thirty-one for 2455. Every one of the 8, respectively 32, possible second-indicator assignments is retained. The resulting 48 and 992 systems have exact integer Farkas witnesses.

For 1060, use the basis $(1, g, X, Y, gY, gX)$. The five nonunit low columns have total multiplicities in the paired $X, gX$ and $Y, gY$ coordinates

$$A = (16, 11, 11, 6, 22), \qquad B = (17, 11, 11, 5, 22).$$

Their difference vectors $u, v$ obey $\sum u_j = \sum v_j = -2$, $\sum u_j^2 \le 12$, $\sum v_j^2 \le 20$, $u \equiv A$, $v \equiv B \pmod 2$. There are $18 \cdot 59 = 1062$ pairs; positivity of the remaining Gram matrix leaves 369, and exact character reciprocity leaves four frames. All 32 indicator choices for each frame are contradicted by 128 integral Farkas witnesses. These three proofs use no indicator parity shortcut and no numerical optimization in replay.

For the general reduction of 1060, positive sphericalization lifts make $g$ act positively on its three free pairs. The dimension character is $(1, 1, x, y, y, x)$, so its field is a quotient of the rank-three $g = 1$ algebra. It contains $\mathbb{Q}(\sqrt{273})$, hence equals that field. Write $x = (a + b\sqrt{273})/2$, $y = (c + d\sqrt{273})/2$. Orthogonality with the conjugate character gives

$$a^2 + c^2 + 4 = 273(b^2 + d^2), \qquad \Delta = 273s + (ab + cd)\sqrt{273}, \quad s = b^2 + d^2.$$

The original trace is 4396 and the other four codegrees exceed one, so $0 < s \le 8$. Thus $b, d \in [-2, 2]$, $a, c \in [-46, 46]$. There are 144 parity-compatible solutions. Integrality of

$\Delta/12$ and $\Delta/(2184+132\sqrt{273})$ leaves 32 assignments and the four values $2184\pm108\sqrt{273}$, $2184\pm132\sqrt{273}$. The first pair fails the quadratic $d$-number test: its norm 1585584 does not divide $4368^2$. The second pair is the FP global-dimension orbit. Galois conjugation and [15, Proposition 8.23] contradict the pseudo-unitary certificate, proving general nonexistence.

For $529, 2455$, use $(1, g, x, y, a, b)$ with $gx = y$, $ga = a$, $gb = b$. The FP dimensions are $(1, 1, d, d, d-1, d+1)$, $d = 16+\sqrt{257}$, and the codegrees are $4, 4, 8, 8, D, D'$ with $D = 2056+128\sqrt{257}$. The original trace is 4136. Gauge the lift of $g$ to dimension one; its fixed objects force its left and right action signs to agree. In 529, $x^* = y$ and dual invariance forces positive action on that pair; in 2455 gauge the pair to achieve it. The signed $g = 1$ and $g = -1$ components have dimensions four and two. The former has a one-dimensional character, so neither component can contain a noncommutative matrix block. Thus the signed algebra is commutative, with $E = \mathbb{Q}(\sqrt{257}) \subset K$ and $[K : \mathbb{Q}] = 2$ or 4. For $\theta = \Delta/8$, induction gives $\theta, \theta/\sqrt{257} \in \mathcal{O}_K$.

If $K = E$, the trace is a positive multiple of 2056. At both split two-adic embeddings, the mod-two table gives $x$ odd and $a = 2a_0, b = 2b_0$. Integrality of $\Delta/8 = (2 + 2x^2 + 4(a_0^2 + b_0^2))/8$ forces $a_0^2 + b_0^2$ odd, hence one modulo four. Therefore $\Delta \equiv 8 \pmod{16}$ at both embeddings, and its trace is divisible by sixteen. The only possible trace is 4112. The four other codegrees are positive. Decreasing any coefficient magnitude $7, 8, 9$ loses at least 24 from the full trace, which would leave at most 4112. Thus the signed table is saturated. The complete oriented sign domains, with positive $g$ action, contain exactly eight associative tables for each ring; every table has a certified unit- and dual-preserving gauge to the original one. Each therefore supplies a spherical section. The nonzero dimension characters lie only in the FP orbit (the others have zero entries), so the pseudo-unitary certificates exclude this case.

If $[K : \mathbb{Q}] = 4$, the canonical trace of $(1 + g)/2$ gives $\mathrm{Tr}_K(1/\theta) = 4$ and $\mathrm{Tr}_K(\theta) \leq 517$. The element $\theta$ generates $K$: if quadratic of norm $n$, these identities give full trace $4n$, whereas $\theta/\sqrt{257}$ integral forces $n \geq 257$; if rational, they give $\theta = 1$, also impossible. Relative trace and integrality give the minimal polynomial

$$P = X^4 - 257uX^3 + 257wX^2 - 4 \cdot 257^2 vX + 257^2 v,$$

where $u = 1, 2$ and

$$v \mid 257^2 u^4, \quad v^2 \mid w^4, \quad 1 \leq v \leq \lfloor 257u^3/64 \rfloor, \quad 1 \leq w \leq \lfloor 3 \cdot 257u^2/8 \rfloor.$$

The bounds are Maclaurin's inequalities, and the divisibilities are the $d$-number test. Exactly 1057 polynomials occur; exact Sturm counts reject 736 for lack of four positive roots. The discriminants of the remaining 321 are neither squares nor 257 times squares. These are the two possibilities for an abelian quartic containing $E$ (biquadratic or cyclic). This excludes the quartic case.

**The pseudo-unitary obstruction for** 8785**.** Put $\phi = (1+\sqrt{5})/2$, $t = 8 + 13\phi$ and $D = 5(1 + t^2)$. The dimensions are $(1, t, t-1, t-1, t+1, t+1)$ and the two smallest nonunit induction dimensions are both $3t + 2$. Their columns have unit coordinate one, the other five coordinates summing to three, and exactly eight nonnegative solutions to the dimension equation. For each of the $8^2$ labeled pairs, dimensions and reciprocity with these two columns give a rational system of rank six for every other column. Exactly ten nonnegative integral frames remain; full row sums and reciprocity are checked.

For a nontrivial self-dual simple of dimension $d$, let $\ell$ be the dimension mass in its induction contributed by the twist-one summands of $I(1)$. By [38, Theorems 2.5 and 2.7], the first twist moment is zero and the second is $D\nu_2(X)$, with $\nu_2(X) \leq 1$. Cauchy's inequality applied to the remaining real twist moments gives

$$2\ell^2 \leq (Dd - \ell)(D(d+1) - 2\ell), \qquad (3d+1)\ell \leq Dd(d+1).$$

For $d = t$, the ten frames have low masses $14473 + 23413\phi$ (six frames), $13641 + 22061\phi$ (two), or $12809 + 20709\phi$ (two). The differences between the left and right sides of the last inequality are respectively

$$257812 + 417144\phi, \quad 184284 + 298168\phi, \quad 110756 + 179192\phi,$$

all positive. This excludes every frame and completes the 8785 proof.

**The rank-three extension** 3748**: categorical reduction.** This tensor is

$$R = A[X]/(X^2 - 1 - QX), \qquad Q = 2 + 2a + b, \quad a^2 = 1 + b,\ ab = a + b,\ b^2 = 1 + a + b.$$

The basis $(1, a, b, X, aX, bX)$ is source order $(0, 1, 5, 2, 3, 4)$. In a signed lift of $A$, write $a^2 = 1 + ub$, $ab = ua + vb$, $b^2 = 1 + va + wb$, with $u, v, w = \pm 1$. Associativity gives $uw = 1$, and a sign gauge makes all signs positive. Thus any categorification of $A$ is spherical, and [38, Theorem 1.1] identifies its fusion category $\mathcal{D}$ as $\mathrm{PSU}(2)_5$ or a Galois conjugate.

Lift this fusion-closed $A$ to sphericalization of the full category, choose $X$, and define the other lifts by $aX, bX$. Its entire $A$ action is positive. The signed relation is $X^2 = 1 + (p + qa + rb)X$, where $p, q \in \{-2, 0, 2\}$ and $r = \pm 1$. Simultaneously changing the three complementary lifts allows $r = 1$. For the nine cases the signed Casimir polynomials are irreducible sextics. Their pairs (trace, constant coefficient) are

$$(189, 3296573), (77, 703493), (245, 6754013),$$
$$(105, 1279733), (105, 434581), (385, 2854789),$$
$$(133, 1279733), (245, 3027661), (637, 5274997).$$

In lexicographic order on $(p, q)$, the first eight violate $P(0) \mid \mathrm{Tr}(P)^6$. Thus the sphericalization $d$-number test leaves only $(p, q, r) = (2, 2, 1)$, giving a spherical section of the full category. Its original global-codegree polynomial

$$P_f = X^6 - 637X^5 + 22295X^4 - 298116X^3 + 1854944X^2 - 5274997X + 5274997$$

is irreducible, so there is a pseudo-unitary Galois conjugate.

Let $\mathcal{M}$ be the complementary $\mathcal{D}$-bimodule category. The actual left-module functor $d \mapsto dX$ maps simples bijectively and is an equivalence. Transport of its right action gives a tensor autoequivalence of $\mathcal{D}$ fixing simple labels; hence $\mathcal{M}$ is an invertible twisted regular bimodule. The result $\mathrm{BrPic}(\mathrm{Ad}(A_6)) = 1$, including Galois conjugates, is [58, Theorem 1.2]. Thus $\mathcal{M}$ is categorically regular, with the unit sent to $X$.

Its relative center therefore has simple basis $B \sqcup Bt$, where $B = A \otimes A = K_0(Z(\mathcal{D}))$ and $t$ forgets to $X$. The central action of $Z(\mathcal{D})$, equivalently naturality of the relative half-braiding, makes $B$ commute with $t$. The unique simple over $X$ is self-dual, and positivity forces

$$t^2 = 1 + Ct, \qquad C \in B_{\geq 0}, \qquad F(C) = 2 + 2a + b.$$

These are the fifteen explicitly enumerated relative-center lifts. The formal-codegree $d$-number test excludes thirteen, leaving only $C = Q \otimes 1$ and $C = 1 \otimes Q$. Each one-sided case contains a fusion subcategory whose forgetful functor to the full category is bijective on simples, hence a tensor equivalence. Its section supplies coherent half-braidings with all objects, giving a central lift of $\mathcal{D}$ to $Z(\mathcal{C})$; on $\mathcal{D}$ it is one of the two nondegenerate braidings. The relative-center construction is [20, Section 2.2]. Consequently [57, Theorem 3.13] applies inside the center:

$$Z(\mathcal{C}) = \mathcal{D} \boxtimes \mathcal{E}.$$

No braiding or tensor factorization of $\mathcal{C}$ has been assumed.

**A six-versus-four Galois contradiction.** Use positive dimensions

$$a = 2\cos(\pi/7), \quad b = a^2 - 1, \quad h = \sqrt{13}, \quad q = 2 + 2a + b, \quad x = (q + bh)/2,$$

and put $d = 1 + a^2 + b^2$, $f = d(1 + x^2)$. The identities $a^3 - a^2 - 2a + 1 = 0$, $q^2 + 4 = 13b^2$ and $x^2 - qx - 1 = 0$ hold. The six numbers $1, a, b, x, ax, bx$ are $\mathbb{Q}$-independent, so dimensions determine forgetful vectors. The centralizer has dimension $f^2/d$.

The induction object $I(1)$ has six multiplicity-one simple summands, all of twist one, by [38, Theorems 2.5 and 2.13]. Its integral multiplicity vector $n$ satisfies normalized $Sn = n$ by [38, Proposition 2.8], so each signed Galois permutation fixes $n$. The six distinct unit Galois dimensions $f/\sigma(f)$, distinct because $P_f$ is irreducible, exhaust these summands. Normalized $S$ is compatible with the Deligne product. The three embeddings $a \mapsto a, a^2 - a - 1, 2 - a^2$ have $\mathcal{D}$ unit Galois labels $1, b, a$, respectively, since their squared dimensions are $d/\sigma(d)$. Dividing by these dimensions identifies six known $\mathcal{E}$ simples with forgetful vectors

$$\begin{aligned} &(1,0,0,0,0,0), \quad (1,0,0,2,2,1),\\ &(0,0,1,1,3,4), \quad (0,0,1,0,0,1),\\ &(0,1,0,1,2,1), \quad (0,1,0,1,1,2). \end{aligned}$$

Their twists are inverse to those of their $\mathcal{D}$ labels. Since $\mathrm{Norm}(d) = 49$, Cauchy's theorem [5, Theorem 3.9] makes all these twists seven-primary. The second object, $e_-$, corresponds to $h \mapsto -h$, and has dimension $x^2$ and twist one.

The induction Gram identity is $\sum_{z \in \mathrm{Irr}\, Z(\mathcal{C})} [Fz][Fz]^t = \sum_i N_i^2$. Subtract the eighteen vectors obtained by multiplying these six known vectors by $1, a, b$. The residual is zero in the first three rows and columns and has lower block

$$T = \begin{pmatrix} 24 & 36 & 44 \\ 36 & 68 & 80 \\ 44 & 80 & 104 \end{pmatrix}.$$

All contributions are nonnegative outer products. Every unknown $\mathcal{E}$ simple therefore forgets to $XU$, where $U = u_0 + u_1 a + u_2 b$, $u_i \geq 0$, not all zero. Its whole $\mathcal{D}$ block contributes $u_0^2 + u_1^2 + u_2^2$ to $T_{00}$, so the sum of these squared norms over unknown simples is 24.

Under $a \mapsto a^2 - a - 1$, $h \mapsto h$, the positive dimension Galois action on unknown vectors is $u \mapsto \pm Hu$, where

$$H = \begin{pmatrix} 0 & -1 & 1 \\ -1 & -1 & 1 \\ -2 & 0 & 1 \end{pmatrix}, \qquad H^3 = -I.$$

This follows directly from $d_{\widehat{\sigma}(y)} = d_{\widehat{\sigma}(1)} |\sigma(d_y)|$: the transformed $U$ is $(b^2/\sigma(b))\sigma(U)$. Injectivity of the dimension map requires all coordinates to have one sign; mixed signs are impossible. A dimension orbit has length one or three, and equal multiplicity along the orbit. A fixed positive vector is $n(1, 2, 1)$, with squared norm $6n^2$. If a three-element orbit had norm sum below six, one vector would be a coordinate unit. Their complete positive-sign orbits have norm sums $12, 12, 6$, respectively. Hence every dimension orbit costs at least six in the Gram budget. Any exact dimension can therefore occur with multiplicity at most $24/6 = 4$.

On the other hand, $\mathrm{Norm}_{\mathbb{Q}(a,h)/\mathbb{Q}}(f^2/d) = 7^4\, 13^6$. Cauchy's theorem forces an $\mathcal{E}$ twist with a nontrivial thirteen-primary component; the corresponding object is unknown. Choose a cyclotomic automorphism fixing all prime-to-thirteen roots and acting by exponent two on the thirteen-primary roots, possible by the Chinese remainder theorem in a common cyclotomic field for the normalized modular representation and its twists. It fixes $a$, negates $h$, and sends $x$ to $-1/x$. Its unit image is $e_-$, so it preserves each unknown dimension: $x^2 |\sigma(xU)| = xU$.

For actual twists the normalized modular Galois theorem [49, Theorem II(iii)], divided by its unit case, reads

$$\theta_{\widehat{\sigma}(y)} = \theta_{\widehat{\sigma}(1)}\sigma^2(\theta_y) = \sigma^2(\theta_y).$$

Exponent four has order six modulo thirteen. Thus the nontrivial thirteen-primary twist forces at least six different objects of the same dimension, also for higher thirteen powers. This contradicts the bound four and completes the proof of Theorem 7.20.

# 8 Complete modular reconstruction for $R^{(6)}_{4,155}$

Write $d = 4 + 3\sqrt{2}$, $D = 72 + 48\sqrt{2}$, and $z = e^{2\pi i/24}$. Section 7.4 gives

$$Z(\mathcal{C}) \simeq \mathcal{S} \boxtimes \mathcal{B}, \qquad \operatorname{rank} \mathcal{B} = 26, \qquad \dim \mathcal{B} = 144d^2,$$

where $\mathcal{S}$ is the semion category. The multiplicative central charge of $\mathcal{B}$ is $z^{-3}$. Thus

$$s = \frac{S_{\mathcal{B}}}{12d}, \qquad t = zT_{\mathcal{B}}$$

obey $s^2 = C$ and $(st)^3 = C$, where $C$ is charge conjugation. Dong–Lin–Ng's congruence and Galois theorem gives a representation of $\mathrm{SL}_2(\mathbb{Z}/24\mathbb{Z})$ with entries in $\mathbb{Q}(z)$; see arXiv:1201.6644, Theorem II and Section 4. All labels in this section are the zero-based labels of the frozen rank-26 restriction list; the vacuum is label 1.

## 8.1 Two inexpensive duality obstructions

**Lemma 8.1** (Determinant obstruction)**.** *If a modular representation has $s^2 = C$ and $(st)^3 = C$, then $\det C = (\det t)^{-6}$. For the retained spectrum of $\mathcal{B}$, the number $r$ of self-dual simples satisfies $r \equiv 26 \pmod 4$.*

*Proof.* Taking determinants gives $\det s = (\det t)^{-3}$. The twist exponents in $T_{\mathcal{B}}$ sum to 258, so $\det t = z^{258+26} = z^{20}$ and $\det C = 1$. An involution on 26 labels with $r$ fixed labels has determinant $(-1)^{(26-r)/2}$. The value $r = 12$ is therefore impossible; the two remaining values are 10 and 14. □

For every $a$ coprime to 24, $a^2 = 1 \pmod{24}$. Hence the signed Galois matrix is

$$G_a = \sigma_a(s)s^{-1} = \rho(aI).$$

It is central, involutory, and commutes with $s, t$. Its simultaneous eigenspaces have rational bases. If $G_a$ acts on such a space by a character $\chi(a)$, each entry of its $s$ block transforms by that character. Write $\chi_i, \chi_2, \chi_3$ for the characters belonging to $i, \sqrt{2}, \sqrt{3}$.

The exact character-table calculation described below forces a 12-dimensional trivial-character block, a six-dimensional $\chi_2$ block, a six-dimensional $\chi_i$ block, and a final two-dimensional block. That last block has character $\chi_i$ when the two odd $2d$ pairs are dual pairs, and $\chi_3$ when both pairs are self-dual.

**Lemma 8.2** (Rational-square obstruction)**.** *The self-dual odd-$2d$ branch is impossible. Consequently $\mathcal{B}$ has exactly ten self-dual simples.*

*Proof.* In the self-dual branch use the equal-norm rational basis $e_{12} - e_{13}, e_{14} - e_{15}$. On this $\chi_3$ space,

$$t = \operatorname{diag}(i, iw), \qquad s = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad w = e^{2\pi i/3},$$

and $a, b, c \in \sqrt{3}\,\mathbb{Q}$. The block is irreducible, so $b \neq 0$. The relation $s^2 = I$ gives $c = -a$. The off-diagonal entry of $sts = t^{-1}st^{-1}$ gives

$$a = \frac{1}{i(iw)(i - iw)} = \frac{1}{\sqrt{3}}, \qquad b^2 = \frac{2}{3}.$$

Thus $(b/\sqrt{3})^2 = 2/9$, impossible for a rational number. This is a field obstruction, not a numerical approximation. □

## 8.2 Exact determination of every $S$ entry

The replay first constructs and verifies the class algebras and character tables of $\mathrm{SL}_2(\mathbb{Z}/8\mathbb{Z})$ and $\mathrm{SL}_2(\mathbb{Z}/3\mathbb{Z})$. Their 210 tensor-product irreducibles exhaust the irreducibles at level 24. There are 718 dimension-26 decompositions with the prescribed $t$ spectrum. Signed Galois permutations and the known nonreal-character rows leave eight. The known $\chi_2$ block has $\mathrm{Tr}(st) = 0$ and leaves four. The preceding rational-square obstruction removes the self-dual branch.

On the mandatory $\chi_i$ six-space, take the pair-difference basis

$$(2,5),\ (3,6),\ (4,7),\ (20,21),\ (16,17),\ (18,19),$$

where $(a, b)$ denotes $e_a - e_b$. The block is

$$s = \frac{i}{2}\begin{pmatrix} 0 & 0 & 1 & -1 & -1 & -1 \\ 0 & 0 & -1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 1 & -1 \\ -1 & 1 & 0 & 0 & 1 & -1 \\ -1 & -1 & 1 & 1 & 0 & 0 \\ -1 & -1 & -1 & -1 & 0 & 0 \end{pmatrix}.$$

The last two rows are known low rows. Their modular relations leave three rational variables; their square relations force them to $0, 1/2, 0$. The additional $\chi_i$ two-space has $s = iI$.

Only the rational 12-dimensional block remains. Its basis is

$$e_1 - e_{23},\ e_0 - e_{22},\ e_2 + e_5,\ e_3 + e_6,\ e_4 + e_7,\ e_{20} + e_{21},$$
$$e_8,\ e_9,\ e_{10},\ e_{11},\ e_{12} + e_{13},\ e_{14} + e_{15}.$$

Let $P$ have these basis vectors as columns, $N = P^{\mathsf{T}}P = \mathrm{diag}(2, 2, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2)$, and $H = P^{\mathsf{T}}sP$. Its known first two rows, together with their square and modular relations, leave ten rational variables. Two further linear equations from the full system force one variable to zero. Expanding the remaining modular relations in the cyclotomic basis then gives 187 rational polynomials of degree at most two in the nine remaining variables. Row reduction on their monomials produces the nine linear consequences

$$\begin{array}{lll} h_{7,11} = -1/3, & h_{7,7} = -1/6, & h_{7,8} = 1/3, \\ h_{7,9} = 1/3, & h_{8,11} = 2/3, & h_{8,8} = -1/6, \\ h_{8,9} = -1/6, & h_{9,11} = -2/3, & h_{9,9} = -1/6. \end{array}$$

Substitution satisfies all the original equations. Thus the solution is unique; the final replay uses rational linear elimination, not a Gröbner calculation. The integer matrix $6H$ and all basis conventions are recorded in `matrix_blocks.json`. Its trace is $\mathrm{Tr}(N^{-1}H) = -2$, selecting the $3 + 9$ representation type and excluding the $3 + 3 + 3 + 3$ alternative.

**Theorem 8.3** (Unique modular candidate)**.** *In the retained normalization and labels, the necessary rank-26 modular datum is unique. Its modular representation has irreducible dimensions* $9, 6, 4, 3, 2, 1, 1$*. Every one of its* $26^3$ *Verlinde coefficients is a nonnegative integer, with maximum 9.*

*Proof.* The preceding block determinations exhaust the 26-dimensional space. For each block its contribution to the original normalized matrix is $PN^{-1}HN^{-1}P^{\mathsf{T}}$. Their sum is therefore forced. The exact replay checks every matrix entry of $s^2 = C$ and $sts = t^{-1}st^{-1}$, all known low rows, and every Verlinde coefficient in $\mathbb{Q}(z)$. A separate implementation verifies all fusion associativity identities and exact diagonalization by the reconstructed $s$, as well as direct Verlinde computations for four selected fusion matrices. These are checks of the forced candidate, not associators for a fusion category. □

### 8.3 What the completed data pass

All 456,976 ordered four-object Vafa equations hold exactly modulo 24. The ordinary second indicators agree with duality: the ten self-dual simples have indicator $+1$ except label 24, which has indicator $-1$. It occurs with multiplicity zero in every $U^* \otimes U$ for simple $U$. The 182 ordinary cyclic spectra for the 26 simples at orders $2, 3, 4, 6, 8, 12, 24$ have nonnegative integral eigenvalue multiplicities. All 676 products also agree with the frozen restriction functor to the original rank-six ring.

The full rank-52 matrix for $Z(\mathcal{C})$ gives generalized-indicator checks for every central target. All 7,176 cyclic spectra for 52 central targets, six original simples and orders 2 through 24 pass, together with 312 degree-one normalization identities. For $\mathcal{B}$ itself, all 17,576 generalized second-indicator blocks and all 17,576 degree-one identities pass. The trace formula is Ng–Schauenburg, arXiv:0806.2493, Corollary 5.6, with Lemmas 2.5–2.7 and Proposition 2.8. In particular a target of twist $\theta$ requires the normalization $E^n = \theta^{-1}$ before cyclic Fourier inversion. These are finite necessary checks. The exponent 24 alone does not turn them into a proof of feasibility at every tensor order.

No known category has been identified with these data. Matching dimensions alone is insufficient, and the near-group constructions discussed next do not provide the required condensation.

## 9 Categorical consequences of the rank-26 reconstruction

The results below give a construction obstruction and restrictions on algebras in the reconstructed fusion ring. They do not establish or exclude a categorification of $R^{(6)}_{4,155}$. For the rank-26 category $\mathcal{B}$, we use the numbering of the accompanying reconstruction: the unit is object 1, the fermion is object 0, and

$$d = 4 + 3\sqrt{2}, \qquad b = 2 + 4d = 18 + 12\sqrt{2}, \qquad A = 23, \quad L = 25,$$

where $\mathrm{FPdim}(A) = b - 1$, $\mathrm{FPdim}(L) = b$, and $\mathrm{FPdim}(\mathcal{B}) = 8b^2$. All assertions about $\mathcal{B}$ are conditional on a positive ribbon categorification of the reconstructed data.

### 9.1 The order-32 near-group centers cannot give the target by dimension-eight condensation

Radchenko–Wheeler construct complex near-group categories of type $(G, |G|)$ for every product of two cyclic groups; unitarity is not asserted. Their primary source is *Real quadratic fields and finite quantum dilogarithms I*, `https://arxiv.org/abs/2609.21892`, version 2, Section 5.4, Theorem 12 and Corollary 6. Thus their construction supplies the three order-32 groups $C_{32}$,

$C_2 \times C_{16}$, and $C_4 \times C_8$. The following argument applies directly to their algebraic construction; it does not import a classification of unitary centers.

**A half-braiding lemma.** In the strict endomorphism realization, write the simple objects as $\alpha_g$ and $\rho$. The displayed identities in the cited Section 5.4 include

$$\alpha_g\rho = \rho, \quad \rho\alpha_g = \mathrm{Ad}(U_g)\rho, \quad U_gU_h = U_{g+h}, \quad \alpha_g(U_h) = \chi(g,h)U_h,$$

where $\chi$ is a nondegenerate symmetric bicharacter. There is also a nonzero split inclusion $T : \rho \longrightarrow \rho^2$. If a central object $Z$ has multiplicity-free pointed underlying object $\bigoplus_{g\in H} \alpha_g$, then every one of its summands is central, with self-braiding $\chi(g,g)^{-1}$.

*Proof.* Its half-braiding with $\alpha_h$ is diagonal, with scalar $a(g,h)$ on label $g$. Write the entry of its half-braiding with $\rho$ from label $g$ to label $k$ as $M_{kg}U_k$. Naturality at $U_h : \rho \to \rho\alpha_h$ and the hexagon give

$$U_hM_{kg}U_k = a(k,h)M_{kg}U_k\alpha_g(U_h).$$

Thus $M_{kg} \neq 0$ implies $a(k,h)\chi(g,h) = 1$ for every $h$. Nondegeneracy of $\chi$ permits at most one nonzero entry in each row of $M$. Since $M$ is invertible, it is monomial, with a permutation $\pi$ of the labels $H$. The hexagon for $\rho^2$ has output label $\pi^2(g)$, whereas naturality at $T$ compares it with a map having output label $\pi(g)$. Both maps are nonzero, being compositions of isomorphisms and split inclusions. Projection onto the tagged direct summands therefore gives $\pi^2(g) = \pi(g)$. Hence $\pi$ is the identity. All half-braidings preserve each summand, and the preceding scalar identity gives $a(g,h) = \chi(g,h)^{-1}$. □

**Proposition.** None of these three order-32 Radchenko–Wheeler categories has a connected étale algebra of dimension eight in its center. Consequently their centers cannot produce the proposed center of $R^{(6)}_{4,155}$ by a single connected-étale condensation.

*Proof.* Use the established two-dimension reduction to pass to a positive spherical Galois representative; a hypothetical algebra transports under the same conjugation. Put

$$r = 16 + 12\sqrt{2} = 4d, \qquad \Delta = 32 + r^2 = 8(72 + 48\sqrt{2}).$$

The formal-codegree decomposition of $I(1)$ has constituent dimensions $1$, $r+1$, and $31$ copies of $r+2$. Indeed the 31 nontrivial group characters have formal codegree 32, while the two remaining characters send $\rho$ to its two quadratic roots. In particular every nonunit constituent of $I(1)$ has dimension greater than eight.

Suppose $E \in Z(\mathcal{C})$ is connected étale of dimension eight. Its underlying object is pointed, since $r > 8$. Adjunction and the preceding constituent bound give

$$\dim \mathrm{Hom}_{\mathcal{C}}(1, F(E)) = \dim \mathrm{Hom}_{Z(\mathcal{C})}(I(1), E) = 1.$$

Thus $F(E)$ is a connected separable algebra in the pointed subcategory. It is simple as a right module over itself, since its endomorphism space is $\mathrm{Hom}_{\mathcal{C}}(1, F(E))$. Each $\alpha_g \otimes F(E)$ is also simple. Free-module adjunction now shows

$$[F(E) : \alpha_g] = \dim \mathrm{Hom}_{F(E)}(\alpha_g \otimes F(E), F(E)) \in \{0, 1\}.$$

Its support is the stabilizer subgroup $H$ of this simple module and has order eight. The lemma splits $E$ into central invertibles. Twist-freeness of a connected étale algebra then forces $\chi(h,h) = 1$ for every $h \in H$.

For completeness the required bicharacter count is elementary and uniform over all nondegenerate choices. On $C_{32}$, $\chi(x,x) = \exp(2\pi icx^2/32)$ with $c$ odd, so the bosons are exactly the four multiples of eight. On $C_2 \times C_{16}$, write diagonal parameters $p/2, q/16$ and off-diagonal

parameter $r_0/2$. Nondegeneracy forces $p = 1$ and $q$ odd: otherwise the order-16 generator maps to a character of smaller order, or the order-two generator is in the pairing kernel or has the same character as eight times that generator. The boson equation is $8a + qb_0^2 = 0 \pmod{16}$; it forces $a = 0$ and $4 \mid b_0$. On $C_4 \times C_8$, write parameters $p/4, q/8, r_0/4$. Here $p, q$ must be odd: if $q$ is even the order-eight character has smaller order, while if $p$ is even the nonzero element $(2, 4r_0)$ is in the kernel. The boson equation

$$2pa^2 + qb_0^2 + 4r_0ab_0 = 0 \pmod 8$$

first forces $b_0$ even, then $a$ even, and finally $4 \mid b_0$. There are again only four bosons. No required subgroup $H$ exists. Finally, the source-center dimension is $\Delta^2 = 64 \dim(\mathcal{C}_{155})^2$; the local-module dimension formula forces a condensation to the target center to use an algebra of dimension eight. □

This excludes all dimension-eight connected étale algebras in these centers, not merely pointed candidates. It does not exclude other source categories, products followed by condensation, or the unresolved near-group type $(8, 16)$.

## 9.2 The only possible connected étale underlying objects in $\mathcal{B}$

**Proposition.** Every connected étale algebra in a positive ribbon categorification of the reconstructed rank-26 data has underlying object

$$1, \qquad 1 + A, \qquad 1 + A + L.$$

This statement does not assert uniqueness of algebra structures.

*Proof.* The twist-one simples are precisely $1, 2, 5, 23, 25$, with $2^* = 5$. The reconstructed fusion rules give

$$2 \otimes 2 = 9 + 10 + 16 + 19, \qquad 2 \otimes 5 = 1 + 8 + 11 + 22.$$

The first product has no twist-one constituent; the second has only the unit among such constituents. A connected étale algebra has a nondegenerate Frobenius pairing. If it contained object 2, it would contain 5, and multiplication of suitable copies of 2 and 5 to the unit would be nonzero. Its multiplication on $2 \otimes 2$ would vanish. Associativity on $2 \otimes 2 \otimes 5$ is then impossible: multiplying the first two factors gives zero, whereas contracting the last two to the unit and multiplying by the first gives a nonzero map. This argument also works with arbitrary multiplicities, by choosing a nonzero entry of the pairing on the multiplicity spaces. Thus neither 2 nor 5 can occur.

Write the remaining candidate as $E = 1 + sA + tL$, with $s, t \geq 0$. Its dimension is $1 - s + (s + t)b$, at most $\sqrt{\mathrm{FPdim}(\mathcal{B})} = 2\sqrt{2}\, b$. If $s + t \geq 3$, its dimension is at least $3b - 2$, which exceeds this bound by four. Hence $s + t \leq 2$. Apart from the three stated objects, the possibilities have dimensions $b + 1$, $2b - 1$, and $2b + 1$, with norms

$$N(b + 1) = 73, \qquad N(2b - 1) = 73, \qquad N(2b + 1) = 217.$$

But

$$N(\mathrm{FPdim}(\mathcal{B})) = N(8b^2) = 82944 = 2^{10}3^4.$$

The number $\mathrm{FPdim}(\mathcal{B})/\mathrm{FPdim}(E)^2$ must be an algebraic integer, being the dimension of the local-module category. Its norm would require $73^2$ or $217^2$ to divide 82944, which is false. □

The standard algebra facts used here—the Frobenius pairing, twist-freeness in the positive setting, and local-module dimension—are discussed in Davydov–Müger–Nikshych–Ostrik, *The Witt group of non-degenerate braided fusion categories*, Section 3, `https://arxiv.org/abs/1009.2117`.

**Fusion subcategories and restriction.** There are exactly four fusion subcategories: Vec, sVec, the rank-16 fermion centralizer, and $\mathcal{B}$. This conclusion follows directly from a few support properties of the full fusion table. Every noninvertible simple other than $4, 7$ has 22 in its product with its dual. For $4, 7$, that product contains 8, whose product with its dual contains 22. The square of 22 contains every even simple except the fermion; it contains 23, and $22 \otimes 23$ contains the fermion as well. Thus every noninvertible even simple generates the entire even category. A subcategory containing it and an odd object is faithfully $C_2$-graded with that even component. Equality of the dimensions of grading components forces it to be all of $\mathcal{B}$. The accompanying integer checker also verifies every one of the 676 restriction-product identities

$$W(i)W(j) = \sum_k N_{ij}^k W(k)$$

in the original rank-six ring. Thus the full reconstructed fusion rules pass the restriction-functor consistency check for every product. The checks and exact support sets are recorded in `full_fusion_categorical_receipt.json`.

### 9.3 The canonical intermediate condensation adds no new near-group category

When $\mathcal{B}$ comes from a categorification $\mathcal{C}$ of the rank-six ring, the restricted adjoint algebra is $E = 1 + A + L$. Its even subalgebra $E_0 = 1 + A$ has dimension $b$. The local category $\mathcal{K} = \mathcal{B}_{E_0}^0$ has dimension eight. The summand $L$, as an $E_0$-module, is an invertible boson $\ell$; the free fermion $f_0 = fE_0$ is distinct, and $f_0\ell \neq \ell$ has the same underlying object $L$. Their twists $1, -1, 1, 1$ give a toric-code factor. Preservation of central charge then gives

$$\mathcal{K} \simeq \mathrm{ToricCode} \boxtimes \mathrm{Semion}^{\mathrm{rev}}.$$

In fact the category of all modules $\mathcal{D} = \mathcal{B}_{E_0}$ is tensor equivalent to $\mathcal{C} \boxtimes \mathrm{Rep}(C_2)$. Here are details preventing an inference from merely fixing simple objects. The boson $\ell$ realizes $\mathcal{D}$ as a $C_2$-equivariantization of $\mathcal{C}$. Its action fixes $X, Y$, since each is the free image of a $d$-dimensional simple of $\mathcal{B}$. It fixes every invertible as well, since their unique central lifts have distinct twists $1, -1, i, -i$. Consequently $\mathcal{D}$ has eight invertibles and four simples of dimension $d$.

The local category $\mathcal{K}$ supplies all eight invertibles, while $Z(\mathcal{D}) \simeq \mathcal{B} \boxtimes \mathcal{K}^{\mathrm{rev}}$ has sixteen. The kernel of the central forgetful map, and hence the character group of the universal grading group $U$, has order two. Equal dimensions of grading components, by comparing coefficients in $a + cd^2 = (16 + 32d)/|U|$, imply $c = 4/|U|$. Thus $|U|$ divides four; such a group is abelian, and therefore $U = C_2$. The adjoint component has four invertibles and two noninvertibles. Its two noninvertibles must map respectively to $X, Y$ under de-equivariantization: two mapping to $X$ would contradict the occurrence of $Y$ in $X^2$, and conversely. Their products force four distinct invertible images. Hence $\mathcal{D}_{\mathrm{ad}} \to \mathcal{C}$ is a tensor equivalence. The central boson $\ell$ lies outside this component and gives the asserted tensor-product decomposition. This canonical construction therefore supplies no exchange action producing a near-group category of type $(8, 16)$.

## 10 Odd cyclic Clifford constraints for the non-self-dual residual

The anticommuting stabilizers in Proposition 5.21 constrain every odd pivotal rotation, including rotations with an arbitrary central target. The result gives a uniform necessary condition for the residual ring $R_{4,155}^{(6)}$. Its complete tested range passes this condition.

**Theorem 10.1** (Odd cyclic Clifford factorization)**.** *Assume the hypotheses of Proposition 5.21 for an object $Y$. Let $n \geq 3$ be odd, let $Z$ be a simple of $Z(\mathcal{C})$, and let $E$ be the generalized*

*pivotal rotation on*

$$V = \mathrm{Hom}_{\mathcal{C}}(F(Z), Y^{\otimes n}), \qquad E^n = \theta_Z^{-1} I.$$

*Choose an $n$th root of $\theta_Z$ and use it to normalize $E$ to $E_0^n = I$. Put $r_n = 2^{(n-1)/2}$ and define integers $c_j$ by*

$$c_j = \#\left\{ (\epsilon_1, \ldots, \epsilon_{(n-1)/2}) \in \{\pm 1\}^{(n-1)/2} : \frac{1}{2} \sum_{a=1}^{(n-1)/2} a\epsilon_a \equiv j \pmod{n} \right\},$$

*where division by two takes place in $\mathbb{Z}/n\mathbb{Z}$. If $m_j$ is the multiplicity of $\zeta_n^j$ as an eigenvalue of $E_0$, then there are nonnegative integers $b_j$ such that*

$$m_j = \sum_{k \in \mathbb{Z}/n\mathbb{Z}} c_{j-k} b_k, \qquad \sum_j b_j = \frac{\dim V}{r_n}. \tag{10.1}$$

*The circulant convolution matrix $(c_{j-k})$ is invertible over $\mathbb{Q}$, so the integers $b_j$, when they exist, are uniquely determined.*

*Proof.* Use the normalized left and right stabilizer actions and pair-insertion endomorphism $a$ from Proposition 5.21. Insertions at adjacent positions anticommute, while insertions on disjoint pairs commute, and every insertion squares to the identity. Naturality of generalized rotation in the $n-1$ unmoved tensor factors transports one ordinary adjacent insertion to the next. The rotation definitions and naturality are those of [33] and Ng–Schauenburg, *Congruence subgroups and generalized Frobenius–Schur indicators*, arXiv:0806.2493, Lemmas 2.5–2.7. Define the cyclic boundary insertion by conjugation with $E$. Since $E^n$ is scalar, conjugation has exact period dividing $n$. Thus the $n$ operators $A_j$, indexed modulo $n$, satisfy

$$A_j^2 = I, \qquad A_j A_k = -A_k A_j \quad (j-k = \pm 1), \qquad A_j A_k = A_k A_j \quad (j-k \neq 0, \pm 1).$$

For $n = 3$, the three distinct pairs all anticommute. For $n \geq 5$, any pair can be rotated away from the cyclic boundary, reducing these identities to the adjacent or disjoint insertion calculations.

The abstract algebra with these relations has the ordered monomials $A_0^{e_0} \cdots A_{n-1}^{e_{n-1}}$, $e_j \in \{0, 1\}$, as a basis. Its commutator form over $\mathbb{F}_2$ is the adjacency matrix of an $n$-cycle. Its radical is the line of the all-ones vector: the radical equations are $e_{j-1} = e_{j+1}$, and $n$ is odd. Hence the algebra is $M_{r_n}(\mathbb{C}) \oplus M_{r_n}(\mathbb{C})$.

For a concrete model, use Clifford generators $\gamma_j$ and put $A_j = i\gamma_j\gamma_{j+1}$, with cyclic indices. Replacing every $A_j$ by $-A_j$ gives the other simple block. The cyclic permutation of the $\gamma_j$ is an orthogonal transformation of determinant one. Its spin lift implements $A_j \mapsto A_{j+1}$ on either block. Choose the unique sign of that lift for which its $n$th power is one, and call the resulting operator $U$. The nontrivial orthogonal rotation angles are $2\pi a/n$, $1 \leq a \leq (n-1)/2$. The exterior-algebra model of the spin representation therefore gives eigenvalues

$$\zeta_n^{\frac{1}{2}\sum_a a\epsilon_a}, \qquad \epsilon_a \in \{\pm 1\},$$

with the exponent interpreted modulo $n$. This includes the sign normalization $U^n = I$, and gives exactly the multiplicities $c_j$. Both simple algebra blocks have this same spectrum.

The operators $E_0$ and $U$ implement the same algebra automorphism. Their quotient $U^{-1}E_0$ consequently acts on the multiplicity spaces, commutes with $U$, and has $n$th power one. Tensoring its cyclic characters with the spin character proves (10.1). A different chosen $n$th root of $\theta_Z$ shifts all cyclic labels, and reversing the rotation reverses them; neither change affects feasibility.

The eigenvalues of the circulant matrix are the traces of the powers of $U$. Every power of an odd cycle is a permutation with only odd cycles, so it has no orthogonal eigenvalue $-1$. The spin-character product of half-angle cosines is therefore nonzero. This proves invertibility. □

**Proposition 10.2** (All-order path divisibility for $R'(m)$)**.** *Under the same stabilizer hypotheses, every simple multiplicity in $Y^{\otimes n}$ is divisible by $2^{\lfloor (n-1)/2 \rfloor}$ for all $n \geq 1$. For the based rings $R'(m)$ of (7.1), the corresponding infinite family of fusion-coefficient divisibilities holds if and only if $m$ is even.*

*Proof.* The $n-1$ ordinary adjacent insertions generate the twisted group algebra whose commutator form is the adjacency matrix of a path with $n-1$ vertices. This matrix has rank $2\lfloor (n-1)/2 \rfloor$ over $\mathbb{F}_2$. Every simple module consequently has dimension $2^{\lfloor (n-1)/2 \rfloor}$. Restricting the insertion action to each simple multiplicity space proves the first assertion.

Write $x = [X]$, $y = [Y]$ in $R'(m)$. Direct multiplication gives

$$x^3 = m(1 + h + g + hg) + 2m^2 x + (2 + 2m^2)y.$$

Thus the divisibility at $n = 3$ requires $m$ even. Conversely, suppose $m$ is even. The coefficients of $x^3$ and $x^4$ are even, and the fusion relations give

$$x^4 = 2mx^3 + 4mx + 2(1 + h), \qquad x^n = 2mx^{n-1} + 4mx^{n-3} + 4x^{n-4} \quad (n \geq 5).$$

If the required divisibilities hold for smaller positive powers, the three terms on the right have respective two-adic divisibilities at least

$$2 + \left\lfloor \frac{n-2}{2} \right\rfloor, \qquad 3 + \left\lfloor \frac{n-4}{2} \right\rfloor, \qquad 2 + \left\lfloor \frac{n-5}{2} \right\rfloor.$$

Each is at least $\lfloor (n-1)/2 \rfloor$. The cases $n = 1, 2, 3, 4$ start the induction. Duality proves the same statement for $y$. □

**Exact scope for $R^{(6)}_{4,155}$.** The anticommuting-stabilizer hypothesis holds for $h$ and each of $X, Y$. The generalized rotation spectra from Section 8 provide the input to (10.1). Exact rational inversion checks both noninvertibles, all 52 central targets, and every odd order $3, 5, \ldots, 23$: all 1144 blocks have nonnegative integral quotient multiplicities. At order three the spin multiplicities are $(0, 1, 1)$, so the quotient is explicitly $b_j = \dim(V)/2 - m_j$. The all-order path condition likewise holds because $m = 4$ is even. These assertions give necessary constraints and exact surviving data; they supply neither associators nor a categorification or exclusion.

The source-bound checker is `evidence/rigidity_20261017/rank6/r155/verify_odd_clifford.py`. It pins the full generalized-rotation certificate and source tensor by SHA256, reconstructs each odd spin spectrum, solves every rational convolution exactly, and rejects nonintegral or negative quotients. Its negative controls run under both ordinary and optimized Python. From the repository root, run

```
python evidence/rigidity_20261017/rank6/r155/verify_odd_clifford.py \
  --repo .
```

The replay verifies these finite character assertions, not the categorical arguments of Theorem 10.1.

# 11 Finite Morita reduction for $R^{(6)}_{8,1660}$

**Proposition 11.1** (A general formal-codegree test for modules)**.** *Let $\mathcal{F}$ be pseudo-unitary with commutative Grothendieck ring. For an indecomposable module category, let $m_\chi$ be the multiplicity of the character $\chi$ in its complexified based module, and let $f_\chi$ be the formal codegree. Then*

$$\sum_\chi \frac{m_\chi}{f_\chi} \leq 1.$$

*If equality holds, the Morita dual has rank $\sum_\chi m_\chi^2$, and its action is faithful and equals the full commutant. In this case its simple action matrices must give a square nonnegative integral Gram factorization of the linking-category matrix below. Failure of that bounded factorization is an obstruction to the based module being a module category.*

*Proof.* Ostrik's module-character reciprocity identifies $m_\chi$ with the multiplicity in the module's Lagrangian algebra of the central simple of dimension $\dim(\mathcal{F})/f_\chi$. Positivity and the dimension of that algebra give the inequality. Equality exhausts its support inside the canonical algebra. Its endomorphism algebra is consequently $\bigoplus_\chi M_{m_\chi}(\mathbb{C})$ and acts faithfully on the module Hom space. The next proof details these identifications and their precise primary source. The linking associativity identity then gives the stated Gram factorization. □

This note gives necessary module-category conditions, not an existence or nonexistence theorem. No braiding on $\mathcal{C}$ or its relative center is assumed. Let $\mathcal{C}$ have simples $1, g, X, gX, \sigma, T = \sigma X$ and rules

$$g^2 = 1, \quad g\sigma = \sigma, \quad \sigma^2 = 1 + g, \qquad X^2 = 1 + (3(1+g) + 4\sigma)X.$$

Put $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$, so $\mathrm{FPdim}\,\mathcal{C} = 4(1+d^2)$. All matrices below use the action on the basis of simple objects of a finite semisimple module category.

**Lemma 11.2.** *Every indecomposable based module over the Ising fusion ring, whose simple action matrices obey duality transpose, is its three-object regular module. Consequently every $\mathcal{C}$-module category restricts to a direct sum of regular Ising module categories at the level of based modules.*

*Proof.* Write $H = N_\sigma$ and $G = N_g$. Then $G$ is a permutation involution, $H = H^{\mathsf{T}} \geq 0$, $GH = H$, and $H^2 = I + G$. An indecomposable component of $H$ has spectral radius $\sqrt{2}$: this follows either from its dimension vector or directly from these identities and $GH = H$. There are no entries of $H$ greater than one. There are no loops: a loop connected to any different vertex has norm at least the golden ratio, while an isolated loop contradicts $H^2 = I + G$. No vertex has degree three or more, and a path with four vertices already has norm greater than $\sqrt{2}$. Thus every connected component is the three-vertex path $A_3$; the possibilities of one or two vertices contradict $H^2 = I + G$. Its two leaves are exchanged by $G$ and its middle vertex is fixed. These are the regular Ising matrices. □

**Theorem 11.3** (Uniform module rank)**.** *Every indecomposable finite semisimple $\mathcal{C}$-module category has exactly six simple objects. Its restriction to $\mathcal{I} = \langle g, \sigma \rangle$ consists of two regular Ising blocks.*

*Proof.* Initially suppose there are $n$ blocks. Order each block as $1, g, \sigma$. The commutant of the two regular Ising matrices consists exactly of

$$J(a,b,c) = \begin{pmatrix} a & b & c \\ b & a & c \\ c & c & a+b \end{pmatrix}.$$

Since $X$ commutes with $g, \sigma$, and is self-dual, its action has form $N_X = P \otimes I + Q \otimes N_g + R \otimes N_\sigma$, where $P, Q, R$ are symmetric nonnegative integral $n$ by $n$ matrices. Set $U = P + Q$, $V = P - Q$. The $X^2$ relation gives

$$V^2 = I, \quad U^2 + 2R^2 = I + 6U + 8R, \quad UR + RU = 6R + 4U. \tag{11.1}$$

The matrix $M = U + \sqrt{2}R$ therefore satisfies

$$M^2 - (6 + 4\sqrt{2})M - I = 0. \tag{11.2}$$

It is irreducible: otherwise a union of Ising blocks would be invariant under all the generators of $K_0(\mathcal{C})$, contradicting indecomposability. Hence its positive Perron eigenvalue $d$ is simple. The polynomial in (11.2) is irreducible over $\mathbb{Q}(\sqrt{2})$: its discriminant is $72+48\sqrt{2}=(4\sqrt{3}+2\sqrt{6})^2$, whose square root does not belong to $\mathbb{Q}(\sqrt{2})$. Thus the characteristic polynomial of $M$ is a power of (11.2); simplicity of the Perron root forces that power to be one. Therefore $n=2$. □

**Theorem 11.4** (Complete based-module list)**.** *Up to interchanging the two Ising blocks, the possible matrices $U, R$ are exactly*

$$U=\begin{pmatrix} a & b \\ b & 6-a \end{pmatrix}, \qquad R=\begin{pmatrix} c & f \\ f & 4-c \end{pmatrix},$$

*with the following four rows. These are based-module possibilities; their associativity constraints have not been solved.*

| *type* | $a$ | $b$ | $c$ | $f$ | $V=P-Q$ |
|---|---|---|---|---|---|
| *I* | *0* | *1* | *0* | *0* | *signed transposition* |
| *II* | *2* | *3* | *2* | *2* | *signed transposition* |
| *III* | *1* | *2* | *0* | *1* | $\mathrm{diag}(\epsilon_1,\epsilon_2)$ |
| *IV* | *1* | *2* | *1* | *2* | $\mathrm{diag}(\epsilon_1,\epsilon_2)$ |

*Here $\epsilon_i\in\{1,-1\}$ and $P=(U+V)/2$, $Q=(U-V)/2$. The two signed transpositions give equivalent modules by swapping the two leaves in one Ising block. Thus there are ten labelled-$X$ based modules, or six after additionally allowing the ring automorphism $X\leftrightarrow gX$.*

*Proof.* Equation (11.2) and $n=2$ give $\operatorname{tr}U=6$ and $\operatorname{tr}R=4$. The matrix $V$ is symmetric integral and $V^2=I$, so it is a signed permutation. It must satisfy $V\equiv U \pmod 2$, with $U\pm V\geq 0$ entrywise. Put $w=a-3$, $s=c-2$. The two equations (11.1) become

$$ws+bf=6, \qquad w^2+b^2+2(s^2+f^2)=18. \tag{11.3}$$

All variables are integral; $|w|\leq 3$, $|s|\leq 2$, and $b,f\geq 0$. Cauchy–Schwarz implies $3\leq s^2+f^2\leq 6$. The only sums of two integral squares in this interval are 4 and 5. For value 4, the squared norm of $(w,b)$ is 10; for value 5, it is 8. Substitution in $ws+bf=6$ gives exactly the four rows and their block exchanges. Parity gives precisely the displayed options for $V$. Conversely these matrices are nonnegative integral and satisfy every fusion relation and transpose relation. Their $X$ action connects the two Ising blocks, so they are indecomposable based modules. □

**Corollary 11.5** (Four small algebra supports)**.** *Every indecomposable $\mathcal{C}$-module category has a connected separable internal-end algebra on one of the following underlying objects:*

$$1, \qquad 1+X+gX+2T, \qquad 1+X \text{ or } 1+gX, \qquad 1+X+T \text{ or } 1+gX+T. \tag{11.4}$$

*Consequently these supports exhaust the possible Morita classes, with no assertion that the nontrivial supports admit algebra structures or that different structures give different Morita classes.*

*Proof.* Choose a leaf $m$ in the first Ising block of each row. The coefficient of a simple $Z$ in $\underline{\mathrm{End}}(m)$ is $(N_Z)_{mm}$. Thus this internal end is $1+P_{11}X+Q_{11}gX+R_{11}T$. Substitution gives (11.4). The standard internal-Hom construction supplies a connected separable algebra and realizes the original module category as its modules. The first row has internal end 1 and is necessarily the regular module category, even categorically. □

The Perron dimension ratio of the second Ising block to the first is, respectively,

$$d, \quad 3-2\sqrt{2}+2\sqrt{3}-\sqrt{6}, \quad \sqrt{2}+\sqrt{3}, \quad (\sqrt{2}+\sqrt{6})/2.$$

The dimension quotients $\operatorname{FPdim}\mathcal{C}/\operatorname{FPdim}Q$ for the three nontrivial supports in (11.4) are, respectively,

$$144-96\sqrt{2}+80\sqrt{3}-56\sqrt{6}, \qquad 24+8\sqrt{6}, \qquad 12+4\sqrt{3}.$$

All are algebraic integers, so elementary dimension divisibility does not exclude these targets. The two choices of the second diagonal sign in types III and IV are not removed by the chosen algebra support. They may encode different algebra/module associativity structures.

**Theorem 11.6** (Full-center saturation and ranks of all Morita duals)**.** *Use the established reduction to the pseudo-unitary form of $\mathcal{C}$. For any indecomposable module category $\mathcal{M}$, the dual fusion category $\mathcal{D}=\operatorname{Fun}_{\mathcal{C}}(\mathcal{M},\mathcal{M})$ has rank six or eight. More precisely,*

| *matrix $V$* | $\operatorname{rank}\mathcal{D}$ | $K_0(\mathcal{D})\otimes\mathbb{C}$ |
|---|---|---|
| $\operatorname{spec}(V)=\{1,-1\}$ | 6 | $\mathbb{C}^6$ |
| $V=I$ *or* $-I$ | 8 | $\mathbb{C}^4\oplus M_2(\mathbb{C})$. |

*The action of $K_0(\mathcal{D})\otimes\mathbb{C}$ on $K_0(\mathcal{M})\otimes\mathbb{C}$ is faithful and is the full commutant of the $K_0(\mathcal{C})$ action.*

*Proof.* Write $D=\operatorname{FPdim}\mathcal{C}$ and let $A_\chi$ denote the six simple summands of the canonical algebra $I_{\mathcal{C}}(1)$. They occur once, because $K_0(\mathcal{C})$ is commutative. Ostrik's Theorem 2.13 gives $\dim A_\chi = D/f_\chi$. Let $L_{\mathcal{M}}$ be the Lagrangian algebra in $Z(\mathcal{C})$ corresponding to $\mathcal{M}$; under the Morita equivalence of centers it is $I_{\mathcal{D}}(1)$. The precise module-character reciprocity needed here is given by Ostrik's Remark 2.19: the bimodule $K_0(\mathcal{M})\otimes\mathbb{C}$ is identified with

$$\operatorname{Hom}_{Z(\mathcal{C})}(I_{\mathcal{C}}(1),L_{\mathcal{M}})$$

over the corner algebras $\operatorname{End}(I_{\mathcal{C}}(1))$ and $\operatorname{End}(L_{\mathcal{M}})$. This follows by applying that remark to the linking multi-fusion category of $\mathcal{C}$ and $\mathcal{M}$. Its twist hypothesis is automatic in the pseudo-unitary case, as stated at the end of the same remark. Thus the multiplicity of $\chi$ in the module character is exactly $[L_{\mathcal{M}}:A_\chi]$.

The four characters with $g=1$ each occur once: each of the two Ising spin evaluations gives a two-dimensional matrix with its two distinct roots. The two characters with $g=-1$, $\sigma=0$, $X=\pm1$ occur with multiplicities $m_+,m_-$ equal to the eigenvalue multiplicities of $V$; hence $m_++m_-=2$. Both have formal codegree 4. Consequently the dimensions of all the already specified constituents of $L_{\mathcal{M}}$ sum to exactly $D$, just as those of $I_{\mathcal{C}}(1)$ do. Since $\dim L_{\mathcal{M}}=D$ and all dimensions are positive, there are no additional constituents. Therefore

$$L_{\mathcal{M}}=\bigoplus_{\chi:g=1}A_\chi \;\oplus\; m_+A_+\oplus m_-A_-, \qquad \dim\operatorname{End}(L_{\mathcal{M}})=4+m_+^2+m_-^2.$$

Ostrik's Corollary 2.16 identifies this endomorphism algebra with the complexified Grothendieck algebra of $\mathcal{D}$, proving the table. Every simple constituent of $L_{\mathcal{M}}$ also occurs in $I_{\mathcal{C}}(1)$. Consequently the action of $\operatorname{End}(L_{\mathcal{M}})$ on the displayed Hom space sees every matrix block and is faithful. Its block decomposition also shows that it is the full commutant. □

**A first forced Morita-dual ring.** For type III with $V=\operatorname{diag}(1,-1)$, the rank-six dual, if the module category exists, has the original fusion ring $R^{(6)}_{8,1660}$. Here is a short exact reconstruction. Write $N_x$ for the six known action matrices on $\mathcal{M}$, and $Q_y$ for the unknown

right action matrices of the dual. Associativity in the linking multi-fusion category and Frobenius reciprocity imply

$$\sum_x N_x[a,b]N_x[c,d] = \sum_y Q_y[c,a]Q_y[d,b]. \tag{11.5}$$

The central fermion of $\mathcal{C}$ gives a dual invertible whose action is $N_g$, so the two matrices $I, N_g$ are fixed. Choose a leaf $a$ in the first Ising block. The diagonal entry of (11.5) at the coordinate $(a,a)$ is 2; after removing $I$, precisely one further simple matrix $Q_Y$ has entry 1 there. It is therefore the corresponding Gram column minus the column of $I \otimes I$. Its $g$-translate is another simple. After removing these four matrices, the residual Gram matrix has rank two. At coordinate $(a, \sigma a)$ its diagonal is 2, so both remaining matrices have entry 1. That Gram column is their sum $v$; if $H$ is the residual Gram matrix, $2H - vv^{\mathsf{T}}$ is the outer product of their difference. This determines them up to exchange. The resulting matrices, called $Q_Z, Q_s$, satisfy

$$s^2 = 1 + g, \qquad sY = Z, \qquad Y^2 = 1 + 3Y + 3gY + 4Z.$$

The exact script `reconstruct_type_iii.py` verifies (11.5), nonnegative integrality, linear independence, and all fusion products. Thus this branch is a possible Morita self-partner, not an exclusion.

**Theorem 11.7** (Complete integral Morita Gram reduction)**.** *The two type-IV based modules with $V = I$ or $V = -I$ cannot be module categories. For the other types, equation* (11.5)*, the central fermion, and the based-ring axioms leave exactly the following action bases. Set*

$$a_0 = 2 + \sqrt{2} + \sqrt{3} + \sqrt{6}, \quad b_0 = 2 + 2\sqrt{2} + \sqrt{3} + \sqrt{6}, \quad c_0 = 5 + 4\sqrt{2} + 3\sqrt{3} + 2\sqrt{6}.$$

| *module type* | *number* | *dimensions of forced dual rings* |
|---|---|---|
| *II* | *1* | $1, 1, \sqrt{2}, d, d, \sqrt{2}d$ |
| *III, mixed signs* | *1* | $1, 1, \sqrt{2}, d, d, \sqrt{2}d$ |
| *III, equal signs* | *2* | $1, 1, d, d, a_0, a_0, b_0, b_0$ |
| *IV, mixed signs* | *2* | $1, 1, \sqrt{2}, d, d, \sqrt{2}d$ *or* $1, 1, 1+\sqrt{3}, a_0, a_0, c_0$ |
| *IV, equal signs* | *0* | *none.* |

*The three occurrences of the first dimension list have the original ring. The second rank-six ring is self-dual with commuting generators $g, S, Y$ satisfying*

$$g^2 = 1, \quad gS = S, \quad S^2 = 1 + g + 2S, \qquad Y^2 = 1 + (1 + g + 2S)Y.$$

*Its maximal structure constant is* 16*. The two rank-eight rings have maximal structure constant* 3 *and are not isomorphic: in one the $b_0$ pair is a dual pair and all other simples are self-dual; in the other the $a_0$ pair is a dual pair and all other simples are self-dual.*

*Proof.* Here is a complete finite calculation with explicit bounds. It is implemented in `reconstruct_dual_actions.py`; its output is `dual_actions_receipt.json`. Flatten action matrices to vectors. Equation (11.5) gives a known nonnegative symmetric 36 by 36 matrix $K = \sum_x N_x \otimes N_x$. Subtract the outer products of the known matrices $I, N_g$. For type III also subtract the two matrices forced in the preceding reconstruction. The residual matrix $H$ has rank $r$, equal to the number of remaining dual simples.

Choose $r$ coordinates whose principal submatrix $B$ is invertible. Any remaining simple column $q$ is determined by its coordinates $v$ there: $q = H_{*,J}B^{-1}v$. Because there are exactly $r$ remaining columns, their square factorization of $B$ implies

$$v^{\mathsf{T}}B^{-1}v = 1, \qquad 0 \le v_i \le \lfloor \sqrt{B_{ii}} \rfloor.$$

Conversely, an orthogonal basis of $r$ such vectors, lifting to nonnegative integral columns, gives the whole Gram factorization. This is a bounded integer ellipsoid enumeration, followed by a clique search for orthogonal bases. The numbers of candidate columns and orthogonal bases for rows II, III mixed, III equal, IV mixed, IV equal are respectively

$$(4,1), \quad (2,1), \quad (10,2), \quad (12,3), \quad (12,0).$$

Testing nonnegative integral products, closure under transpose, and the unit coefficient duality rule leaves respectively $1,1,2,2,0$ fusion action bases. Faithfulness proved above makes their multiplication the actual dual fusion ring. The receipt records every resulting matrix and every structure constant, so the counts and identifications require no numerical optimizer. The signs $V \mapsto -V$ amount to relabelling $X$ and $gX$, which leaves $K = \sum_x N_x \otimes N_x$ unchanged. □

**Corollary 11.8** (A Morita second-indicator exclusion)**.** *The rank-eight ring in which the $a_0$ pair is nonself-dual cannot occur as a Morita dual of $\mathcal{C}$. In the other rank-eight possibility every self-dual simple has second indicator $+1$.*

*Proof.* For a pseudo-unitary category $\mathcal{F}$, put $W_2(\mathcal{F}) = \sum_Z \nu_2(Z) \dim Z$. The center indicator formula gives $\tau_2(Z(\mathcal{F})) = \dim(\mathcal{F}) W_2(\mathcal{F})$. Positive spherical structures and the twist are preserved by Morita equivalence of centers, so $W_2$ is a Morita invariant here. The established indicators of $\mathcal{C}$ give

$$W_2(\mathcal{C}) = 2 - \sqrt{2} + (2+\sqrt{2})d = 12 + 6\sqrt{2} + 6\sqrt{3} + 4\sqrt{6}.$$

If the $a_0$ pair is nonself-dual, the possible indicator sum has form $u + s_d d + s_b b_0$, where $u \in \{0,2\}$ and $s_d, s_b \in \{-2,0,2\}$. Comparing $\sqrt{3}$ and $\sqrt{6}$ coefficients forces $s_d = s_b = 2$. The resulting $\sqrt{2}$ coefficient is 8, not 6, a contradiction. This argument does not need a prior choice of the indicator of the nontrivial invertible. If instead the $b_0$ pair is nonself-dual, the same coefficient comparison gives $s_d = s_a = 2$ and then $u = 2$. Indeed $W_2(\mathcal{C}) = 2 + 2d + 2a_0$. □

**Corollary 11.9** (No ordinary generalized-near-group Morita model)**.** *No category with this ring can be Morita equivalent to a generalized near-group category in which invertibles act transitively on all noninvertible simples. The same holds after any finite chain of finite group extensions, equivariantizations and inverse such operations starting from an ordinary generalized-near-group category.*

*Proof.* For a generalized-near-group category the noninvertibles have common dimension $z$, and $z^2 = k + mz$ for integers $k > 0, m \geq 0$. Its global dimension lies in $\mathbb{Q}(z)$, of degree at most two. Morita equivalence preserves FP dimension; extensions and equivariantizations multiply it by positive integers, and their inverses divide by integers. In contrast $\mathrm{FPdim}\,\mathcal{C} = 144 + 96\sqrt{2} + 80\sqrt{3} + 56\sqrt{6}$ has degree four: none of the three nontrivial automorphisms of $\mathbb{Q}(\sqrt{2},\sqrt{3})$ fixes it. □

**Remaining gap.** The displayed nimreps do not supply their module associators. The full-center and Gram arguments now leave only three possible Morita-dual based rings: the original rank-six ring, the displayed rank-six $\mathrm{Ad}(E_6)$ extension, and one rank-eight ring. Their existence, and the associators of the candidate modules, remain unproved. The present note excludes module/dual alternatives but does not exclude or construct $R^{(6)}_{8,1660}$.

**Background sources.** The internal-Hom/algebra description of semisimple module categories is standard; see V. Ostrik, *Module categories, weak Hopf algebras and modular invariants*, arXiv:math/0111139. For module traces in the pseudo-unitary case and their symmetric Frobenius algebras, see G. Schaumann, *Traces on module categories over fusion categories*, especially

Section 5 and Section 6, arXiv:1206.5716. The matrix reduction and its four-row solution above are proved here and do not require the module-trace theorem. The full-center argument uses V. Ostrik, *Pivotal fusion categories of rank 3*, Theorem 2.13, Corollary 2.16 and especially Remark 2.19, arXiv:1309.4822, pp. 6–8. The center indicator formula is from S.-H. Ng and P. Schauenburg, *Frobenius–Schur indicators and exponents of spherical categories*, Theorem 4.1, arXiv:math/0601012.

### 11.1 A signed Galois–fermion cut for $R^{(6)}_{8,1660}$

The following necessary condition removes one further labelled profile from the frozen 1670-profile frontier. It leaves 1669 profiles, in branches of sizes $163, 753, 753$; it does not resolve categorifiability.

Work in the established pseudo-unitary Galois form, with $\mathcal{M} = Z(\mathcal{C})$, $T = \sigma X$, and restriction vectors in the order $(1, g, X, gX, \sigma, T)$. Let $h$ be the central fermion and let $\widehat{\alpha} = Z(\alpha)$ be the previously constructed involution. For $a \in (\mathbb{Z}/48\mathbb{Z})^{\times}$, write $\pi_a$ for the Galois permutation and $\delta_a(Z) \in \{\pm 1\}$ for its sign, so that for the normalized modular matrix $S$,

$$\sigma_a(S_{L,Z}) = \delta_a(Z) S_{L,\pi_a Z}, \qquad \theta_{\pi_a Z} = \theta_Z^{a^2}.$$

**Lemma 11.10** (The signed restriction action)**.** *For every central simple $Z$ with restriction vector $v$,*

$$v_{\pi_5 Z} = \delta_5(Z) B_5 v, \qquad B_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 2 & 2 & -1 & -2 & -2 & 2 \\ 2 & 2 & -2 & -1 & -2 & 2 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 2 & 2 & -2 & -2 & -4 & 3 \end{pmatrix}.$$

*The sign is the one making the vector nonnegative. Moreover $\pi_5$ commutes with tensoring by $h$ and with $\widehat{\alpha}$, and preserves monodromy charge with $h$.*

*Proof.* The six constituents $L_\chi$ of $I(1)$ correspond to the six characters of $K_0(\mathcal{C})$, with formal codegrees $f_\chi$. The known induction rows are

$$S_{L_\chi,Z} = \frac{\chi(FZ)}{f_\chi}.$$

On $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $\sigma_5$ negates both square roots. For all six characters and all six basis elements, direct arithmetic gives

$$\sigma_5\left(\frac{\chi(v)}{f_\chi}\right) = \frac{\chi(B_5 v)}{f_\chi}.$$

The character table is invertible, proving the asserted restriction formula. The verifier checks these 36 identities exactly in the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$, and checks $B_5^2 = 1$.

Write $\epsilon_Z$ for the charge. Since $S_{h,Z}/S_{1,Z} = \epsilon_Z \in \{\pm 1\}$, the column Galois identity shows that $\pi_5$ preserves charge. Also $S_{hY,Z} = \epsilon_Z S_{Y,Z}$ and $d_{hY} = d_Y$ imply $\delta_5(hY) = \delta_5(Y)$; applying Galois to the row identity and using invertibility of $S$ gives $\pi_5(hY) = h\pi_5(Y)$. Finally, the permutation matrix of a braided autoequivalence commutes with $S$ and is rational, so it commutes with the signed Galois matrix $S^{-1}\sigma_5(S)$. Preservation of dimensions also preserves its signs, proving commutation with $\widehat{\alpha}$. □

**Lemma 11.11** (Four-divisibility of the spin fixed locus)**.** *In a restriction fiber*

$$v = (0, 0, b, c, 1, 3), \qquad b + c = 4,$$

*the number of $h$-fixed simples is divisible by four.*

*Proof.* The signed formula gives $B_5 v = -(0,0,c,b,1,3)$, so $\gamma = \pi_5 \widehat{\alpha}$ preserves the fiber. Its simples have primitive sixteenth-root twists; therefore $\theta_{\gamma Z} = \theta_Z^{25} = -\theta_Z$. The preceding lemma implies that $\gamma$ preserves the $h$-fixed locus. Duality preserves that locus, the restriction vector and the twist. It acts freely: a self-dual simple with this restriction would have to have indicator $-1$ from the odd $\sigma$ multiplicity and indicator $+1$ from the odd $T$ multiplicity. Thus each twist occurs an even number of times in the fixed locus. The bijection $\gamma$ gives equal multiplicities for the two twists $\theta, -\theta$, proving divisibility by four. No claim that $\gamma$ is a tensor functor is used. □

Here is the complete exclusion of the zero-based raw profile 133 in low frame zero, character labelling $(0,5,4,1,3,2)$. Remove the four forced even low/high pairs. The remaining 18 high simples have the following restrictions:

| symbol | restriction | multiplicity |
|---|---|---|
| $A$ | $T$ | 2 |
| $A'$ | $2X + 2gX + 3T$ | 2 |
| $B$ | $X + gX + 2T$ | 2 |
| $C$ | $4X + 4gX + 4T$ | 2 |
| $\Sigma$ | $2X + 2gX + \sigma + 3T$ | 8 |
| $P$ | $2X + 2gX + 2T$ | 1 |
| $Q$ | $4X + 4gX + 8T$ | 1 |

All these fibers are $h$-invariant. Let $e_U, o_U, t_U$ be their numbers of even $h$-pairs, odd $h$-pairs and odd $h$-fixed simples. The signed Galois map exchanges $A, A'$, so all three orbit counts agree for this pair. Put $a = e_A = e_{A'}$, $b = e_B$, $c = e_C$, and $z = e_\Sigma$. Then $a, b, c \in \{0,1\}$, $e_P = e_Q = 0$, and the previously proved duality parity gives $z \in \{0,2,4\}$. The even-dimension equation is exactly

$$3a + b + 8c + 3z = 15.$$

Its only solutions are

$$(a,b,c,z) = (1,0,0,4), \qquad (0,1,1,2).$$

For total remaining orbit counts $E, O, H$, the already established rank identity in this low frame is $O = 18 - 3E - 3$. The first solution gives $E = 6$ and $O = -3$, impossible. The second gives $E = 4$ and $O = 3$. The fibers $B, C$ are entirely even, and $P, Q$ have multiplicity one. Consequently

$$2o_A + o_\Sigma = 3, \qquad 0 \le o_A \le 1, \quad 0 \le o_\Sigma \le 2.$$

Thus $o_A = o_\Sigma = 1$, and $t_\Sigma = 8 - 2e_\Sigma - 2o_\Sigma = 2$, contradicting the lemma.

The portable verifier `verify.py` replays the frozen frontier, checks the signed identities, and supplies directly checked orbit-count witnesses for the other 1669 profiles. It also independently checks this two-case proof for profile 133. The broader replay imposes Galois equality of all three $h$-orbit counts, spin-fiber and spin-fixed four-divisibility, and the prior dimension, rank, involution and AMV/sixteenth-indicator conditions. Its receipt is `certificate.json`; no optimizer or full center fusion reconstruction is used. Passing this necessary test does not assert existence of spins, modular data, or a categorification.

## 12 Further constraints on the two unresolved rings

The two rings $R^{(6)}_{4,155}$ and $R^{(6)}_{8,1660}$ remain unresolved over $\mathbb{C}$. We now refine the necessary data without changing either categorification verdict. For $R^{(6)}_{8,1660}$, graded induction packets reduce the 1669 profiles of Section 11.1 to $1136 = 120 + 508 + 508$. We also prove that every tensor functor from a hypothetical categorification of either ring is fully faithful. Both centers have no

nontrivial Tannakian subcategory, and neither Morita class contains a braided representative. We also record exactly which analytic and small braid-group tests give no further exclusion.

## 12.1 Fixed inputs

We use the exact fusion data and established reductions of Sections 7.5 and 11. For $R^{(6)}_{8,1660}$, the ordered basis is

$$(1, g, X, gX, \sigma, T), \qquad T = \sigma X,$$

with

$$g^2 = 1, \quad g\sigma = \sigma, \quad \sigma^2 = 1 + g, \qquad X^2 = 1 + \big(3(1+g) + 4\sigma\big)X.$$

In its positive spherical Galois form,

$$d = \mathrm{FPdim}(X) = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}, \quad (d_i) = (1, 1, d, d, \sqrt{2}, \sqrt{2}d),$$

and the second indicators are $\nu_2(\sigma) = -1$ and $\nu_2 = +1$ for every other simple. Sections 7.5 and 11 establish this Galois reduction, the complete based-module list used below, and the unique central fermion $h$ with forgetful image $g$. Its center twist orders divide 48. A retained profile specifies necessary restriction multiplicities, not a center or a categorification.

## 12.2 A tensor-quotient obstruction

**Theorem 12.1.** *Let $\mathcal{C}$ be the positive spherical Galois form of a hypothetical categorification of $R^{(6)}_{8,1660}$. Every dominant pivotal tensor functor $F : \mathcal{C} \to \mathcal{D}$ to a pivotal fusion category is an equivalence. Consequently, a unitary categorification has no proper unitary tensor quotient.*

*Proof.* Regard $\mathcal{D}$ as a $\mathcal{C}$-module category through $F$. Dominance makes it indecomposable. The complete based-module analysis in Section 11 gives six simple module objects, arranged in two regular Ising blocks. With the two leaves followed by the middle vertex in each block,

$$N_X = P \otimes I + Q \otimes N_g + R \otimes N_\sigma, \quad P + Q = \begin{pmatrix} a & b \\ b & 6-a \end{pmatrix}, \quad R = \begin{pmatrix} c & f \\ f & 4-c \end{pmatrix}.$$

Up to exchanging blocks, the possibilities are

| Type | $a$ | $b$ | $c$ | $f$ | Second-to-first leaf dimension ratio $r$ |
|---|---|---|---|---|---|
| I | 0 | 1 | 0 | 0 | $3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$ |
| II | 2 | 3 | 2 | 2 | $3 - 2\sqrt{2} + 2\sqrt{3} - \sqrt{6}$ |
| III | 1 | 2 | 0 | 1 | $\sqrt{2} + \sqrt{3}$ |
| IV | 1 | 2 | 1 | 2 | $(\sqrt{2} + \sqrt{6})/2$ |

All four ratios exceed one. The signed choices of $P - Q$ do not change the two entries used here.

Tensor functors preserve Frobenius–Perron dimensions. Thus $F\sigma$ is simple, since its dimension is $\sqrt{2} < 2$. Moreover $Fg \neq 1$: otherwise $(F\sigma)^2 = 2 \cdot 1$, contradicting the multiplicity-one unit coefficient in the product of a simple with its dual. Hence $1, Fg, F\sigma$ form an Ising subcategory of $\mathcal{D}$. The unit of $\mathcal{D}$ must be a leaf, since $Fg$ fixes a middle vertex but $Fg \otimes 1 \neq 1$. It must be a leaf of the first block: normalization at a leaf of the second block would give a simple dimension $1/r < 1$. Therefore the target dimension vector is $(1, 1, \sqrt{2}, r, r, \sqrt{2}r)$.

A pivotal tensor functor preserves the transpose sign of a nondegenerate self-duality. A negative-sign self-dual simple can occur in a positive-sign self-dual object only with even multiplicity: its multiplicity space carries a nondegenerate alternating form. The simple $F\sigma$ has negative sign, while $F(X)$ and $F(T)$ carry positive-sign self-dualities. Their multiplicities

of $F\sigma$ are respectively $c$ and $a$. Thus $a, c$ must both be even, excluding types III and IV. This argument does not assert the generally false equality $\nu_2(FX) = \nu_2(X)$ for nonsimple $FX$.

For type II, rational bounds on the three square roots give

$$\frac{148}{125} < r < \frac{1189}{1000} < \frac{6}{5}.$$

Hence $1 < r$ and $r^2 < 2$. This is impossible for a simple object $Y$ in a fusion category: an invertible has dimension one, while a noninvertible has $YY^*$ containing the unit and at least one other simple, so $\mathrm{FPdim}(Y)^2 \geq 2$.

Only type I remains. At its possible unit leaf the internal end is 1. If $R_F$ is the right adjoint, then $R_F(1) = 1$, and the projection formula gives $R_F F(Z) \cong Z \otimes R_F(1) \cong Z$. Thus $F$ is fully faithful and, by dominance, an equivalence. A unitary tensor functor preserves the canonical positive pivotal structures, giving the stated corollary. □

This argument uses pivotal compatibility. Section 12.7 below removes that hypothesis by proving that it is automatic on the positive Galois form and its dominant image. Nontrivial module categories and the retained Morita-dual based rings remain possible. The exact verifier checks all twelve raw signed choices (ten after leaf relabelling), the fusion identities, the entries $a, c$, and the rational dimension bounds. An independent audit rederives the based-module possibilities and the proof.

### 12.3 Graded induction packets

Put $\mathcal{M} = Z(\mathcal{C})$, and grade its simples by monodromy with the fermion $h$. Write the canonical induced algebra as $A = I_{\mathcal{C}}(1) = A_+ \oplus A_-$. If $R$ is the restriction matrix, with columns $v_Z$ in the fixed basis above, the projection formula gives

$$N_A = R^{\mathsf{T}} R.$$

The matrices $P = N_{A_+}$ and $Q = N_{A_-}$ are respectively its same-charge and opposite-charge blocks. Since $\mathcal{M}$ is braided,

$$PQ = QP. \tag{12.1}$$

Writing $H_+ = R_+ R_+^{\mathsf{T}}$ and $H_- = R_- R_-^{\mathsf{T}}$, the even-to-odd block of this identity says

$$v^{\mathsf{T}}(H_+ - H_-)w = 0 \quad (v \text{ an even column},\ w \text{ an odd column}). \tag{12.2}$$

This identity applies generally to such a grading and known induced object.

For this ring the even low columns, constituting $A_+$, are

$$(1,0,0,0,0,0), \quad (1,0,3,3,0,4),$$
$$(1,0,1,1,0,2), \quad (1,0,2,2,0,2).$$

The odd low columns, constituting $A_-$, are either

$$(1,1,2,4,0,4),\ (1,1,4,2,0,4) \qquad \text{(frame 0)},$$

or two copies of $(1,1,3,3,0,4)$ (frame 1). Their sums are

$$u_+ = (4,0,6,6,0,8), \quad u_- = (2,2,6,6,0,8), \quad u_+ - u_- = 2(1-g).$$

There are four forced even $h$-partners of the even low columns; their restrictions interchange coordinates $1, g$ and $X, gX$. All other columns have zero in the first two coordinates. The coefficient of $A_+^2 - A_-^2$ at an even simple with restriction $v$ is $v \cdot (u_+ - u_-) = 2(v_0 - v_1)$; both sides have zero coefficients on odd simples. Therefore

$$A_+^2 - A_-^2 = 2(A_+ - hA_+), \qquad P^2 - Q^2 = 2(1 - N_h)P. \tag{12.3}$$

This uses only the already forced low columns and their partners.

**The two possible even Grams**

The fixed total Gram is

$$H = H_+ + H_- = \begin{pmatrix} 6 & 2 & 12 & 12 & 0 & 16 \\ 2 & 6 & 12 & 12 & 0 & 16 \\ 12 & 12 & 142 & 138 & 16 & 192 \\ 12 & 12 & 138 & 142 & 16 & 192 \\ 0 & 0 & 16 & 16 & 8 & 24 \\ 16 & 16 & 192 & 192 & 24 & 280 \end{pmatrix}.$$

After the four forced pairs, write a remaining high column as $(0, 0, b, c, z, a)$, put $q = b + c$ and $\delta = b - c$, and sum one contribution per even $h$-pair with multiplicity $e$. The $z$-containing types have $(z, q, a) = (1, 4, 3)$; all other types have $z = 0$. The exact even-sector dimension balance of Section 7.5 is

$$\sum e(q^2 + 2a^2 + 2z^2) = 180, \qquad \sum e(2qa) = 120. \tag{12.4}$$

In particular the $X, T$ and $gX, T$ entries of $H_+$ equal $36 + \sum eqa = 96$. Thus

$$H_+ = \begin{pmatrix} 4 & 0 & 6 & 6 & 0 & 8 \\ 0 & 4 & 6 & 6 & 0 & 8 \\ 6 & 6 & B & C & E & 96 \\ 6 & 6 & C & B & E & 96 \\ 0 & 0 & E & E & F & G \\ 8 & 8 & 96 & 96 & G & J \end{pmatrix}.$$

Apply (12.2) to the even columns $(1, 0, 1, 1, 0, 2)$ and $(1, 0, 3, 3, 0, 4)$, and to the average odd low column $(1, 1, 3, 3, 0, 4)$. The resulting equations are

$$3(B + C) + 960 + 4J = 1940, \qquad 9(B + C) + 2304 + 8J = 4684.$$

They give $J = 140$ and $B + C = 140$. The forced pairs contribute 48 to $J$ and 56 to $B + C$. Subtracting those contributions and using (12.4) yields

$$\boxed{\sum eq^2 = 84, \quad \sum eqa = 60, \quad \sum ea^2 = 46, \quad \sum ez^2 = 2.} \tag{12.5}$$

Consequently $E = 8, F = 4, G = 12$.

The matrix $H_+ - H_- = 2H_+ - H$ now vanishes outside the antisymmetric coordinate lines $1 - g$ and $X - gX$. Its two blocks are respectively

$$2\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad (2B - 142)\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

For an even asymmetric high column, the diagonal entry of (12.3) gives $(2B - 142)\delta^2 = 2\delta^2$, so $B = 72$. For an odd asymmetric column it gives $-(2B - 142)\delta^2 = 2\delta^2$, so $B = 70$. Thus asymmetric columns cannot occupy both charges. In frame 0 the odd low columns themselves force $B = 70$ and exhaust the total asymmetric norm, so all high columns are symmetric. In frame 1 the low columns are symmetric but $H_{XX} - H_{X,gX} = 4$, forcing an asymmetric high column. We obtain exactly

$$H_+(\Delta) = \begin{pmatrix} 4 & 0 & 6 & 6 & 0 & 8 \\ 0 & 4 & 6 & 6 & 0 & 8 \\ 6 & 6 & 70 + \Delta/2 & 70 - \Delta/2 & 8 & 96 \\ 6 & 6 & 70 - \Delta/2 & 70 + \Delta/2 & 8 & 96 \\ 0 & 0 & 8 & 8 & 4 & 12 \\ 8 & 8 & 96 & 96 & 12 & 140 \end{pmatrix}, \quad \Delta = \sum e\delta^2. \tag{12.6}$$

Here $\Delta = 0$ in frame 0 and $\Delta \in \{0, 4\}$ in frame 1. In frame 1 all asymmetric high fibers are odd when $\Delta = 0$, and even when $\Delta = 4$.

**Proposition 12.2.** *Exactly* 533 *of the* 1669 *labelled profiles retained in Section 11.1 fail the graded-packet conditions above. Consequently at most* 1136 *profiles remain.*

The proof is a finite exact exhaustion of the signed-Galois and fermion-orbit count options of Section 11.1, adding (12.5) and the uniform asymmetric charge. It retains both low-frame-one $h$-actions. The branch counts are

| Branch | Input | Excluded | Retained |
|---|---|---|---|
| Frame 0 | 163 | 43 | 120 |
| Frame 1, first labelling | 753 | 245 | 508 |
| Frame 1, second labelling | 753 | 245 | 508 |
| Total | 1669 | 533 | 1136 |

The bounded integer recursion uses no optimizer. Every retained witness is expanded to its full restriction-column multiset and checked against (12.1), (12.3), and (12.6). A second implementation independently rebuilds the orbit options, reproduces every rejection, and checks the complete column multiplicities and both matrix identities. Both implementations bind their inputs to exact source files. This establishes only a necessary profile reduction, not center fusion rules or associators.

## 12.4 A uniform obstruction to group constructions

**Proposition 12.3.** *Neither residual ring can have a categorification whose center contains a nontrivial Tannakian fusion subcategory. Consequently no category Morita equivalent to either residual is a nontrivial faithful finite-group extension or a nontrivial finite-group equivariantization.*

*Proof.* First work in the established positive pseudo-unitary Galois form. A useful general observation is the following. Suppose a positive braided category $\mathcal{M}$ has only the unit as an invertible object of twist one, and the rational norm of $\mathrm{FPdim}(\mathcal{M})$ has at most two prime divisors. If $\mathrm{Rep}(G) \subset \mathcal{M}$ is Tannakian, categorical Lagrange gives

$$\mathrm{FPdim}(\mathcal{M})/|G| \quad \text{an algebraic integer.}$$

Taking its field norm shows that $|G|$ has at most those two prime divisors. Burnside's theorem makes $G$ solvable. If $G \neq 1$, a nontrivial one-dimensional character of $G$ gives a nonunit invertible of twist one in $\mathcal{M}$, a contradiction. Here the positive spherical structure restricts to the canonical one on $\mathrm{Rep}(G)$.

For $\mathcal{M} = Z(\mathcal{C})$ the established complete invertible-twist inventories are $\{1, -1, i, -i\}$ for $R^{(6)}_{4,155}$, up to conjugation, and $\{1, -1\}$ for $R^{(6)}_{8,1660}$. Put $D = \mathrm{FPdim}(\mathcal{C})$. The exact norms are

| ring | field | Norm($D$) |
|---|---|---|
| $R^{(6)}_{4,155}$ | $\mathbb{Q}(\sqrt{2})$ | $576 = 2^6 3^2$ |
| $R^{(6)}_{8,1660}$ | $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ | $147456 = 2^{14} 3^2$. |

Since $\mathrm{FPdim}\, Z(\mathcal{C}) = D^2$, the observation applies to both centers.

Every faithful $G$-grading supplies a Tannakian $\mathrm{Rep}(G)$ in the center; so does every $G$-equivariantization. See [16, Propositions 8.23.1 and 8.23.11]. Braided centers are Morita invariant, giving the construction restrictions. Finally, the previously proved Galois reductions

apply to every complex categorification of these rings. A hypothetical braided Tannakian embedding and its fiber functor transport under that same conjugation. Thus a Tannakian subcategory for the original category would give one for the positive form, which was excluded. □

This prohibits the stated group constructions as a last operation, including after Morita equivalence. It does not prohibit inverse operations such as de-equivariantization, taking a neutral component, or general connected-étale condensation.

## 12.5 No braided Morita model for the second residual

**Proposition 12.4.** *No complex categorification of $R^{(6)}_{8,1660}$ is Morita equivalent to a braided fusion category.*

*Proof.* The exhaustive Morita reduction of Section 11 applies to *every* indecomposable finite semisimple module category. The uniform module-rank theorem gives two regular Ising blocks, the complete based-module list has the four types above, and full-center saturation makes each dual action faithful. The bounded linking-Gram factorization and the second-indicator comparison then leave only three dual based rings: the original ring, a rank-six ring containing

$$g^2 = 1, \quad gS = S, \quad S^2 = 1 + g + 2S, \qquad \mathrm{FPdim}(S) = 1 + \sqrt{3}, \tag{12.7}$$

and one noncommutative rank-eight ring. This reduction is not restricted to pivotal tensor quotients or to selected algebra structures.

The original ring already has no braiding: its Ising subcategory would be nondegenerate, so braided decomposition would split off a rank-two factor, contradicting $N^{\sigma X}_{XX} = 4$. A noncommutative based ring cannot underlie a braided category. It remains to exclude a braiding on the rank-six alternative.

Work in the positive form. Monodromy with the invertible $g$ gives scalar charges compatible with fusion. In the rank-three subcategory (12.7), the occurrence of $S$ in $S^2$ forces its charge $\mu$ to obey $\mu = \mu^2$, hence $\mu = 1$; the occurrence of $g$ in $S^2$ also gives charge one for $g$. Thus $g$ is transparent there. Balancing and $gS = S$ force $\theta_g = 1$. In the canonical positive ribbon structure, $\langle g \rangle$ is therefore Tannakian $\mathrm{Rep}(C_2)$, rather than a transparent fermion.

De-equivariantize this subcategory by $B = 1 + g$. For the free-module functor $F$,

$$\dim \mathrm{End}_B(F(S)) = \dim \mathrm{Hom}(S, S \otimes B) = 2.$$

Hence $F(S)$ is a sum of two distinct simple modules. Its underlying object is $2S$, so each summand has underlying object $S$. Each therefore has module-category Frobenius–Perron dimension $(1 + \sqrt{3})/2$. Its quadratic norm is $-1/2$, so it is not an algebraic integer, a contradiction.

The positive structure needed here passes to every Morita dual: both global dimension and Frobenius–Perron dimension are Morita invariant, so their equality is preserved [15, Theorem 2.15 and Corollary 8.14]. Finally a hypothetical braiding and Morita context on an arbitrary complex form transport under the established Galois conjugation. This proves the asserted unrestricted Morita-model obstruction. □

The corresponding no-braided-Morita theorem for $R^{(6)}_{4,155}$ is established in Section 9. Neither proposition removes another retained based ring: direct exact closure of all supports of $i \otimes i^*$ gives the whole ring in each of the six stored dual tensors. In particular, the term "$\mathrm{Ad}(E_6)$ extension" for the rank-six alternative does not denote a nontrivial group grading. The three candidate Morita rings remain unresolved.

## 12.6 Tests that add no rank-six exclusion

For $R^{(6)}_{4,155}$, let $\mathcal{B}$ hypothetically realize its retained rank-26 modular datum. The group $B_3$ acts on $\mathrm{Hom}(b, a^3)$. Ng–Rowell–Wen's generalized second-indicator formula [35, Theorem 3.1] determines the generator spectrum. The full twist acts by $\theta_b/\theta_a^3$. On an irreducible constituent of dimension $r \le 4$, with generator eigenvalues $\lambda_j$ and full-twist scalar $z$, Tuba–Wenzl [61, Propositions 2.5–2.6 and Theorem 2.9] gives the necessary relations

| $r$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| relation | $z = \lambda_1^6$ | $z = -(\lambda_1\lambda_2)^3$ | $z = (\lambda_1\lambda_2\lambda_3)^2$ | $z^2 = (\lambda_1\lambda_2\lambda_3\lambda_4)^3$. |

A Jordan–Hölder series forces the whole spectrum to partition accordingly, without assuming complete reducibility. All 70 nonzero blocks of dimension at most four pass the exact exponent test modulo 48. Passing such a partition constructs neither a braid representation nor a category; the dimension-four test does not impose all irreducibility conditions.

For both residual rings the primary-three matrix

$$Q_3 = \sum_i d_i^{-1} N_i \otimes N_i \otimes N_i$$

is positive semidefinite by exact simultaneous character diagonalization. All $6^3 = 216$ character-triple eigenvalues are checked in each case: there are respectively 7 and 15 distinct values, with zero and otherwise strictly positive values. Huang–Liu–Palcoux–Wu's Theorem 3.9 [62] implies that *all* primary-$n$ criteria hold; their Proposition 3.13 also gives the Schur product criterion. This exhausts that family of necessary positivity tests for these two rings; it does not establish unitarity.

The support-limit consequence of Lim's Minkowski criterion requires, whenever $N_{ij}^{\ell} > 0$, for every subset $S$ of simple labels,

$$\frac{1}{d_\ell} \sum_{n \in S} N_{\ell k}^{n} d_n \le \frac{1}{d_j} \sum_{\substack{m : N_{jk}^{m} > 0 \\ \exists n \in S :\ N_{im}^{n} > 0}} N_{jk}^{m} d_m. \tag{12.8}$$

All nonempty proper subsets and all applicable quadruples are exhausted: 17856 checks for $R^{(6)}_{4,155}$ and 24924 for $R^{(6)}_{8,1660}$. They reduce to 54 and 215 distinct integer linear forms in the positive dimensions, respectively, and all are nonnegative by exact algebraic root isolation. Empty and full subsets are automatic. This proves the entire $p = \infty$ inequality for arbitrary nonnegative vectors by integrating the level-set indicators in (12.8). The following stronger certificate also settles every finite exponent.

### 12.6.1 Exact exhaustion of the finite-exponent Minkowski family

**Proposition 12.5.** *Both residual tensors satisfy every inequality of [63, Corollary 3.4], for every triple of their six simple labels, every real $1 \le p < \infty$, and every nonnegative input vector.*

*Proof.* Put $P_{uv}(w) = N_{uv}^{w} d_w/(d_u d_v)$, and for fixed $i, j, k$ put $\mu_l = P_{ij}(l)$ and $\nu_m = P_{jk}(m)$. These are probability vectors. Dividing Lim's inequality by $d_i^{1/p} d_k$ gives

$$\left[\sum_l \mu_l \left(\sum_n P_{lk}(n) a_n\right)^p\right]^{1/p} \le \sum_m \nu_m \left[\sum_n P_{im}(n) a_n^p\right]^{1/p}.$$

It suffices to give a nonnegative array $q_{lmn}$ with the three marginals

$$\sum_n q_{lmn} = \mu_l \nu_m, \qquad \sum_m q_{lmn} = \mu_l P_{lk}(n), \qquad \sum_l q_{lmn} = \nu_m P_{im}(n).$$

Indeed, after discarding zero-weight labels, set $f(l,m) = \sum_n q_{lmn} a_n/(\mu_l \nu_m)$. The second marginal identifies the left side with $\|\sum_m \nu_m f(\cdot,m)\|_{L^p(\mu)}$. Weighted Minkowski bounds this by $\sum_m \nu_m \|f(\cdot,m)\|_{L^p(\mu)}$. Conditional Jensen, using the first marginal, then bounds each norm by $[\sum_n P_{im}(n) a_n^p]^{1/p}$, by the third marginal.

The portable certificates give exact algebraic arrays for all eight triples of noninvertible labels in $R^{(6)}_{4,155}$ and all 64 such triples in $R^{(6)}_{8,1660}$, with respectively 168 and 952 nonzero entries. If a label is invertible, either $\mu$ or $\nu$ is a point mass and a coupling is obtained directly from the other conditional distribution. The verifier checks all three marginals also in these cases. In total it checks all 216 triples and 23328 marginal equations per ring, as well as exact nonnegativity by real-root isolation and equality with the original census tensors. Floating-point optimization was used only to discover supports; the supplied proof uses exact coefficients and identities. □

This exhausts Corollary 3.4 on the two given six-element bases. It does not assert exhaustion of other commuting-square constructions or of tests on possible Morita duals, and it supplies no categorification. The proof and replay are in `evidence/rank6_laptop_20261002/all_p_minkowski/`.

## 12.7 Automatic pivotality and tensor-quotient rigidity

The pivotal compatibility assumed in Theorem 12.1 is automatic after passage to the positive Galois form and its dominant image. We give the general argument before applying it to both residual rings.

**Lemma 12.6** (Pseudo-unitary images and canonical pivotal structures)**.** *Let $F\colon \mathcal{C} \to \mathcal{D}$ be a tensor functor between complex fusion categories.*

1. *If $F$ is dominant and $\mathcal{C}$ is pseudo-unitary, then $\mathcal{D}$ is pseudo-unitary.*
2. *If $\mathcal{C}$ and $\mathcal{D}$ are pseudo-unitary, then $F$ preserves their canonical spherical structures.*

*Proof.* For (i), regard $\mathcal{D}$ as a left $\mathcal{C}$-module category via $X \cdot M = F(X) \otimes M$. Dominance implies that this module category is indecomposable: its component containing $\mathbf{1}$ contains every simple summand of every $F(X)$, hence every simple object of $\mathcal{D}$. Its dual $\mathcal{E} = \mathrm{Fun}_{\mathcal{C}}(\mathcal{D},\mathcal{D})$ is therefore a fusion category. It has the same categorical and Frobenius–Perron dimensions as $\mathcal{C}$, by [16, Corollaries 9.3.3 and 7.16.7 and Proposition 9.3.9]. Thus $\mathcal{E}$ is pseudo-unitary.

Right tensor multiplication defines a fully faithful tensor functor $\mathcal{D}^{\mathrm{rev}} \to \mathcal{E}$ (with composition as the tensor product in $\mathcal{E}$). Indeed, a $\mathcal{C}$-module natural transformation between $-\otimes Y$ and $-\otimes Z$ has component $\mathrm{id}_{F(X)} \otimes f$ at $F(X)$, where its component at $\mathbf{1}$ is $f\colon Y \to Z$. For a direct summand $M$ of $F(X)$, naturality with the splitting maps forces its component to be $\mathrm{id}_M \otimes f$. Dominance proves the assertion for every $M$, and conversely every $f$ gives such a transformation. Restricting the positive spherical structure of $\mathcal{E}$ to this fusion subcategory gives a positive spherical structure on $\mathcal{D}^{\mathrm{rev}}$, and hence on $\mathcal{D}$. Its dimensions equal Frobenius–Perron dimensions, so $\mathcal{D}$ is pseudo-unitary.

We first record a finite-spectrum observation for (ii). Every monoidal natural automorphism $u$ of a tensor functor between fusion categories has finite order. To see this, choose representatives $X_i$ of the finitely many source simples. The union of the spectra of $u_{X_i}$, and the maximal size of their Jordan blocks, are finite. Naturality with direct-sum decompositions gives the same spectrum set and Jordan bound for $u_X$ for every source object $X$. If $\lambda$ is an eigenvalue of $u_X$, a nonzero eigensubobject has nonzero tensor powers, so monoidality shows that $\lambda^n$ is an eigenvalue of $u_{X^{\otimes n}}$ for every $n \geq 1$. Thus $\lambda$ is a root of unity.

There cannot be a Jordan block of size $r > 1$. Such a block occurs on a multiplicity space of a simple target object $Y$, and its $n$th tensor power occurs on a multiplicity space tensored

with the nonzero object $Y^{\otimes n}$. The $n$th tensor power of a size-$r$ Jordan block with nonzero eigenvalue has a block of size $n(r-1)+1$ in characteristic zero. Explicitly, after removing its eigenvalue, write the block as $\exp(N)$, with $N^{r-1} \neq 0$ and $N^r = 0$. Its tensor power is the exponential of the sum of $n$ commuting copies of $N$; the power $n(r-1)$ of that sum is nonzero, by the multinomial formula, and its next power vanishes. This contradicts the uniform Jordan bound. Therefore all the $u_{X_i}$ are diagonalizable with root-of-unity eigenvalues. A common multiple of their eigenvalue orders is the order of $u$.

Now write $j_{\mathcal{C}}$ and $j_{\mathcal{D}}$ for the canonical spherical structures, whose existence and uniqueness are [16, Proposition 9.5.1]. The canonical comparison between $F$ and double duals identifies $F(j_{\mathcal{C},X})$ with a monoidal natural isomorphism $F(X) \to F(X)^{**}$. Comparing this with $j_{\mathcal{D},F(X)}$ gives a monoidal natural automorphism $u$ of $F$. Preservation of evaluation and coevaluation by $F$ gives

$$\mathrm{tr}_{\mathcal{D}}(u_X) = \dim_{\mathcal{C}}(X) = \mathrm{FPdim}(X) = \mathrm{FPdim}(F(X)) = \dim_{\mathcal{D}}(F(X)).$$

Here preservation of Frobenius–Perron dimension is [16, Proposition 4.5.7]. By the observation just proved, $u_X$ is diagonalizable with eigenvalues of modulus one. Its categorical trace is the sum of these eigenvalues weighted by the positive dimensions of the corresponding eigensubobjects. The displayed equality forces every eigenvalue to be 1. Hence $u_X = \mathrm{id}_{F(X)}$ for every $X$, which is exactly pivotal compatibility. □

**Theorem 12.7** (Tensor-quotient rigidity of the residual rings)**.** *Let $\mathcal{C}$ be a complex fusion category with Grothendieck ring $R^{(6)}_{4,155}$ or $R^{(6)}_{8,1660}$. Every tensor functor from $\mathcal{C}$ to a complex fusion category is fully faithful. Equivalently, every dominant tensor functor from $\mathcal{C}$ is a tensor equivalence.*

*Proof.* First suppose $F\colon \mathcal{C} \to \mathcal{D}$ is dominant. Fusion categories and tensor functors between them are defined over algebraic number fields [16, Corollary 9.1.8]; choose a common such field. Extend the Galois conjugation of Proposition 7.1 for $R^{(6)}_{4,155}$, or the sphericalization argument in Section 7.5 for $R^{(6)}_{8,1660}$, to this field. It transports $F$ to a dominant tensor functor $F^{\tau}\colon \mathcal{C}^{\tau} \to \mathcal{D}^{\tau}$, with $\mathcal{C}^{\tau}$ pseudo-unitary and with the same based ring. Lemma 12.6 makes $\mathcal{D}^{\tau}$ pseudo-unitary and $F^{\tau}$ pivotal for the canonical structures. An equivalence after conjugation is an equivalence before conjugation. Thus we may work in the positive spherical form.

For $R^{(6)}_{8,1660}$, Theorem 12.1 now makes $F$ an equivalence.

It remains to consider $R^{(6)}_{4,155}$. Write $U : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ for the forgetful functor and use the established decomposition

$$\mathcal{Z}(\mathcal{C}) \simeq \mathcal{S} \boxtimes \mathcal{B}, \qquad \theta_s = i, \qquad U(s \boxtimes \mathbf{1}) = g,$$

where $s$ is the nontrivial semion. The basis of $K_0(\mathcal{C})$ is $\mathbf{1}, h, g, hg, X, Y$, with $gX = Y$ and $hX = X$. For clarity denote the simple of $\mathcal{B}$ with numerical label $j$ by $B_j$; thus $B_1 = \mathbf{1}$. The complete list of twist-one simples of the center, with their forgetful images, is

| central simple | forgetful image |
|---|---|
| $\mathbf{1} \boxtimes B_1$ | $\mathbf{1}$ |
| $\mathbf{1} \boxtimes B_2$ | $Y$ |
| $\mathbf{1} \boxtimes B_5$ | $X$ |
| $\mathbf{1} \boxtimes B_{23}$ | $\mathbf{1} + 2X + 2Y$ |
| $\mathbf{1} \boxtimes B_{25}$ | $\mathbf{1} + h + 2X + 2Y$ |
| $s \boxtimes B_{18},\ s \boxtimes B_{19}$ | $\mathbf{1} + X + Y$ |
| $s \boxtimes B_{24}$ | $\mathbf{1} + h + 2X + 2Y.$ |

(12.9)

Indeed the twist-one labels in $\mathcal{B}$ are $1, 2, 5, 23, 25$, while the twist-$(-i)$ labels are $18, 19, 24$. Before tensoring by $s$, the last three restrict respectively to $g + X + Y$, $g + X + Y$, and $g + hg + 2X + 2Y$.

Let $R$ be the right adjoint of $F$. Its induced central algebra $E$ has underlying algebra $R(\mathbf{1})$, and

$$\dim \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, U(E)) = \dim \mathrm{Hom}_{\mathcal{D}}(F(\mathbf{1}), \mathbf{1}) = 1.$$

The induced central-algebra construction of Bruguières–Natale [64, Section 5] identifies $\mathcal{D}$ with the category of modules over $U(E)$. In characteristic zero this induced central algebra is separable: the underlying module category is semisimple, and the bimodule splitting of multiplication can be averaged to a morphism in the center. Thus $E$ is connected étale in $\mathcal{Z}(\mathcal{C})$. Positivity of the spherical structure gives $\theta_E = \mathrm{id}_E$; see Schopieray [65, Lemma 2.2.4]. The displayed connectedness in $\mathcal{C}$, together with (12.9), therefore leaves only

$$E \simeq \mathbf{1} \oplus n(\mathbf{1} \boxtimes B_2) \oplus n(\mathbf{1} \boxtimes B_5), \qquad n \geq 0.$$

Here the two multiplicities agree because $B_2^* = B_5$ and an étale algebra has a nondegenerate Frobenius pairing.

Put $Z = \mathbf{1} \boxtimes B_2$. The reconstructed fusion rules give

$$B_2^2 = B_9 + B_{10} + B_{16} + B_{19}, \qquad B_2 B_5 = B_1 + B_8 + B_{11} + B_{22}.$$

Thus multiplication on any two copies of $Z$ in $E$ vanishes, and multiplication on a copy of $Z$ and a copy of $Z^*$ has image in the unit summand. If $n > 0$, nondegeneracy of the Frobenius pairing supplies copies $Z \hookrightarrow E$ and $Z^* \hookrightarrow E$ for which this latter multiplication is nonzero. Associativity on $Z \otimes Z \otimes Z^*$ is then impossible: multiplying the first two factors gives zero, whereas multiplying the last two and then the first gives a nonzero multiple of $\mathrm{id}_Z \otimes \mathrm{ev}_Z$. Consequently $n = 0$ and $R(\mathbf{1}) \simeq \mathbf{1}$. The projection formula gives $RF(V) \simeq V \otimes R(\mathbf{1}) \simeq V$; hence $F$ is fully faithful and, by dominance, an equivalence.

Finally, an arbitrary tensor functor factors through the full fusion subcategory generated by its image. Its first factor is dominant and hence, by what we have proved, an equivalence. Its second factor is a fully faithful inclusion. This proves the statement for every tensor functor. □

This theorem gives a restriction on hypothetical categorifications; it neither constructs nor excludes either one. The two general categorification questions remain open.

## 12.8 Certificate scope and remaining questions

The accompanying quotient, packet-Gram, braid-group, and analytic-exhaustion verifiers retain their exact receipts and input bindings; the independent packet audit is retained separately. The corresponding Python entry points are `verify_tensor_quotient_obstruction.py`, `verify_packet_gram.py`, `verify_braid3.py`, `verify_analytic_exhaustion.py`, and `verify_morita_structure.py`. The low-profile, signed-Galois, indicator, and based-module certificates used in the preceding sections remain the fixed prerequisites. The packet, quotient and structural certificates are retained under `evidence/residual_20261002/rank6/`.

Neither ring has a categorical model or an unrestricted nonexistence proof from this work. In particular Theorem 12.7 excludes proper tensor quotients, not categorifications themselves. The graded-packet identities reduce the necessary center-profile count for $R^{(6)}_{8,1660}$ from 1669 to 1136; the classification totals remain 25362 exclusions, 57 realizations, and 2 unresolved rings.

## 13 Twist-one support at multiplicity eight

The equality case of the dimension-block bound identifies every twist-one simple in the hypothetical center of $R^{(6)}_{8,1660}$. This gives a shorter proof of tensor-quotient rigidity. The ring remains unresolved: the argument neither constructs a categorification nor proves nonexistence.

**Proposition 13.1.** *Let $\mathcal{C}$ be a pseudo-unitary categorification of $R^{(6)}_{8,1660}$, with its canonical positive spherical structure. The twist-one simple objects of $Z(\mathcal{C})$ are exactly the six simple summands of $I_{\mathcal{C}}(1)$.*

*Proof.* Put $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$ and $D = 4(1 + d^2) = 144 + 96\sqrt{2} + 80\sqrt{3} + 56\sqrt{6}$. Then

$$\frac{1}{D} = \frac{6 - 2\sqrt{3} - \sqrt{6}}{48}.$$

The dimension and global-dimension fields are both $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, so the dimensional grading has order one. For the characters $\chi_3, \chi_6$ of $\mathrm{Gal}(K/\mathbb{Q})$ corresponding to $\sqrt{3}, \sqrt{6}$, the Fourier coefficients of $1/D$ are $-\sqrt{3}/6$ and $-\sqrt{6}/12$. Their squared absolute values are, respectively, $1/12$ and $1/24$, equal to the reciprocals of their conductors. Thus both attain equality in the dimension-block bound of [52, Theorem 5.1].

Here is the equality consequence we use. Let $L_\gamma$, for $\gamma \in \mathrm{Gal}(K/\mathbb{Q})$, be the four summands of $I(1)$ indexed by the conjugates of the dimension character, and set $w_\chi = \sum_\gamma \chi(\gamma) e_{L_\gamma}$. Write $s = S/D$ for the normalized modular matrix of $Z(\mathcal{C})$, and let $P$ be the orthogonal projection onto the twist-one eigenspace. The proof of the cited theorem gives

$$\|P s w_\chi\|^2 \le \frac{4}{\mathfrak{f}(\chi)}.$$

The four dimension-block coordinates already contribute exactly $4|\widehat{(1/D)}(\chi)|^2 = 4/\mathfrak{f}(\chi)$. Consequently every other twist-one simple $Z$ has zero $\chi$-Fourier coefficient in $\dim(Z)/D$. Applying this to $\chi_3, \chi_6$ gives

$$\theta_Z = 1, \quad Z \notin \{L_\gamma\} \quad \Longrightarrow \quad \frac{\dim Z}{D} \in \mathbb{Q}(\sqrt{2}). \tag{$*$}$$

This uses unitarity of the modular representation, not a unitary structure on $\mathcal{C}$.

Suppose that $Z$ is twist-one and not a summand of $I(1)$. Adjunction gives $[FZ : 1] = 0$. Its $g$-multiplicity is also zero: the established central fermion $h$ gives $I(g) = hI(1)$; the four even low constituents have $h$-translates of twist $-1$, while the two odd low constituents are permuted among themselves by $h$. Therefore write

$$FZ = bX + c\,gX + z\sigma + aT, \qquad T = \sigma X.$$

Since $\dim Z = z\sqrt{2} + (b + c + a\sqrt{2})d$, direct expansion of $(*)$ gives

$$b + c = 4z, \qquad a = 3z.$$

If $z = 0$, this is the zero restriction vector, impossible for a nonzero object. If $z > 0$, the Ising projection established in Proposition 7.11 forces the twist of $Z$ to have primitive order 16, again a contradiction. Finally, all six summands of $I(1)$ have twist one, and each has multiplicity one because $K_0(\mathcal{C})$ is commutative. □

**Corollary 13.2.** *Every tensor functor from a complex categorification of $R^{(6)}_{8,1660}$ to a complex fusion category is fully faithful.*

*Proof.* Replace the target by the fusion subcategory generated by the image, so that the functor is dominant. Apply the unrestricted sphericalization argument in Section 7.5 and Lemma 12.6 to pass to the positive Galois form and give source and target their canonical spherical structures; the functor preserves these structures. Its induced central algebra $A$ is a connected étale algebra of twist one in $Z(\mathcal{C})$, and its underlying algebra is connected: if $R$ is the right adjoint, then

$$\dim \mathrm{Hom}_{\mathcal{C}}(1, R(1)) = \dim \mathrm{Hom}_{\mathcal{D}}(1, 1) = 1.$$

By the proposition every constituent of $A$ lies in $I(1)$. Every such simple restricts with unit multiplicity one. Since the unit already occurs in $A$, underlying connectedness forces $A = 1$. The projection formula gives $RF(V) = V \otimes R(1) = V$, so $F$ is fully faithful, and hence an equivalence onto its image. Transporting back under the Galois conjugation proves the assertion. □

**Finite modular representation compatibility.** The exact twist-one support also determines the whole compression $PsP$ from the six known low rows. Its spectrum is

$$\{1, 1, -1/2, 0, -\sqrt{3}/6, -\sqrt{6}/12\}$$

in the first low branch; in the second, one eigenvalue 1 is replaced by 0. The complete irreducible-character tables of $\mathrm{SL}_2(\mathbb{Z}/48)$ give exact compatible compression spectra. The diagonal torus acts on the twist-one space with multiplicities $(3, 1, 1, 1)$ for the quadratic characters of discriminants $1, 8, 12, 24$. Imposing these multiplicities, the compressed spectra, the forced Ising-center subrepresentation, and zero high twist-one multiplicities strengthens the existing necessary system. Nevertheless, first-branch profile 10 admits an integer witness of center rank 37: all 1536 equations and 1447 variable bounds pass exact replay. The portable data are in `evidence/theoretical_20261016/rank6/r1660_modular_witness/`. This witness satisfies the stated linear necessary conditions, but Section 18 excludes its modular realization by a half-integral Verlinde coefficient. The other refinements of the retained profiles remain open.

## 14 Further necessary tests at multiplicity four

The categorifiability of $R^{(6)}_{4,155}$ remains unresolved. Its three nontrivial invertibles, in source labels $3, 4, 5$, fix respectively two, zero and zero simple labels. The involution-module test with at most two fixed labels therefore covers all three; the $1 + h$ module action is analyzed in Section 7.4.

For the retained rank-26 centralizer candidate, the fermion fixes six labels. The expanded anti-invariant test admits an explicit six-dimensional integer representation satisfying all fusion identities, transpose duality, and fixed-block entry bounds and parities. One generator has characteristic polynomial

$$(x^2 + 9)(x^2 + 1)(x^2 - 1).$$

Splitting the six fixed labels reconstructs a nonnegative integer based module of rank 22; all 676 fusion identities are verified exactly. The portable witness and replay are recorded under `evidence/theoretical_20261016/rank6/r155_center_module/`. This is a passing necessary condition, not a module-category construction. Section 15 further checks compatibility with the Morita self-duality forced by the fermion; this condition also passes and leaves the associativity constraints unresolved.

The separate reflection refinement likewise supplies passing necessary partitions for all 226 local braid blocks of dimensions 6–308, covering 80 normalized spectral profiles. Neither test constructs braid representations or associators, and neither changes the two-case rank-six residual list.

## 15 A fermion forces a prescribed Morita self-duality

The rank-22 based module in the preceding section must satisfy an additional identity that uses the braiding of the rank-26 candidate. This condition prescribes its Morita dual, instead of allowing an arbitrary dual Gram factorization.

**Lemma 15.1** (Fermion Azumaya crossing)**.** *Let $\mathcal{B}$ be a braided fusion category containing a fermion $f$, and let $A = \mathbf{1} \oplus f$ be its group algebra as an algebra object. For simple right $A$-modules $m_a$, put*

$$M_i[a,b] = \dim \mathrm{Hom}_{\mathcal{B}_A}(i \otimes m_a, m_b), \qquad i \in \mathrm{Irr}(\mathcal{B}).$$

*Then, for all module labels $a, b, c, d$,*

$$\sum_i M_i[a,b]M_i[c,d] = \sum_i M_i[c,a]M_i[d,b]. \tag{15.1}$$

*Proof.* Inside $\langle f \rangle \simeq \mathrm{sVec}$, the algebra $A$ is the one-generator Clifford algebra. Its Azumaya map $A \otimes A^{\mathrm{op}} \longrightarrow \underline{\mathrm{End}}(A)$ is an isomorphism: the identity, left multiplication by the odd generator, graded right multiplication by that generator, and their composite form a basis of the four-dimensional endomorphism algebra. The braided embedding into $\mathcal{B}$ preserves this isomorphism. Thus $A$ is Azumaya, and both alpha-induction functors $\mathcal{B} \longrightarrow {}_A\mathcal{B}_A$ are tensor equivalences. Here we use the Azumaya criterion in Davydov–Nikshych, *The Picard crossed module of a braided tensor category*, Section 3.2, Definition 3.5 and the paragraph following it (arXiv:1202.0061).

Under either equivalence, the right action on $\mathcal{B}_A$ has the same multiplicity matrices as the left action; the braiding identifies the corresponding products. Associativity in the Morita linking category, applied to $(m_a m_b^*)m_c = m_a(m_b^* m_c)$, therefore gives

$$\sum_i M_i[b,a]M_i[c,d] = \sum_i M_i[b,c]M_i[a,d].$$

Exchange $a, b$, and use $M_{i^*} = M_i^{\mathsf{T}}$ while reindexing the right-hand sum by $i \mapsto i^*$. This proves (15.1). □

**Application to $R^{(6)}_{4,155}$.** For the retained rank-26 modular candidate, reconstruct the 26 action matrices from the six-dimensional anti-invariant witness. All $22^4 = 234256$ identities (15.1) hold exactly. The replay uses integer arithmetic and binds both frozen inputs by SHA256; it is recorded in `evidence/unified_20261016/rank6/verify_fermion_crossing.py`. From the repository root, run

```
python evidence/unified_20261016/rank6/verify_fermion_crossing.py --repo .
```

Thus the known based module is compatible with the prescribed Azumaya Morita self-duality. The remaining problem is realization by coherent module associators. This passing identity constructs no module category and proves no categorification or exclusion; both rank-six cases remain unresolved. Braidedness is essential in the lemma: commutativity of a Grothendieck ring alone does not identify the two Morita actions.

# 16 A forced third indicator at multiplicity eight

The exact twist-one support yields a further necessary condition for $R^{(6)}_{8,1660}$. It removes two incompatible choices of saturated irreducible modular representations. The rank-37 witness passes this test; §18 gives a further obstruction to it. All indicators below refer to the canonical positive spherical structure on the positive Galois representative.

**Proposition 16.1.** *Let $T = \sigma X$ and $\omega = e^{2\pi i/3}$. Every pseudo-unitary categorification of $R^{(6)}_{8,1660}$ satisfies*

$$\nu_3(T) = 2, \qquad \nu_3(X) = \nu_3(gX) \in \{2+\omega, 2+\omega^2\}.$$

*It also satisfies*

$$\nu_5(X) = \nu_5(gX) = \nu_5(T) = 2, \qquad \nu_7(X) = \nu_7(gX) = 3, \quad \nu_7(T) = 4.$$

*Proof.* Use the normalized modular matrices $s, t$ of the center and the four vectors $e_{L_\gamma}$ from Proposition 13.1. For $r = 3, 6$ set $w_r = \sum_\gamma \chi_r(\gamma) e_{L_\gamma}$, where $\chi_r$ corresponds to $\sqrt{r}$. Then $\|w_r\|^2 = 4$ and $\langle e_1, w_r\rangle = 1$. The twist-one torus multiplicities $(3,1,1,1)$ make $\mathbb{C}w_r$ the entire $\chi_r$ eigenspace. The compressed $s$ eigenvalues there are $-\sqrt{3}/6$ and $-\sqrt{6}/12$.

The exact character tables of $\mathrm{SL}_2(\mathbb{Z}/48) \cong \mathrm{SL}_2(\mathbb{Z}/16) \times \mathrm{SL}_2(\mathbb{Z}/3)$ now give only the following irreducible possibilities, in the frozen local table order:

$$W_3 : \ (33,4), (36,5), \qquad W_6 : \ (54,4), (61,5).$$

The replay exhausts all 532 pairs, checking the torus multiplicities, central sign and the compression polynomials from the preceding section. Every eligible summand carrying $\chi_r$ has one-dimensional $t$-fixed space. Thus $w_r$ belongs to one multiplicity-one irreducible $W_r$, and the orthogonal projection of $e_1$ onto $W_r$ is $w_r/4$. Writing $P_r$ for projection onto $W_r^t$, put

$$m_r(n) = \mathrm{Tr}_{W_r}(P_r s^{-1} t^n s).$$

The preceding projection identity gives $\langle s w_r, t^n s e_1\rangle = m_r(n)$.

For a central dimension $z$, let $\ell_r(z)$ be the rational $\sqrt{r}$ coordinate of $z/D$. The low-character formula gives $(sw_r)_A = 4\sqrt{r}\,\ell_r(\dim A)$. The center indicator formula therefore yields

$$\sum_i \ell_r(\dim X_i)\nu_n(X_i) = \frac{m_r(n)}{4\sqrt{r}}.$$

In source order $(1, g, X, gX, \sigma, T)$, the scaled weights are

$$24\ell_3 = (-1,-1,1,1,-1,-1), \qquad 48\ell_6 = (-1,-1,-1,-1,-2,2).$$

Put $a_n = 2\sqrt{3}\, m_3(n)$ and $b_n = 2\sqrt{6}\, m_6(n)$. For odd $n$, Proposition 7.15 and the vanishing odd indicators of $g, \sigma$ give

$$\nu_n(X) = \frac{3 + 2a_n + b_n}{2}, \qquad \nu_n(T) = 2 + a_n + b_n.$$

These coefficients are exact character averages. At local modulus $m = 16, 3$, use

$$\frac{1}{m}\sum_{j=0}^{m-1} \mathrm{char}_W\left(\begin{pmatrix} 1-jn & j \\ -n & 1 \end{pmatrix}\right),$$

the trace of $P_W s^{-1} t^n s$; multiply the local values. At $n = 3$, the four choices give

| $W_3$ | $W_6$ | $\nu_3(X)$ | $\nu_3(T)$ |
|---|---|---|---|
| $(33,4)$ | $(54,4)$ | $2+\omega$ | $2$ |
| $(33,4)$ | $(61,5)$ | $-3\omega^2$ | $-4\omega^2$ |
| $(36,5)$ | $(54,4)$ | $-3\omega$ | $-4\omega$ |
| $(36,5)$ | $(61,5)$ | $2+\omega^2$ | $2$ |

But $\dim \mathrm{Hom}(1, X^3) = 3$, so third rotation has trace $a + b\omega + c\omega^2$ with $a, b, c \geq 0$ and $a + b + c = 3$. A real part of $3/2$ forces $a = 2$ and $b + c = 1$, excluding the two mixed rows, whose imaginary parts have magnitude $3\sqrt{3}/2$. The averages at $n = 5, 7$ give the other stated values. □

**Scope and replay.** The two surviving pairs are complex conjugates; no reality assertion for third indicators of self-dual objects is used. The specified rank-37 necessary-system witness uses $(36,5), (61,5)$ and passes these indicator constraints. It is excluded by the separate Verlinde argument in §18. The fifth and seventh invariant-space dimensions for $(X, T)$ are $(402, 2272)$ and $(55401, 626784)$; the displayed indicators satisfy the elementary cyclic-spectrum bounds. The source-bound standard-library replay and independent proof audit are in `evidence/unified_20261016/rank6/higher_indicators/`. Neither a categorification nor a general exclusion follows, and both rank-six cases remain unresolved.

# 17 All indicators and phase-wise induction masses at multiplicity eight

The two saturated modular summands in Proposition 16.1 determine the complete indicator sequences, including even orders. Fourier inversion then fixes the dimension-weighted restriction mass at each central twist. These are necessary conditions on a hypothetical category; the ring remains unresolved.

Write $r = \sqrt{2}$, $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$, and $D = 4(1 + d^2)$. We work with the positive spherical Galois representative and source order $(1, g, X, gX, \sigma, T)$, where $T = \sigma X$. Let $\zeta = e^{2\pi i/48}$. The two saturated choices are

$$(W_3, W_6) = ((33,4),(54,4)), \qquad ((36,5),(61,5)),$$

in the local character-table order of §16. They are complex conjugates. For each choice put

$$a_n = 2\sqrt{3}\,\mathrm{Tr}_{W_3}(P_3 s^{-1} t^n s), \qquad b_n = 2\sqrt{6}\,\mathrm{Tr}_{W_6}(P_6 s^{-1} t^n s),$$

where $P_j$ projects onto the one-dimensional $t$-fixed line of $W_j$.

**Theorem 17.1** (Complete indicator sequences)**.** *For every positive integer $n$, put*

$$g_n = \begin{cases} 1 & 2 \mid n, \\ 0 & 2 \nmid n, \end{cases} \qquad s_n = \begin{cases} \sqrt{2}\cos(3n\pi/8) & 2 \mid n, \\ 0 & 2 \nmid n. \end{cases}$$

*Then*

$$\nu_n(X) = \nu_n(gX) = \frac{3 + 3g_n + 4s_n + 2a_n + b_n}{2},$$
$$\nu_n(T) = 2 + 2g_n + 3s_n + a_n + b_n.$$

*Together with $\nu_n(1) = 1$, $\nu_n(g) = g_n$ and $\nu_n(\sigma) = s_n$, these give all six indicator sequences. Every sequence is periodic of period dividing 48.*

*Proof.* The projection and central-indicator identities in the proof of Proposition 16.1 hold for every $n$; only its subsequent simplification used oddness. In particular,

$$\sum_i \ell_j(\dim X_i)\nu_n(X_i) = \frac{\mathrm{Tr}_{W_j}(P_j s^{-1}t^n s)}{4\sqrt{j}} \qquad (j = 3, 6).$$

Saturation makes the orthogonal projection of $e_1$ onto $W_j$ equal to $w_j/4$, so no unknown complementary summand contributes to this identity. The tensor involution of Proposition 7.15 gives equal indicators for $X$ and $gX$ at every order. The invertible $g$ has trivial pointed associator and indicator $+1$ at even orders. For the positive Ising subcategory with $\nu_2(\sigma) = -1$, the modular indicator formula [32, Theorem 7.5] gives $s_n$: using an Ising braiding with $\theta_\sigma = e^{3\pi i/8}$, the four fusion pairs contributing to $\sigma$ yield zero at odd orders and $\sqrt{2}\cos(3n\pi/8)$ at even orders. This calculation depends only on the underlying positive pivotal Ising category.

Substitute these indicators into the two coefficient rows

$$24\ell_3 = (-1, -1, 1, 1, -1, -1), \qquad 48\ell_6 = (-1, -1, -1, -1, -2, 2).$$

Writing $x = \nu_n(X) = \nu_n(gX)$ and $u = \nu_n(T)$ gives

$$2x - u = a_n + 1 + g_n + s_n, \qquad -2x + 2u = b_n + 1 + g_n + 2s_n.$$

Solving gives the stated formulas. Every actual central twist has order dividing 48 by Proposition 7.11; the central-indicator formula therefore proves the periodicity as well. □

**Theorem 17.2** (Central phase masses)**.** *For $k \in \mathbb{Z}/48$ and a base simple $X_i$, define*

$$M_i(k) = \sum_{\theta_Z = \zeta^k} \dim(Z)[F(Z) : X_i], \qquad Z \in \mathrm{Irr}\, Z(\mathcal{C}).$$

*These masses are determined by*

$$M_i(k) = \frac{D}{48}\sum_{n=1}^{48} \nu_n(X_i)\zeta^{-kn}. \tag{17.1}$$

*For the choice $((33, 4), (54, 4))$, the complete twist support is*

$$\{0, 8, 9, 12, 14, 15, 20, 24, 26, 30, 32, 33, 36, 39, 42, 44\}.$$

*For the conjugate choice, negate these exponents modulo 48. There are exactly two simples in each of the four phases $9, 15, 33, 39$. All eight have the same restriction*

$$F(Z) = 2X + 2gX + \sigma + 3T. \tag{17.2}$$

*These are precisely the central simples whose restrictions contain $\sigma$.*

*Proof.* The center formula [32, Theorem 4.1] and adjunction give

$$D\nu_n(X_i) = \sum_k M_i(k)\zeta^{kn}.$$

Discrete Fourier inversion gives (17.1). The exact character averages determining $a_n, b_n$ are

$$\frac{1}{m}\sum_{j=0}^{m-1} \mathrm{char}_W \begin{pmatrix} 1 - jn & j \\ -n & 1 \end{pmatrix} \quad (m = 16, 3),$$

multiplied across the two local factors. Evaluation of these finite sums and the Fourier transform in $\mathbb{Q}(\zeta)$ gives the displayed support. Every mass outside it is zero. Positive dimensions and faithfulness of $F$ imply that no central simple can lie at a phase with all six masses zero.

For each $k = 9, 15, 33, 39$, the nonzero coordinates relevant here are

$$\frac{(M_X(k), M_\sigma(k), M_T(k))}{d} = (8\sqrt{3} + 8\sqrt{6},\ 4\sqrt{3} + 4\sqrt{6},\ 12\sqrt{3} + 12\sqrt{6}).$$

The $gX$ coordinate equals the $X$ coordinate, and every other phase has $M_\sigma(k) = 0$. Proposition 7.11 shows that a $\sigma$-containing column is $(0, 0, b, c, 1, 3)$ with $b + c = 4$, and its dimension divided by $d$ is $2\sqrt{3} + 2\sqrt{6}$. Hence exactly two such simples occur in each of the four phases.

Such a simple cannot be self-dual. A self-duality form on its restriction would have to have sign $-1$ because the multiplicity of $\sigma$ is odd, and sign $+1$ because the multiplicity of $T$ is odd. Duality preserves both twists and restrictions, so the two $\sigma$-containing objects in a phase are a dual pair with identical columns. Their fixed total $b + c = 4$ exhausts the entire displayed $M_X + M_{gX}$ mass. Positivity therefore leaves no $X$ or $gX$ contribution from any other object in that phase. The separate $X$ and $gX$ masses now give $b = c = 2$. These two columns also exhaust the $\sigma$ and $T$ masses, while the unit and $g$ masses are zero. Faithfulness and positivity exclude every additional simple in the phase, proving (17.2) and the exact fiber size. □

**Corollary 17.3** (Homogeneous phase Gram constraints)**.** *Use the first conjugate choice in Theorem 17.2. At any phase in the following table, write the restriction columns as* $(0, 0, b, c, 0, a)$ *and put* $q = b + c$*. The phase-wise sums satisfy*

| $k$ | $\sum q^2$ | $\sum qa$ | $\sum a^2$ |
|---|---|---|---|
| $8, 32$ | 40 | 24 | 16 |
| 12 | 4 | 4 | 4 |
| $14, 26$ | 16 | 16 | 16 |
| 20 | 72 | 40 | 24 |
| $30, 42, 44$ | 8 | 8 | 8 |
| 36 | 36 | 20 | 12 |

*Moreover* $\sum bq = \sum cq$ *and* $\sum ba = \sum ca$ *in every listed phase. Every individual column in phases* $12, 14, 26, 30, 42, 44$ *has* $q = a$*.*

*Proof.* The unit can occur in a restriction only at twist one. Since $I(g) = hI(1)$ for the central fermion $h$, its constituents have twists 1 or $-1$; thus the listed phases have no $g$ coordinate either. Theorem 17.2 removes the $\sigma$ coordinate. Each remaining column has dimension $d(q + \sqrt{2}a)$. Consequently

$$M_X/d = \sum bq + \sqrt{2}\sum ba, \quad M_{gX}/d = \sum cq + \sqrt{2}\sum ca, \quad M_T/d = \sum aq + \sqrt{2}\sum a^2.$$

The exact masses have $M_X = M_{gX}$. Comparing the rational and $\sqrt{2}$ coefficients gives the equalities and the table. In its homogeneous rows, $\sum(q - a)^2 = 0$. Every summand is a nonnegative integer, so each is zero. □

**Exact verification and scope.** The source-bound evidence in `evidence/rigidity_20261017/rank6/r1660/` regenerates the local character averages, all 48 indicator values, and all 288 phase masses for each conjugate choice. An independent SymPy computation uses polynomial remainders modulo $\Phi_{48}$ and rational linear solves; it checks every Fourier inversion, every inverse transform, the full Ising indicator sequence, and independence of the dimension-field basis. The checks certify the finite arithmetic in these arguments, not categorical existence or nonexistence. In particular, no unknown modular complement is assumed to coincide with the rank-37 witness.

**Corollary 17.4** (The full phase-mass restriction domain)**.** *The* 1136 *labelled restriction profiles satisfying the preceding Galois–fermion and graded-induction constraints have exactly* 640 *profiles compatible with all the phase Gram identities above. Their three branch sizes are* $68, 286, 286$.

*Proof.* In each low frame, the six unit-induced columns exhaust all six phase-zero masses in (17.1). Their four forced even fermion partners exhaust all six phase-24 masses. These are exact equalities in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$; positivity excludes any additional column at either phase. Remove these four high partners. Theorem 17.2 requires exactly eight spin columns $(0, 2, 2, 1, 3)$ in the high-column convention that omits the unit coordinate. For each remaining column type $(b, c, a)$, its five phase-moment contributions are

$$(b(b+c),\ c(b+c),\ ba,\ ca,\ a^2).$$

Allocate the available multiplicities among the ten phases in Corollary 17.3, with target vector $(Q/2, Q/2, H/2, H/2, A)$ when the displayed triple is $(Q, H, A)$. All variables are nonnegative integers bounded by the available column multiplicities. The retained witnesses give every passing allocation. An independent exact coefficient-convolution calculation constructs all possible phase monomials; meet-in-the-middle support products then test whether their product has the required global multiplicity vector. It reproduces precisely the three stated counts. Both implementations use exact integers and exhaust their finite domains. This tests necessary restriction data; it imposes no existence of associators or full modular fusion matrices. □

# 18 A Verlinde obstruction to the rank-37 witness

The preceding necessary conditions do not suffice for modular data. We now exclude the particular rank-37 restriction, twist, and representation witness specified below. This does not exclude $R^{(6)}_{8,1660}$ or every refinement of its retained profiles.

**Proposition 18.1.** *The specified rank-*37 *witness cannot be the modular data of a center: it forces a fusion coefficient to equal* $1/2$.

*Proof.* Use the source order $(1, g, X, gX, \sigma, T)$, where $T = \sigma X$. The witness has a low simple $A$ and exactly two simples $B_+, B_-$ of twist $\zeta_{48}^4$, with

$$F(A) = 1 + X + gX + 2T, \qquad F(B_\pm) = X + gX + 2T.$$

Among its active modular irreducibles, precisely $W = (61, 5)$ and $H = (0, 1)$ contain that twist, each with multiplicity one. Here $W$ has dimension 12, simple twist spectrum, and a one-dimensional twist-one space; $H$ is one-dimensional. These assertions follow directly from the frozen local character tables.

Let $e_W = w_6/2$ be the normalized twist-one vector of $W$, as in §16. Reciprocity gives equal nonzero coordinates of $se_W$ at the two repeated restriction rows. Hence

$$b = \frac{e_{B_+} + e_{B_-}}{\sqrt{2}} \in W, \qquad h = \frac{e_{B_+} - e_{B_-}}{\sqrt{2}} \in H.$$

For a center simple $E$ with restriction row $v_E$, put

$$a_E = |\langle e_E, se_W \rangle|^2 = \frac{L_6(v_E)^2}{96}, \qquad L_6 = (-1, -1, -1, -1, -2, 2).$$

The last equality follows from $sw_6 = 4\sqrt{6}\,\ell_6$ and $48\ell_6(F(E)) = L_6(v_E)$. Write

$$r_E = \frac{s_{A,E}}{s_{0,E}}, \qquad q_t = \frac{\sum_{\theta_E = \zeta_{48}^t} r_E a_E}{\sum_{\theta_E = \zeta_{48}^t} a_E}.$$

Since every twist eigenspace of $W$ is one-dimensional, $q_t$ is the expectation of the Verlinde spectral diagonal on that line. The exact transition probabilities are

$$p_t = \mathrm{Tr}_W(P_4 s P_t s^{-1}).$$

Expanding the two projectors gives character averages. For a local factor of modulus $m$, with local target phase $c$, the average is

$$\frac{1}{m^2} \sum_{j,k \bmod m} \zeta_m^{-cj-tk} \,\mathrm{char}_W(T^j S T^k S^{-1}).$$

The local phases of 4 are $(12, 1)$; CRT reconstructs a global phase as $3a + 16b \pmod{48}$. Exact evaluation gives

| $t$ | 0 | 16 | 6 | 22 | 12 | 28 |
|---|---|---|---|---|---|---|
| $p_t$ | 1/12 | 1/24 | 1/6 | 1/12 | 1/12 | 1/24 |
| $q_t$ | 15 | 3 | −1 | −1 | −1 | −1 |

| $t$ | 18 | 34 | 24 | 40 | 36 | 4 |
|---|---|---|---|---|---|---|
| $p_t$ | 1/6 | 1/12 | 1/12 | 1/24 | 1/12 | 1/24 |
| $q_t$ | −1 | −1 | 15 | 3 | −1 | −1 |

and therefore $\langle b, N_A b\rangle = \sum_t p_t q_t = 2$. Here $N_A = s\,\mathrm{diag}(r_E)s^*$ by Verlinde.

On the one-dimensional summand $H$, the matrix $s$ acts by a scalar. Moreover $r_{B_+} = r_{B_-} = -1$. Thus $N_A h = -h$, and its mixed matrix coefficients with $b$ vanish. Consequently

$$N_{A,B_+}^{B_+} = \langle e_{B_+}, N_A e_{B_+}\rangle = \frac{2-1}{2} = \frac{1}{2},$$

contradicting integral fusion multiplicities. □

**Scope and replay.** The proof uses no numerical modular completion. The standard-library checker in `evidence/unified_20261016/rank6/verlinde_obstruction/` binds the source tensor, the two certified local character tables, and the specified witness. It checks all 1536 equations and 1447 variable bounds of the linear necessary system, which pass, and then obtains the coefficient 1/2 exactly. Its normal and optimized receipts and an independent categorical audit are included. Other representation and twist refinements are not covered by this proposition; both rank-six categorification questions remain unresolved.

# 19 Even cyclic Clifford constraints and cross-rank consequences

Both rank-six questions remain open. We extend the stabilizer-rotation criterion to even orders, and identify precisely two consequences of the existing structural reductions.

## 19.1 Even cyclic characters

Use the hypotheses of Proposition 5.21: an order-two invertible $h$ admits the algebra $1+h$, a simple $Y$ is fixed on both sides, and its normalized left and right $h$-actions anticommute. On $V = \mathrm{Hom}(FZ, Y^{\otimes n})$, normalize generalized rotation to an operator $E_0$ with $E_0^n = 1$. The pair-insertion operators $A_j$, indexed cyclically, satisfy

$$A_j^2 = 1, \qquad A_j A_{j+1} = -A_{j+1} A_j, \qquad A_j A_k = A_k A_j \quad (k \neq j \pm 1, j), \qquad E_0 A_j E_0^{-1} = A_{j+1}.$$

The boundary relations follow from the same rotation naturality as in the odd-order proof; normalization of $E_0$ does not change conjugation.

**Proposition 19.1** (Even cyclic character criterion)**.** *Let $n = 2m \geq 4$ and $r = 2^{m-1}$. The eigenvalue multiplicity vector of $E_0$ is a nonnegative integral sum of cyclic shifts of the following three multisets of exponents modulo $n$:*

$$\left\{ -\frac{1}{2} \sum_{a \in S} a : S \subset \{1, 3, \ldots, 2m-1\},\ |S| \text{ even} \right\},$$
$$\left\{ \frac{1 - \sum_{a \in S} a}{2} : S \subset \{1, 3, \ldots, 2m-1\},\ |S| \text{ odd} \right\},$$
$$\bigcup_{S \subset \{1, \ldots, m-1\}} \left\{ -\sum_{a \in S} a,\ m - \sum_{a \in S} a \right\}.$$

*Their dimensions are respectively $r, r, 2r$. The divisions by two are ordinary integer divisions of even numerators.*

*Proof.* The adjacency pairing of the even cycle over $\mathbb{F}_2$ has two-dimensional radical, generated by its alternating vectors. The insertion algebra is consequently a direct sum of four copies of $M_r(\mathbb{C})$. Rotation fixes two blocks and exchanges two.

For the fixed blocks, represent the generators by adjacent Clifford bilinears with antiperiodic boundary. The rotation has eigenangles $(2a+1)\pi/n$, $0 \leq a < m$. Its two half-spin representations have weights indexed by even and odd subsets, respectively. Rescaling their implementers to have $n$th power one gives the first two multisets. Any other rescaling adds a constant modulo $n$ to every exponent.

For the exchanged blocks, use periodic boundary. The orthogonal transformation has angles $2\pi a/n$, $1 \leq a < m$, and eigenvalues $1, -1$. A pin implementer has opposite eigenvalue pairs; after the same normalization these give the third multiset. Its shifts have only $m$ distinct characters because shifting by $m$ preserves it.

Each fixed block has $n$ possible scalar implementers. The exchanged pair has $m$ irreducibles induced from its rotation-square stabilizer. Their squared dimensions sum to

$$2nr^2 + m(2r)^2 = n2^n,$$

the dimension of the insertion algebra extended by cyclic rotation. Thus these representations exhaust its irreducibles. Decomposing $V$ proves the assertion. □

For $R6_4_155$, the frozen generalized-rotation certificate supplies orders $4, 6, \ldots, 24$, both noninvertible objects, and all 52 central targets. Every one of the 1144 resulting multiplicity vectors has an exact nonnegative integral character decomposition. The independent integer replay also rejects a mutated witness. Hence the criterion gives no exclusion or associator construction for this ring.

### 19.2 Pointed automorphisms and the rank-eight alternative

The earlier central-lift theorem for $R6_4_155$ distinguishes the four pointed simples by their unique central twists $1, i, -1, -i$. A tensor autoequivalence preserves the unique pivotal structure and these lifts, and therefore fixes every pointed simple class. The vanishing and uniqueness assertions transport through the established Galois reduction, so this conclusion applies to arbitrary complex categorifications. An invertible normalizer in an extension consequently centralizes the Klein-four pointed subgroup. In particular it cannot give the specified $D_8$ pointed extension. This excludes a rank-nine Morita branch, not $R6_4_155$ itself.

The surviving rank-eight alternative in the established $R6_8_1660$ Morita list is exactly $R8_3_941$. The basis permutation from that alternative to the census basis is

$$(0, 3, 6, 7, 4, 5, 2, 1).$$

All 512 structure constants agree. The statement remains conditional: the nonpointed algebra producing this module category has not been forced to exist. The guaranteed pointed algebra on the rank-eight side does not supply its inverse.

Complete proofs, pinned inputs, exact character decompositions and the cross-rank identification are in the rank-six folders of `evidence/unresolved_retry_20261024/`. The numerical laptop search is an incomplete heuristic campaign and supplies neither a model nor an exclusion. No new rank-six classification decision is asserted.

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**AI disclosure.** GPT-6 Astra was used as a research assistant. This manuscript and its accompanying research data are AI-generated. They should be treated with caution: they have not been independently verified by a human mathematician or formalized in Lean.